

AY27 8670 Prerequisite Coursebook

Site: [Marine Corps eLearning Ecosystem - Moodle](#)
Course: 8670: Prerequisite AY27 Master
Book: AY27 8670 Prerequisite Coursebook

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Lesson 1: MCDP 7 – Learning

Lesson Hours: 1.00

1. Introduction

The purpose of this lesson is to describe the Marine Corps' learning philosophy and explain why learning is critically important to the profession of arms. The lesson begins by exposing the student to the nature of learning and how learning enables Marines to thrive and succeed in combat and competition. The lesson then delves into the characteristics of the culture, the individual learner, and the learning environment that support and encourage successful learning. The lesson concludes by describing how leaders foster these characteristics while continuing to learn themselves.

2. Educational Objectives

- a. Identify what learning means to Marines in the context of warfighting.
- b. Identify the difference between education and training and why this matters.
- c. Identify the principles of learning.
- d. Identify the responsibilities of individual Marines, instructors, and leaders regarding learning.
- e. Identify the characteristics of a successful Marine learner.
- f. Identify the characteristics of a positive learning environment.

3. Key Terms

- Learning
- Education
- Training
- Intellectual Edge
- Principles of Learning
- Growth Mindset
- Intrinsic Motivation
- Surfaces and Gaps
- Learner-Centric Model
- Self-Directed Learning

4. Preparation

All quizzes, discussion forum posts, and assignments are executed in the Moodle course site.

- a. Lesson Reading: [MCDP 7 – Learning](#) (20 Feb 2020) (81 Pages).
- b. Other Requirements: None.
- c. Supplemental References (optional): [MCDP 7 Audio File](#) (audio file will open in marines.mil).

Lesson 2: Fundamentals of National Defense

Lesson Hours: 2.0

1. Introduction

The U.S. Marine Corps is a key instrument in the execution of U.S. national security strategy. As an expeditionary force-in-readiness, the Marine Corps has consistently been called upon to implement elements of our national security strategy and its supporting national defense strategy and national military strategy. While the Marine Corps is not a strategy-making organization in the sense of designing a national military strategy or even drafting strategies to fight specific wars, the effective execution of strategy requires an understanding of both its intent and context. In order to perform our responsibilities to the Nation, Marines must possess an understanding of the concepts that compose national defense.

2. Educational Objectives

- a. Identify the three categories of strategic interests.
- b. Identify the purpose of national strategy and its relationship to military strategy.
- c. Identify the purpose of the national military strategy.
- d. Identify the purpose of the Chairman's Risk Assessment.
- e. Identify the instruments of national power.
- f. Identify four strategic uses of military force.
- g. Identify the three broad categories associated with competition continuum.
- h. Identify facets of the nature, character, and theory of war.
- i. Identify the levels of war and their interrelationships.
- j. Identify the responsibilities of the President of the United States and the Secretary of Defense.
- k. Identify the purpose of geographic combatant commanders.
- l. Identify the responsibilities of the Chairman, Joint Chiefs of Staff.
- m. Identify the purpose of unified action and its relationship to unity of effort.
- n. Identify the purpose of interorganizational cooperation.

3. Key Terms

- Nature of war
- Changing character of war
- Strategy
- Instruments of national power
- Strategic uses of military force
- Competition continuum
- Levels of war
- Unified commands
- Unified action
- Interorganizational cooperation

4. Preparation

All quizzes, discussion forum posts, and assignments are executed in the Moodle course site.

- a. Lesson Reading:
 - [*Fundamentals of National Defense*](#) (31 pages).
 - [*National Security Strategy*](#) (4 Dec 2025) (33 pages).
 - [*National Military Strategy \[Unclassified Summary\]*](#) (8 pages).
- b. Other Requirements: None.

c. Supplemental References (optional):

- [Congressional Research Service, *The National Security Council: Background and Issues for Congress* \(19 Oct 2022\).](#)
- [*National Strategy for Homeland Security* \(2007\).](#)
- [*2022 Homeland Security Priorities* \(2022\).](#)
- [JP 1 w/Change 1, *Doctrine for the Armed Forces of the United States* \(12 Jul 2017\).](#)
- [JP 3-08, *Interorganizational Cooperation* \(12 Oct 2016\).](#)
- [JDN 1-19, *Competition Continuum* \(3 Jun 2019\).](#)
- [CJCSM 3105.01B, *Joint Risk Analysis Methodology* \(22 Dec 2023\).](#)
- [MCDP 1 w/Change 1, *Warfighting* \(4 Apr 2018\).](#)

Lesson 2 Reading

FUNDAMENTALS OF NATIONAL DEFENSE

References:

- a) *National Security Strategy* (4 Dec 2025)
- b) Congressional Research Service, *The National Security Council: Background and Issues for Congress* (19 Oct 2022)
- c) *National Military Strategy* [Unclassified Summary] (2022)
- d) JP 1 w/Change 1, *Doctrine for the Armed Forces of the United States* (12 Jul 2017)
- e) JP 3-08, *Interorganizational Cooperation* (12 Oct 2016)
- f) JDN 1-19, *Competition Continuum* (3 Jun 2019)
- g) CJCSM 3105.01B, *Joint Risk Analysis Methodology* (22 Dec 2022)
- h) MCDP 1 w/Change 1, *Warfighting* (4 Apr 2018)

1. Introduction

The United States (U.S.) Armed Forces fulfill unique and crucial roles, defending the U.S. against all adversaries while serving the Nation as a bulwark and the guarantor of its security and independence. The U.S. Armed Forces function within the American system of civil-military relations and serve under the civilian control of the President, the Commander-in-Chief. Furthermore, U.S. Armed Forces embody the highest values and standards of American society and the profession of arms. In order for the U.S. Armed Forces to execute their responsibilities to the Nation, they must possess the strategic skills and understanding necessary to participate effectively in the strategic environment of the 21st century. This includes an understanding of how the U.S. Armed Forces integrate with the other instruments of National power as well as a fundamental understanding of the application of force—frequently referred to as “war”—to achieve the Nation’s goals.

2. National Strategic Direction

Prior to discussing how U.S. Armed Forces support the achievement of U.S. National objectives, it should be understood what these goals are, how they are developed, and how they are accomplished through strategy.

a. National and Strategic Interests

At the highest levels, a nation expresses its wants, needs, and concerns as national interests. National interests include **domestic as well as international** issues. National interests normally involve four main areas: survival and security, political and territorial integrity, economic stability and well-being, and stability. Interests are central to a discussion of strategy because interests signal a nation’s desires and intentions to other nations.

Strategic interests are defined as the perceived needs and aspirations of a state or non-state actor in relation to its **international** environment (a subset of national interests). Strategic interests are frequently described using three categories:

- **Peripheral**— A strategic interest that, if threatened, poses a moderate risk to a key partner or region, with low hazard to the actor’s population, territory, or civil society.
- **Important**— A strategic interest that, if threatened, poses a catastrophic or significant hazard to the actor’s population, territory, or civil society.
- **Vital**— A strategic interest that, if threatened, poses an existential hazard to the actor’s population, territory, or civil society.

Vital interests are distinguished from other interests by the fact that nations are usually unwilling to compromise on them and are often prepared to resort to conflict in support of them.

b. National Objectives

National and strategic interests are often vague or consist of highly generalized abstractions. While these interests underpin national strategy, the specifics of the strategy must focus on more concrete *ends*. The specific goals and aims of national strategy are often referred to as objectives. Objectives are the *ends* a nation must achieve to promote, protect, or attain its interests. Objectives tend to be more tangible than interests because they normally describe specific activities or conditions that must be attained. Objectives provide the departure point for national strategy in that they describe what a state is actually trying to do.

c. Strategy Theory

Once national objectives are determined, strategies must be developed to make progress toward achieving those goals. Strategy, broadly defined, is the process of interrelating *ends* (i.e., objectives) and *means* (i.e., resources). When this process is applied to a particular set of *ends* and *means*, the product, being the strategy, is a specific *way* (i.e., a method) of using specified means to achieve a distinct *end*. The term "strategy" is thus both a process and a product.

In its simplest expression, strategy determines what needs to be accomplished, the methods to accomplish it, and the resources required by those methods. Mathematically, we might articulate this as:

$$\text{Strategy} = \text{ends} + \text{ways} + \text{means}$$

Or expressed in a different way, a comprehensive and effective strategy answers four basic questions:

- What are the desired ends?
- What are the ways to get there?
- What means or resources are available?
- What are the risks associated with the strategy?

Arthur F. Lykke Jr. describes a need to balance *ends*, *ways*, and *means*, which he likens to the three legs of a stool (the stool itself representing the strategy).

- A strategy is balanced and entails little risk if the selected way (method) is capable and has sufficient *means* (resources) to obtain the desired *end* (objective).
- If either the *ways* or *means* legs are too short (due to inadequacies), or the *end* leg is too long (the goals are unrealistic), the strategy is out of balance, and the risk is high.
- To balance a strategy, the legs must be adjusted. Desired *ends* can be scaled back to fit within the available *means* or *means* can be increased to fully support the selected *way(s)*. When the *means* are inadequate, planners must consider alternative *ways*.

Because all of these "balancing" choices are strategic decisions, this balancing act is the heart of strategic art.¹

Strategy formulation must consider strategic environment factors (e.g., geography, character, and relationship of various political entities, their interests, the emphasis they place on those interests, available resources, etc.). These factors present themselves differently in each strategic interaction and exert considerable influence on a particular strategic situation. Additionally, these factors may change during execution, necessitating revision of the strategy.

d. Instruments of National Power

One way a nation's strategic planners identify potential *ways* and *means* is to examine what is referred to as the "instruments of national power." This is a conceptual grouping of national activities and resources into four categories: diplomatic, information, military, and economic (DIME).

(1) Diplomatic

Diplomacy is the principal instrument of national power for using diplomatic and international relations activities to engage with other states and foreign groups to advance U.S. values, interests, and objectives, and to elicit foreign support for U.S. military operations. Diplomacy is the principal means of organizing coalitions and alliances, which may include states and non-state entities as partners, allies, surrogates, and/or proxies. Diplomacy may include support to sympathetic groups opposing a hostile regime or occupying power, outreach to dispersed populations, the use of international organizations to achieve strategic objectives, or the arbitration or mediation of political disputes between international actors.

(2) Information

Information includes and aggregates numerous social, cultural, cognitive, technical, and physical attributes that act upon and impact knowledge, understanding, beliefs, world views, and, ultimately, actions of an individual, group, system, community, or organization. Information is an increasingly significant instrument of national power and a strategic resource critical to national security. Previously considered in the context of traditional nation-states, the concept of information as an instrument of national power now extends to non-state actors to include private citizens and media groups, as well as terrorists and transnational organized criminal organizations who are using information to further their causes and undermine those of other groups to include nations and multinational organizations.

Information will be discussed in detail in the 8672 *Operations in the Information Environment* sub-course.

(3) Military

The military instrument is the use of, or threat to use, force to achieve national objectives. Military power is the sum of a nation's weapons and equipment, trained manpower, organizations, doctrines, industrial base, and sustainment capacity. The military instrument can be employed in a variety of ways that are short of combat, such as training allies, establishing presence, or acting as a show-of-force. However, the main use of military power is in conflict.

While the military instrument is often the main effort during war, the nature and objectives of the particular conflict must be examined to determine the appropriate relationship between the use of military force and the application of the other instruments of national power.

Military professionals naturally concentrate on the military means of strategy, but they should also be conscious of the other means that can be exploited and must be defended in the larger political struggle. Most importantly, they must understand that military force is an inappropriate tool for the solution of most political difficulties. Force is at best a necessary means for clearing obstacles to more peaceful solutions. This appreciation of the role of force is a vital component of military professionalism, for military leaders have a responsibility to ensure that political leaders understand both the capabilities and the limitations of the military instrument.

(4) Economic

The economic instrument uses the application of material resources to achieve national objectives. Nations employ economic means to protect their own industries and markets, to improve the quality of life of their people, to stabilize the economy and government of friends and allies, and to deter destabilizing and hostile actions by other nations. Specific economic means include regulation of trade practices, loans and loan guarantees, monetary and investment policies, foreign aid, subsidies, and technology transfers. In a free society (i.e., not centrally directed), the economic instrument also includes the leveraging of resources managed by private citizens.

As with the diplomatic instrument, the economic instrument generally has primacy over the military instrument during peace and is often used before military force during a crisis; changes in trade or monetary policy, economic sanctions, or some type of embargo are frequently the first steps taken in an effort to influence an adversary's behavior. However, economic considerations continue to be at the forefront of any conflict, and the use of economic measures to support the friendly war effort and to undermine the adversary's ability to resist continue throughout the course of a war.

e. Integrating the Instruments of National Power

Because the instruments of national power contain the *ways* and *means* to achieve national objectives, national strategy is the art and science of developing and using the diplomatic, economic, and informational powers of a nation together with its military, during peace and war, to secure national objectives. National strategy connotes a global perspective, but it requires coordination of all the elements of national power at the regional or theater level as well.

In the U.S., the Executive Branch, under the President's direction, is responsible for integrating the instruments of national power to advance and defend U.S. values, interests, and objectives. The President does this with the advice and assistance of the National Security Council (NSC) through the issuance of policy and commitment statements and through the interagency coordination process.

(1) Policy and Commitments

Specific policies and commitments describe the ways in which *ends* are to be achieved. Policy is a method(s) or pattern(s) of actions designed to attain specific objectives (i.e., policy is the *ways* by which strategy reaches its objectives). Policy can represent a broad course of action or intent. Commitments are expressions of a nation's intention to use specific instruments of national power to achieve objectives. Whereas policy might express general intent, a course of action, or restraints on action, commitments pledge nations to take specific actions at specific times and places.

Note: Policy should not be confused with politics. Politics is the art or science concerned with guiding or influencing governmental policy (i.e., the interaction between people to decide what objectives should be and how they should be achieved), while policy refers to the general goals set by a government or large governing body and the course of action adopted to achieve those goals (i.e., the actual objectives and methods). Keep in mind that policy is produced through a political process and thus is almost always the result of compromises among various policymakers and may be communicated with nuanced differences to separate audiences.

The articulation of national interests, objectives, policies, and commitments linked to use of the instruments of national power is sometimes referred to as *grand strategy*, *grand national strategy*, or, in the U.S., the *National Security Strategy (NSS)*.

(2) National Security Strategy

The NSS is a congressionally mandated document (originating in the Goldwater-Nichols Department of Defense Reorganization Act of 1986), prepared and published by the Executive Branch (signed by the President), that encompasses national defense, foreign relations, and economic relations and assistance; and aims, among other objectives, at providing a favorable foreign relations position and a defense posture capable of defeating hostile action.

Note: On 4 Dec 2025, the White House released the latest National Security Strategy. This document is required reading for this lesson.

(3) Subordinate Strategies and Related Documents

Grand strategies or national security strategies are implemented by subordinate strategies: diplomatic strategies, economic strategies, military strategies, and so forth. This lesson will focus primarily on military strategies. Military strategy applies the military instrument of national power towards the accomplishment of the political objectives of the overall national strategy. The departure point for military strategy, therefore, is the objectives of the national strategy. From there, military strategy must identify a military goal or objective that will lead to the accomplishment of the political objective. The military objective then provides the basis for the identification of specific ways to accomplish that objective. The selection of one of these courses of action and its further development results in a strategic concept that embodies the key components of the chosen military strategy. The military strategy is not developed in isolation from the other instruments of national power. The military objectives and strategy must also be compatible, if not tangibly linked, with the diplomatic, economic, and informational objectives and strategies.

(a) The National Strategy for Homeland Security (NSHS). The NSHS, signed by the President, provides National direction to secure the homeland through a comprehensive framework for organizing the efforts of federal, state, local, and private organizations whose primary functions are often unrelated to national security.

(b) The Unified Command Plan (UCP). The UCP is a classified document that assigns missions; planning, training, and operational responsibilities; force structure; and geographic areas of responsibilities to the various combatant commanders. It is prepared by the Chairman of the Joint Chiefs of Staff (CJCS) and reviewed and updated every two years by the President.

(c) National Defense Strategy (NDS). The NDS is the U.S. Department of Defense's (DoD) capstone strategic guidance. It translates and refines the NSS into broad military guidance for military planning, military strategy, force posturing, force constructs, force modernization, etc. It is produced by the U.S. Office of the Secretary of Defense (OSD) and is signed by the Secretary of Defense (SecDef). The NDS is expected to be produced every four years and is generally publicly available.

Note: The 2022 NDS was published on 27 October 2022 and is covered in detail in the *National Defense Strategy* lesson. On 2 May 2025, the SecDef directed the development of the 2025 NDS. At the time this lesson was finalized, the 2025 NDS had not been released.

Note: On 5 Sep 2025, President Donald J. Trump signed Executive Order 14347 to restore the historic name "Department of War" as a secondary title for the Department of Defense. The order authorized the Secretary of Defense, the Department of Defense, and subordinate officials to use secondary titles such as "Secretary of War" and "Department of War" in official correspondence, public communications, ceremonial contexts, and non-statutory documents within the executive branch, and instructs the Secretary of War to recommend actions, to include legislative and executive actions, required to permanently rename the U.S. Department of Defense to the U.S. Department of War. To avoid awkward language, the EWSDEP courseware will continue to use the statutory titles used in current references until such legislative and executive actions are completed.

(d) The Guidance for the Employment of the Force (GEF). The GEF includes classified written guidance from the SecDef to the CJCS for the preparation and review of contingency plans for specific missions. This guidance includes the relative priority of the plans, specific force levels, and supporting resource levels projected to be available for the period of time for which such plans are to be effective.

(e) The Defense Planning Guidance (DPG). The DPG is a classified document, signed by the SecDef, that establishes DoD's force development planning and resource priorities to prevail in the Nation's current operations and to develop a balanced joint force to meet future contingencies. The purpose of the DPG is to provide guidance in the form of goals, priorities, and objectives, including fiscal constraints, for the development of each Military Department's *Program Objective Memorandum* and *Budget Estimate Submissions*.

(f) The National Military Strategy (NMS). The NMS is an overarching military strategic framework that implements the substantial body of policy and strategy direction provided in the NSS, NDS, DPG, and other documents. The NMS is a classified document. It is written by the Joint Chiefs of Staff (JCS) with input from the Service Chiefs and the combatant commanders. It is signed by the CJCS.

The latest NMS was issued in 2022. An unclassified summary of the NMS is required reading for this lesson.

(g) CJCS Risk Assessment. Every year, the CJCS must provide a risk assessment to the SecDef and Congress about the strategic risks to national interests and military risks in executing the NMS (in accordance with Title 10, U.S.C.). The CJCS continually considers risk when fulfilling the primary roles (assess, advise, direct, execute) within the Joint Strategic Planning System (JSPS). Specifically, the CRA provides a risk baseline that informs their assessment and advisory actions throughout the year. This risk assessment includes the following:

- The strategic risks to United States interests and the military risks in executing the missions of the NMS.
- The levels of risk distinguishing between the concepts of probability and consequences, including an identification of what constitutes “significant” risk in the judgement of the CJCS.
- An assessment of the risks in the NMS by category and level and the ways in which risk might manifest itself, including how risk is projected to increase, decrease, or remain stable over time; and for each category of risk, an assessment of the extent to which current or future risk increases, decreases, or is stable as a result of budgetary priorities, tradeoffs, or fiscal constraints or limitations as currently estimated and applied in the most current future.
- An assessment of the risk associated with the assumptions or plans of the NMS about the contributions of support of other departments and agencies of the U.S. Government, alliances, allies, friendly nations, and contractors.
- An assessment of the critical deficiencies and strengths in force capabilities (including manpower, logistics, intelligence, and mobility support) identified during the preparation and review of the contingency plans of each unified combatant command, and an assessment of the effect of such deficiencies and strengths for the NMS.

(4) Interagency and Interorganizational Coordination

Achieving national strategic objectives requires the effective and efficient use of all the instruments of national power in a coordinated and synchronized fashion. This is referred to as *unified effort*. At the President’s direction, military power is integrated with the other instruments of national power to advance and defend U.S. values, interests, and objectives. To accomplish this integration, DoD agencies interact with non-DoD agencies, gaining a mutual understanding of the capabilities, limitations, and consequences of military and civilian actions. They also identify the ways in which military and civilian capabilities best complement each other. The NSC and the Homeland Security Council (HSC) play key roles in the integration of all instruments of national power by facilitating mutual understanding and cooperation. Further, the unified actions of military and civilian organizations in sharing information, cooperating, and striving together to accomplish a common goal is the essence of interorganizational coordination that makes *unity of effort* possible.

Meeting the challenges of current and future operations requires the concerted effort of all instruments of U.S. national power, plus foreign governmental agencies and military forces and civilian organizations. Within the Executive Branch alone, achieving unity of effort is often complicated by organizational “stove-piping,” crisis-driven planning, and divergent organizational processes and cultures. These differences have certain benefits but are not well-suited for addressing the range of challenges that cut across available organizational expertise. Problems arise when each agency interprets NSC and HSC policy guidance differently, sets different priorities for execution, and does not act in concert. These issues are exacerbated by the competing interests and practices of participating foreign governments and military forces, intergovernmental organizations, non-government organizations, and private sector entities.

Consequently, there is a need to conduct integrated planning to effectively employ the appropriate instruments of national power. A comprehensive approach seeks to stimulate a cooperative culture within a collaborative environment while facilitating a shared understanding of the situation. In its simplest form, a comprehensive approach should invigorate existing processes and strengthen interorganizational relationships. This approach should forward the respective goals of all parties, ensuring stakeholders do not negate or contradict the efforts of others.

In joint operations, *unity of effort* is ensured by establishing *unity of command*. Unity of command is based on the designation of a single commander with the authority to direct and coordinate the efforts of all assigned forces in pursuit of a common objective. Commanders exercise military command and control to ensure that military operations are planned and conducted in accordance with the guidance and direction received from the President and SecDef in coordination with other authorities (i.e., alliance or multinational leadership). In operations involving interorganizational partners and other stakeholders, where the commander may not control all elements, they seek cooperation and build consensus to achieve unity of effort. Unity of command can be difficult to achieve when working with multiple stakeholders. To compensate, commanders concentrate on working with the participating national forces to obtain unity of effort. Consensus building is the key element to unity of effort.

While unity of command and the exercise of command and control apply strictly to military forces and operations, *unified action* among all interorganizational participants is necessary to achieve *unity of effort* in military operations involving engaged civilian organizations and foreign military forces or military participation in civilian-led operations. *Unified action* is the DoD doctrinal term that represents a comprehensive approach. It begins with national strategic direction from the President and includes a wide scope of actions, including interorganizational coordination techniques, information sharing, collaborative planning, and the synchronization of military operations with the activities of all the civilian stakeholders.

Achieving unity of effort requires the application of a comprehensive approach that includes coordination, consensus building, cooperation, collaboration, compromise, consultation, and de-confliction among all the stakeholders toward an objective. An inclusive approach of working closely with stakeholders is often more appropriate than a military command and control focused approach. Taking an authoritative, military approach may be counterproductive to effective interorganizational relationships, impede unified action, and compromise mission accomplishment. Gaining unity of effort is never settled and permanent; it takes constant effort to sustain interorganizational relationships.

Case Study:

Employment of the Elements of National Power in the Cuban Missile Crisis of 1962

The following historical vignette provides an example of how the instruments of power are used in a coordinated fashion to achieve the strategic objectives.

Background

Fidel Castro's defeat of Fulgencio Batista and his appointment to as Premier and President of Cuba in 1959 gave rise to new international tensions in the Caribbean. The Eisenhower administration established diplomatic relations with Castro during the revolution because of the discord and corruption within the Batista regime. The U.S. extended an arm of diplomacy to Castro due to his repeated statements during a 1959 visit to the U.S. that his revolutionary movement was not communist in its beliefs. What was unknown at the time was that Raul Castro, Fidel's younger brother, was secretly meeting with members of the Czechoslovakian and Polish governments in an effort to procure weapons for Cuban military forces. These secret meetings opened a dialogue between the Warsaw Pact communists and members of the Cuban Communist Party. The Kremlin leadership believed they had the obligation to support revolutionary movements around the globe. Once notified about the request for weapons and advisory support from Castro's Cuba, the Soviets decided the risk of assisting a new communist satellite in Latin America was worth challenging the U.S.

The U.S., now being led by President John F. Kennedy, severed diplomatic and consular relations with Cuba in January 1961. Later that year, at its conference in Uruguay, the Organization of the American States (OAS) voted to exclude Cuba "from participation in the inter-American system." The measure was approved, and an additional resolution was also adopted prohibiting OAS members from selling arms to Cuba. A U.S. Senate resolution regarding Cuba authorized the use of force, if necessary, to curb Cuban aggression and subversion in the western hemisphere. The resolution stated that the U.S. was determined "to prevent the creation or use of an externally supported offensive military capability endangering the security of the U.S." and to "support the aspirations of the Cuban people for a return to self-determination." In the House of Representatives, a foreign aid appropriations bill had approved amendments designed to terminate aid to any country that used merchant ships to transport arms or goods of any type to Cuba. These policies existed before the confirmation of the missile sites and worsened with the placement of the Soviet missiles.

After examining American threats against Cuba and his own country's nuclear imbalance vis-a'-vis the United States in the spring of 1962, Soviet Premier Nikita Khrushchev decided, with the unanimous support of the Presidium of the Central Committee, to ship medium- and intermediate-range ballistic missiles to Cuba. Khrushchev resented the stationing of American ballistic missiles targeted against the USSR on North Atlantic Treaty Organization (NATO) bases, and he wished to place the U.S. under a similar threat. Stationing Soviet missiles in Cuba would help to compensate for the imbalance between Soviet and American ICBMs, gain leverage for Soviet policy goals in Europe, and deter the U.S. from invading Cuba. After receiving permission from President Fidel Castro to deploy the weapons, surface-to-air missiles, and medium-range ballistic missile (MRBM) support equipment, left the Soviet Union for Cuba in July, with the first R-12 MRBMs arriving on September 9, 1962. American reconnaissance flights (see Figure 1) were unable to photograph their installation for ten days due to clouds obscuring Cuba; thus, the Soviets gained the opportunity to slip the missiles onto the island undetected. The U-2 overflights were part of a broad continuing effort to monitor and contain Cuba, so Soviet efforts at maintaining secrecy were unlikely to remain successful.

Missile Discovery

In the early morning hours of 14 October 1962, a recently converted CIA Lockheed U-2 aircraft completed its preparation for take-off from Edwards Air Force Base in California. The airplane, piloted by Air Force Major R. Heyser, flew for nearly five hours to reach its designated target area off the southern tip of Florida. The purpose of the flight was to conduct a high-altitude photo-reconnaissance mission to obtain aerial photographs of San Cristobal, Cuba. Major Heyser piloted his specially designed plane over the western portion of the island at the prescribed altitude of 72,500 feet above sea level. He kept his aircraft steady and level as was required to produce clear and decipherable photographs. In a span of twelve minutes, the camera mounted on the underside of the U-2 took 3,000 frames of data to be analyzed by the CIA's National Photographic Interpretation Center. Upon Heyser's landing at McCoy Air Force Base near Orlando, Florida, two Air Force Generals retrieved the precious cargo and transported it to Washington, D.C. for analysis. Less than forty-eight hours later, the reaction to the analysts' interpretation initiated a foreign policy showdown between the two superpowers possessing nuclear arsenals capable of permanently altering civilization.

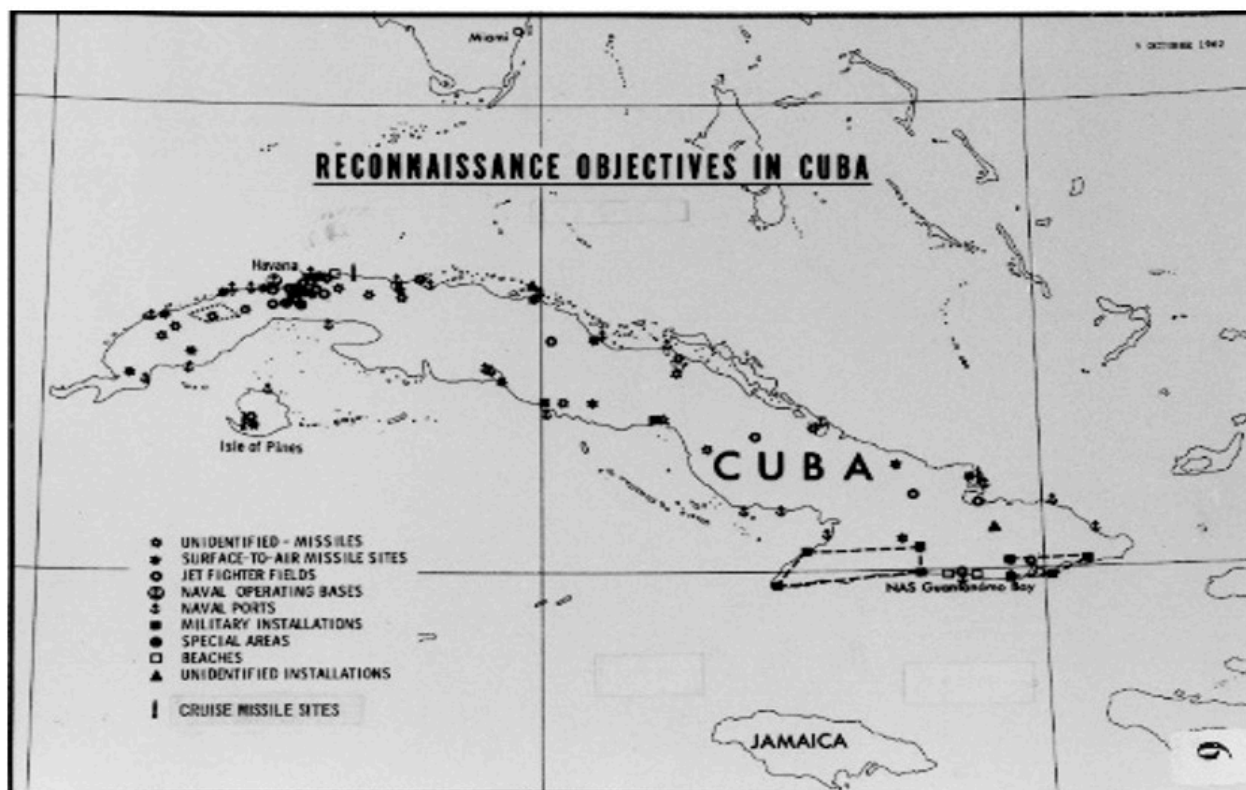


Figure 1: Reconnaissance Objectives.²

Military Instrument

On October 16, Kennedy convened the decision-making body called the Executive Committee of the National Security Council (EXCOM) to analyze the problems the missiles presented and options for removing them. He basically replaced the full NSC meetings with EXCOM. All of the EXCOM members were initially convinced that the only options available involved offensive military action. Their convictions were based upon a firm belief that the Soviet Union would only respect and respond to the use of force and that they would not initiate WWII with their reactions. Kennedy was convinced that if he would respond militarily in Cuba, the Soviets would use that action as justification to invade Berlin. The EXCOM recommended a variety of options ranging from surgical air strikes on the missile sites to a comprehensive series of attacks and invasions throughout the island. When the initial meeting adjourned, the group agreed to move forward with some form of military retaliation. Secretary of Defense Robert McNamara and the Joint Chiefs began the process of troop alerts and issuance of initial guidance to the military to prepare for future actions.

Later that same day, a second EXCOM meeting was held, and it was decided that the missiles would not be operational for a few more days. This provided the committee with additional time before having to act. Further deliberation quieted the assertive members known as the "Hawks" and facilitated discussion of additional alternatives. The "Hawks" were those members of the EXCOM who desired a swift military response and held little patience for other means of determining a resolution. Supplementary options included a naval blockade of the island, troop movements to increase the forces stationed in the southeast, reinforcement of the garrison at Guantanamo, and increased covert support of the U.S. backed guerrillas.

Over the course of the next few meetings, Secretary McNamara and the Joint Chiefs presented Kennedy with an extensive discussion on decision points related to potential actions and reactions involving military power. The discussions included a risk assessment of the potential outcomes, risks, and gains of various military responses. The group agreed that military action against the missile sites would only be effective against non-operational sites. Once the missile sites were operational, the risk that aggressive action on the part of the U.S. would be met by Soviet retaliatory launches increased significantly. The detailed requirements that arose during the planning for limited surgical air strikes developed into a massive list of military assets that Kennedy was not ready to commit to the crisis. Strikes against the missile sites no longer seemed an appropriate reaction but instead were seen as carrying too great a risk of escalation.

After continued deliberations, it was decided that a quarantine (blockade) of Cuba executed by military forces would provide the proper level of response as well as additional time to monitor the Soviet actions. The blockade concept continued to gain support among the members of the EXCOM. The blockade was considered a strong enough action to promote a reaction of war from the Soviet Union. Although Kennedy had determined that an aggressive military response would not benefit the situation, he did acknowledge the significance it had to the crisis. He repeatedly had the EXCOM plan for a military response as the crisis developed, with the hope of not having to employ it. He continued to

outwardly portray that a military response remained a feasible and readily available option should the situation deteriorate. His decision to utilize the military to reinforce the diplomatic efforts displayed a creative understanding of the application of the instruments of national power to the crisis.

American naval forces also generated some subtle but critical side effects beyond the mere function of a blockade. The preponderance of naval forces around Cuba meant that there were enough ships and aircraft to enforce the quarantine, which meant that Soviet ships carrying any additional missiles, warheads, and support equipment were not going to make it to Cuba. The dominance of the U.S. Navy in this instance thus generated certainty within this aspect of the confrontation, which meant that Kennedy and Khrushchev had a couple of fewer variables to worry about. Composed of two attack aircraft carriers, an antisubmarine aircraft carrier, three cruisers, and thirty-two destroyers, the Navy task force was strong but not over the top. The three cruisers, eighteen of the destroyers, and a few additional ships comprised the actual quarantine barrier, which lay in an arc from east to north, 500 miles from Cuba, although ships could intercept and follow Soviet ships farther out.

Diplomatic Instrument

The Kennedy administration was conscious of the need to establish legitimacy with the world community and especially its allies in dealing with the missile issue. They realized that without the political support and backing of world opinion and their allies, their planned actions would be viewed as unjustifiable. During the morning EXCOM meeting on 18 October, Defense Secretary McNamara presented three options to confront the Soviet Union. The first of those was a diplomatic option that consisted of an open discourse between the primary decision-makers involved. His other options supported the views and thoughts of the aggressive military leaders. Due to the passions resident within the EXCOM, the group spent many more hours in debate over the military options and placed the diplomatic concerns aside. Kennedy realized that the engagement of formal diplomacy remained one of the responsibilities and duties as the chief executive of the U.S. Since a majority of the EXCOM was focused primarily on military action, he seemed determined to actively seek diplomatic engagement himself.

Kennedy decided to employ a balanced scheme of diplomacy to bring the crisis to a peaceful close. In addition to the different departments' efforts, he initiated diplomacy at a leader-to-leader level by sending Khrushchev personal letters. These letters opened a dialogue between the leaders in which both expressed their sincere hope to avoid catastrophic consequences if the situation in Cuba was not resolved diplomatically. Kennedy continued to stress the need for control and prudence through his numerous communiques with the Soviet leader. Kennedy used this means of diplomacy in an effort to appeal to Khrushchev's emotions and common fears of the enormity of the situation.

Kennedy's use of third-party diplomacy proved successful once both the Soviets and U.S. determined the proper candidates to conduct this style of negotiations. The U.S. employed members of the U.N., significant members of the press, and noteworthy embassy personnel to engage in deliberations during the crisis. These methods of deliberation enabled the discussion of multiple options that could not be debated openly due to public opinion. Kennedy was convinced that peaceful means could be established that would bring an end to the crisis.

Informational Instrument

Premier Khrushchev told a gathering at the Cuban embassy in Moscow: "Alarming news is coming from Cuba at present, news that the most aggressive American monopolists are preparing a direct attack on Cuba." Soviet Ambassador Dobrynin assured U.S. Ambassador to the U.N., Adlai Stevenson, that "only defensive weapons are being supplied" to Cuba. Robert Kennedy met with Ambassador Dobrynin and was informed that Premier Khrushchev assured President Kennedy that there would be no surface-to-surface missiles or offensive weapons placed in Cuba. Robert Kennedy related the conversation to the president and suggested issuing a statement making it clear that the U.S. would not tolerate the introduction of offensive weapons into Cuba.

President Kennedy released a statement which revealed that surface to air missiles and substantially more military personnel than previously estimated had been detected in Cuba. Kennedy did concede, however, that: "There is no evidence of any organized combat force in Cuba from any Soviet Bloc country; of military bases provided to Russia; of a violation of the 1934 treaty relating to Guantanamo; of the presence of offensive ground-to-ground missiles; or other significant offensive capability...Were it otherwise, the gravest issues would arise."

President Kennedy, at a news conference, stated that Fidel Castro, "in a frantic effort to bolster his regime," was attempting to "arouse the Cuban people by charges of an imminent American invasion." The president reiterated that new movements of Soviet technical and military personnel to Cuba did not constitute a serious threat and that "unilateral military intervention on the part of the United States cannot currently be either required or justified." Nevertheless, he again warned that if Cuba "should ever attempt to export its aggressive purposes by force...or become an offensive military base of significant capacity for the Soviet Union, then this country will do whatever must be done to protect its own security and that of its allies."

President Kennedy was able to achieve a foreign policy victory during the Cuban Missile Crisis due to his keen understanding of the principle of tailored instruments of national power and the discernment he demonstrated while applying them throughout the deliberations. He properly identified and leveraged the influential instruments against both the Soviet Union and Cuba. He furthered the national interests, challenged communism within the Western Hemisphere, retained prominence in Europe, and peacefully resolved the Cuban Missile Crisis.

Kennedy had at his disposal the sum of the instruments of national power of arguably the world's most powerful nation in 1962. During the deliberations, it became apparent that the instruments of the powerful military and strong economy that the U.S. possessed would not be decisive in the context of this event. Although the influence of these two instruments remained significant, they were considered by Kennedy as the lesser desired instruments for this occasion. He determined that the instrument of diplomacy was the most influential in the situation, even at its tensest moments. As the crisis developed, the military and informational instruments became useful as a means of reinforcement to the diplomatic efforts he had employed.

Economic Instrument

MCDP 1-1, Strategy discusses the instruments of national power in Chapter 2, "Strategy: Ends and Means." That chapter is introduced with a quote from President John F. Kennedy's address to the graduates of the U.S. Naval Academy Class of 1961:

"You [military professionals] must know something about strategy and tactics and logistics, but also economics and politics and diplomacy and history. You must know everything you can about military power, and you must also understand the limits of military power.

*You must understand that few of the problems of our time have been solved by military power alone."*³

While economics is the implicit foundation of all other instruments of national power, economic activity generates the revenue (either through taxes, tariffs, or creating the necessary credit rating to facilitate borrowing) for the state to fund the military, diplomatic, and informational instruments, the economic instrument of national power is treated as separate and distinct from the national economy.

Viewing the Cuban Missile Crisis through the lens of the instruments of national power requires a look at the first strategic communication signals of the Kennedy administration. After the often-quoted preamble of President Kennedy's January 20, 1961, inaugural address, the president conveyed the following pledge to the Western Hemisphere:

*"To our sister republics south of our border, we offer a special pledge—to convert our good words into good deeds—in a new alliance for progress—to assist free men and free governments in casting off the chains of poverty. But this peaceful revolution of hope cannot become the prey of hostile powers. Let all our neighbors know that we shall join with them to oppose aggression or subversion anywhere in the Americas. And let every other power know that this Hemisphere intends to remain the master of its own house."*⁴

*As mentioned in the Background section, the Kennedy administration worked through the OAS. Established fourteen years before the crisis, the OAS is the world's oldest regional organization and originally consisted of 21 members (Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, Cuba, Dominican Republic, Ecuador, El Salvador, Guatemala, Haiti, Honduras, Mexico, Nicaragua, Panama, Paraguay, Peru, United States, Uruguay, and Venezuela). Before Cuba's suspension, OAS representatives convened in August 1961 at Punta del Este (Uruguay). The U.S. Secretary of the Treasury, C. Douglas Dillon, led the U.S. delegation that negotiated the Alliance for Progress. As described in the Oxford Research Encyclopedia, the Alliance for Progress pledged \$20 billion over the remainder of the decade in "grants, loans and direct private investments" in exchange for the recipients' pledge "to work for equality and social justice by promoting agrarian reform and progressive income taxes."*⁵

*Cuba's delegate at Punta del Este, Ernesto "Che" Guevara, encouraged the nineteen other prospective recipients to reject the terms of the Alliance for Progress as "the United States comes with the sack of gold ... in one hand, and the barrier to isolate Cuba in the other."*⁶

*The nineteen recipients of Alliance for Progress funds were unanimous in their support for the Kennedy administration's stance in the Cuban Missile Crisis. On the eighth day of the crisis (October 23, 1962), within six hours of the U.S. proposal of a "resolution authorizing OAS members to use force, individually or collectively, to impose a quarantine on Cuba," all member states voted to adopt the resolution (except Uruguay, which initially abstained but voted to affirm the resolution the following day).*⁷

By the evening of the eighth day of the crisis (October 23, 1962), President Kennedy was able to include the following language in his proclamation regarding "INTERDICTION OF THE DELIVERY OF OFFENSIVE WEAPONS TO CUBA":

"WHEREAS the Organ of Consultation of the American Republics meeting in Washington on October 23, 1962, recommended that the Member States, in accordance with Articles 6 and 8 of the Inter-American Treaty of Reciprocal Assistance, take all measures, individually and collectively, including the use of armed force, which they may deem necessary to ensure that the Government of Cuba cannot continue to receive from the Sino-Soviet powers military material and related supplies which may threaten the peace and security of the Continent and to prevent the missiles in Cuba with offensive capability from ever becoming an active threat to the peace and security of the Continent:

*NOW, THEREFORE, I, JOHN F. KENNEDY, President of the United States of America, acting under and by virtue of the authority conferred upon me by the Constitution and statutes of the United States, in accordance with the aforementioned resolutions of the United States Congress and of the Organ of Consultation of the American Republics, and to defend the security of the United States, do hereby proclaim that the forces under my command are ordered, beginning at 2:00 P.M. Greenwich time October 24, 1962, to interdict, subject to the instructions herein contained, the delivery of offensive weapons and associated materiel to Cuba."*⁸

3. Strategic Uses of the Military Force

We will now transition from strategic direction to how the military contributes to achieving strategic objectives.

a. The Military Instrument in Support

For the U.S., the DoD reinforces traditional U.S. tools of diplomacy by providing military options to ensure the President and our diplomats negotiate from a position of strength. The credible use or threat of force gives strength to, and in some cases, enables, the diplomatic process. Usually, the DoD is in a supporting role to other U.S. agencies and organizations to pursue a given strategy. The chief of mission, normally the U.S. ambassador, and the corresponding country team are normally in charge of diplomatic-military activities in a country.

In very rare instances, when other instruments of national power prove insufficient, the military becomes the nation's primary instrument. In either case, the military facilitates and supports the application of the other instruments. There are four strategic uses of military force: assurance, deterrence, coercion, and compellence.

(1) Assurance

Assurance is using the military instrument to demonstrate commitment and support to U.S. allies and partners. Assurance often takes the form of security cooperation, combined exercises, and forward stationing of U.S. forces. It can also take the form of stabilization missions to provide security and meet the basic human needs of populations in a conflict-affected area, and include foreign humanitarian assistance missions as a demonstration of commitment to the international order and support to those in need.

(2) Deterrence

Deterrence is the prevention of undesired action by the existence of a credible threat of unacceptable counteraction, and/or the belief that the costs of action outweigh the perceived benefits. The joint force deters by maintaining joint readiness, demonstrating the ability to project power globally, stationing forces forward, and conducting multinational security cooperation and training, when necessary.

(3) Coercion

Coercion is the use, or threat of use, of violent military force to change enemy or adversary behavior. It is frequently used in conjunction with diplomacy. Coercion differs from deterrence in that the adversary has already taken an action contrary to U.S. interests. Rather than maintaining the status quo, coercion is the application of military force to increase the pressure on an enemy or adversary through punishment or denial. Coercion by punishment seeks to increase costs to force an enemy or adversary to choose to stop pursuing a particular course of action. Coercion by denial uses military means to prevent or convince an adversary that they are unable to achieve their objective at a reasonable cost. Coercion provides flexibility; however, it relies on an accurate estimation of the enemy's or adversary's will. Ultimately, the choice remains with the enemy or adversary to continue the course of action if the will is high enough.

(4) Compellence

Compellence is the violent application of military force used to project our will on the enemy by eliminating their resistance. If an adversary values an objective that threatens our national security above all else, then it is imperative to remove those in power or use military force to destroy that which gives the enemy the will to resist. To compel an enemy, there must be the national will to win and the means available to do it decisively. Compellence requires strong national commitment. The two basic methods to compel an enemy are annihilation and attrition. Rarely employed in isolation from the other, annihilation is the complete destruction of the adversary's means, and attrition is the process of gradually reducing their strength or effectiveness through sustained attack or pressure. Annihilation and attrition will be discussed in detail in the 8671 *Doctrine* sub-course.

b. Competition Continuum

The four strategic uses of the military instrument of power, outlined above, are implemented through campaigns, operations, and activities that vary in purpose, scale, risk, and intensity, and occur across a competition continuum (Figure 2). The competition continuum describes three broad categories of strategic relationships among the actors in the operational environment: cooperation, adversarial competition below armed conflict, and armed conflict/war to distinguish relationships between actors and clarify options for the use of military force within each context. During campaigns and operations, the joint force will likely cooperate with allies and partners, compete against adversaries, and fight enemies concurrently in the same operational environment.



Figure 2: Notional Competition Continuum.

(1) Cooperation

Cooperation includes interactions among actors in the international system to build mutual security relationships, the development of multinational partner military and security capabilities for self-defense and multinational operations, and the provision for security forces with steady-state and contingency access to the military and security capabilities of allies and partners. Although interests will rarely perfectly align, cooperative relations that promote shared security objectives underpin the international order, enhance collective security, help to ensure access to global commons, enable burden-sharing, facilitate adversarial competition, and deter or mitigate armed conflict.

(2) Competition

Adversarial competition below armed conflict exists when two or more states or non-state adversaries have incompatible interests, but neither seeks armed conflict. Adversaries may use military activity just below the violence threshold of armed conflict to secure their national interests. Adversarial competition below armed conflict can either be defensive or offensive. Defensive actions protect friendly force advantages while imposing costs on adversaries, ensuring that it is unlikely they will move toward armed conflict. Offensive actions leverage friendly force advantages while eroding adversary's advantages. Internationally, competition takes the form of information activities, cyberspace operations, limited special operations, and large-scale operations that demonstrate a "show of force," to name just a few. To counter this activity, the joint force typically supports other instruments of national power to achieve U.S. national objectives. The joint force supports the national effort through such things as foreign internal defense, cyberspace operations, and information and messaging activities. These activities are generally in line with the deterrence use of military force.

(3) Armed Conflict

All state and non-state actors have interests; inevitably, some of those interests conflict with the interests of others. Conflict can arise as a result of a threat (or perceived threat) to any strategic interest. Armed conflict occurs when an actor uses violence as the primary means to satisfy its interests. While conflict is always related to some national interest or objective, it is normally the outgrowth of a specific policy or commitment. Although there is no single, legal definition for the term, armed conflict typically implies war that occurs internationally or within states (non-international) between state or non-state actors. International law distinguishes armed conflict from disturbances (riots, violent protests, etc.) by the intensity of the conflict, organization of the actors involved, and the capacity of each actor to sustain armed violence. For the DoD, armed conflict involves two strategic uses of military force: coercion and compellence. Armed conflict varies in intensity and may include actions and activities from across the range of military operations (see Figure 3).

The competition continuum and range of military operations will be covered in detail in the 8671 *Doctrine* sub-course.

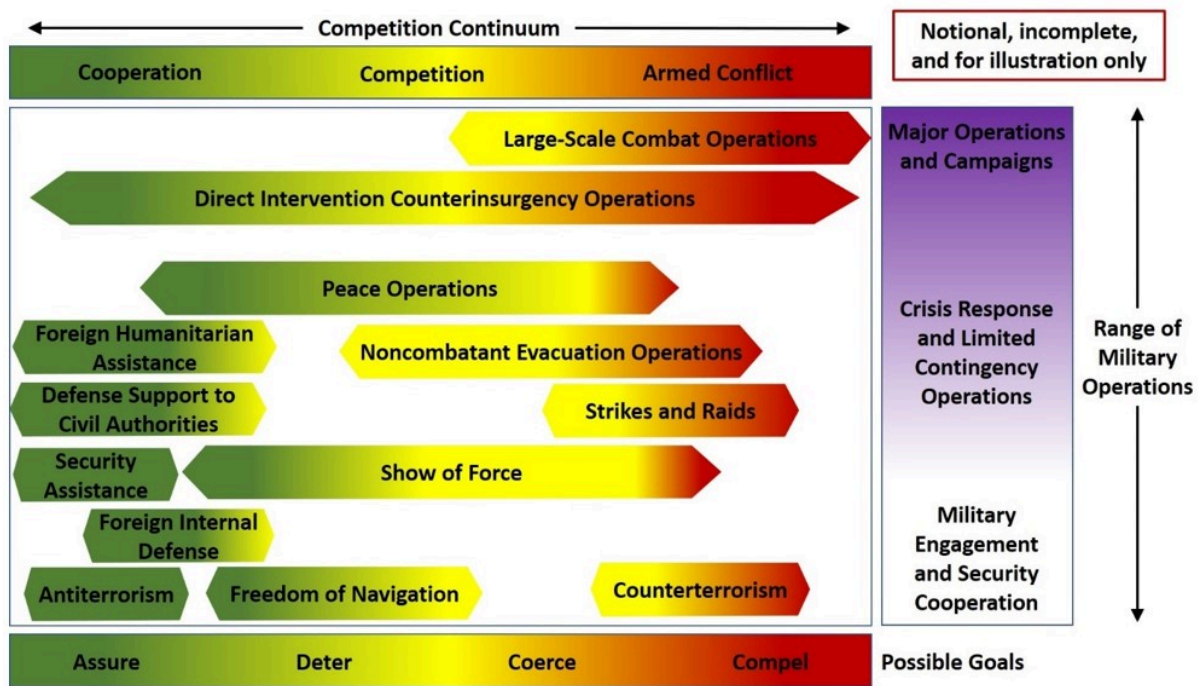


Figure 3: Competition Continuum, Range of Military Operations, and the Strategic Uses of the Military.

4. War

War is generally thought of as a state of open and direct armed conflict between different countries, state-like entities, or armed organizations to achieve political objectives. Wars may be fought within a country by a central government or occupying power opposed by insurgent, separatist, or resistance groups. Wars may be fought in semi-autonomous regions by armed groups that do not recognize national borders. The nineteenth-century Prussian general and strategic theorist, Carl von Clausewitz, described war “as continuation of policy by other means.” Political activity at the international level is based on state and non-state actors maximizing their relative power to ensure the security of their strategic interests. Political activity can take many forms, including diplomatic, informational, military, and/or economic action that is well within the norms and rules of peaceful international interaction. Use of violence, however, is a bright line in the spectrum of action, beyond which the involved parties are, at some level, waging war, which has a nature and character all its own.

Note: The term “war” is used here to describe activities beyond those of only “legally declared war” through an act of Congress. A more encompassing term that describes this concept may be “warfare.” The nature, character, and theory of war will be discussed in detail in the 8671 *Doctrine* sub-course.

a. The Nature of War

War is a violent clash of interests between or among organized groups characterized by the use of military force. War is fundamentally a human endeavor, its nature inextricably linked to human nature, which is unchanging. The fact that war entails the use of violence (or the expression of human passion through violent means) to attain its ends gives it fundamental, constant qualities that define it as a distinct social phenomenon. The use of violence against other human beings arouses a level of emotion—passion—in both combatants and bystanders that is orders of magnitude greater than that aroused by any other social interaction. That passion works continuously to undermine and distort the critical analysis and lucid decision making needed to prosecute war in a manner that effectively achieves its objectives. As in all social interactions, uncertainty, unpredictability, chance, and human fallibility permeate war. Moreover, the danger and physical stress inherent in war magnify the effects of these characteristics dramatically. Within this fog and friction, participants in war can rarely determine the best course of action empirically. Judgment and intuition must replace certainty to adapt to and accommodate the unexpected. Finally, war’s natural tendency is to escalate as each side tries to achieve material and psychological supremacy. Absent some controlling factor on one or both sides, war will escalate to maximum effort, maximum use of force, and maximum ends—what Clausewitz defined as “absolute war.”

(1) The Human Dimension

Because war is a clash between opposing human wills, the human dimension is central in war. The human dimension is the combination of the cognitive, social, and physical aspects of humans, both individuals and groups. It is the human dimension that infuses war with its intangibles, including intuition, fear, genius, and moral factors. War is shaped by human nature and is subject to the complexities, inconsistencies, and peculiarities that characterize human behavior. Since war includes acts of nationally directed violence, it will invariably be shaped by human factors and emotions. Moreover, war is an extreme trial of physical and moral strengths, and tests institutional and individual sources of endurance and resilience.

(2) Friction

The interplay of human factors, unknown contingencies, and chance is what Clausewitz referred to as friction. The very essence of war as a clash of opposing human forces and wills creates much of this friction. War's interactive nature, the duality of two (or more) independent and animate forces seeking advantage, creates the context to best understand the existence of friction as a central element in the phenomenon known as war. This sort of friction refers to all the factors and obstacles that combine to make even the smallest actions in war seem difficult, if not impossible. Anyone who has tried to start a convoy, foot march, embarkation, or operation on time, only to be made late by a stream of minor delays, has experienced this friction firsthand.

(3) Uncertainty

Related to friction is the reality of uncertainty. War takes place in an atmosphere of uncertainty, or Clausewitz's fog of war. The reality of unknown and possibly unknowable relevant elements is a pervasive fundamental of human conflict. Despite significant investment in technologies to understand and anticipate our opponents and other variables in the operating environment, the existence of friction and random chance mean we must accept that we cannot reduce uncertainty to zero. Commanders must be cognizant of the fact that the enemy is constantly trying to increase our uncertainty through deception and other means. We must learn to fight in an environment of uncertainty, which we can do by developing simple, flexible plans; planning for branches and contingencies; developing standing operating procedures; and fostering initiative among subordinates throughout the joint force. We will find success in war, as we have in the past, depending in large part on our ability to learn from operational experience and adapt to a constantly changing situation.

b. Changing Character of War

While war's fundamental nature remains constant, its character—its face or form, what it looks like—shifts, sometimes dramatically, over both time and space. War's variety is almost limitless, and the character of war at any point in time and in any area reflects the nature of the societies waging it. It reflects political, military, economic, social, cultural, and technological dynamics at work in the societies at war at that time, and it manifests itself in who within each society participates (or refuses to participate), why they do so, and how they do so, especially regarding employing force (Figure 4). For instance, due to our industrial and technical history, the U.S. traditionally has organized, trained, and equipped for "high-end" conflict. This has utility against developed countries, but unfortunately, this doctrine has less utility against undeveloped adversaries that are ideologically driven. A critical job of senior national and military leaders is to determine the character of the war that might occur. As Clausewitz warns, "The first, the supreme, the most far-reaching act of judgment that the statesman and commander have to make is to establish the kind of war on which they are embarking; neither mistaking it for, nor trying to turn it into, something that is alien to its nature." That judgment should inform the political decision to fight or to take action that could lead to war.

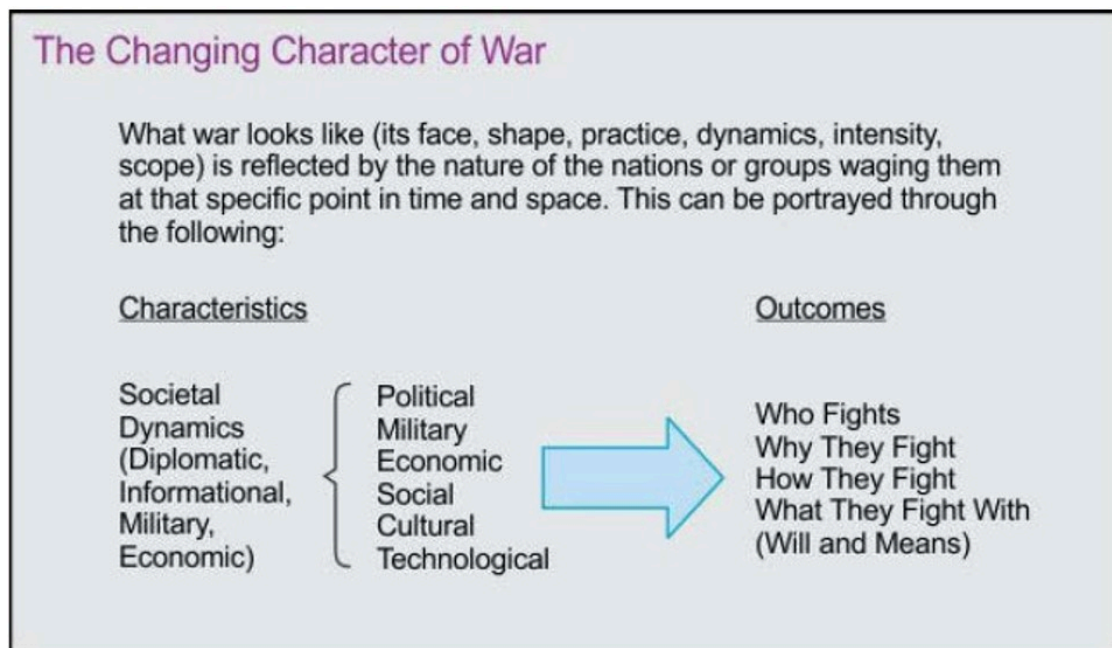


Figure 4: Changing Character of War.

c. Theory of War

What matters ultimately in war is strategic success: attainment of political aims and the protection of national interests. The decision to commit military force to war involves both policy and politics. The policy objectives to be achieved should be the foremost determinants of a war's conduct. The central tenet in the theory of war is that **war must serve policy**.

Given that war does not exist for its own sake, but as Clausewitz stated, "an extension of policy...", senior military leaders must be attentive to policy objectives. As the policy objectives of the application of military force may vary across a continuum of competition, so should the application of violence vary in accordance with those aims. The policy aim that is the motive for war must also be the foremost driver for the formulation of a military objectives and the conduct of a war. Policy objectives must be balanced with the means by which to achieve them. It is important to recognize that many strategic situations encountered by a nation cannot be solved by military means. In addition, war tends to take its own course as it unfolds. War is not an inanimate instrument, but an animate force which may likely have unintended consequences that may change the situation for the warring nations.

To say that war is an extension of politics and policy is not to say that war is strictly a rational phenomenon; it also contains social, cultural, legal, and psychological elements, which can be spurred by irrational motives. These can also exert a strong influence on the conduct of war as well as on war's usefulness for solving policy-based problems. When the policy objectives are extreme, such as the annihilation of an enemy nation, then the military objectives toward destruction will align with the policy objective, resulting in few policy restrictions on the military conduct of war. On the other hand, the more limited the policy objectives, the higher probability of policy restrictions on the application of military force. Commanders must recognize that since all military actions must serve policies, these policy restrictions on military action may be perfectly correct. At the same time, military leaders have a responsibility to advise government leadership when the limitations imposed on military action jeopardize their ability to achieve supporting military objectives and, therefore, policy objectives.

History shows that national leaders, both political and military, who fail to understand this relationship sow the seeds for ultimate failure - even when their armed forces achieve initial battlefield success. Battlefield brilliance seldom rescues a bad strategy.

To begin to comprehend warfare, one must understand the strategic interests and will that drive it. Warfare is a violent clash of opposing wills. Will is the volition and determination to achieve a goal - in this sense, an objective related to strategic interests. In war, achieving the strategic objective depends upon ways, means, and will. In warfare, the means employed are military capabilities and resources. Warfare is never, however, limited to purely military means. While means are largely a fixed commodity and easy to measure, the ways by which those means are employed can change. Will varies greatly depending upon the objective and is difficult to measure. Generally, objectives related to vital interests elicit the most amount of will, whereas those associated with peripheral interests elicit the least. Military leaders must account for national will in proposing acceptable military options to national leadership. In addition, leaders must be particularly cautious of situations where an asymmetry of will exists.

Asymmetry of will occurs when one side, usually the side with fewer means, has a vital interest threatened, while the other side has a lesser interest at stake. Since interest and will are related, this leads to an asymmetry of will that can tilt the probability of success in the "weaker" opponent's favor, regardless of the means available.

Any discussion of ends and means in war must begin with two basic points. First, war is an expression of politics. The ends or goals of any party waging war - even though those goals may be social, economic, religious, or ideological in nature - are by definition political goals. Second, wars are fought by political entities that have unique characteristics and often very dissimilar goals and resources. To understand any conflict, one must appreciate the ways in which the means and ends of the participants may vary.

d. Levels of War

Military activities take place at several interrelated levels, which form a hierarchy. These levels are strategic, operational, and tactical (Figure 5). There are no finite limits or boundaries between these levels, but they help commanders design and synchronize operations, allocate resources, and assign tasks to the appropriate command.



Figure 5: Levels of War.

(1) Strategic Level

The highest level is the strategic level. Activities at the strategic level focus directly on policy objectives. Strategy applies to peace as well as war. There is a distinction between national strategy, which coordinates and focuses all the instruments of national power to attain the policy objectives, and military strategy, which is the application of military force to secure the policy objectives. Military strategy, thus, is subordinate to national strategy. Military strategy can be thought of as the art of winning wars and securing peace. Strategy involves establishing goals, assigning forces, providing assets, and imposing conditions on the use of force in theaters of war. Strategy derived from political and policy objectives must be clearly understood to be the sole authoritative basis for all operations.

(2) Tactical Level

The lowest level of war is the tactical level. The tactical level of war is the level of war at which battles and engagements are planned and executed to achieve military objectives assigned to tactical units or task forces. Activities at this level focus on the ordered arrangement and maneuver of combat elements in relation to each other and to the adversary to achieve combat objectives. It is important to understand tactics within the context of the levels of war. The strategic and operational levels provide the context for tactical operations. Without this context, tactical operations are just a series of disconnected and unfocused actions. Strategic and operational success is a measure of how one or more battles link to winning a major operation or campaign. In turn, tactical success is a measure of how one or more engagements link to winning a battle.

- **Engagement.** An engagement is a small, tactical conflict, usually between opposing, lower echelon maneuver forces. They are usually short, executed in terms of minutes, hours, or days. Engagements can result from one side's deliberate offensive movement against an opponent or from a chance encounter between two opponents, such as a meeting engagement.
- **Battle.** A battle consists of a set of related engagements that lasts longer and involves larger forces than an engagement. They occur when adversaries commit to fight to a decision at a particular time and place for a significant objective. Battles can affect the course of a campaign or major operation.

(3) Operational Level

The operational level of war sups the strategic and tactical levels. It is the use of tactical results to attain strategic objectives. The operational level includes deciding when, where, and under what conditions to engage the adversary in battle—and when, where, and under what conditions to refuse battle in support of higher aims. Actions at this level imply a broader dimension of time and space than actions at the tactical level. As strategy deals with winning wars and tactics with winning battles and engagements, the operational level of war is the art and science of winning campaigns.

- **Campaign.** A campaign is a series of related military operations aimed at accomplishing a strategic or operational objective within a given time and space. A campaign is usually aimed at achieving a particular strategic result within a specific geographic theater. A war or other sustained conflict sometimes consists of a single campaign or several campaigns. If there is more than one campaign, these can be conducted either in sequence or simultaneously if there is more than one theater of war.

It is important to understand that levels of command, sizes of units, types of equipment, or types of forces or components are not associated with a particular level of war. National assets, such as intelligence and communications satellites, previously considered principally in a strategic context, are an important adjunct to tactical operations. Commanders consider actions strategic, operational, or tactical based on their effect or contribution to achieving strategic, operational, or tactical objectives.

(4) Strategic – Operational Connection

As previously mentioned, the levels of war are interrelated. Strategic, operational, and tactical commanders, forces, and events are continually interacting with each other. The political end state envisioned by policy makers determines the strategic goals of all military actions. It is also important to point out that the relationship between strategy and operations runs both ways. Strategy shapes the design of the campaign, but strategy must also adapt to operational circumstances and events that occur during that campaign.

Strategy guides operations in three ways: it establishes aims, allocates resources, and imposes restraints and constraints on military action. Together with the nature and actions of the adversary and the characteristics of the area of operations, strategic guidance defines the parameters within which operations are conducted.

First, strategy translates policy objectives into military terms by establishing the strategic aim. What political effect must our military forces achieve? What enemy assets must our tactical forces seize, neutralize, threaten, or actually destroy to either bend the enemy to our will or break the enemy completely? The operational commander's principal task is to determine and pursue the sequence of actions that will most directly accomplish the military strategic mission. It is important to keep in mind that the military strategic aim is but one part of a broader national strategy. Strategists must be prepared to modify aims in the light of actual developments, as they reevaluate costs, capabilities, and expectations. While required to pursue the established aim, the operational commander is obliged to communicate the associated risks to superiors. When aims are unclear, the commander must seek clarification and convey the impact—positive or negative—of continued ambiguity.

Second, strategy provides resources, both tangible resources such as material and personnel, and intangible resources such as political and public support for military operations. When resources are insufficient despite all that skill, talent, dedication, and creativity can do, the operational commander must seek additional resources or request modification of the aims.

Third, strategy, because it is influenced by political and social concerns, places conditions on the conduct of military operations. These conditions take the form of restraints and constraints. Restraints prohibit or restrict certain military actions, such as the prohibition imposed on MacArthur against bombing targets north of the Yalu River in Korea in 1950 or the United States' policy not to make first use of chemical weapons in World War II. Restraints may be constant (as the laws of armed conflict), or situational (such as rules of engagement). Constraints, on the other hand, obligate the commander to certain military courses of action, such as making the decision that the Arab members of the Coalition conducting Operation DESERT STORM (1991) should be the liberators of Kuwait City.

(5) Tactical – Operational Connection

Just as strategy shapes the design of the operational level of war, while simultaneously adapting to operational circumstances and events, so operations must interact with tactics. Operational plans and directives that are rooted in political and strategic aims establish the necessary focus and goals for tactical actions. Operational planning provides the context for tactical decision-making. Without this operational coherence, war at the tactical level is reduced to a series of disconnected and unfocused tactical actions. Just as operations must serve strategy by combining tactical actions so as to achieve the strategic aim most effectively and economically, operations must also serve tactics by creating the most advantageous conditions for tactical actions.

(6) Interdependence of the Levels of War

It is important to understand that a specific echelon of command is not necessarily concerned with only one level of war. A theater commander's concerns are clearly both strategic and operational. A MAGTF commander's responsibilities will be operational in some situations and largely tactical in others and may actually span the transition from tactics to operations in still others. A commander's responsibilities within the hierarchy depend on the scale and nature of the conflict and may shift up and down as the war develops.

Actions at one level can often influence the situation at other levels. Harmony among the levels tends to reinforce success, while disharmony tends to negate success. Obviously, failure at one level tends to naturally lessen success at the other levels. With today's constant 24-hour media coverage and easy access to the Internet by our adversaries for propaganda, a tactical-level plan and resulting action can have severe operational or strategic implications. For example, an action by one Marine in the operating environment at the tactical level could potentially cause significant disruption to operational and strategic-level planning. The following vignette provides a good example of a tactical action having strategic implications.

In November 2004, after 20 months of combat operations in Iraq, U.S. Forces led a counterinsurgency fight against insurgents in the city of Fallujah. This marked a return to conventional style operations of systematically clearing a city, causing some of the deadliest fighting of the Iraq War. Insurgents, counterinsurgents, and civilians were injured and killed during this bloody event despite a considerable effort to mitigate loss of life.

During the initial days of the operation, embedded journalist Kevin Sites videotaped a Marine shooting an unarmed Iraqi combatant inside a mosque in Fallujah. Although the journalist felt some trepidation about releasing the video, his obligation to his profession prevailed. The footage quickly emerged in the U.S. media promoting calls for action by the U.S. public. The United States Marine Corps hastily issued a press release stating that the matter was under investigation and launched a preliminary inquiry into the matter.

The insurgency, although preoccupied in fierce action against the counterinsurgency in Fallujah, used the footage to rally support among the Sunni population in the area. In an attempt to further discredit the newly elected Iraqi Prime Minister, who was Shiite, the insurgency relayed a message of a Shia led government, backed by infidel invaders, attacking unarmed men in a mosque. This message rallied support for the insurgency and additional fighters migrated to Fallujah. Additionally, the insurgent's exploitation of the mosque shooting contributed to a successful and extensive insurgent recruiting campaign throughout Iraq.⁹

5. National Security Structure

The national security structure provides the strategic direction that guides employment of the military instrument of national power as part of a global strategy. Continuing assessments of the current and future strategic environment inform the national strategy, goals, and objectives. The national security structure is established in public law.

- The operational chain of command is clearly established from the President through the SecDef to the combatant commanders.
- The Service Chiefs are responsible for organizing, training, and equipping Service forces. While the Service Chiefs are members of the JCS, the combatant commanders are responsible for the planning and execution of joint operations.
- The CJCS is the *principal* military advisor to the President, NSC, and the SecDef. The CJCS outranks all other officers of the Armed Forces but **does not exercise military command** over the combatant commanders, JCS, or any of the Armed Forces.
- The Joint Staff is under the exclusive direction of the CJCS. It is organized along conventional staff lines to support the CJCS and the other members of the JCS of Staff in performing their duties. The Joint Staff does not function as an overall Armed Forces general staff and has no executive authority.

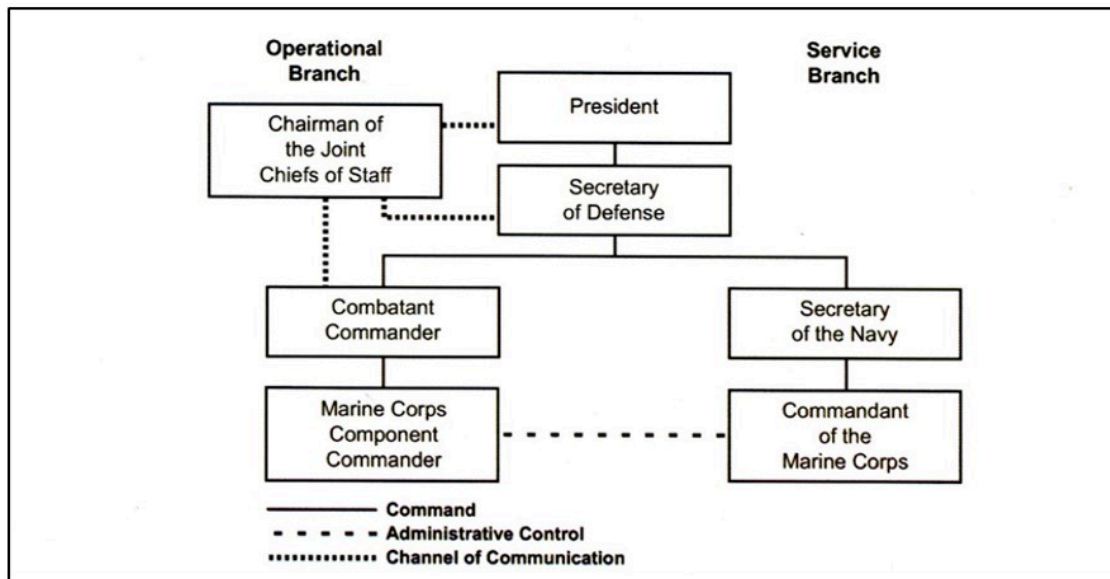
a. The President and the Secretary of Defense

The President and the SecDef, or their duly deputized alternates or successors, exercise authority over the Armed Forces through combatant commanders (and through the Secretaries of the Military Departments and the Service Chiefs for those forces not assigned to the combatant commanders). The President and the SecDef translate policy into national strategic military objectives.

In general, as discussed above, the President frames strategic context by defining national interests and goals in documents, such as the NSS, presidential policy directives, executive orders, and other strategic documents.

The President and SecDef exercise authority and control of the armed forces through a single chain of command with two distinct branches (Figure 6). The first branch runs from the President, through the SecDef, directly to the combatant commanders (CCDRs) for missions and forces assigned to their commands. The second branch, used for purposes other than operational direction of forces assigned to the combatant commands, runs from the President through the SecDef to the Secretaries of the Military Departments.

Combatant commanders are the vital link between those who determine national security policy and strategy and the military forces that conduct military operations designed to achieve national strategic objectives.



GCCs are the vital link between those who determine national security policy and strategy and the military forces or subordinate joint force commanders (JFCs) that conduct military operations within their geographical areas of responsibility (AORs) (Figure 7). GCCs are responsible for a large geographical area, requiring single responsibility for effective coordination of the operations within that area. Directives flow from the President and SecDef through CJCS to the GCCs, who plan and conduct the operations that achieve national, alliance, or coalition strategic

objectives. GCCs provide guidance and direction through strategic estimates, command strategies, and plans and orders for the employment of military force. As military force may not achieve national objectives, it must be coordinated, synchronized, and, if appropriate, integrated with other government agencies, intergovernmental organizations, nongovernmental organizations, multinational forces, and elements of the private sector.

Using their strategic estimate(s), GCCs develop strategies that translate national and multinational direction into strategic concepts or courses of action to meet strategic and joint operation planning requirements. GCCs' plans provide strategic direction; assign missions, tasks, forces, and resources; designate objectives; provide authoritative direction; promulgate rules of engagement or rules for the use of force; establish constraints and restraints; and define policies and concepts of operations to be integrated into subordinate or supporting plans. GCCs also exercise authority for force protection over all DoD personnel (including their dependents) assigned, attached, transiting through, or training in the GCC's AOR. The exception is for those for whom a chief of mission retains security responsibility. In a foreign country, the chief of mission is responsible to the President for directing, coordinating, and supervising all U.S. government elements in the host nation, except for those elements under the command of a combatant commander. GCCs are responsible for coordinating with chiefs of mission in their geographic area of responsibility across the range of military operations and for negotiating a memorandum of agreement with the chiefs of mission in designated countries to support military operations. The current GCCs are:

- U.S. Africa Command (AFRICOM), Stuttgart, Germany
- U.S. Central Command (CENTCOM), MacDill Air Force Base, Tampa, Florida
- U.S. European Command (EUCOM), Stuttgart, Germany
- U.S. Indo-Pacific Command (INDOPACOM), Camp H.M. Smith, Hawaii
- U.S. Northern Command (NORTHCOM), Peterson Air Force Base, Colorado
- U.S. Southern Command (SOUTHCOM), Miami, Florida

(2) Functional Combatant Commands

Functional component commanders (FCCs) provide support to and may be supported by GCCs and other FCCs as directed by a higher authority. FCCs are responsible for a large functional area requiring single responsibility for effective coordination of the operations therein. These responsibilities are normally global in nature. The President and SecDef direct what specific support and to whom such support will be provided. When an FCC is the supported CCDR and operating within GCC's AORs, close coordination and communication between them is paramount. The current FCCs are:

- U.S. Cyber Command (CYBERCOM), Fort Meade, Maryland
- U.S. Strategic Command (STRATCOM), Offutt Air Force Base, Nebraska
- U.S. Special Operations Command (SOCOM), MacDill Air Force Base, Tampa, Florida
- U.S. Transportation Command (TRANSCOM), Scott Air Force Base, Illinois

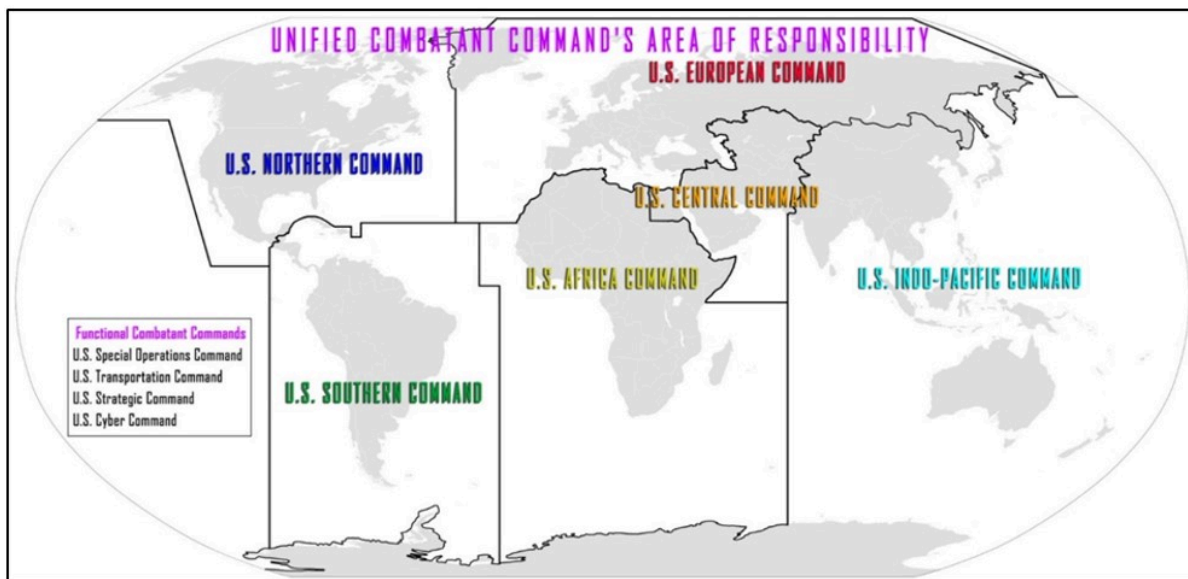


Figure 7: Geographic Combatant Commander's Area of Responsibility

f. Unified Action

Unified action in military usage is a broad term referring to the synchronization, coordination, and/or integration of the activities of governmental and nongovernmental entities with military operations to achieve unity of effort. The geographic combatant commanders are the vital link between those who determine national security policy and strategy and the Service components that conduct military operations within the geographic combatant commanders' areas of responsibility. Directives flow from the President and SecDef, through the CJCS, to the geographic combatant commanders. The geographic combatant commanders plan and conduct the operations that achieve national, alliance, or coalition strategic objectives. They provide guidance and direction through strategic estimates, command strategies, theater campaign plans, contingency plans, operation plans, and operation orders for the employment of military force, which may occur in cooperation with other government agencies, intergovernmental organizations, nongovernmental organizations, multinational forces, and elements of the private sector.

Notes

¹ Arthur F. Lykke Jr., ed., *Military Strategy: Theory and Application*. Carlisle, PA: U.S. Army War College, 1998.

² The National Security Archive at The George Washington University.

³ Theodore C. Sorenson et al., *Let the Word Go Forth: The Speeches, Statements, and Writings of John F. Kennedy* (New York: Delacorte Press, 1988), p. 243; quoted in MCDP 1-1 *Strategy*, p. 36.

⁴ John F. Kennedy January 20, 1961: *Inaugural Address*, available at <https://millercenter.org/the-presidency/presidential-speeches/january-20-1961-inaugural-address>; accessed 12 July 2019.

⁵ Stephen R. Gabe *Alliance for Progress*, The Oxford Research Encyclopedia, available at <https://oxfordre.com/latinamericanhistory/view/10.1093/acrefore/9780199366439.001.0001/acrefore-9780199366439-e-95>; accessed 12 July 2019.

⁶ Stella Krepp, *Cuba and the OAS: A Story of Dramatic Fallout and Reconciliation*, available at <https://www.wilsoncenter.org/blog-post/cuba-and-the-oas-story-dramatic-fallout-and-reconciliation>; accessed 12 July 2019.

⁷ James M. Lindsay, Council On Foreign Relations' The Water's Edge *TWE Remembers: The OAS Endorses a Quarantine of Cuba (Cuban Missile Crisis, Day Eight)*; available at <https://www.cfr.org/blog/twe-remembers-oas-endorses-quarantine-cuba-cuban-missile-crisis-day-eight>; accessed 12 July 2019.

⁸ John F. Kennedy Presidential Library and Museum, *Interdiction of the Delivery of Offensive Weapons to Cuba*, available at <https://microsites.jfklibrary.org/cmc/oct23/doc4.html>; accessed 12 July 2019.

⁹ *Rules of Engagement*, Master of Military Studies Research Paper, submitted by Major W.C. Reaves, U.S. Marine Corps, USMC Command and Staff College, April 2013. (Approved for public release; distribution is unlimited.)

Lesson 3: National Defense Strategy

Lesson Hours: 1.00

1. Introduction

As discussed in the previous lesson, the *National Defense Strategy* (NDS) translates and refines the *National Security Strategy* into broad military guidance for military planning, military strategy, force posturing, force constructs, and force modernization. Following an overview of the strategic environment, this lesson introduces Department of Defense priorities and the strategic approach that will be used to achieve these priorities.

Note: On 1 May 2025, the Secretary of Defense issued a memorandum directing the development of the 2025 NDS. At the time of this lesson's publishing, however, the 2025 NDS has not been released, and the 2022 NDS remains in effect.

2. Educational Objectives

- a. Identify the aspects of the increasingly complex security environment.
- b. Identify pacing challenges, acute threats, threats to the U.S. homeland, and other persistent threats.
- c. Identify emerging technological advancements, "gray zone" activities, and global challenges.

- d. Identify the Department of Defense priorities that are required to maintain our security and prosperity.
- e. Identify how the Department of Defense will respond to competitor strategies.
- f. Identify the force attributes required to strengthen and sustain deterrence, and to prevail in conflict.
- g. Identify the ways in which enduring advantage will be built into U.S. forces.

3. Key Terms

- Strategic competition
- Acute threats
- Persistent threats
- Complex escalation dynamics
- Gray zone activities
- Transboundary challenges
- Integrated deterrence
- Campaigning
- Force planning
- Advantage

4. Preparation

All quizzes, discussion forum posts, and assignments are executed in the Moodle course site.

a. Lesson Reading:

- [*2022 National Defense Strategy of the United States of America*](#) (27 Oct 2022) (31 pages). Note: The *2022 Nuclear Posture Review* and *2022 Missile Defense Review* are contained within this document but are only included for completeness (they are not required to be read).
- [*Secretary of Defense Memorandum, National Defense Strategy*](#) (1 May 2025) (1 page).
- [*2025 Interim National Defense Guidance*](#), Secretary of Defense (1 May 2025) (6 slides).

a. Supplemental References (Optional):

- *2022 Nuclear Posture Review* (27 Oct 2022) - included in lesson reading document.
- *2022 Missile Defense Review* (27 Oct 2022) - included in lesson reading document.
- [*Strategy for Countering Weapons of Mass Destruction*](#) (2023).

Lesson 4: Department of War

Lesson Hours: 2.00

1. Introduction

In accordance with Title 10, *United States Code (USC)*, the Department of Defense (DOD) supports the military operations and activities (i.e., core missions) of the U.S. Armed Forces required to achieve the strategic objectives of the *National Security Strategy*, *National Defense Strategy*, and *National Military Strategy*. This lesson introduces those core missions and the organization and unique functions of the Department of the Army, the Department of the Air Force (including the U.S. Air Force and U.S. Space Force), the U.S. Special Operations Command, and the U.S. Coast Guard. The Department of the Navy (including the U.S. Navy and the U.S. Marine Corps) will be covered in the subsequent lesson.

2. Educational Objectives

- a. Identify the functions of the Department of Defense.

- b. Identify the functions of the Secretary of Defense.
- c. Identify the common functions of Military Departments.
- d. Identify the organization of the Department of the Army.
- e. Identify the unique functions assigned to the Department of the Army.
- f. Identify the differences between the various U.S. Army brigade combat teams.
- g. Identify the organization of the Department of the Air Force.
- h. Identify the unique functions assigned to the Department of the Air Force.
- i. Identify the cornerstone responsibilities and core competencies of the U.S. Space Force.
- j. Identify the spacepower disciplines associated with the U.S. Space Force.
- k. Identify the various organizations within the U.S. Special Operations Command.
- l. Identify the operations that are conducted by the U.S. Special Operations Command.
- m. Identify the primary responsibilities of the U.S. Coast Guard.
- n. Identify the specific functions performed by the U.S. Coast Guard when providing direct or cooperative support to the Department of Defense.

3. Key Terms

- Department of Defense
- Secretary of Defense
- Department of the Army
- U.S. Army
- Department of the Air Force
- U.S. Air Force
- U.S. Space Force
- U.S. Special Operations Command
- U.S. Coast Guard

4. Preparation

All quizzes, discussion forum posts, and assignments are executed in the Moodle course site.

- a. Lesson Reading: [Department of War](#) (57 pages).
- b. Other Requirements: None.
- c. Supplemental References (optional):
 - [JP 1, Doctrine for the Armed Forces of the United States](#) (2017).
 - [JP 1, Vol. 2, The Joint Force](#) (2020).
 - [DoDD 5100.01, w/Change 1 Functions of the Department of Defense and Its Major Components](#) (17 Sep 2020).
 - [The U.S. Military's Force Structure: A Primer](#), Congressional Budget Office (May 2021).
 - [Defense Primer: Organization of U.S. Ground Forces](#), Congressional Research Service, (10 Oct 2024).
 - [The U.S. Army's Command Structure](#), <https://www.army.mil/info/organization> (Dec 2025).
 - [ATP 3-91, Division Operations](#) (Oct 2014).
 - [Comprehensive Plan for the Organizational Structure of the U.S. Space Force](#), Department of the Air Force Report to Congressional Committees (Feb 2020).
 - [The Official Home Page of the U.S. Air Force](#), <https://www.af.mil> (Dec 2025).
 - [SCP, Spacepower - Doctrine for Space Forces](#) (June 2020).
 - [United States Space Force](#), <https://www.spaceforce.mil/> (Dec 2025).
 - [Fact Book 2025, United States Special Operations Command](#), United States Special Operations Command (2025).
 - [USSOCOM](#), <https://www.socom.mil/> (Dec 2025).
 - [Air Force Special Operations Command](#), <https://www.afsoc.af.mil/> (Dec 2025).

- [United States Army Special Operations Command](https://www.army.mil/usasoc), <https://www.army.mil/usasoc> (Dec 2025).
- [Navy Special Warfare Command](https://www.socom.mil/navsoc), <https://www.socom.mil/navsoc> (Dec 2025).

Lesson 4 Reading

DEPARTMENT OF WAR

References:

- a) JP 1, *Doctrine for the Armed Forces of the United States* (2017)
- b) JP 1, Vol. 2, *The Joint Force* (2020)
- c) DoDD 5100.01 w/ Change 1, *Functions of the Department of Defense and Its Major Components* (17 Sep 2020)
- d) *The U.S. Military's Force Structure: A Primer*, Congressional Budget Office (May 2021)
- e) *Defense Primer: Organization of U.S. Ground Forces*, Congressional Research Service, (10 Oct 2024)
- f) *The U.S. Army's Command Structure*, <https://www.army.mil/info/organization> (Dec 2025)
- g) ATP 3-91, *Division Operations* (Oct 2014)
- h) *Comprehensive Plan for the Organizational Structure of the U.S. Space Force*, Department of the Air Force Report to Congressional Committees (Feb 2020)
- i) The Official Home Page of the U.S. Air Force, <https://www.af.mil> (Dec 2025)
- j) SCP, *Spacepower - Doctrine for Space Forces* (June 2020)
- k) *United States Space Force*, <https://www.spaceforce.mil/> (Dec 2025)
- l) *Fact Book 2025, United States Special Operations Command*, United States Special Operations Command (2025)
- m) *USSOCOM*, <https://www.socom.mil/> (Dec 2025)
- n) *Air Force Special Operations Command*, <https://www.afsoc.af.mil/> (Dec 2025)
- o) *United States Army Special Operations Command*, <https://www.army.mil/usasoc> (Dec 2025)
- p) *Navy Special Warfare Command*, <https://www.socom.mil/navsoc> (Dec 2025)

1. Background

World War II and its aftermath furnished the impetus for the unification of the Military Departments under a single cabinet-level secretary. Anticipating the needs of a peacetime military organization, an in-depth review by congressional, executive, and military groups began even before the end of the war. Service interests that reflected the opinions of experienced wartime military and civilian leaders with vastly different views of the postwar future influenced the studies. Issues that dominated the search for a consensus included retention of air power in the Navy, maintenance of a separate Marine Corps, and the responsibilities of the new Department of the Air Force.



Figure 1: U.S. Department of War Seal.

The *National Security Act of 1947* was monumental legislation. After almost 50 years that included overseas wartime experience beginning with the Spanish-American War, a modern military organization came into existence. Unification of the Services under a single department was law and the powers of the Secretary of Defense (SecDef) were identified but subject to broad interpretation. The roles and missions of the military Services were defined by Executive Order but would not be statutorily defined until 1958. The act created the National Military Establishment (NME) under the leadership of a civilian secretary and created secretaries for the new Departments of the Army, Navy, and Air Force.

In 1949, the *National Security Act* was amended to change the name of the NME to the Department of Defense (DoD) and to recognize it as an executive department. Further, it changed the role of the Services to Military Departments within DoD. The *DoD Reorganization Act of 1958* strengthened the Secretary of Defense's direction, authority, and control over the department and clarified the operational chain of command from the President and SecDef to the combatant commanders.

The role of the SecDef has changed since the position was established in 1947. Originally, the SecDef had only general authority over the NME, an authority shared with the civilian secretaries of the Military Departments. In 1949, the position was strengthened with the appointment as head of an executive department, reduction of the role of Military Department heads (removing Cabinet-level status), and the assumption of budgeting responsibilities. **Today, SecDef is the principal assistant to the President for all matters relating to the DoD.** SecDef has nearly plenary authority, direction, and control of the entire department. Moreover, the *Goldwater-Nichols DoD Reorganization Act of 1986* makes clear SecDef's position in the operational chain of command.

The *Goldwater-Nichols DoD Reorganization Act of 1986* further refined the organization and division of responsibilities within the DoD and features the following:

- The operational chain of command is clearly established from the President through the SecDef to the combatant commanders.
- Service Chiefs (Chief of Staff of the Army, Commandant of the Marine Corps, Chief of Naval Operations, and Chief of Staff of the Air Force) are responsible for organizing, training, and equipping Service forces. While the Service Chiefs are members of the Joint Chiefs of Staff (JCS), the combat commanders are responsible for the planning and execution of joint operations.
- The Chairman of the Joint Chiefs of Staff (CJCS) is the **principal military advisor** to the President, National Security Council, and the SecDef.
- The Joint Staff is under the exclusive direction of the CJCS. It is organized along conventional staff lines to support the CJCS and the JCS in performing their duties.

On 5 Sept 2025, President Donald J. Trump signed Executive Order 14347 to restore the historic name "Department of War" as a secondary title for the Department of Defense. The Order authorizes the Secretary of Defense, the Department of Defense, and subordinate officials to use secondary titles such as "Secretary of War" and "Department of War" in official correspondence, public communications, ceremonial contexts, and non-statutory documents within the executive branch, and instructs the Secretary of War to recommend actions, to include legislative and executive actions, required to permanently rename the U.S. Department of Defense to the U.S. Department of War.

Note: To avoid confusion and awkward language, the EWSDEP courseware will continue to use the statutory titles used in current references until such legislative and executive actions are completed.

2. Introduction

This lesson will focus on the functions of the DoD, the SecDef, the Departments of the Army, Air Force, the U.S. Special Operations Command (USSOCOM), and the U.S. Coast Guard.

The SecDef is the principal assistant to the President in all matters relating to the DoD. All functions in DoD and its component agencies are performed under the authority, direction, and control of SecDef.

The DoD is composed of the Office of the SecDef (OSD), the Military Departments, the JCS, the Joint Staff, the combatant commands (CCMDs), the Inspector General, agencies/bureaus, field activities, and such other offices and commands established or designated by law, by the President, or by SecDef. For purposes other than the operational direction of the combatant commands, the chain of command runs from the President to the SecDef to the Secretaries of the Military Departments, and as prescribed by the Secretaries, to the commanders of Military Service forces. The functions of the heads of these offices shall be as assigned by SecDef according to existing law.

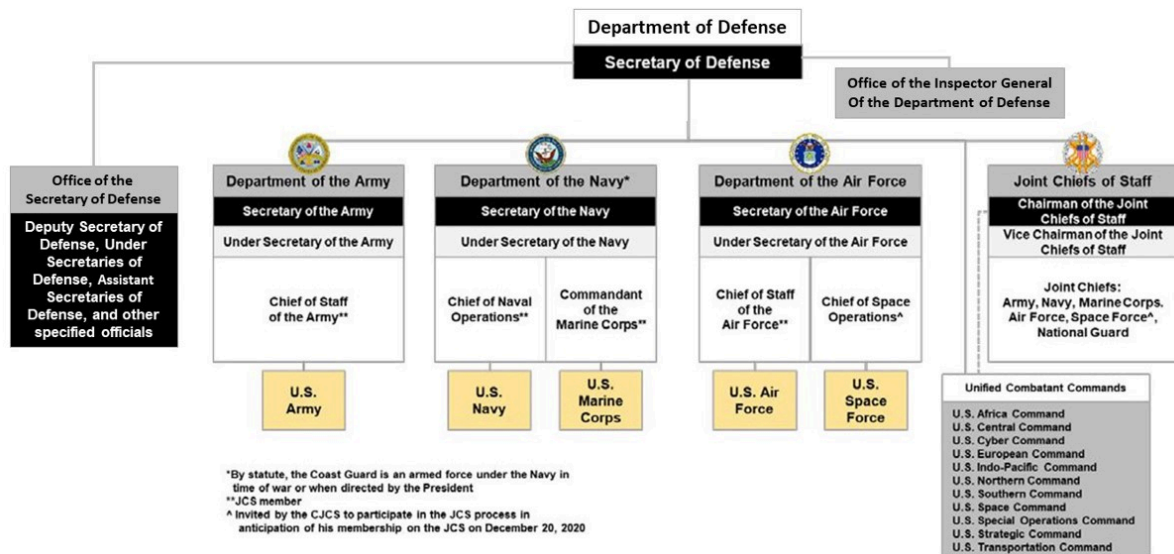


Figure 2: DoD Organizational Structure.

3. Functions of the DoD

As prescribed by higher authority, DoD will maintain and employ Armed Forces to:

- Support and defend the Constitution of the U.S. against all enemies, foreign and domestic.
- Ensure, by timely and effective military action, the security of the U.S., its territories, and areas vital to its interests.
- Uphold and advance the national policies and interests of the U.S.

The functions and responsibilities assigned to the Secretaries of the Military Departments, the Services, the CJCS, the Joint Staff, and the combatant commands are carried out in such a manner as to achieve the following:

- Provide the best military advice to the President and SecDef.
- Effective strategic direction of the Armed Forces.
- Employment of the Armed Forces as a joint force.
- Integration of the Armed Forces into an effective and efficient team.
- Prevention of unnecessary duplication or overlapping capabilities among the Services by using personnel, intelligence, facilities, equipment, supplies, and services of all Services such that military effectiveness and economy of resources will thereby be increased.
- Coordination of Armed Forces operations to promote efficiency and economy and to prevent gaps in responsibility.
- Effective multinational operations and interorganizational and nongovernmental organization (NGO) coordination.

4. Functions of the SecDef

The SecDef is a cabinet member and Head of the DoD, an Executive Department. Pursuant to section 113 of Title 10, *United States Code*, the SecDef is the principal assistant and advisor to the President on Defense matters and serves as the leader and chief executive officer of the Department. The SecDef:

- Exercises authority, direction, and control over the DoD.
- Performs the functions and duties of the office and exercises its powers through, or with the aid of, such persons in, or organizations of, the DoD as he or she designates.
- Provides policy guidance to the Heads of DoD and OSD components for the preparation and review of the program recommendations and budget proposals of their respective components. Such guidance includes:
 - National security objectives and policies.
 - The priorities of military missions.

- Resource levels projected to be available for the period of time for which such recommendations and proposals are to be effective.
- Strategic direction and policy, program, and resources priorities in support of the President's *National Security Strategy* and *Unified Command Plan*. Such guidance will be promulgated in the *National Defense Strategy*, *Quadrennial Defense Review*, and *Quadrennial Roles and Missions Review*, as well as other documents, as appropriate.
- Provides, with the advice and assistance of the Chairman of the Joint Chiefs of Staff, guidelines to direct the effective planning, preparation, resourcing, and execution of military operations to the Secretaries of the Military Departments and to the Commanders of the Combatant Commands.
- Keeps the Secretaries of the Military Departments informed regarding military operations and activities of the DoD that directly affect their respective responsibilities.
- Provides policy guidance to the Department regarding support to civil authorities in an incident of national significance or a catastrophic event, for homeland defense, and for defense support to civil authorities.
- Provides policy guidance to the Department regarding detection and monitoring of all aerial, maritime, land, space, and cyberspace threats to the national security of the United States.

5. Common Functions of the Military Departments

Subject to the authority, direction, and control of the SecDef, the Secretaries of the Departments of the Army, Navy, and Air Force (and in certain cases, the Commander, USSOCOM) are responsible for, and have the authority necessary to conduct, all affairs of their respective Departments, including:

- Recruiting
- Organizing
- Supplying
- Equipping (including research and development)
- Training
- Servicing
- Mobilizing
- Demobilizing
- Administering (including the morale and welfare of personnel)
- Maintaining
- Construction, outfitting, and repairs of military equipment
- Construction, maintenance, and repair of buildings, structures, and utilities, as well as the acquisition, management, and disposal of real property and natural resources

6. Common Functions of the Military Services

The Army, the Navy, the Air Force, the Space Force, the Marine Corps, and the Coast Guard, (when transferred to the Department of the Navy in accordance with sections 2, 3, and 145 of Title 14, *United States Code*) to include the Active and Reserve Components of each, under their respective Secretaries, shall provide conventional, strategic, and special operations forces to conduct the range of operations as defined by the President and the SecDef. Further, they shall perform the following common functions:

- Develop concepts, doctrine, tactics, techniques, and procedures; and organize, train, equip, and provide land, naval, air, space, and cyberspace forces, in coordination with the other Military Services. This includes Combatant Commands, U.S. government (USG) departments and agencies, and international partners, as required, that enable joint force commanders to conduct decisive operations across the spectrum of conflict to achieve the desired end state.
- Determine Military Service force requirements and make recommendations concerning force requirements to support national security objectives and strategy and to meet the operational requirements of the Combatant Commands.
- Recommend to the JCS the assignment and deployment of forces to the Combatant Commands established by the President through the SecDef.
- Monitor and assess Military Service operational readiness and capabilities of forces for assignment to the Combatant Commands and plan for the use of the intrinsic capabilities of the other Military Services and USSOCOM that may be made available.

- Develop doctrine, tactics, techniques, and procedures for employment by Military Service forces and:
 - Assist the CJCS in the development of joint doctrine.
 - Coordinate with the CJCS, the Combatant Commands, the other Military Services, USG departments and agencies, partner security forces, and non-governmental organizations in the development of the doctrine, tactics, techniques, and procedures necessary for participation in and/or command of joint, interorganizational, and multinational operations.
 - Coordinate with the Commander, USSOCOM, in the development of the doctrine, tactics, techniques, and procedures employed by Military Service forces when related to special operations.
- Provide for training for joint operations and joint exercises in support of Combatant Command operational requirements, including the development of Military Service joint training requirements, policies, procedures, and publications.
- Provide logistical support for Military Service and all forces assigned to joint commands, including procurement, distribution, supply, equipment, and maintenance, unless otherwise directed by the SecDef.
- Organize, train, and equip forces to contribute unique service capabilities to the joint force commander to conduct the following functions across all domains, including land, maritime, air, space, and cyberspace:
 - Intelligence, surveillance, reconnaissance (ISR), and information operations, to include electronic warfare and military information support operations to provide situational awareness and enable decision superiority across the range of military operations.
 - Offensive and defensive cyberspace operations to achieve cyberspace superiority in coordination with the other Military Services, Combatant Commands, and USG departments and agencies.
 - Special operations in coordination with USSOCOM and other Combatant Commands, the Military Services, and other DoD Components.
 - Personnel recovery operations in coordination with USSOCOM and other Combatant Commands, the Military Services, and other DoD Components.
 - Counter weapons of mass destruction.
 - Building partnership capacity/security force assistance operations.
 - Forcible entry operations.
 - Missile Defense.
 - Other functions as assigned, such as Presidential support and antiterrorism.
- Organize, train, and equip forces to conduct support to civil authorities in the U.S. and abroad, to include support for disaster relief, consequence management, mass migration, disease eradication, law enforcement, counter-narcotics, critical infrastructure protection, and response to terrorist attack, in coordination with the other Military Services, Combatant Commands, National Guard, and USG departments and agencies.
- Operate organic land vehicles, aircraft, cyber assets, spacecraft or space systems, and ships or craft.
- Conduct operational testing and evaluation.
- Provide command and control.
- Provide force protection.
- Consult and coordinate with the other Military Services on all matters of joint concern.

7. Department of the Army

The Department of the Army includes land combat, service forces, and such aviation, water transport, and space and cyberspace forces organized, trained, and equipped primarily for prompt and sustained land combat, and to support the other Military Services and joint forces. The Army is responsible for the preparation of land forces necessary for the effective prosecution of war and military operations short of war, except as otherwise assigned. The Army is the Nation's principal land force and promotes national values and interests by conducting military engagement and security cooperation; deterring aggression and violence; and compelling enemy behavioral change or compliance, should deterrence fail. The Army contributes forces through a rotational, cyclical readiness model that provides a predictable and sustainable supply of modular forces to the Combatant Commands, and a surge capacity for unexpected contingencies.

a. The Functions of the Army



Figure 3: U.S. Army Service Mark.

In addition to the common military service functions, the Army, within the Department of the Army, shall develop concepts, doctrine, tactics, techniques, and procedures; and organize, train, equip, and provide forces with expeditionary and campaign qualities to perform the following specific functions:

- Conduct prompt and sustained combined arms combat operations on land in all environments and types of terrain, including complex urban environments, to defeat enemy ground forces, and seize, occupy, and defend land areas.
- Conduct air and missile defense to support joint campaigns and assist in achieving air superiority.
- Conduct airborne and air assault, and amphibious operations. The Army has primary responsibility for the development of airborne doctrine, tactics, techniques, and equipment.
- Conduct Civil Affairs Operations (CAO).
- Conduct riverine operations.
- Occupy territories abroad and provide for the initial establishment of a military government pending transfer of this responsibility to another authority.
- Interdict enemy sea, space, air power, and communications through operations on or from the land.
- Provide logistics to joint operations and campaigns, including joint over-the-shore and intra-theater transport of time-sensitive, mission-critical personnel and materiel.
- Provide support for space operations to enhance joint campaigns in coordination with the other Military Services, Combatant Commands, and USG departments and agencies.
- Conduct authorized civil works programs, to include projects for improvement of navigation, flood control, beach erosion control, and other water resource developments in the United States, its territories, and its possessions, and conduct other civil activities prescribed by law.
- Provide intra-theater aeromedical evacuation.
- Conduct reconnaissance, surveillance, and target acquisition.
- Operate land lines of communication.

b. Organization

The Army is composed of two distinct and equally important components: the active component and the reserve components. The reserve components are the U.S Army Reserve and the Army National Guard. Regardless of component, the Army conducts both operational and institutional missions. The operational Army consists of numbered armies, corps, divisions, brigades, and battalions that conduct full spectrum operations around the world. The institutional Army supports the operational Army. Institutional organizations provide the infrastructure necessary to raise, train, equip, deploy, and ensure the readiness of all Army forces. Figure 4 shows the Army command structure.

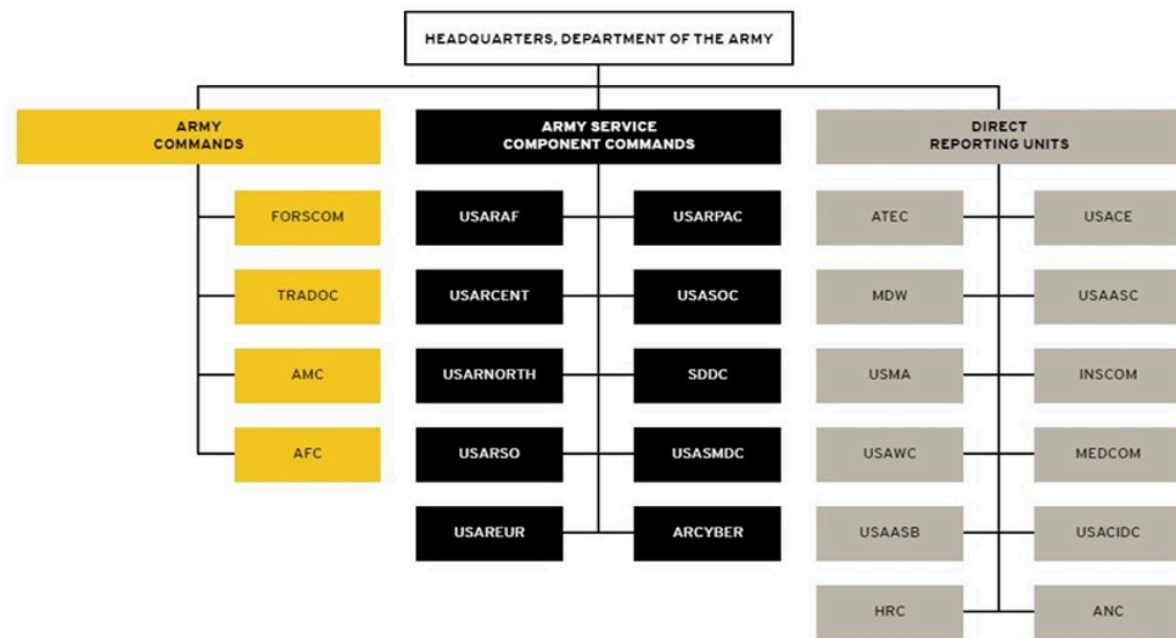


Figure 4: Army Command Structure.

(1) Army Commands

- **(a) U.S. Army Forces Command (FORSCOM), Fort Bragg, NC.** FORSCOM trains, mobilizes, deploys, sustains, transforms, and reconstitutes assigned conventional forces, providing relevant and ready land power to combatant commanders. The Army is also establishing a command element within FORSCOM with the intent to conduct training and readiness oversight of the Security Force Assistance Brigades (SFABs).
- **(b) U.S. Army Training and Doctrine Command (TRADOC), Fort Eustis, VA.** TRADOC recruits, trains, and educates the Army's Soldiers; develops leaders; supports training in units; develops doctrine; establishes standards; and builds the future Army.
- **(c) U.S. Army Material Command (AMC), Redstone Arsenal, AL.** AMC provides superior technology, acquisition support, and logistics to ensure dominant land force capability for Soldiers, the U.S., and our Allies.
- **(d) U.S. Army Futures Command (AFC), Austin, TX.** AFC modernizes the Army for the future; integrates the future operational environment, develops and delivers future force requirements, designs future force organizations, and delivers materiel capabilities.

(2) Army Service Component Commands

- U.S. Army Africa (USARAF), Vicenza, Italy
- U.S. Army Central (USARCEN), Shaw Air Force Base, SC
- U.S. Army North (USARNORTH), Fort Sam Houston, TX
- U.S. Army South (USARSOUTH), Fort Sam Houston, TX
- U.S. Army Europe (USAREUR), Wiesbaden, Germany
- U.S. Army Pacific (USARPAC), Fort Shafter, HI
- U.S. Army Special Operations Command (USASOC), Fort Bragg, NC
- Military Surface Deployment and Distribution Command (SDDC), Scott Air Force Base, IL
- U.S. Army Space and Missile Defense Command/Army Forces Strategic Command (USASMDC/ARSTRAT), Huntsville, AL
- U.S. Army Cyber Command (USARCYBER), Fort Belvoir, VA

Note: For the purpose of this lesson, the U.S. Army Direct Reporting Units are not defined.

c. Distribution of Personnel

Of the nearly one million military personnel serving in the Army as a whole, roughly half are in support units, and a third are in combat units. The rest belong to units that perform various overhead functions, such as recruiting, training, and equipping combat units. The Army's reserve component is slightly larger than its active component, with 54 percent of the service's total personnel.

Since the 1970s, the Army has interpreted the DoD's Total Force Policy—which involves treating a service's various components as a single force—by concentrating combat units in the active component and support units in the reserve component. Since 2017, the Army has planned to have an average of 59 percent of its combat personnel in the active component and 75 percent of its support personnel in the reserve component. The practical effect of that distribution is that the Army has enough support units in its active component to conduct relatively small operations on its own, but larger combat operations usually require it to mobilize a significant number of reservists to provide support for the active-component combat units—as occurred during the occupation of Iraq.

d. Command Levels and Units

(1) Composition

The Army's combat units at any command level contain two to five subordinate units of a similar type, plus additional supporting units. For example, an infantry brigade has:

- Two or three infantry battalions.
- One cavalry squadron.
- One battalion each of special troops, artillery, engineers, and logistics.

Similarly, an infantry battalion has:

- Three infantry companies.
- One heavy weapons company.
- One headquarters company.

That pattern is repeated at lower levels (a company consists of platoons, and platoons consist of squads) and at higher levels (a division consists of brigade combat teams, and a corps consists of divisions), as detailed below. However, some command levels have different names depending on the type of unit; for instance, cavalry squadrons are at the same command level as infantry battalions.

Most support units are assigned to higher command levels, which give them specific missions. A Brigade Combat Team (BCT) does not include the support units that the Congressional Budget Office attributes to it in this analysis—those units are division-, corps-, or theater-level assets that would be deployed to support the BCT and without which the BCT could not function.

(2) Command Levels

The Army is generally organized as hierarchies of units, with each type of unit commanded by a noncommissioned or commissioned officer of a specific rank. Those units are described here from smallest to largest.

(a) Squad/Section. A squad is commanded by a sergeant and has 4 to 12 personnel. A section is a group of vehicles, generally two in number.

(b) Platoon. A platoon is commanded by a second lieutenant and includes varying numbers of subordinate squads or sections. It has 16 to 50 personnel. Heavy platoons have four armored vehicles (such as tanks or infantry fighting vehicles, depending on the type of platoon).

(c) Company/Troop/Battery. A company is commanded by a captain and includes two to five subordinate platoons (usually three or four). It has about 60 to 200 personnel. Heavy companies have 14 armored vehicles. Cavalry companies are called troops; artillery companies are called batteries.

(d) Battalion/Squadron. A battalion is commanded by a lieutenant colonel and usually includes three to five combat companies and one support company. It has about 400 to 1,000 personnel. Heavy battalions have 58 armored vehicles. Cavalry battalions are called squadrons.

(e) Brigade Combat Team/Functional Support Brigade/Regiment/Group. A brigade is commanded by a colonel and is generally configured as either a BCT or a functional support brigade (FSB). A BCT has about 4,400 to 4,700 personnel, depending on whether it is an armored, Stryker, or infantry BCT. An FSB has about 3,000 to 5,000 personnel, depending on its type (of which there are 20). Cavalry units of brigades are called regiments; some types of support brigade size are called groups.

(f) Division. A division is commanded by a major general and includes two to five BCTs (usually four), an aviation brigade, an artillery brigade, an engineer brigade, and a logistics brigade. Divisions have about 12,000 to 16,000 personnel.

(g) Corps. A corps is commanded by a lieutenant general and includes two to five divisions and numerous support brigades and commands. Corps have about 40,000 to 100,000 personnel.

(h) Army. An army is the highest command level in a given theater of operations and typically has 100,000 to 300,000 personnel. It is an element of a joint command structure—the Army's component is commanded by a general. An operational theater is established to support one or more corps (usually two) and includes numerous support brigades and support commands.

(3) Unit Size

A possible source of confusion when discussing Army units is that although combat units generally have a fixed set of subordinate units assigned to them, many support units do not have such a fixed composition. Instead, they are intended to have units assigned to them as the need arises. For example, a combat brigade typically has more than 4,000 personnel assigned to it, but a support brigade might have only about 100 personnel. That difference does not indicate a large variation in size between the two types of brigades; rather, it reflects the fact that the support brigade does not have permanently assigned subordinate units. Support brigades are perhaps better thought of as brigade headquarters, which are company-sized units of about 100 personnel that provide command and control for subordinate support units. Thus, it is important to note whether a given Army unit includes or does not include subordinate units. Similarly, descriptions of the total number of brigades in the Army can be misleading because of differences between BCTs and other types of brigades.

Another possible source of confusion involves differing ways to count the number of personnel in a unit. The size and organization of Army units are based on an official template, the Army's Table of Organization and Equipment for that type of unit. However, actual Army units do not always conform to their template for a variety of reasons—they may not include all of the subordinate organizations, they may be manned at a higher or lower level than 100 percent, or they may be transitioning from one template to another. In recent years, for example, the Army has transitioned many of its BCTs from an older template, with two subordinate maneuver battalions, to the current design, with three subordinate maneuver battalions. When discussing the size of BCTs, this report uses the personnel numbers in the Army's official templates.

(4) Ground Units

The Army is responsible for providing the bulk of U.S. ground combat forces. To that end, the service is organized primarily around BCTs. These are large combined-arms formations designed to contain 4,400 to 4,700 soldiers each and include infantry, artillery, engineering, and other types of units. The Army has 30 BCTs in the active component and 26 in the National Guard (there are none in the Army Reserve). The vast majority of the Army's support units exist to support combat operations by BCTs, and the vast majority of the Army's administrative units exist to create, train, and maintain BCTs and their support units.



Figure 5: Army Stryker Brigade Combat Team.

The current organization of the Army into BCTs is a change from historical practice. Before the mid-2000s, when the service launched a "modularity" initiative, the Army was organized for nearly a century around divisions (which involved fewer but larger formations, with 12,000 to 18,000 soldiers each). During that period, units in Army divisions could be separated into ad hoc BCTs (typically, three BCTs per division), but those units were generally not organized to operate independently at any command level below the division. In the current structure, BCTs are permanently organized for independent operations, and division headquarters exist to provide command and control for operations that involve multiple BCTs. Today, the Army has three types of brigade combat teams, which are roughly analogous to heavy, medium, and light

forces; armored BCTs have large numbers of the heaviest armored vehicles, Stryker BCTs have large numbers of lightly armored vehicles (called Stryker vehicles), and infantry BCTs have few armored vehicles. The Army maintains a mix of BCTs so it can use the type of unit most appropriate for a given military operation.

The Army is distinct not only for the number of ground combat forces it can provide but also for the large number of armored vehicles in its inventory and for the wide array of support units it contains. Those support units include units with significant firepower, such as artillery brigades (which have missile launchers as well as traditional cannon artillery), aviation brigades (which have attack, reconnaissance, utility, or cargo helicopters), and other combat arms (such as Patriot missile launchers to defend against other missiles and aircraft). Army support units include many other types of specialized units, such as construction engineers, military intelligence, military police, and the Army's extensive logistics apparatus. Many of those types of units are responsible for supporting not just Army units in the field but all of the other services in a combat operation. For example, the Army is generally responsible for all theater logistics functions, port operations, and enemy prisoner-of-war detention operations.

Although each type of BCT has its own strengths and weaknesses, the Army's ground forces overall are exceptionally powerful combat units that are generally considered capable of defeating any conventional ground forces—such as other national armies—that they might be expected to fight. The United States has not suffered a serious defeat from other conventional ground forces since 1950 when the Chinese military intervened in the Korean War. Since then, the U.S. Army has consistently been able to overwhelm opponents who have attempted conventional operations against it.

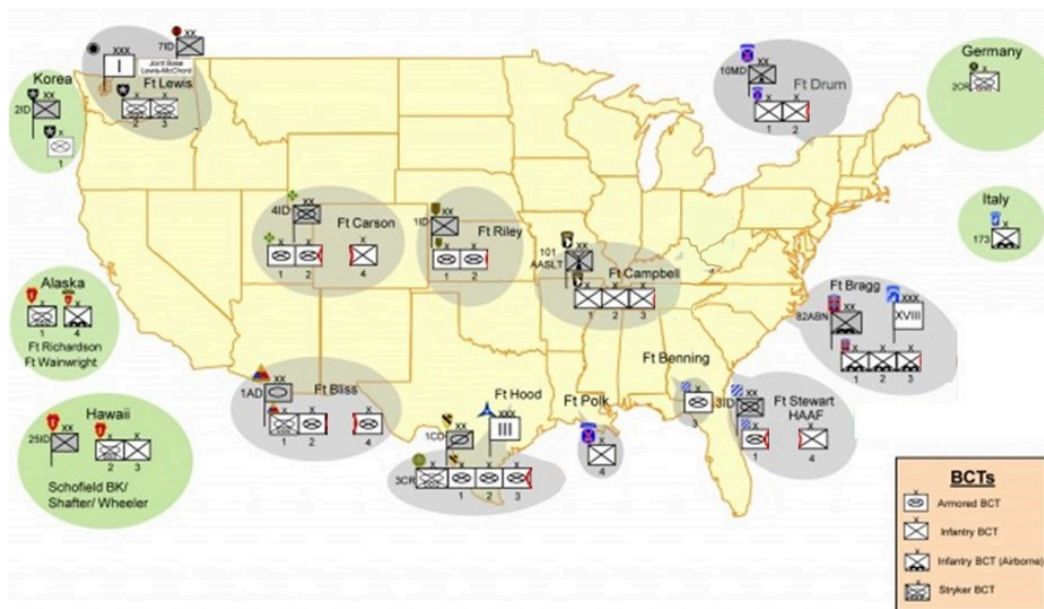


Figure 6: U.S. Army Ground Unit Locations.

(a) Armored BCTs (ABCT). ABCTs are large tactical formations that operate fairly independently. They are designed to include about 4,700 personnel and are equipped with the heaviest and most powerful armored combat vehicles in the U.S. inventory: M1 Abrams series tanks, M2/M3 Bradley series infantry vehicles/scout vehicles, M109 series self-propelled howitzers, and numerous M2- and M113-derived support vehicles. Vehicles such as those—which run on tracks for off-road mobility and are heavily armored to protect against attack—are not assigned to all elements of an armored BCT. Each BCT also has several hundred wheeled vehicles that generally are not armored. Nevertheless, armored BCTs are, by a large margin, the most heavily armed and armored variety of U.S. ground forces.

ABCTs are descendants of the heavy divisions that were intended to defend Europe in the event of a large-scale attack by Soviet forces during the Cold War. Although in recent years the Army has not focused specifically on the ability to destroy opponents' armored vehicles, armored BCTs still have strong anti-armor capability, particularly when supplemented with Army helicopters and other U.S. airpower.

Their equipment and organization have historically been oriented toward high-intensity combat with conventional armored opponents. More recently, the U.S. relied extensively on heavy divisions during Operations DESERT STORM and IRAQI FREEDOM, but it did not use any heavy forces in the invasion of Afghanistan in 2001 (Operation ENDURING FREEDOM). In Iraq, heavy BCTs often operated in a modified configuration without their heavy vehicles, which made them better suited to counterinsurgency and urban operations—an example of the way the Army adapts its units to meet the needs of each operation. See Figure 7.

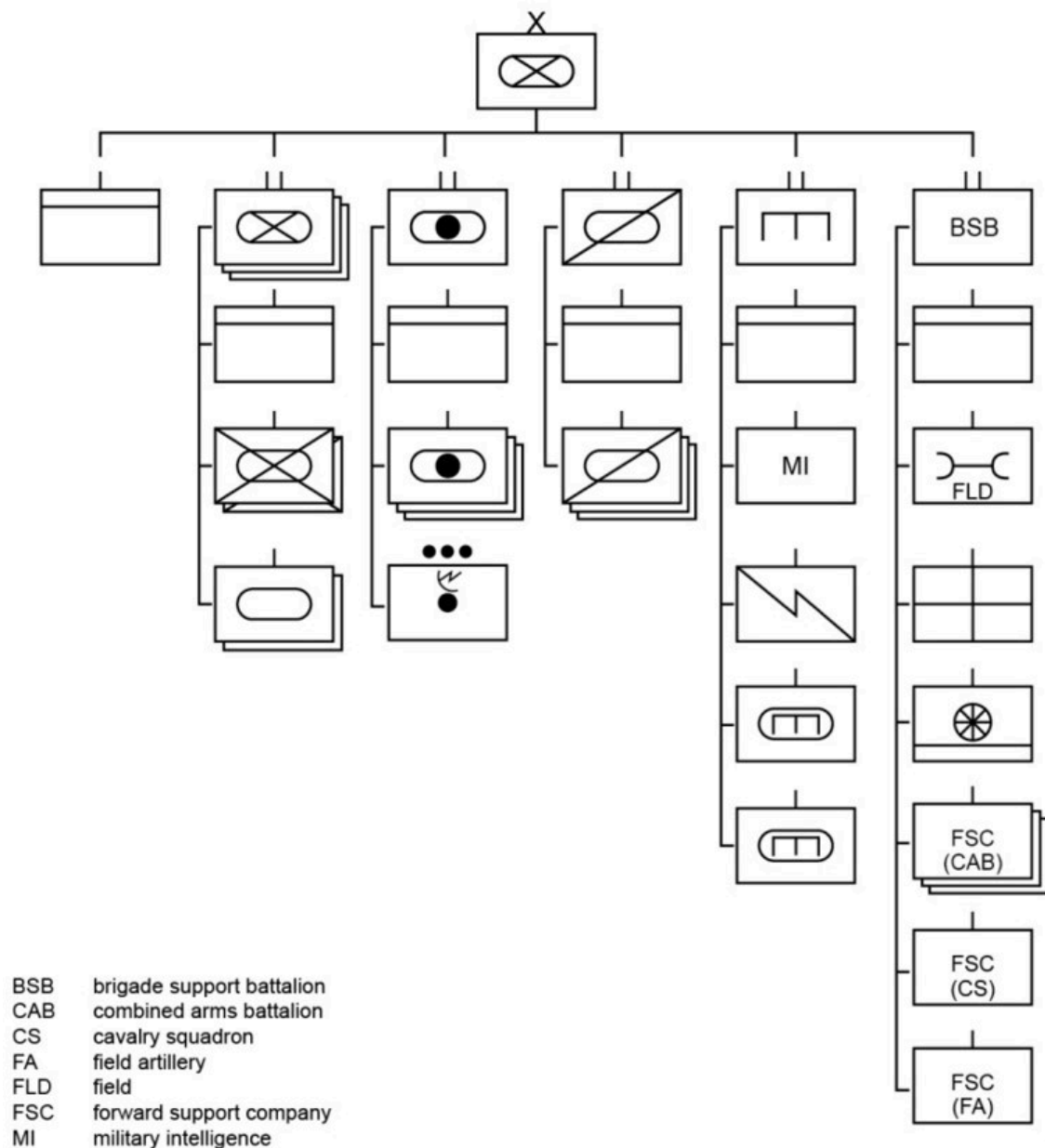


Figure 7: U.S. Army ABCT.

(b) Stryker BCTs (SBCT). Like armored brigade combat teams (BCTs), Stryker BCTs are large tactical formations that can operate relatively independently. However, Stryker BCTs are designed to have about 200 fewer personnel than armored BCTs (approximately 4,500) and are equipped with medium weight, wheeled armored vehicles of the Stryker family, and several hundred unarmored wheeled vehicles. SBCTs provide the Army with more infantry in armored personnel carriers than any other type of brigade combat team.

SBCTs were created as part of a 1999 initiative to transform the Army into a more mobile and responsive force. The *Stryker* family of vehicles was intended to provide a medium-weight force that would be easier to deploy rapidly than heavy forces, but that would have more combat power and ability to move around the battlefield than light forces. Plans at the time called for making Stryker vehicles small and light enough to fit on C-130 transport aircraft. However, combat experience in Iraq has led the Army to improve the armor of most of its vehicles, and Stryker vehicles have become much too heavy to be transported on C-130s. The Stryker force has proved helpful in fighting unconventional forces, such as those in Iraq and Afghanistan. Such operations require large numbers of infantry personnel and benefit when all of those personnel have access to armored transport vehicles—both traits that Stryker BCTs possess. Similarly, the infrastructure in Afghanistan is too poor for the tanks and fighting vehicles of armored BCTs to operate there, but the lighter-weight Stryker vehicles can operate in parts of that country. See Figure 8.

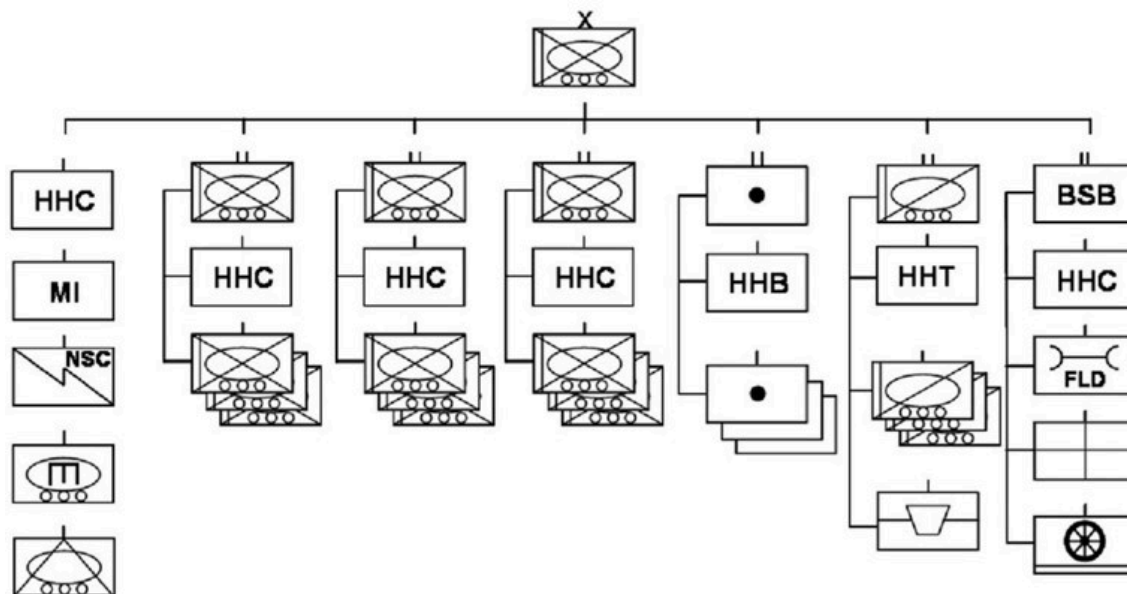


Figure 8: U.S. Army SBCT.

(c) Infantry BCTs (IBCT). IBCTs—the most numerous type of BCT, also commonly called light BCTs—are relatively independent tactical formations that are designed to include approximately 4,400 personnel. Most of those personnel are expected to engage in combat on foot, although each IBCT also has several hundred wheeled, generally unarmored, vehicles assigned to it for transport. Unlike ABCT or SBCTs, IBCTs come in some specialized variants. For example, airborne units (such as the brigades of the 82nd Airborne Division) are specially trained and equipped to drop by parachute from fixed-wing aircraft, and air-assault units (such as the brigades of the 101st Air Assault Division) are given special training and additional supporting helicopters to conduct assaults from rotary-wing aircraft. Because they have the least equipment weight, IBCTs are considered the easiest to deploy of all types of brigade combat teams.

IBCTs are designed to be capable of deploying rapidly to distant locations. However, because they have no armored vehicles and few vehicle-mounted weapons, the Army's light forces lack the protection and combat power of heavy forces. Nevertheless, ICTs have significant firepower, and they are capable of calling on the same array of support assets—such as artillery, attack helicopters, and air strikes—as any other type of BCT. In addition, IBCTs can often operate more effectively than armored forces in such difficult locations as cities, forests, or mountains, where they can derive substantial defensive benefits from the terrain. For those reasons, IBCTs are considered suitable for a wide variety of operations. See Figure 9.

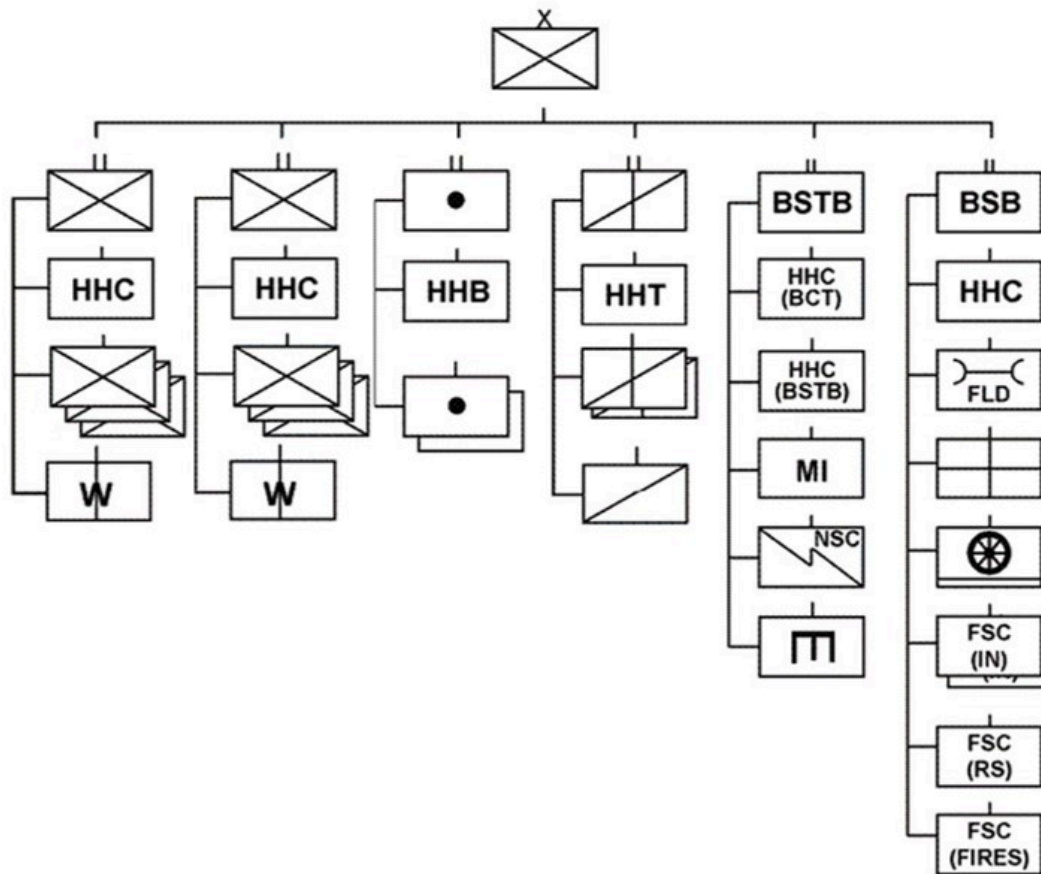


Figure 9: U.S. Army IBCT.

(5) Army Aviation Units

Until the end of World War II, the Army used various types of fixed-wing combat aircraft. After the war, however, the Air Force was spun off as a separate service from the Army. Since then, inter-service agreements have prohibited the Army from using fixed-wing aircraft for combat (although it continues to use them for other purposes, such as reconnaissance and transport). Instead, the Army's aviation brigades rely on helicopters.

In most respects, aviation brigades are similar to other types of supporting forces, but they merit separate treatment because of their visibility and cost, the Army's occasional use of them as independent forces, and the ease of distinguishing them from other supporting forces. The Army maintains aviation brigades in its active and reserve components.

The Army's aviation brigades provide important forms of support in almost all operations involving Army forces. Those brigades include attack helicopters (AH-64 Apaches to attack targets on the ground) and utility and cargo helicopters (UH-60 Blackhawks and CH-47 Chinooks to transport soldiers, equipment, and supplies). For light-infantry forces operating in poor terrain with limited infrastructure—such as portions of Afghanistan—helicopter transportation is often the only practical method of deploying troops to and from combat operations.

(6) Army Special Forces Units

The Army's special-operations forces include the 75th Ranger Regiment, the 160th Special Operations Aviation Regiment, and seven special-forces groups. Those units are trained, equipped, and overseen by the USSOCOM.

Note: These forces will be covered by a separate reading on SOCOM.

8. Department of the Air Force

The Department of the Air Force shall organize, train, equip, and provide air, space, and cyberspace forces for the conduct of prompt and sustained combat operations, military engagement, and security cooperation in defense of the Nation, and to support the other Military Services and joint forces. The Department of the Air Force is composed of the Office of the Secretary of the Air Force, and air, space, and cyberspace forces, both combat and support within the U.S Air Force and U.S Space Force, not otherwise assigned. Figure 10 shows the organization of Department of the Air Force. The senior civilian leadership includes the Secretary of the Air Force and the Under Secretary of the Air Force. The senior military leadership includes the Chief of Staff of the Air Force and Chief of Space Operations.

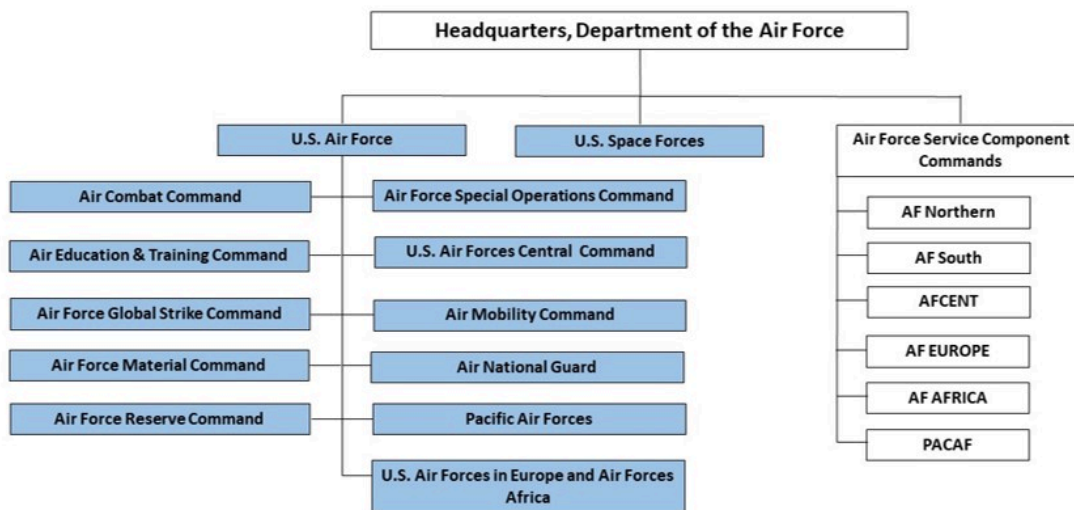


Figure 10: Department of the Air Force Command Structure.

a. U.S. Air Force



Figure 11: U.S. Air Force Seal.

The Air Force is the Nation's principal air and space force and is responsible for the preparation of forces necessary for the effective prosecution of war. The Air Force will provide the Nation with global vigilance, global reach, and global power in the form of in-place, forward-based, and expeditionary forces possessing the capacity to deter aggression and violence by state, non-state, and individual actors to prevent conflict, and, should deterrence fail, prosecute the full range of military operations in support of U.S. national interests.

(1) Functions of the U.S. Air Force

In addition to the common military service functions, the Air Force, within the Department of the Air Force, shall develop concepts, doctrine, tactics, techniques, and procedures and organize, train, equip, and provide forces to perform the following specific functions:

- Conduct nuclear operations in support of strategic deterrence, to include providing and maintaining nuclear surety and capabilities.
- Conduct offensive and defensive operations, to include appropriate air and missile defense, to gain and maintain air superiority and air supremacy as required, to enable the conduct of operations by U.S. and allied land, sea, air, space, and special operations forces.
- Conduct global precision attack, to include strategic attack, interdiction, close air support, and prompt global strike.

- Provide timely, global integrated intelligence, surveillance, and reconnaissance (ISR) capability and capacity from forward deployed locations and globally distributed centers to support worldwide operations.
- Conduct offensive and defensive operations to gain and maintain space superiority to enable the conduct of operations by U.S. and allied land, sea, air, space, and cyberspace forces. Conduct space operations to enhance joint campaigns, in coordination with the other Military Services, Combatant Commands, and USG departments and agencies.
- Provide rapid global mobility to employ and sustain organic air and space forces and other Military Service and USSOCOM forces, as directed, to include airlift forces for airborne operations, air logistical support, tanker forces for in-flight refueling, and assets for aeromedical evacuation.
- Provide agile combat support to enhance the air and space campaign and the deployment, employment, sustainment, and redeployment of air and space forces and other forces operating within the air and space domains, to include joint air and space bases, and for the Armed Forces other than which is organic to the individual Military Services and USSOCOM in coordination with the other Military Services, Combatant Commands, and USG departments and agencies.
- Conduct global personnel recovery operations, including theater-wide combat and civil search and rescue, in coordination with the other Military Services, USSOCOM, and DoD Components.
- Conduct global integrated command and control for air and space operations.

(2) Organization

The U.S. Air Force includes an active component, a reserve component which has two parts—the Air Force Reserve and the Air National Guard—and all federal civilians employed by the Air Force. It is the smallest of the three military departments in terms of both number of personnel and operation and support budget.

The Air Force operates a fleet of aircraft of widely varying sizes that are designed to accomplish a broad array of missions. Types of aircraft unique to the Air Force include long-range bombers, large transport aircraft, and large tanker aircraft. The Air Force also operates a large number of fighter and attack aircraft; aircraft that provide capabilities for airborne command and control, intelligence, reconnaissance, and surveillance (ISR), and electronic warfare (EW); and helicopters and tilt-rotor aircraft for combat rescue and special-operations missions. In addition, the Air Force operates a fleet of unmanned air systems (drones) that can carry equipment for ISR and EW missions as well as weapons to attack ground targets. Because the Air Force's aircraft are expected to operate mainly from established air bases, their designs do not have to give up performance capabilities in exchange for specialized adaptations, such as the ones that enable the Navy's aircraft to operate from aboard ships.

The Air Force is also responsible for most of the military's space systems that provide important support to the entire DoD (such as Global Positioning System satellites). Combat units in the Air Force are generally organized as squadrons of aircraft. Those squadrons vary widely in size—with anything from 8 to 24 aircraft being common—as well as in types of aircraft. Such variation makes it difficult to provide a single measure of force structure for the Air Force similar to an Army brigade combat team or a Navy carrier strike group. The Air Force also includes support units (the vast majority of which are used to support combat operations by aircraft squadrons) and administrative units (almost all of which exist to create or maintain the service's combat units and support units).

Of the nearly half a million military personnel serving in the Air Force as a whole, 29 percent are in support units, and 37 percent are in combat units. The rest belong to units that perform various overhead functions, such as training and maintenance. More than the other services, the Air Force integrates the personnel from its active and reserve components very tightly—in many cases, it is misleading to treat the Air Force as composed of separate active and reserve component units. Many Air Force units are “multi-compo” (multiple component) units, made up of personnel and equipment from both the active and the reserve components. In other cases, equipment assigned to one component may be operated by personnel from the other component. About one-third of the Air Force's aircraft are assigned to the reserve component, which more closely resembles the Army's practice than that of the Navy or Marine Corps. The Air Force's reserve component is also unusual in that its pilots, unlike reservists in the other services, are frequently more experienced than their active-component counterparts. For example, about one-quarter of the Air Force's fleet of C-17 cargo aircraft is assigned to the reserve component. However, cargo aircraft are commonly crewed by personnel from both the active and the reserve components, so it would not be accurate to treat one-quarter of C-17 squadrons as being in the reserve component and the other three-quarters as being in the active component (in actuality, about 90 percent of the personnel assigned to C-17 squadrons are reserve-component personnel).

(3) Command Levels and Units

Today's Air Force typically does not operate with formations larger than squadrons. In the past, the service relied more heavily on wings (groups of three squadrons with 24 aircraft per squadron). It also experimented with a larger formation, called an air expeditionary force, composed of several different types of squadrons. Currently, however, the Air Force generally deploys a group of squadrons organized for a specific mission with higher-level commands, such as wings, used to provide command and control for the deployed squadrons. As noted above, squadron sizes vary greatly, making counts of squadrons a somewhat misleading measure of force structure. Currently, larger aircraft, such as cargo lifters and bombers, are generally grouped into smaller squadrons, whereas tactical aircraft tend to be grouped into larger

squadrons. However, squadron sizes are not standardized even for specific types of aircraft. For example, although fighter aircraft are often described as organized into squadrons of 24 aircraft, the Air Force actually organizes F-16s in squadrons of 15, 18, or 24 aircraft. The U.S. Air Force has the following numbers of flying squadrons:

- 100 Tactical Aviation Squadrons.
- 9 Bomber Squadrons
- 43 Airlift Squadrons.
- 36 Air Refueling Squadrons.
- 30 Unmanned Air System Squadrons.

(4) Air Force Commands

(a) Air Combat Command (ACC), Langley AFB, VA. ACC's mission is to support global implementation of the national security strategy. ACC operates fighter, bomber, reconnaissance, battle-management, and electronic-combat aircraft. It also provides command, control, communications, and intelligence systems and conducts global information operations.

(b) Air Education and Training Command (AETC), Joint Base San Antonio, TX. AETC recruits, trains, and educates quality people for the aerospace force and the nation. AETC provides basic military training, initial and advanced technical training, flight training, and professional and degree-granting professional education.

(c) Air Force Global Strike Command (AFGSC), Barksdale AFB, LA. The mission of the AFGSC is nuclear and conventional global strike, a key component of strategic deterrence. AFGSC aligns intercontinental ballistic missiles (ICBMs) and dual-mission capable bomber forces under a single command that is responsible for organizing, training, and equipping ICBM forces, B-2 and B-52 bomber forces, and other deterrence capabilities in support of combatant commanders.

(d) Air Force Material Command (AFMC), Wright-Patterson AFB, OH. AFMC develops, acquires, and sustains the aerospace power needed to defend the U.S. This is accomplished through management, research, acquisition, development, testing, and maintenance of existing and future weapons systems and their components.

(e) Air Force Reserve Command (AFRC), Robbins AFB, GA. AFRC provides citizen airmen to defend the U.S. and protect its interests through aerospace power.

(f) Air Mobility Command (AMC), Scott AFB, IL. AMC provides airlift, air refueling, special air mission, and aeromedical evacuation for U.S. forces.

(g) Air National Guard (ANG), Joint Base Andrews, MD. The ANG is the Air Force component of the National Guard.

(h) Air Force Special Operations Command (AFSOC), Hurlburt Field, FL. AFSOC provides combat search and rescue (CSAR), agile combat support, information warfare, precision aerospace fires, psychological operations, and specialized aerospace mobility and refueling to combatant commanders. AFSOC will be discussed further within the USSOCOM portion of the reading.

(5) Air Force Service Component Commands

- Air Forces Africa (AF Africa), Ramstein Air Base, Germany
- U.S. Air Forces in Europe (AF Europe), Ramstein Air Base, Germany
- U.S. Air Forces Central Command (AFCENT), Shaw ADB, SC
- U.S. Air Force Northern Command (AFNORTH), Tyndall AFB, FL
- U.S. Air Force Southern Command (AFSOUTH), Davis-Monthan AFB, AZ
- Pacific Air Forces (PACAF), Joint Base Pearl Harbor-Hickam, HI

(6) Strengths and Limitations of U.S. Air Forces

Each type of aircraft has its own strengths and weaknesses, but overall, Air Force squadrons are exceptionally powerful units. Very few other countries' air forces have sufficient combat power to consider challenging U.S. control of the air; in many of the conflicts that the U.S. has engaged in over the past few decades, opponents have chosen to safeguard their air forces by keeping them grounded for the duration of the conflict. In addition, few nations currently have ground-based air defenses capable of seriously hindering U.S. air operations. The U.S. has faced only limited competition from hostile fighter aircraft since 1950 (when China intervened in the Korean War), and it has been able to overcome every opposing country's air-defense system. In the majority of U.S. conflicts since World War II, U.S. air forces have been able to operate essentially at will, either from the beginning of the conflict or a short time thereafter, once the opponent's air defenses had been destroyed.

The U.S. has historically had a lower threshold for using air and naval forces in combat than for using ground forces. Although flexibility and response time have made aircraft carriers a commonly used option for conducting aerial attacks in small interventions, Air Force aircraft have played a role in almost every U.S. conflict since the service was created. Through international agreements, the U.S. has access to an extensive network of air bases around the world. In addition, the Air Force's tanker fleet is capable of extending the range of Air Force aircraft to allow attacks on almost any possible hostile country. Air Force squadrons can also be deployed more quickly than ground forces, and their ability to fly at high speeds to distant locations allows them to put virtually any location at risk of attack (provided that its air defenses have been sufficiently degraded or can be avoided).

(7) Tactical Aircraft

Tactical aircraft, which make up the majority of the Air Force's combat fleet, consist of relatively small aircraft designed to engage in air-to-air combat (fighters), to strike targets on the ground (attack aircraft), or both (multirole aircraft, which the Air Force designates as fighters). Tactical aviation accounts for about 33 percent of the Air Force's total operation and support funding.

In the past, most types of tactical aircraft were highly specialized for either air-to-air or air-to-ground combat. Today, those two forms of combat are still the main roles for the Air Force's tactical aviation fleet, but the most numerous type of aircraft in the fleet is a multirole aircraft (the F-16). Only a small portion of the tactical aviation fleet consists of purely attack aircraft (A-10s). Moreover, the Air Force's newest air-to-air fighter (the F-22) was designed with some ground-attack capability. The emphasis on multirole aircraft is likely to continue in the future with the introduction of the F-35, which was designed primarily to attack ground targets but has air-to-air capability as well. Despite their versatility, multirole fighters are most likely to be used for specific missions according to their individual strengths. For example, F-22 fighters are considered best suited to perform the most difficult air-to-air combat missions, and F-16s and F-35s are best suited to carry out ground-attack missions.

(a) A-10 Thunderbolt II. Nicknamed the "*Warthog*," A-10 attack aircraft have almost no air-to-air combat ability; they were designed mainly to provide air support for friendly ground forces (by attacking hostile ground forces engaged in combat). The A-10 is noteworthy for its large cannon, a 30-millimeter (mm) Gatling gun designed for attacking armored combat vehicles. (By comparison, other types of Air Force tactical aircraft have a 20 mm Gatling gun.) A-10s have good visibility from the cockpit and can fly relatively slowly, factors that give pilots an excellent view of the battlefield they are supporting. However, in recent years, the DoD proposed retiring the A-10 fleet, arguing that those aircraft cannot withstand modern air defenses and are too expensive to maintain in the force.



Figure 12: Fairchild Republic A-10 *Thunderbolt II*.

(b) F-15 Eagle. F-15 fighter aircraft come in several versions, including the C model ("*Eagle*"), intended mainly for air-to-air combat, and the E model ("*Strike Eagle*"), intended mainly for ground-attack missions. Until the introduction of the F-22, the F-15C was the Air Force's primary vehicle for achieving air superiority in a theater of operations; it is still considered a highly capable fighter plane. The F-15E model is a relatively large strike aircraft—by the standards of tactical aviation—with a fairly long range and large capacity for carrying bombs and extra fuel.



Figure 13: McDonnell Douglas F-15C Eagle.

(c) F-16 Fighting Falcon. F-16 fighters are the most numerous aircraft in the Air Force's tactical aviation fleet. Originally designed as a low-cost air-to-air fighter that could operate only during daylight hours, the F-16 has evolved into a very effective multirole fighter that can operate at any time of the day. F-16s are relatively small and lightweight, with a correspondingly limited range and payload capacity. Part of the F-16 fleet has been upgraded with specialized equipment for attacking and suppressing enemy air-defense systems.



Figure 14: General Dynamics F-16C Fighting Falcon.

(d) F-22 Raptor. F-22 fighters are the Air Force's newest aircraft designed specifically for air-to-air combat. They incorporate "stealth" design characteristics that make them difficult to observe with radar, and they are generally considered the most capable air-to-air combat aircraft being fielded by any nation. The F-22 was initially designed with limited ground-attack capability, but the Air Force has been modifying the aircraft to improve that capability.



Figure 15: Lockheed Martin F-22 *Raptor*.

(e) F-35A Lightning. The F-35A, the Air Force's variant of the *Joint Strike Fighter*, is currently in production and has been in service since 2017. The Air Force plans to buy 1,763 F-35s to replace the A-10 and F-16 as the Air Force's main tactical strike platform. The largest improvement it provides is stealth; it will give the Air Force a large fleet of hard-to-observe strike aircraft. The F-35A will also be capable of air-to-air combat, although not to the same degree as the F-22. Capabilities that the F-35A will not offer are a cannon comparable to that of the A-10 and the slow flying speed useful for finding and attacking ground targets.



Figure 16: Lockheed Martin F-35A *Lightning II*.

The Air Force's tactical aircraft have been used extensively in almost every conflict in which the U.S. has taken part since the 1940s. Likewise, most potential scenarios for future conflicts are likely to include the heavy use of tactical aviation. In general, tactical aircraft are responsible for securing U.S. control of the air (by destroying an opponent's air forces and air defenses) and for supporting U.S. war efforts by attacking ground targets. In a few cases, such as the enforcement of "no-fly zones," securing U.S. control of the air is the sole mission. That mission is overwhelmingly the responsibility of Air Force tactical aviation.

(8) Bomber Aircraft

The Air Force's bomber fleet has two main roles: delivering nuclear weapons and performing strikes with conventional weapons. Historically, the Air Force viewed the delivery of nuclear weapons as the primary purpose of long-range bombers, with conventional strikes as a secondary role. However, events since the collapse of the Soviet Union have generally increased the emphasis on conventional strike missions for the bomber fleet. One of the Air Force's three types of long-range bombers, the B-1B, is no longer capable of delivering nuclear weapons and is now devoted entirely to conventional strike missions. In addition, many of the Air Force's B-52s are slated for conversion to a conventional-only configuration to comply with the New START arms control treaty.

Bombers account for about 10 percent of the Air Force's total operation and support funding. Unlike tactical aviation, bombers are large aircraft that can travel long distances and loiter above an area for an extended period without refueling (characteristics referred to as endurance) and can deliver a large payload of munitions. Those capabilities make bombers especially well-suited to performing strike missions—their long range allows them to be based relatively far from the theater of operations (freeing up space in closer air bases for shorter-range aircraft); their loitering time lets them remain in an area longer, allowing them to respond more rapidly to requests from ground forces for air support; and their large load of munitions enables them to provide substantial air support before needing to return to bases to rearm.

The enormous weapons payload of the bomber fleet allows it to contribute a very substantial share of the U.S. military's capability to strike targets, despite its relatively small numbers. For example, a B-1B can carry 84 500-pound bombs in a single sortie, whereas an F-16 could carry 12, although an F-16 typically flies more sorties per day and thus could deliver those 12 bombs more often. However, the Air Force can capitalize on bombers' large payloads only on missions in which enough targets can be identified to use the number of weapons carried.

(a) B-52 Stratofortress. B-52s are the oldest of the Air Force's bombers, dating to the 1960s. The Air Force plans to keep them in service at least through 2040. B-52s have the ability to carry a great variety of weapons and have the longest unrefueled endurance of the Air Force's bomber fleet. Because of their age, however, B-52s would probably have trouble penetrating modern air-defense systems and thus are best suited to operating in undefended airspace or to delivering cruise missiles from outside defended airspace.



Figure 17: Boeing B-52H Stratofortress.

(b) B-1B Lancer. The B-1B fleet is younger than the B-52 fleet, having been built in the 1980s. Although B-1Bs were designed to deliver nuclear weapons, the U.S. modified them to remove that capability in order to comply with arms control treaties. Today, B-1Bs are intended only to perform conventional strikes. Although they incorporate some features that make them harder to observe than B-52s, they are not considered as capable of surviving in hostile airspace as the more recent B-2s. Nevertheless, the Air Force has used B-1Bs to conduct air strikes in hostile airspace in recent operations—the B-1B fleet delivered more bombs in Operation IRAQI FREEDOM than any other type of aircraft—albeit often with support from other aircraft.



Figure 18: Rockwell B-1B Lancer.

(c) B-2 Spirit. B-2s are the newest and most modern U.S. bombers. Built in the late 1980s and the 1990s, they are notable for the extensive stealth design features that help them penetrate hostile airspace undetected, and they are considered more difficult to target and attack than other U.S. bombers. However, unlike with other bombers, the Air Force is reluctant to deploy B-2 squadrons to bases overseas, preferring to have them conduct strikes directly from their base in Missouri. Two reasons, according to the Air Force, are the planes' demanding maintenance requirements (associated with the special radar-absorbing coating on the outside of the aircraft) and the need for atmospherically controlled hangars. Nevertheless, the B-2 can be deployed overseas, if necessary, and has been on occasion. In practice, flying most missions from U.S. bases means that B-2 sorties are extremely long and demanding, which limits the number of sorties that the small B-2 fleet (16 aircraft) can conduct to those in which stealth is most essential.



Figure 19: Northrop Grumman B-2 Spirit.

Air Force bombers have been employed with increasing frequency in modern U.S. conflicts. Their use was relatively limited in Operation DESERT STORM—B-52s delivered cruise missiles during the initial wave of strikes and conducted some bombing missions afterward—but at the time, the Air Force still saw bombers as primarily dedicated to nuclear missions. Since then, with the collapse of the Soviet Union, bombers have been used in larger roles in more conflicts. For example, the B-1B fleet was first employed for conventional air strikes during the 1990s enforcement of no-fly zones over Iraq; later it was used during operations in Kosovo, in Operations ENDURING FREEDOM and IRAQI FREEDOM, and in the subsequent occupations of Afghanistan and Iraq. The B-2 fleet was first employed for conventional strikes in Kosovo and was also used during Operations ENDURING FREEDOM and IRAQI FREEDOM. B-52s have often been mentioned as being particularly useful during the occupations of Afghanistan and Iraq because their large fuel load allows them to remain on station, waiting for requests for fire support, for long periods.

(9) Cargo Aircraft

The Air Force's fleet of cargo aircraft exists to "airlift" personnel and equipment between or within theaters of operations. Inter-theater transport is generally conducted by the larger, longer-range, and more expensive C-5 and C-17 aircraft. Intra-theater transport is usually performed by the smaller, shorter-range, and less expensive C-130 aircraft, although the C-17 was designed to operate from shorter runways, making it an option for transport missions between theaters as well. Cargo aircraft account for about 16 percent of the Air Force's total operation and support funding.



Figure 20: Lockheed C-130E *Hercules*.



Figure 21: Lockheed C-5 *Galaxy*.



Figure 22: McDonnell Douglas/Boeing C-17 *Globemaster III*.

To supplement its airlift capabilities, the Air Force runs a program called the Civil Reserve Air Fleet (CRAF). Under that program, U.S. civilian air carriers that operate certain models of aircraft receive preferential access to air transport contracts with the DoD; in return, those carriers allow the Air Force to use their aircraft for military transport missions in times of conflict. The CRAF program ensures that the Air Force has a large reserve of transport aircraft available in situations in which it may need more airlift capability than its own fleet can provide. Most eligible U.S. civilian airlines participate in the CRAF program, which generally gives the Air Force access to an additional 400 intertheater transport aircraft and 100 intra-theater transport aircraft (although the numbers vary over time). Because CRAF aircraft are designed for civilian use, they are not suitable for certain military missions, such as transporting the largest armored vehicles. But for some purposes, such as carrying passengers, CRAF aircraft are frequently a better alternative in times of conflict than the Air Force's transport aircraft.

The primary advantage of moving cargo and passengers by air is that it is much faster than transport by sea. In many scenarios for possible conflicts, the use of air transport would let U.S. forces reach a theater of operations within a day, rather than the weeks that sea transport might require. In addition, aircraft can move supplies to almost any portion of the globe, whereas many theaters of operations (such as Afghanistan) are far from the sea and would require additional land transportation to move personnel and cargo from ports to the theater. Even in an ongoing operation, the speed and responsiveness of air transport can be extremely valuable in providing logistics support—for example, being able to bring in crucial supplies on a day's notice is preferable to needing a month's notice.

The Air Force's cargo aircraft have been employed extensively in every U.S. conflict in the modern era. Notable examples include the use of those aircraft to rapidly deploy elements of the 82nd Airborne Division to Saudi Arabia in 1990 after the Iraqi invasion of Kuwait and the parachuting of special-forces personnel into Afghanistan in 2001 during the early phases of U.S. operations there. The U.S. military has relied especially heavily on air transport throughout its operations in Afghanistan because that country is landlocked, with the closest access to seaports being in neighboring Pakistan.

Most of DoD's potential scenarios for future conflicts envision heavy reliance on air transport. DoD has set several goals over the years for the amount of air transport capability it needs. The analytic measure generally used to assess the capacity of the airlift fleet is ton-miles per day (the ability to transport 1 ton of cargo 1 mile every day). That measure can be difficult to translate into numbers of aircraft because it depends greatly on the characteristics of a given scenario. In general, however, because the U.S. military's ability to transport cargo to a theater of operations is more likely to be limited by the infrastructure in that theater than by the number of aircraft in the Air Force's inventory, a larger inventory of cargo aircraft would allow the U.S. to support more operations simultaneously or to reduce reliance on CRAF aircraft. Conversely, a smaller inventory of cargo aircraft would either lessen the Air Force's ability to support large operations in multiple theaters simultaneously or require greater reliance on CRAF aircraft.

(10) Tanker Aircraft

The tanker fleet exists primarily to refuel the Air Force's other aircraft while they are in flight. Although the fleet was originally established to refuel strategic bombers on long-range nuclear strike missions into the Soviet Union, tankers have proved valuable for refueling tactical aircraft in almost every U.S. operation of the post-Cold War era. In addition, all of the Air Force's tankers are capable of transporting cargo as a secondary mission. Tanker aircraft account for about 14 percent of the Air Force's total operation and support funding.

Without aerial refueling, tactical aircraft would typically have ranges of only a few hundred miles, so they would have to be based close to their areas of operations, would have less ability to loiter in a location for very long during a mission, and in some cases would have to reduce the weight of the weapons they carried. By contrast, with aerial refueling, the endurance (range and loitering time) of tactical aviation is limited largely by pilots' endurance, and aircraft can be fully loaded with weapons. Those differences increase the utility of tactical aircraft during a conflict in various ways:

- In many theaters, infrastructure constraints limit how many tactical aircraft the U.S. can deploy near an area of operations. Aerial refueling expands the number of bases from which tactical aircraft can reach a given area, allowing the U.S. to use more tactical aircraft in a conflict than it could otherwise.
- An aircraft's fuel consumption increases when it carries a heavy load of weapons; aerial refueling can reduce the need to make trade-offs between the number of weapons an aircraft can carry and the distance it can carry them.
- In many types of missions, it is beneficial for tactical aircraft to be able to loiter, on call, until needed so they can respond more rapidly to requests from ground forces for air support. Aerial refueling can enhance the U.S. military's effectiveness in those types of missions by allowing tactical aircraft to loiter for longer periods.
- In some large theaters, tactical aircraft would be unable to reach distant targets at all without aerial refueling. Bombers are larger than tactical aircraft and have longer ranges, but aerial refueling offers some of the same benefits to bomber missions. For example, B-2 bombers require specialized basing infrastructure that makes them difficult to deploy overseas. But with aerial refueling, B-2 bombers can strike targets anywhere in the world from their base in Missouri.

One limitation of the current aerial refueling fleet is that its tankers are large and slow, with few defenses. During a conflict in which the U.S. had not yet neutralized an opponent's fighter aircraft, tankers would be vulnerable to attack. In practice, however, the U.S. has not faced any major aerial threats since the end of the Cold War, so that limitation has not been significant.

Another drawback of the U.S. tanker fleet results from the use of two different, incompatible methods of aerial refueling. The Navy and Marine Corps employ "probe and drogue" refueling systems on their tankers, fixed-wing aircraft, and rotary-wing aircraft, whereas the Air Force employs a "boom" refueling system on its tankers, tactical aircraft, and bombers. Many Air Force tankers are also equipped to allow for probe-and-drogue refueling, so they can refuel tactical aircraft from the Navy and Marine Corps during operations. However, the need to accommodate both systems in joint operations requires the Air Force to equip some tankers to make them capable of both methods—at a higher cost than would be necessary otherwise—and to coordinate to ensure that the correct types of tankers are assigned to support the correct types of aircraft.

The Air Force's tanker aircraft have been used extensively in every major U.S. conflict since the 1960s. Tankers were especially important in operations such as the invasion of Afghanistan, in which the U.S. had very limited access to air bases near the area of operations, so aerial refueling was vital to enable the Air Force's tactical aircraft and the Navy's carrier aircraft to attack targets in the theater. Many of the DoD's potential scenarios for future conflicts also envision heavy reliance on aerial refueling.

Although the Air Force's tanker fleet is large, it tends to be quite old. The bulk of the fleet consists of KC-135s built in the 1950s and 1960s. (Until the end of the Cold War and Operation Desert Storm in 1991, the Air Force mainly saw tankers as useful for supporting a nuclear attack on the Soviet Union rather than for supporting tactical aviation in ongoing conflicts.) Leaders of the Air Force have often stated that KC-135s are too old and need to be replaced immediately, but many analysts have suggested that those tankers are in good enough shape to continue serving for many years. Consequently, the major issue relating to the future of the tanker fleet is not its size but the speed with which the Air Force should replace the KC-135 with the new KC-46, which is undergoing operational testing.



Figure 23: Boeing KC-135 Stratotanker.

(11) Unmanned Aircraft Systems (UASs)

Unmanned air systems account for about 6 percent of the Air Force's total operation and support funding and are expected to decline as the Air Force retires its MQ-1s.



Figure 24: General Atomics MQ-1 Predator.

The Air Force's unmanned aircraft are used primarily for surveillance. In addition, MQ-1s and MQ-9s can be armed with a few missiles or small bombs to conduct limited strike operations. An example of that capability is the U.S.'s well-publicized use of unmanned aircraft to kill suspected terrorists in Pakistan, Yemen, Somalia, and other countries.



Figure 25: General Atomics MQ-9 Reaper.

Today's drones have several advantages: They are generally less expensive to buy than manned aircraft, they can fly very long missions without being limited by the endurance of human aircrews, and they can operate without putting a pilot at risk of injury, capture, or death. Disadvantages of drones include their vulnerability to air defenses and the lack of a human onboard to address split-second issues in ways that might not be possible by a remote operator. Not all of those factors are inherent to unmanned systems; rather, they have resulted from the state of available technology and from specific choices about what capabilities the military needed during the past decade and a half—the span over which most of today's drones were purchased.

In their current configuration, most of the Air Force's unmanned aircraft are intended to operate mainly in undefended airspace and would generally not be capable of surviving engagements with modern air defenses. Thus, they would have limited utility in a high-intensity conventional conflict; they are most useful in low-intensity and unconventional conflicts, such as the occupations of Iraq and Afghanistan and counterterrorism missions.

The U.S. has had small numbers of unmanned aircraft for many decades, but the widespread deployment of highly capable unmanned air systems is a fairly recent phenomenon. The MQ-1 and RQ-4 were developed in the 1990s and fielded in the 2000s, and the MQ-9 was developed in the 2000s and fielded in the 2010s. Despite their recent introduction, those unmanned aircraft have been used heavily in recent operations, particularly in the war on terrorism and the occupations of Iraq and Afghanistan. Although efforts to arm unmanned surveillance aircraft began before the invasions of Afghanistan and Iraq, the current widespread practice of arming drones to attack ground targets appears to have evolved from their extensive use in those conflicts. Mounting weapons on an unmanned surveillance aircraft has proved to be particularly useful in counterinsurgency and counterterrorism operations because it has enabled DoD to attack small, mobile targets as soon as they are detected and identified without having to summon another aircraft to carry out the attack (such "fleeting" targets would often be lost before the strike aircraft could arrive). For missions requiring substantial firepower, however, the strike capacity offered by today's drones, though useful, is minor compared with that of tactical aircraft or bombers.

b. U.S. Space Force



Figure 26: U.S. Space Force Seal.

In direct response to the launch of the *Sputnik 1* Soviet spacecraft, pioneers in the U.S. military orchestrated a series of milestones that propelled America's status as the preeminent spacefaring Nation. Launched from Cape Canaveral Air Force Station on 31 January 1958, the U.S. Army spacecraft *Explorer 1* was the first American spacecraft to orbit the Earth. Two months later, the U.S. Navy successfully orbited *Vanguard 1*, demonstrating the first solar powered spacecraft. That same year, the Advanced Research Project Agency launched and successfully demonstrated the world's first communications spacecraft. These milestones paved the way for Project *Corona*, a collaboration between the U.S. Air Force and the Central Intelligence Agency. Launched on 18 August 1960 as part of Project *Corona*, the *Discoverer 14* spacecraft returned surveillance imagery of over 1,650,000 square miles of Soviet Union territory. Through these successive accomplishments, the U.S. military established that humanity could control and exploit space in pursuit of prosperity and security.

Since these early milestones, the U.S. Army, U.S. Navy, and U.S. Air Force have each developed space capabilities that enhance landpower, seapower, airpower, and cyberpower, respectively. This decentralized context has shaped U.S. military thinking about space for the last 60 years. A product of this institutional structure is that current military theory and doctrine almost exclusively focus on space as an adjunct to other forms of military power without capturing the direct and independent impact space has on U.S. prosperity and security. The doctrine presented in the following chapters elevates spacepower as a distinct formulation of military power on par with landpower, seapower, airpower, and cyberpower.

The *National Defense Authorization Act* for Fiscal Year 2020 established the U.S. Space Force as a new branch of the Armed Forces within the Department of the Air Force. Establishing the U.S. Space Force is a strategic imperative for our Nation. The U.S. relies on space for its military power projection and economic vitality, and the DoD (DoW) must be prepared to address growing threats from potential adversaries. Establishing the Space Force will elevate the role of space in national defense and transform how DoD organizes, trains, and equips space forces in order to prepare for new security challenges in an era of great power competition.

(1) Space Operations

Space is a unique physical domain, contiguous only with the air domain, but interconnected with all domains. This shapes how humanity accesses and operates in space. Our ability to conduct operations that access, exploit, and defend space rests on simultaneous action across all domains, but also requires serious consideration of the space domain's unique physical, network, and cognitive dimensions.

(a) Physical Dimension. The *physical dimension* of the space domain encompasses the orbital environment and the spacecraft operating within the domain. This dimension starts in the upper reaches of Earth's atmosphere, intersecting and extending beyond the physical location required for sustained orbital flight. Gravity shapes the invisible terrain of the space domain. The perpetual force of gravity makes constant natural motion the defining characteristic of the space domain's physical dimension. No object in orbit occupies a fixed position. Even a geostationary spacecraft "fixed" over an equatorial longitude travels over 164,000 miles in a 24-hour period.

Once placed in orbit, most spacecraft never return to Earth. Military spacecraft are forward employed for the duration of their lifespan. This makes onboard expendables and consumables (e.g., fuel) and spacecraft reliability the primary determinants of mission duration. Terrestrial perspectives on warfare use attributes like transit time, range, and endurance to describe the reach of military operations. The orbital perspective of military power describes the reach of military operations based on access windows, revisit rate, mission lifespan, survivability relative to threat systems, and the tradeoffs between time, position, and total energy.

Space was once a sanctuary from attack, but the emergence, advanced development, and proliferation of a wide range of demonstrated counterspace weapons by potential adversaries has reversed this paradigm. Today, space, like all other domains, is realized to be contested due to the increasing threat to orbiting assets by adversary weapons systems. There is no forward edge of the battle area behind which military spacecraft can reconstitute and recover. Spacecraft remain in orbit through peace and war where they are potentially at risk from adversary counterspace capabilities and the hostile space environment.

(b) Network Dimension. The *network dimension* of space operations allows users to command, control, and exploit space capabilities through a physical and logical architecture that collects, transmits, and processes data around the world and across the domain. Because of these dependencies, cyberspace operations within this network dimension are a crucial and inescapable component of military space operations and represent the primary linkage to the other warfighting domains. These dependencies can also create avenues of enemy attack that offer lower costs and a higher chance of success than orbital warfare within the space domain only.

Nodes and links are the fundamental components of the network dimension.

- **Nodes.** Nodes are elements of the space architecture capable of creating, processing, receiving, or transmitting data. Mission ground systems, control antennas, user equipment, space observation sites, and spacecraft payloads are examples of key space domain nodes.
- **Links.** transport data between nodes. In addition to terrestrial networks, the electromagnetic spectrum (EMS) is a vital link for all space architectures. Orbital spacecraft downlink their data, receive commands, and transmit telemetry through the EMS. Additionally, active

and passive EMS applications allow sensors to monitor, detect, track, and characterize resident space objects. Because of the prevalence of remote operations, the EMS is the primary conduit through which the control and exploitation of the space domain is achieved.

(c) Cognitive Dimension. The space domain's cognitive dimension encompasses the perceptions and mental processes of those who transmit, receive, synthesize, analyze, report, decide, and act on information coming from and to the space domain. Every domain of human affairs includes a cognitive component. The prevalence of remote operations amplifies the importance of the space domain's cognitive dimension. While space is a physical location, operations are executed and interpreted through virtual stimuli. Ultimately, space systems are tools that extend the ability of an individual or group to perform tasks in, from, and to the space domain. Space systems are not static systems; they are designed, employed, and exploited by thinking agents. All these associated processes include cognitive components that shape and define human activity in the space domain.

A complete understanding of military spacepower must encompass the domain's physical, network, and cognitive dimensions, among others. These dimensions are not sequential segments in a serial architecture. Instead, this model of space operations recognizes the simultaneous and interrelated influence all three dimensions have on military spacepower. These dimensions are tightly coupled, and space operations must deliberately consider and affect all three to realize the benefits of spacepower.

(2) Employment of Space Forces

Military space forces conduct prompt and sustained space operations, accomplishing three cornerstone responsibilities.

(a) Preserve Freedom of Action. The U.S.'s ability to project and employ national power is predicated on access to space. Therefore, unfettered access and freedom to operate in space is a vital national interest. Assuring freedom of operation in space is a fundamental role of the U.S. military and, specifically, the U.S. Space Force. Preserving Freedom of Action describes a strategic condition where a nation or sovereign actor has the relative level of control or ability required to accomplish all four components — diplomatic, informational, military, and economic — of their implicit or explicit space strategy. Freedom of action becomes an operational imperative in peace and war, and space security becomes a critical mission across the conflict continuum.

Freedom of action requirements are assessed relative to national strategy. At the political level, military freedom of action may be of limited value should the U.S. lose the ability to employ the other three instruments of power through the space domain. This insight demands that the military concept of preserving freedom of action in space also supports the other components of national spacepower. Because of this holistic approach, military space forces must maintain a strategic perspective and appreciate domain requirements that extend beyond the use of military force.

The terms space parity, space superiority, and space supremacy describe relative degrees of advantage between two or more adversaries.

- **Space parity** describes any condition where no force derives a relative advantage over another at a given time.
- **Space superiority** is a relative degree of control in space of one force over another that would permit the conduct of its operations without prohibitive interference from the adversary while simultaneously denying their opponent freedom of action in the domain at a given time.
- **Space supremacy** implies that one side could conduct operations with relative impunity while denying space domain freedom of action to an adversary. Space supremacy is not always desirable, or attainable against a peer adversary, and should not be the unconditional goal of military spacepower. A rival may wish for their adversary to maintain some space capabilities to reduce the probability of a strategic miscalculation or larger escalation in the conflict and to allow certain elements of national infrastructure to continue without disruption, such as medical services and communication between friendly and adversary decision-makers to broker a de-escalation.

(b) Enable Joint Lethality and Effectiveness. The ability to control and exploit the space domain is an essential component of modern warfare. Military spacepower allows for the rapid dissemination of information on a global scale. Information can be collected and delivered to austere environments without terrestrial infrastructure. Capabilities such as precision attack, maneuver warfare, strategic warning, and global power projection are fundamental to modern warfare. These capabilities must be protected, and military space forces must rely on military operations in the other domains to do so.

Given the vital and interdependent nature of military spacepower within the Joint Force, military space forces must comprehensively and effectively integrate space capabilities into Joint training, planning, and operations. Maximizing Joint lethality and effectiveness requires a cadre of military space forces that are deliberately prepared to integrate spacepower across the range of national and Joint operations. Said differently, the U.S. Space Force must be Joint-smart from its inception, and it must help produce a space-smart Joint Force.

(c) Provide Independent Options. Because nations can generate and apply national power from space, actions in the domain can directly affect a nation's decision calculus. Therefore, a central tenet of military spacepower is the ability to independently achieve strategic effects. In this capacity, military spacepower is more than an adjunct to landpower, seapower, airpower, and cyberpower. Across the conflict continuum, military spacepower provides national leadership with independent military options that advance the nation's prosperity and security. Military space forces achieve national and military objectives by operating in, from, and to the space domain.

Operations in the space domain provide a diverse set of options for national leadership. Actions as benign as the launch or repositioning of a space capability can assure international partners or signal U.S. resolve to strategic competitors. Any nation or non-state actor that depends on space as a source of diplomatic, informational, military, or economic power is vulnerable to military spacepower's coercive potential.

The coercive value of military spacepower is not limited to great power competition. Military spacepower may still have a coercive impact on nations or actors who lack significant space dependencies by projecting power from the space domain. The ability to legally transcend the most remote and protected national boundaries provides a unique opportunity to enable lethal and non-lethal effects against terrestrial targets. Additionally, the specter of global vigilance incentivizes adversaries to adapt their behavior and expend time, energy, and resources to mask or obscure sensitive events. By strengthening our ability to detect and attribute malign action, or using space-based National Technical Means for arms control verification, military spacepower makes deterrence more credible. Military spacepower's ability to enhance the lethality and effectiveness of the Joint Force provides an asymmetric advantage that contributes to conventional deterrence.

Finally, military space forces can shape the security environment through space domain collaboration and security cooperation. Building a partner's spacepower capacity provides non-violent opportunities for strategic effects. These relationships promote U.S. interests by encouraging our partners to act in support of U.S. objectives. By strengthening our alliances and military partnerships, security cooperation makes deterrence more credible and effective. Through these actions, military space forces can reinforce the confidence our partners place in their relationship with the U.S.

(3) Core Competencies

Core Competencies represent the broad portfolio of capabilities military space forces need to provide successfully or efficiently to the Nation. The U.S. Space Force executes five Core Competencies:

(a) Space Security. U.S. prosperity and economic security increasingly rely on the peaceful use of space. Space security protects these interests by establishing conditions for the safe and secure access to space for civil, commercial, Intelligence Community (IC), and multinational partners. Space Security is a presence mission that helps assure partners that the U.S. military is positioned to monitor and protect their interests. Ultimately, Space Security seeks to encourage partners, not compel an adversary; however, if necessary, Space Security includes protecting these mission partners from dangerous or illicit actions. In this regard, combat forces provide a deterrent role for Space Security. Space Security may also include sharing information and domain awareness, developing self-protection capabilities, coordinating anomaly resolution support, maneuver de-confliction, EMS monitoring, launch vehicle ridesharing, protecting lines of communication and national space commerce, and building partner capacity through combined training and exercises.

(b) Combat Power Projection. Combat power projection ensures freedom of action in space for the U.S. and its Allies and, when necessary to defend against threats, denies an adversary freedom of action in space. Combat power – the force available and, when necessary, employed to protect, defend, or defeat threats – is the purest form of military power and serves a distinct role in the pursuit of strategic objectives. Military space forces project combat power for defensive and offensive purposes. Defensive operations enhance control by protecting and preserving U.S. freedom of action in the space domain. When warranted, offensive operations are designed to achieve a relative advantage by negating an adversary's ability to access or exploit the space domain and are therefore essential to achieving space superiority.

- **Defensive operations** protect and preserve friendly space capabilities before, during, or after an attack. Defensive operations are further divided into active and passive actions. Active defensive operations encompass actions to destroy, nullify, or reduce the effectiveness of threats holding friendly space capabilities at risk. Although this may entail reactive operations after an adversary has initiated an attack, active defense also includes proactive efforts to seize the initiative once an attack is imminent. While active defense directly attempts to interdict the employment of capability against friendly space missions, passive defense attempts to improve survivability through system and architectural attributes. Passive defense measures include spacecraft maneuverability; self-protection; disaggregation; orbit diversification; large-scale proliferation; communication, transmission, and emissions security; camouflage, concealment, and deception; and system hardening across all three segments of the space architecture.
- **Offensive operations** target an adversary's space and counterspace capabilities, reducing the effectiveness and lethality of adversary forces across all domains. Offensive operations seek to gain the initiative and may neutralize adversary space missions before they can be employed against friendly forces. Offensive operations are not limited to adversary counterspace systems and can also target the full spectrum of an adversary's ability to exploit the space domain, which includes targets in the terrestrial and cyber domains.

(c) Space Mobility and Logistics (SML). SML is the movement and support of military equipment and personnel into the space domain, from the space domain back to Earth, and through the space domain. The ability to control and exploit the space domain always begins with physical access to orbit. SML starts with the ability to launch military equipment into the proper orbit in a safe, secure, and reliable manner. During conflict, space launch must be dynamic and responsive, providing the ability to augment or reconstitute capability gaps from multiple locations. Today, SML is largely uncontested, though the history of warfare highlights that this condition will not last. Military forces must therefore prepare to defend physical access to the space domain – a key focus of defensive operations and the need for military space forces to be prepared to project combat power.

(d) **Information Mobility.** Information Mobility is the timely, rapid, and reliable collection and transportation of data across the range of military operations in support of tactical, operational, and strategic decision making. Information Mobility includes point-to-point communications; broadcast communications; long-haul communication links; protected strategic communications; machine-to-machine interfaces; position, navigation, and timing; nuclear detonation detection; missile warning; and intelligence, surveillance, and reconnaissance.

Global power projection introduces information collection and transportation requirements, which space is uniquely postured to deliver. Information Mobility extends lines of communication into the most desolate and remote areas of human activity. Military users capitalize on this perspective to share information beyond their line-of-sight, synchronizing power projection on a global scale and across all warfighting domains.

(e) Space Domain Awareness (SDA). SDA encompasses the effective identification, characterization, and understanding of any factor associated with the space domain that could affect space operations and thereby impact the security, safety, economy, or environment of our Nation. SDA leverages the unique subset of intelligence, surveillance, reconnaissance, environmental monitoring, and data sharing arrangements that provide operators and decision makers with a timely depiction of all factors and actors — including friendly, adversary, and third party — impacting domain operations. Furthermore, SDA must be predictive, synthesizing facts and evidence into an assessment of possible and probable future outcomes.

Fundamentally, SDA is a big data challenge. The U.S. Space Force must have the ability to collect, synthesize, fuse, and make sense of extremely large volumes of data from all sources to ensure the U.S. Space Force's ability to have domain awareness. As a digital Service, the U.S. Space Force must leverage its personnel, Allies, civil and industry partners, and big data toolsets to reveal patterns, trends, and associations, especially relating to human behavior and interactions that provide the required SDA. SDA extends across the physical, network, and cognitive dimensions of space operations. Operating in the physical environment of space requires a timely awareness of space weather, lighting conditions, and gravitational topology. In addition to these natural phenomena, military space forces must also maintain awareness of spacecraft orbiting in the domain. This includes active spacecraft and debris. Moreover, when tracking active spacecraft, SDA captures more than orbital trajectory. Complete SDA also includes mission related details such as missions, intentions, system capabilities, patterns-of-life, and the status of consumables and expendables.

(4) Spacepower Disciplines

Seven spacepower disciplines have emerged as necessary components of military spacepower theory: orbital warfare, space electromagnetic warfare, space battle management, space access and sustainment, military intelligence, engineering/acquisition, and cyber operations. As space warfare develops and evolves, additional disciplines will certainly emerge. However, these initial disciplines are the skills the U.S. Space Force needs when developing its personnel to become the masters of space warfare. The combined integration of the seven spacepower disciplines arms the military space forces with the intellectual framework required to perform our core competencies.

(a) Orbital Warfare. Knowledge of orbital maneuver as well as offensive and defensive fires to preserve freedom of access to the domain. Skill to ensure U.S. and coalition space forces can continue to provide capability to the Joint Force while denying that same advantage to the adversary.

(b) Space Electromagnetic Warfare. Knowledge of spectrum awareness, maneuver within the spectrum, and non-kinetic fires within the spectrum to deny adversary use of vital links. Skill to manipulate physical access to communication pathways and awareness of how those pathways contribute to enemy advantage.

(c) Space Battle Management. Knowledge of how to orient to the space domain and skill in making decisions to preserve mission, deny adversary access, and ultimately ensure mission accomplishment. Ability to identify hostile actions and entities, conduct combat identification, target, and direct action in response to an evolving threat environment.

(d) Space Access and Sustainment. Knowledge of processes, support, and logistics required to maintain and prolong operations in the space domain. Ability to resource, apply, and leverage spacepower in, from, and to the space domain.

(e) Military Intelligence. Knowledge to conduct intelligence-led, threat-focused operations based on the insights. Ability to leverage the broader Intelligence Community to ensure military spacepower has the ISR capabilities needed to defend the space domain.

(f) Engineering and Acquisition. Knowledge that ensures military spacepower has the best capabilities in the world to defend the space domain. Ability to form science, technology, and acquisition partnerships with other national security space organizations, commercial entities, Allies, and academia to ensure the warfighters are properly equipped.

(g) Cyber Operations. Knowledge to defend the global networks upon which military spacepower is vitally dependent. Ability to employ cyber security and cyber defense of critical space networks and systems. Skill to employ future offensive capabilities.

9. USSOCOM

a. Special Operations Forces (SOF)



Figure 27: USSOCOM Emblem.

SOF are intended to be versatile forces, capable of conducting a wide range of missions, including those that other military units would not be suited for. Among their multiple roles, the most important are considered to be direct action, special reconnaissance, foreign internal defense, and security force assistance. The last two activities involve helping friendly governments improve their military capabilities (often in order to defeat insurgencies hostile to the U.S.); those missions generally require the largest commitments of SOF personnel and time. Thus, special-operations forces could be described as an exceptionally well-trained and well-equipped set of trainers for foreign militaries—capable, when needed, of performing combat roles as well.

SOF have numerous limits on their use, which relate to the extremely difficult missions they are often assigned. For example, direct-action missions generally require very good intelligence, as well as a situation in which a small force, operating with the benefit of surprise, can achieve a highly valuable objective. Even so, direct-action missions have a mixed record of success; USSOCOM was created in the 1980s largely in response to the failure of special forces (special forces will be discussed later in the reading) to rescue U.S. hostages in Iran.

When special-operations forces are performing their more common role of training foreign militaries, their effectiveness is limited by their host countries' willingness and ability to make use of that training. In general, it is difficult to assess how well a foreign country would combat an insurgency with or without the assistance of U.S. special-operations forces. Insurgencies are generally ended not through military force but through negotiated settlements; however, having a strong military often helps a government persuade insurgents to negotiate and strengthens the government's position during the negotiations.

Another limitation associated with using SOF is that, because they often assist countries that have relatively unstable or unpopular governments, their work risks associating the U.S. with the actions of those countries' militaries, as happened in El Salvador in the 1980s.

Many of the missions for which special-operations forces are intended—as well as many of their past and current operations—are classified. A common complaint of both the SOF and intelligence communities is that because of the classified nature of their work, their failures are more visible than their successes, giving the public a distorted view of their value.

SOF have participated in all major U.S. combat operations since USSOCOM was created. In most cases, their participation was not central to the outcome of those combat operations (largely because their role was limited to providing reconnaissance or carrying out small missions within the larger operation). However, in Operation Enduring Freedom in Afghanistan, SOF units played a leading role in the initial phases of ground combat by assisting Afghan rebel forces by calling in air strikes; conventional U.S. ground forces arrived only after the Taliban had lost control of much of the country. Since the invasion, SOF have been used extensively in and around Afghanistan, achieving a notable success with the direct-action mission of killing Osama bin Laden but experiencing more mixed results when employed as light infantry (as at Tora Bora). SOF have also been widely used for activities other than major combat operations. Some of the largest commitments of U.S. special-operations forces for foreign internal defense and security-force assistance have occurred in El Salvador, Colombia, Iraq, Afghanistan, the Philippines, and, more recently, the Horn of Africa and Trans-Saharan Africa. None of the foreign governments that received such assistance have been militarily overthrown by insurgents or terrorists, although some remain unstable. However, the government of Mali was overthrown by members of the country's military twice since U.S. assistance began, weakening the government in its fight against insurgents and exposing the U.S. to criticism about the effectiveness of its training. Some SOF commitments have also opened the U.S. to criticism because of the actions of the foreign militaries it has assisted (particularly those in Latin America). USSOCOM and other DoD sources frequently describe SOF as crucial for antiterrorism missions. In essence, such missions are the same as traditional SOF missions except that the adversaries are terrorist

groups rather than insurgents or other countries' militaries. Many of the SOF operations in countries mentioned above were antiterrorist missions. Special-operations forces have also participated in a wide variety of smaller missions, such as helping to evacuate noncombatants during a crisis or providing humanitarian assistance or disaster relief.

b. USSOCOM Mission

USSOCOM develops and employs fully capable SOF to conduct global Special Operations and activates, as part of the joint force to support persistent, networked, and distributed global combatant commands' operations and campaigns against state and non-state actors to protect and advance U.S. policies and objectives.

c. USSOCOM Operations

- Civil Affairs Operations.
- Counterinsurgency.
- Counterterrorism.
- Countering Weapons of Mass Destruction (WMD).
- Direct Action.
- Foreign Humanitarian Assistance.
- Foreign Internal Defense.
- Hostage Rescue and Recovery.
- Military Information Support Operations.
- Security Force Assistance.
- Special Reconnaissance.

d. Title 10, USC Responsibilities

Although USSOCOM is a functional combatant command, it has responsibilities very similar to those of the Military Departments. These include:

- Develop special operations strategy, doctrine, and tactics.
- Prepare and submit budget proposals for special operations forces.
- Exercise authority, direction, and control over special operations expenditures.
- Train assigned forces.
- Conduct specialized courses of instruction.
- Validate requirements.
- Establish requirement priorities.
- Ensure interoperability of equipment and forces.
- Formulate and submit intelligence support requirements.
- Monitor special operations officers' promotions, assignments, retention, training, and professional military education.
- Ensure special operations forces' combat readiness.
- Monitor special operations forces' preparedness to carry out assigned missions.
- Develop and acquire special operations-peculiar equipment, materiel, supplies, and services.
- Command and control of U.S.-based special operations forces.
- Provide special operations forces to the geographic combatant commanders.
- Activities specified by the President or SecDef.

e. Relationship Between USSOCOM and the Military Services

Each military service recruits personnel for its special operations units, provides their initial training, and pays their salaries. USSOCOM provides those units with specialized training and equipment. USSOCOM also develops doctrine and strategy for special-operations units and is responsible for ensuring that all U.S. special-operations forces can be used in a unified way by a combatant commander (as opposed to having separate special-operations communities in each service that operate in their own ways and focus on their own limited missions).

SOF include any unit that falls under the USSOCOM. Army Special Operations Command, Naval Special Warfare, Air Force Special Operations Command, and Marine Special Operations Command are all included under this umbrella.

(1) U.S. Army Special Operations Command (USASOC)

USASOC is among the most diverse organizations in the U.S. military, bringing a broad range of competencies and disciplines to support geographic combatant commanders and ambassadors/chiefs of mission worldwide. Established at Fort Bragg, North Carolina, on 1 December 1989, to enhance the readiness of Army SOF, USASOC also functions as the Army component of the USSOCOM. USASOC mission is to prepare for any conflict and arm the joint force with the world's premiere Army SOF. Since 11 September 2001, few elements of the U.S. military have been more involved in the War on Terror than USASOC. In Operation ENDURING FREEDOM, Operation IRAQI FREEDOM, and numerous other hotspots around the world, USASOC Soldiers have been among the first forces to deploy in support of U.S. and coalition force objectives. Some of these Soldiers are now on their 13th or 14th combat rotation. With an allocated strength of 33,000 personnel, USASOC commands active-duty forces and is responsible for training and equipping Special Forces and support units in the Army National Guard.

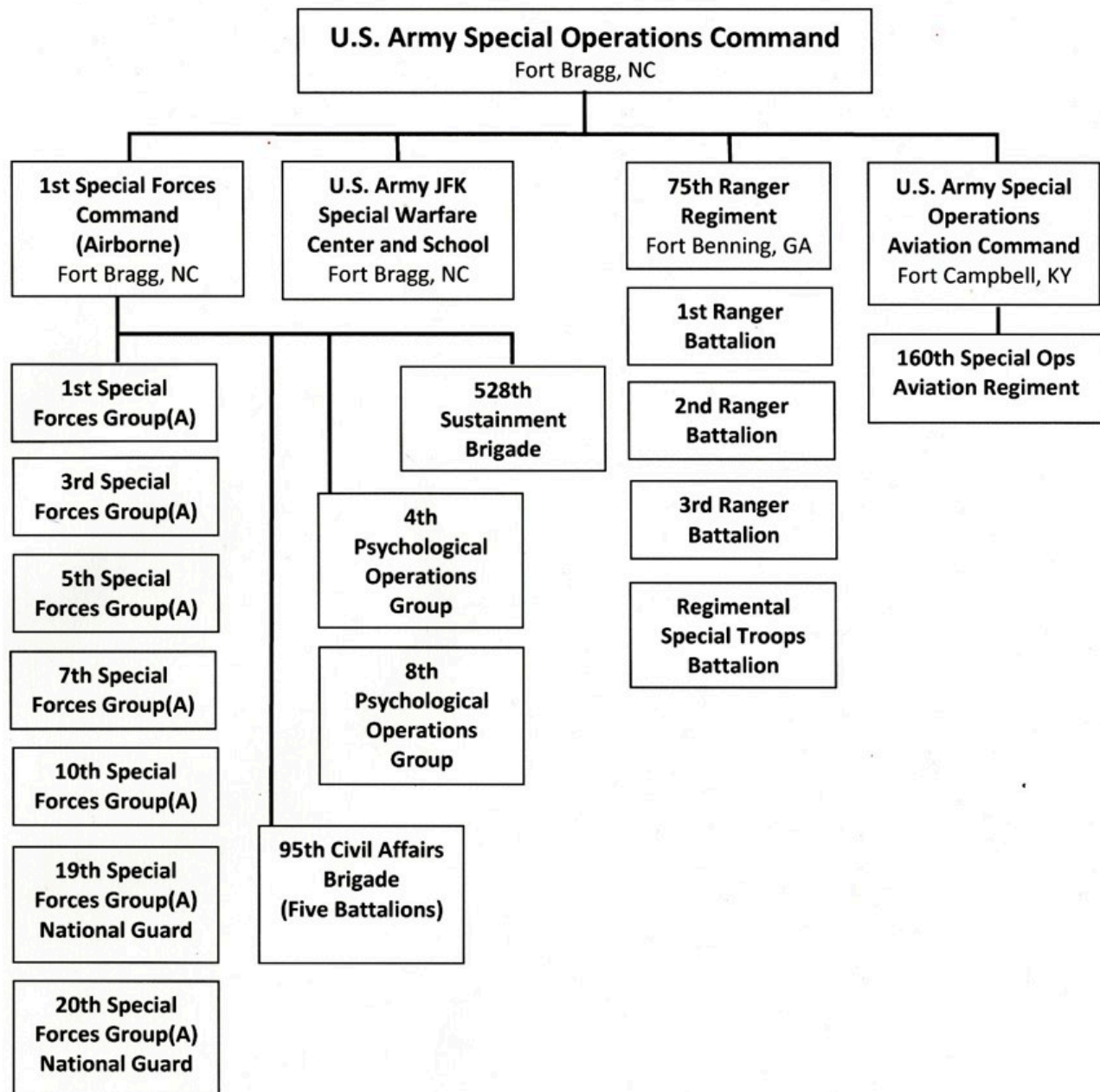


Figure 28: U.S. Army Special Operations Command.

The U.S. Army major commands under USASOC are:

(a) 1st Special Forces Command (Airborne). 1st Special Forces Command (Airborne) organizes, equips, trains, validates, and deploys forces to conduct special operations across the spectrum of conflict in support of USSOCOM, regional commanders, American ambassadors, and other governmental agencies through an indigenous approach, precision targeting operations, crisis response, and developing understanding and wielding influence. 1st SFC (A) is a division-sized element with corps-like responsibilities that consists of 11 subordinate brigades and more than 22,000 personnel. The command is stationed and deployed in more than 70 countries. 1st SFC (A) is composed of four distinct regiments, which include Special Forces (discussed below), Civil Affairs, Psychological Operations, and Sustainment elements. These regiments enable Army Special Operations Forces to provide a holistic and varied approach to emerging threats.

(b) U.S. Army John F. Kennedy Special Warfare Center and School (SWCS). The SWCS at Fort Bragg, N.C., is the Army's Special Operations Center of Excellence, managing and resourcing professional growth for Soldiers in the Army's three distinct special-operations branches: Special Forces, Civil Affairs, and Psychological Operations. SWCS builds the only force specifically trained and educated to shape foreign political and military environments in order to prevent war. Special-operations Soldiers do this by working with host nations, regional partners, and indigenous populations in a culturally attuned manner that allows them to bridge language barriers, open lines of communication, and connect with key political and military leaders in a way that is both immediate and enduring.

(c) U.S. Army Special Operations Aviation Command - Airborne (USASOAC[A]). USASOAC(A) includes the 160th Special Operations Aviation Regiment - Airborne (SOAR) (A). 160th SOAR (A)'s mission is to organize, equip, train, resource, and employ Army special operations aviation forces worldwide in support of contingency missions and ground force commanders. Known as *Night Stalkers*, these Soldiers are recognized for their proficiency in nighttime operations. They are highly trained and ready to accomplish the very toughest missions in all environments, anywhere in the world, day or night, with unparalleled precision.

The 160th SOAR (A) is composed of a regiment headquarters and five battalions. The regiment headquarters is collocated with the 1st and 2nd battalions and the Special Operations Aviation Training Battalion at Fort Campbell, KY; 3rd Battalion is located at Hunter Army Airfield, GA; and 4th Battalion is located at Joint-Base Lewis-McChord, WA. This strategic organizational structure postures the regiment to support special operations forces, mission and training requirements well into the future.

Each warfighting battalion also has a strategic composition of light, medium, and heavy helicopters, all highly modified and designed to meet the unit's unique mission requirements. Currently, 1st Battalion has one AH-6 *Little Bird* helicopter company, one MH-6 *Little Bird* helicopter company, and two companies of MH-60 *Black Hawk* helicopters; 2nd Battalion has two MH-47 *Chinook* helicopter companies; and 3rd and 4th Battalions each have two MH-47 *Chinook* helicopter companies and one MH-60 *Black Hawk* helicopter company. Each battalion also has a Headquarters and Headquarters Company and a maintenance company.

(d) 75th Ranger Regiment. The Regiment's four battalions, located throughout the U.S., can deploy anywhere in the world for no-notice missions. Their capabilities include direct action raids in limited visibility, adverse weather, varied terrain, and complex operating environments to capture or kill designated targets and/or seize terrain and strategic installations. Capable of infiltrating by land, sea, or air, the 75th Ranger Regiment is trained on a wide variety of mobility platforms and operates fully integrated with supporting agencies and other Special Operations Forces as required. Typical Ranger missions include:

- Airfield Seizures.
 - Establish air-landing capabilities for follow-on forces.
 - Establish trans-load sites for precious cargo.
 - Conducted by using both special operations and light infantry tactics, techniques, and procedures.
- Special Operations Raids.
 - Strategic asset—high payoff targets.
 - Destruction or recovery missions.
 - Routinely operate within restrictive rules of engagement.
- Urban Combat.
 - Advanced Military Operations in urban terrain techniques and breaching capabilities.
 - Precision marksmanship.
 - Advanced combatives (hand-to-hand) training.



Figure 29: U.S. Army Special Forces Branch Insignia.

(e) Special Forces. The DoD has traditionally distinguished between “special forces” (SF) and SOF. SF refer to a very specific unit: the 1st Special Forces Regiment, one of the four regiments within the USASOC. The members of the regiment are most often referred to by their distinctive headgear, the Green Beret.

The Army's SF are capable of a wide variety of missions, to include:

- Unconventional warfare.
- Foreign internal defense (FID).
- Special reconnaissance.
- Direct Action (DA).
- Counterterrorism.
- Counter-proliferation.
- Information operations.

As an example of a classic unconventional warfare mission that happened in recent history, after the terror attacks of 11 September 2001, small elements SF embedded with indigenous fighters from Afghanistan’s Northern Alliance and led them into battle. Within a matter of weeks, they had effectively neutralized the Taliban threat – accomplished not with brigades and divisions of soldiers, but with only a couple dozen Special Forces soldiers. This is the capability that the 1st Special Forces Regiment brings to the table and makes them unique in the larger USSOCOM picture.

The 1st Special Forces Regiment is organized into Special Forces Groups (SFG), each focused on a specific geographic area or area of responsibility.

The regular (Active Duty) SFGs are:

- 1st Special Forces Group (Airborne) - Pacific.
- 3rd Special Forces Group (Airborne) - sub-Saharan Africa.
- 5th Special Forces Group (Airborne) - Middle East, Persian Gulf, Central Asia & the Horn of Africa.
- 7th Special Forces Group (Airborne) - Latin America, Central America, and the Caribbean.
- 10th Special Forces Group (Airborne) - Europe.

The 2nd Army National Guard (ANRG) SFGs:

- 19th Special Forces Group (Airborne) - South East Asia & Pacific.
- 20th Special Forces Group (Airborne) - Latin America, Central America, and the Caribbean.

Each Special Forces Group consists of a Headquarters and Headquarters Company (HHC), a Group Support Battalion, and four Special Forces Battalions (SFB). Each SFB has:

- 1 Operational Detachment-Charlie (ODC) - the Battalion Headquarters.
- 3 SF Companies (SFC) - each company has:
 - 6 Operational Detachment-Alpha (ODA). ODAs are usually 12-person units, led by a captain. The ODA is the primary fighting force of the SF.
 - 1 Operational Detachment-Bravo (ODB). ODBs usually have around 12 personnel, who provide command and control, training, intelligence, and counterintelligence support to ODAs.
- 1 Battalion Support Company. The Battalion Support Company consists of signalers, mechanics, riggers, cooks, intelligence, personnel services, chaplain, legal, and others, who support the Special Forces Battalion.

Note: One of the SFCs within an SFG is designated as a Commander's In-Extremis Force (CIF) Company. The CIF Company is focused on Direct Action missions and may deploy separately to support geographic combatant commanders directly.

Where the conditions for direct action are not present, SF can function as highly trained light infantry, although that role is often considered a waste because it does not capitalize on the unique capabilities of special forces.



Figure 30: Navy Special Warfare Command Seal.

(2) Naval Special Warfare Command (NavSpecWarCom)

Established April 16, 1987, NavSpecWarCom is the U.S. Navy's special operations force and the maritime component of USSOCOM. The Naval Special Warfare (NSW) community is organized around eight sea, air, land (SEAL) teams, one SEAL Delivery Vehicle (SDV) team, three special boat teams, and supporting commands that deploy forces worldwide. The community is composed of approximately 9,200 personnel, including more than 2,700 Special Warfare Operators, 700 Special Warfare Boat Operators (SWCC), 700 reserve personnel, 4,000 support personnel, and more than 1,100 civilians.

(a) Mission. The mission of NavSpecWarCom is to man, train, equip, educate, deploy, resource, and sustain forces to conduct direct action and special reconnaissance, support advise-and-assist programs, and build partner capability, in or out of the maritime environment, by employing tailored capabilities in support of military commanders, chiefs of mission, interagency, and foreign partners and allies.

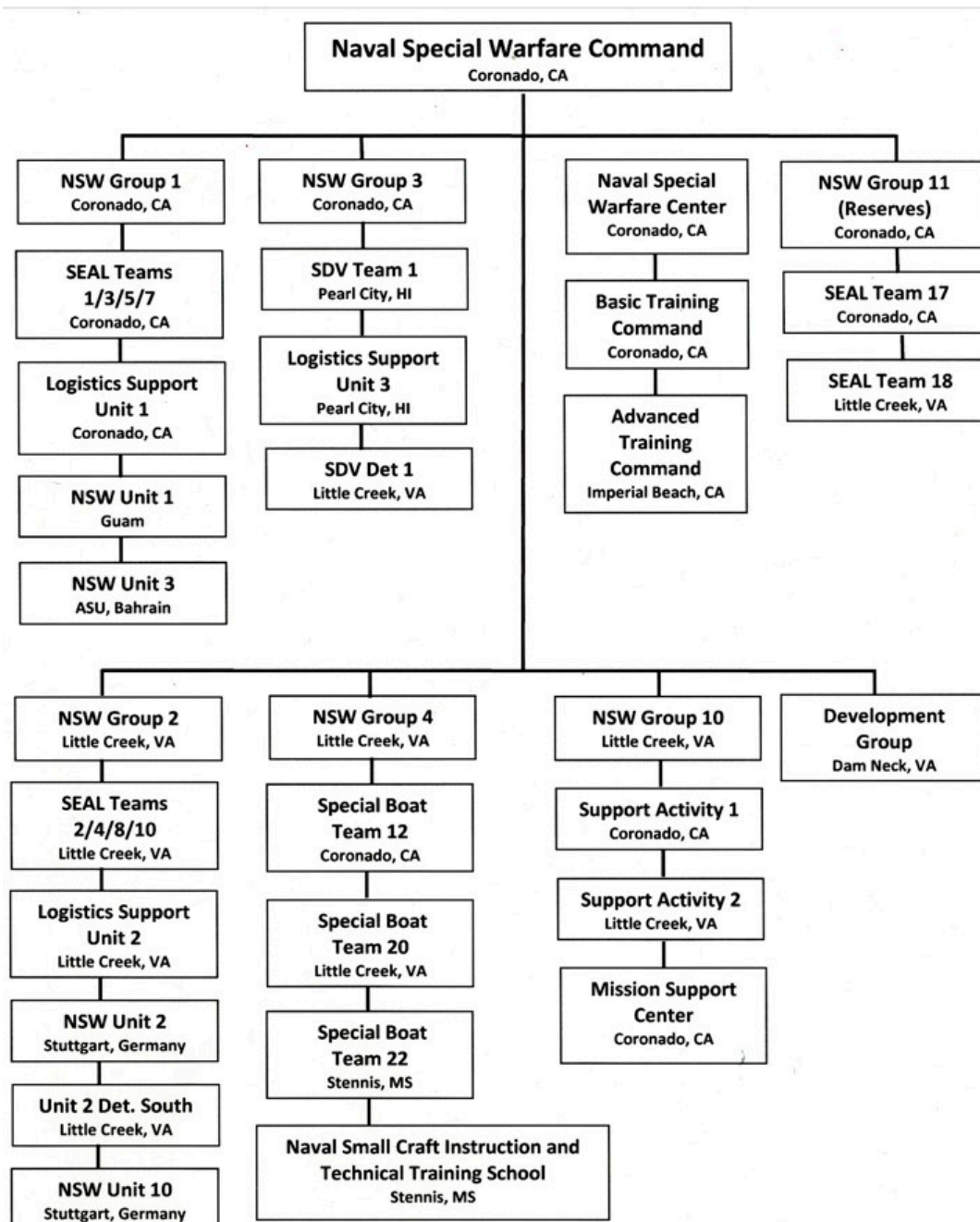


Figure 31: Naval Special Warfare Command.

(b) Organizations and Units.

- The **SEAL team** is the heart of the NSW force; a multipurpose combat force organized and trained to conduct a variety of special operations missions in all environments. SEALs conduct clandestine missions, infiltrating their objective areas by fixed- and rotary-wing aircraft, Navy surface ships, combatant craft, submarines, and ground mobility vehicles.
- **Special Boat Teams** are manned by Special Warfare Combatant-craft Crewmen (SWCC) who operate and maintain state-of-the-art surface craft to conduct coastal patrol and interdiction and support special operations missions. Focusing on infiltration and exfiltration of SEALs and other SOF, SWCCs provide dedicated rapid mobility in shallow water areas where larger ships cannot operate. They also bring to the table a unique SOF capability: Maritime Combatant Craft Aerial Delivery System—the ability to deliver combat craft via parachute drop.

If SEALs and SWCC are considered to be the action arms of NSW, then the enablers are the backbone of the organization. SEALs rely heavily on the services of technicians such as mobile communications teams, tactical cryptologic support, and explosive ordnance disposal specialists. Enablers contribute heavily toward the success of the special warfare operations mission.



Figure 32: Air Force Special Operations Command Shield.

(3) Air Force Special Operations Command (AFSOC)

AFSOC provides Air Force special operations forces for worldwide deployment and assignment to regional unified commands. The command's SOF are composed of highly trained, rapidly deployable Airmen, conducting global special operations missions ranging from precision application of firepower to infiltration, exfiltration, resupply, and refueling of SOF operational elements.

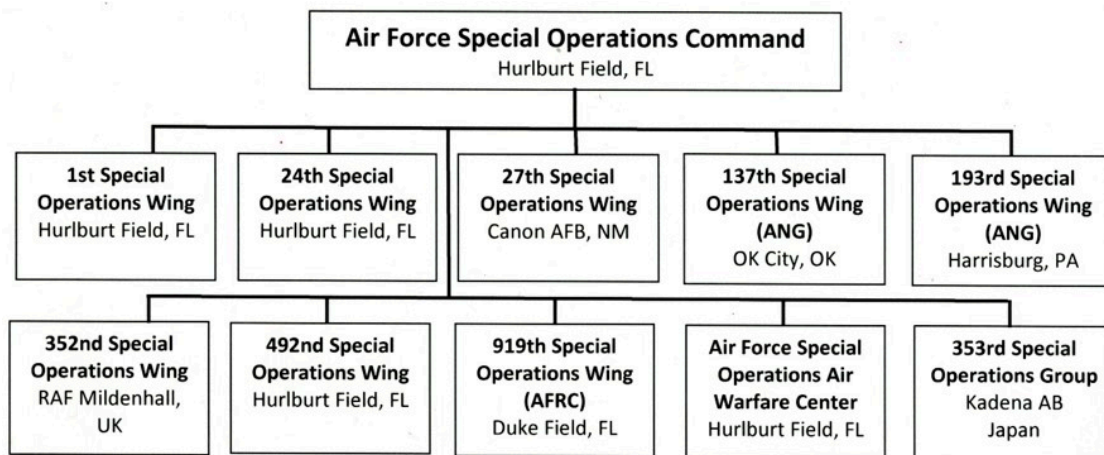


Figure 33: Air Force Special Operations Command.

(a) Mission. The command's core missions include battlefield air operations, agile combat support, aviation foreign internal defense, information operations/military support operations, precision strike, specialized air mobility; command and control; and intelligence, surveillance, and reconnaissance.

AFSOC's unique capabilities include airborne radio and television broadcast for psychological operations, as well as aviation foreign internal defense instructors to provide other governments with military expertise for their internal development. The command's special tactics squadrons combine combat controllers, tactical air control party members, special operations weathermen, and pararescuemen with other service SOF to form versatile joint special operations teams.

(b) Personnel and Resources. AFSOC has more than 20,800 active-duty, Air Force Reserve, Air National Guard, and civilian personnel. The command's active duty and reserve component flying units operate fixed and rotary-wing aircraft, including the CV-22B *Osprey*, AC-130 gunships, EC-130 *Commando Solo*, MC-130 variants, MQ-9 *Reaper*, U-28, C-145A, and C-146A *Wolfhound*.

(c) Organization and Units. The command's forces are organized under five active-duty wings, one Reserve wing, two National Guard wings, one overseas group, and several direct reporting units.

- The 1st Special Operations Wing (SOW) at Hurlburt Field, FL, and the 27th SOW at Cannon AFB, NM, are composed of specialized aircraft to support special operations worldwide. Together, the wings operate AC-130U/J/W, MC-130H/J, CV-22 and U-28A, MQ-9, and

C-146A aircraft.

- The 24th SOW at Hurlburt Field provides special tactics forces for rapid global employment to enable airpower success. The 24th SOW is the only Special Tactics wing in the Air Force.
- The 352nd SOW at Royal Air Force Mildenhall, England, is the Air Force component for Special Operations Command Europe. The 352nd SOW plans and performs specialized operations using advanced aircraft, tactics, air refueling techniques, and special tactics operators to transport, resupply, and support military special forces. The unit has six squadrons and two different types of aircraft: the MC-130J *Commando II* and the CV-22B *Osprey*.
- The 492nd SOW at Hurlburt Field organizes, trains, educates, and equips forces to conduct special operations missions; leads Air Force Irregular Warfare activities; executes special operations test and evaluation and lessons learned programs; and develops doctrine, tactics, techniques, and procedures for Air Force SOF.
- The 353rd Special Operations Group (SOG) at Kadena Air Base, Japan, is the Air Force component for Special Operations Command Pacific. Its mission is air support of joint and allied special operations forces in the Pacific. The 353 SOG is composed of five squadrons and operates the MC-130J *Commando II*.
- Air Force Reserve SOWs. The 919th SOW at Duke Field, FL, is the only special operations wing in the Air Force Reserve and provides operations and maintenance personnel to support the aviation foreign internal defense and combat aviation advisor programs for Air Force Special Operations Command through the use of the C-145A aircraft. The wing conducts U-28, C-145A, and AC-130U formal training unit flight instruction through a classic association with the 492nd SOW. The wing performs Remotely Piloted Aircraft operations in the MQ-9 aircraft and provides operations and maintenance personnel to conduct the Non-Standard Aviation program in the C-146 aircraft through a classic association with the 27th SOW located at Cannon AFB, NM.
- Air National Guard Units. The 137th SOW at Will Rogers Air National Guard Base, OK, maintains and operates the MC-12, designed to augment downrange intelligence, surveillance, and reconnaissance capabilities. The 193rd SOW at Harrisburg International Airport, PA, provides the only airborne psychological operations platform in the DoD with the EC-130 *Commando Solo*.



Figure 34: U.S. Marine Corps Forces, Special Operations Command.

(4) U.S. Marine Corps Forces, Special Operations Command (MARSOC)

In October 2005, the SecDef directed the Marine Corps to form a service component USSOCOM to provide forces to the Commander, USSOCOM. On 23 February 2006, MARSOC's headquarters was formally established.

(a) Mission. MARSOC trains, organizes, equips, and deploys task organized, scalable, and responsive U.S. Marine Corps SOF worldwide in support of CCDRs and other agencies when directed by the commander of USSOCOM.

(b) Personnel and Organization.

- MARSOC currently consists of approximately 3,000 Marines, Sailors, Soldiers, and civilian employees, with an ultimate growth target of 3,255.
- Headquarters. A U.S. Marine Corps Major General commands MARSOC with a support staff that is designed to be compatible with the functional areas within USSOCOM and Headquarters Marine Corps. MARSOC headquarters is responsible for identifying Marine special

operations requirements; developing Marine SOF tactics, techniques, procedures, and doctrine; and executing assigned missions in accordance with designated conditions and standards.

- MARSOC includes three subordinate commands, all located aboard Camp Lejeune, NC.

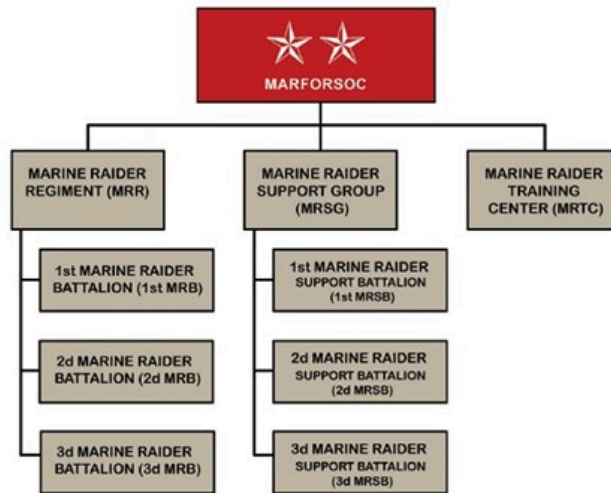


Figure 35: Marine Corps Forces Special Operations Command.

(c) Units.

- Marine Raider Regiment (MRR), which consists of three Marine Raider Battalions (MRB):
 - 1st MRB - Aligned to U.S. Indo-Pacific Command.
 - 2d MRB - Aligned to U.S. Central Command.
 - 3d MRB - Aligned to U.S. Africa Command.
- Marine Raider Support Group (MRSG), which consists of three Marine Raider Support Battalions (MRSB), which provide direct command and control, logistics, and intelligence support to their respective MRB.
- Marine Raider Training Center (MRTC), which consists of multiple branches focused on the assessment, selection, and basic and advanced training of MARSOF personnel.

(d) Employment. Since its inception, MARSOC has maintained a high operational tempo of special operations, deploying units that ranged in size from Marine Special Operations Teams (MSOTs) and reinforced Marine Special Operations Companies (MSOC) to Marine Special Operations Task Forces (MSOTF), in support of the geographic combatant commanders.

Over the years, MARSOC has conducted missions in over 30 different countries. Many of these missions have been focused on strengthening partner counter-terrorism capabilities, assisting with counter narco-terrorism efforts, and providing subject matter expertise and training to partner forces. In 2009, MARSOC deployed its first Special Operations Task Force (SOTF) HQ, built around a Marine Special Operations Battalion, to Afghanistan. Since then, MARSOC has deployed 9 SOTF HQs in support of CENTCOM requirements. In 2016, MARSOC also deployed a Combined Joint Special Operations Task Force (CJSOTF) HQ, led by a Marine Colonel, to Iraq in support of Operation INHERENT RESOLVE. MARSOC's second CJSOTF HQ completed its deployment to Iraq at the end of 2017.

From 2009-2014, MARSOC primarily focused on operations in Afghanistan. Starting in 2015, MARSOC operationally reoriented to better support the geographic combatant commanders in the post-Operation ENDURING FREEDOM environment. MARSOC's forces are now regionally focused in order to provide persistently forward deployed, reinforced MSOCs to U.S. Central Command, U.S. Pacific Command, and U.S. Africa Command.

The MARSOC training continuum produces reinforced MSOCs capable of full spectrum special operations, with particular emphasis on skills related to partner nation capacity building. The three battalions of the MRR and MRSG have been regionally oriented to effectively produce this capability in support of these combatant commands. In addition, each forward deployed MSOC (Rein) is also attuned to the unique regional requirements of their deployment area, including language capability and any specific regional tactical capabilities.



Figure 36: U.S. Coast Guard Seal.

10. U.S. Coast Guard

The Coast Guard is the principal Federal agency responsible for maritime safety, security, and environmental stewardship in U.S. ports and waterways. In this capacity, the Coast Guard protects and defends more than 100,000 miles of U.S. coastline and inland waterways, and safeguards an Exclusive Economic Zone (EEZ) encompassing 4.5 million square miles stretching from North of the Arctic Circle to South of the equator, from Puerto Rico to Guam, encompassing nine time zones—the largest EEZ in the world. As one of the five Armed Services of the U.S., the Coast Guard is the only military branch within the Department of Homeland Security. In addition to its role as an Armed Service, the Coast Guard is a first responder and humanitarian service that provides aid to people in distress or impacted by natural and man-made disasters, whether at sea or ashore. The Coast Guard is a member of the Intelligence Community and is a law enforcement and regulatory agency with broad legal authorities associated with maritime transportation, hazardous materials shipping, bridge administration, oil spill response, pilotage, and vessel construction and operation.

The over 56,000 members of the Coast Guard operate a multi-mission, interoperable fleet of 243 Cutters, 201 fixed and rotary-wing aircraft, and over 1,600 boats. Operational control of surface and air assets is vested in two Coast Guard geographical areas (Pacific and Atlantic), nine Coast Guard Districts, and 35 Sectors located at strategic ports throughout the country. Six Mission Support Logistics and Service Centers provide services for operational assets and shore facilities. Coast Guard program oversight, policy development, and personnel administration are carried out at Coast Guard Headquarters located on the St. Elizabeth's campus in Washington, DC.

On an average day, the Coast Guard:

- Conducts 45 search and rescues
- Saves 10 lives
- Saves over \$1.2M in property
- Seizes 874 pounds of cocaine and 214 pounds of marijuana
- Conducts 57 waterborne patrols of critical maritime infrastructure
- Interdicts 17 illegal migrants
- Escorts 5 high-capacity passenger vessels
- Conducts 24 security boardings in and around U.S. ports
- Screens 360 merchant vessels for potential security threats prior to arrival in U.S. ports
- Conducts 14 fisheries conservation boardings
- Services 82 buoys and fixed aids to navigation
- Investigates 35 pollution incidents
- Completes 26 safety examinations on foreign vessels
- Conducts 105 marine inspections

- Investigates 14 marine casualties involving commercial vessels
- Facilitates movement of \$8.7B. worth of goods and commodities through the Nation's Maritime Transportation System

a. Force Laydown

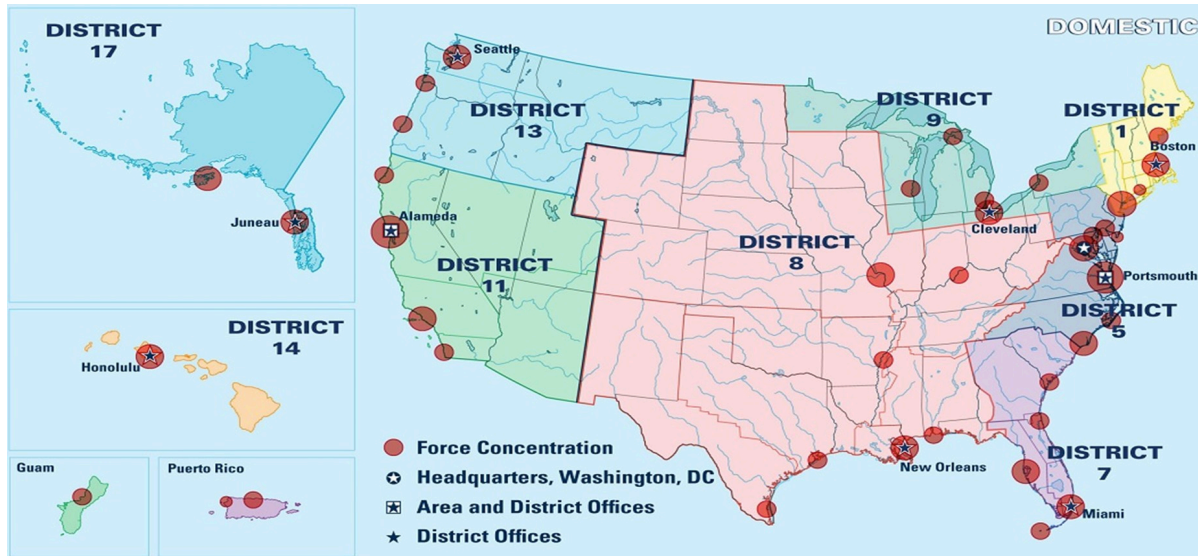


Figure 37: Domestic Coast Guard Stations.

The Coast Guard is divided into two geographic commands: the Atlantic Area and Pacific Area. The areas are further divided into nine district commands. The Atlantic Area includes five district commands covering the Eastern U.S., the Atlantic Ocean, the Great Lakes and the Gulf of Mexico. The Pacific Area includes four district commands covering the Western U.S. and the Pacific Ocean.



Figure 38: International Coast Guard Force Concentrations.

The Coast Guard's permanent international units are depicted above. Patrol Forces Southwest Asia (PATFORSWA) is the Coast Guard's largest unit outside of the U.S. PATFORSWA supported the U.S. Navy's Fifth Fleet in Operation ENDURING FREEDOM. The Coast Guard has two commands (Activities Europe and Activities Far East) that support the Coast Guard's Maritime Safety and Security mission, conducting marine inspection, marine investigation, and international port security operations to ensure the integrity of the Maritime Transportation System. The

Coast Guard also has personnel assigned to eight DoD Combatant Commands and often has a presence on all seven continents and the world's oceans, projecting national sovereignty with our icebreakers, national security cutters, high-endurance cutters, aviation assets, and deployable specialized forces.

b. National Defense Support

The Coast Guard is a unique Military Service residing within the Department of Homeland Security while simultaneously providing direct support to the DoD under its inherent authorities under Titles 10 and 14 of the *USC*. In addressing the Coast Guard when it is not operating in the Department of the Navy, this issuance is descriptive in nature and does not purport to be either directive or regulatory. As directed by the President, and in accordance with *Memorandum of Agreement between the Department of Defense and Department of Homeland Security on the use of Coast Guard Capabilities and Resources in Support of the National Military Strategy*, the Department of the Navy shall coordinate with the Department of Homeland Security regarding Coast Guard military functions in time of limited war or defense contingency, without transfer of Coast Guard authority to the Secretary of the Navy. As directed, the Department of the Navy will provide intelligence, logistical support, and specialized units to the Coast Guard, including designated ships and aircraft, for overseas deployment required by naval component commanders, maritime search and rescue, integrated port security, and coastal defense of the U.S. The Coast Guard shall maintain a state of readiness to function as a specialized Military Service in the Department of the Navy in time of war or national emergency. If specified in a declaration of war by Congress or if directed by the President, the Coast Guard shall operate as a Military Service in the Department of the Navy, and shall continue to do so until the President transfers the Coast Guard back to the Department of Homeland Security by Executive Order pursuant to Section 3 of Title 14, *USC*.

The Coast Guard shall develop concepts, doctrine, tactics, techniques, and procedures and organize, train, equip, and provide forces to perform the following specific functions when providing direct or cooperative support to the DoD:

- Conduct coastal sea control and maritime and air interception and interdiction operations.
- Conduct maritime homeland security and counterterrorism operations.
- Provide for port operations, security, and defense.
- Provide maritime operational threat response.
- Conduct counter-illicit trafficking operations.
- Conduct military environmental response operations.
- Conduct theater security cooperation operations.
- Conduct search and rescue operations.
- Conduct ice operations.
- Provide for marine safety, including aids to navigation.

The Coast Guard will coordinate with the DoD, including the Department of the Navy, to provide specialized Coast Guard units, or obtain Navy units, including designated ships and aircraft, for deployment as requested by Military Service component or joint commanders.

Lesson 5: Department of the Navy

Lesson Hours: 3.00

1. Introduction

The purpose of this lesson is to present the organization and functions of the Department of the Navy and the organization and capabilities of its two Military Services, the U.S. Navy and U.S. Marine Corps. The lesson begins by describing the organization of the Department of the Navy and rapidly transitions to a presentation of the functions, organization, capabilities and platforms, and command and control concept of the U.S. Navy. The lesson concludes with a brief presentation of the functions and overall organization of the U.S. Marine Corps. A more detailed discussion of the U.S. Marine Corps will be had in the 8672 *Functional Employment of the MAGTF* course.

2. Educational Objectives

- a. Identify the organization of the Department of the Navy.

- b. Identify the unique Service functions assigned to the U.S. Navy.
- c. Identify the U.S. Navy operating forces organizations.
- d. Identify the areas of responsibility for each of the U.S. numbered fleets.
- e. Identify the purpose and limitations of aircraft carriers.
- f. Identify the organization and purpose of a carrier strike group.
- g. Identify the purpose and limitations of surface combat ships.
- h. Identify the organization and purpose of a surface action group.
- i. Identify the purpose and limitations of attack submarines.
- j. Identify the purpose and limitations of amphibious ships.
- k. Identify the organization of an amphibious ready group.
- l. Identify the organization and purpose of an expeditionary strike group.
- m. Identify the composition and functions of the U.S. Marine Corps as prescribed in Title 10.
- n. Identify the Marine Corps functions specified by DoD Directive 5100.1.
- o. Identify the Marine Corps functions specified by the Marine Corps Manual.
- p. Identify the purpose of Fleet Marine Forces.
- q. Identify the common elements of all MAGTFs.
- r. Identify the different types of MAGTFs and their specific organizational structures.

3. Key Terms

- Department of the Navy
- Operating Forces
- Type Commands
- Carrier Strike Group (CSG)
- Surface Action Group (SAG)
- Amphibious Ready Group (ARG)
- Expeditionary Strike Group (ESG)
- Fleet Marine Forces
- Marine Air Ground Task Force (MAGTF)
- Command element
- Ground combat element
- Aviation combat element
- Logistics combat element
- Marine Expeditionary Force (MEF)
- Marine Expeditionary Brigade (MEB)
- Marine Expeditionary Unit (MEU)
- Special-Purpose MAGTF (SPMAGTF)

4. Preparation

All quizzes, discussion forum posts, and assignments are executed in the Moodle course site.

- a. Lesson Reading: [Department of the Navy](#) (64 pages).
- b. Other Requirements: None.
- c. Supplemental References (optional):
 - [DODD 5100.01, Functions of the Department of Defense and Its Major Components](#) (17 Sep 2020).

- [OPNAVINST 3501.316C, Force Composition of Afloat Navy and Naval Groups](#) (10 Nov 2017).
- [The U.S. Military's Force Structure: A Primer, "Chapter 3, Department of the Navy,"](#) Congressional Budget Office (May 2021).
- [Navy Force Structure and Shipbuilding Plans: Background and Issues for Congress](#), Congressional Research Service (31 Mar 2025).
- [MCDP 1-0 w/changes 1-3, Marine Corps Operations](#), (29 Mar 2019).
- [MCRP 1-10.1, Organization of the United States Marine Corps](#) (23 Jul 2020).
- [39th Commandant's Planning Guidance](#), General Eric M. Smith, U.S. Marine Corps, Commandant of the Marine Corps (Aug 2024).
- [Force Design Update](#), General Eric M. Smith, U.S. Marine Corps, Commandant of the Marine Corps (Oct 2025).

Lesson 5 Reading

DEPARTMENT OF THE NAVY

References:

- a) DODD 5100.01, *Functions of the Department of Defense and Its Major Components* (17 Sep 2020)
- b) OPNAVINST 3501.316C, *Force Composition of Afloat Navy and Naval Groups* (10 Nov 2017)
- c) *The U.S. Military's Force Structure: A Primer, "Chapter 3, Department of the Navy,"* Congressional Budget Office (May 2021)
- d) *Navy Force Structure and Shipbuilding Plans: Background and Issues for Congress*, Congressional Research Service (31 Mar 2025)
- e) MCDP 1-0 w/changes 1-3, *Marine Corps Operations*, (29 Mar 2019)
- f) MCRP 1-10.1, *Organization of the United States Marine Corps* (23 Jul 2020)
- g) *39th Commandant's Planning Guidance: 38th Commandant of the Marine Corps*, General Eric M. Smith, U.S. Marine Corps, Commandant of the Marine Corps (Aug 2024)
- h) *Force Design Update*, General Eric M. Smith, U.S. Marine Corps, Commandant of the Marine Corps (Oct 2025)

1. Introduction

The Department of the Navy (DoN) is composed of naval, land, air, space, and cyberspace forces, both combat and support, not otherwise assigned, to include those active and reserve component organic forces and capabilities necessary to operate and support the Navy and Marine Corps, the other Military Services, and joint forces. The Navy and Marine Corps provide the Nation's principal maritime force. They employ the global reach, persistent presence through forward-stationed and rotationally-based forces, and operational flexibility to secure the Nation from direct attack; secure strategic access and retain global freedom of action; strengthen existing and emerging alliances and partnerships; establish favorable security conditions; deter aggression and violence by state, non-state, and individual actors; and, should deterrence fail, prosecute the full range of military operations in support of U.S. national interests.

The DoN is the second-largest military department by number of military personnel (the Department of the Army is the largest) and has the second-largest operation and support budget. Because of its sizable acquisition funding, however, it has a larger total budget than any other Military Department.

The DoN's forces are distinctive not only for their number and variety of units, but also for the way in which different types of forces routinely work closely together. The Army and Air Force each essentially focus on a single type of military power (ground combat or air combat), but the Navy and Marine Corps routinely integrate ships with aircraft (as in Carrier Strike Groups), ships with ground combat units (as in Amphibious Ready Groups), and aircraft with ground combat units (as in Marine Expeditionary Forces). Although all U.S. forces are expected to be able to operate jointly with other services, the routine and habitual integration of different types of military power within the DoN goes beyond typical joint operations. For example, the Marine Corps has fewer artillery units to support its ground combat units than the Army does, in part because the Corps prefers to provide additional firepower (fire support) for its combat units by using its attack aircraft—aircraft that may well be based on Navy ships. In contrast, the Army has traditionally structured itself on the assumption that it must have substantial artillery capability in case Air Force aircraft are not available to provide fire support.

2. Functions and Organization of the Department of the Navy

The functions of the DoN are those common to all the Military Departments as presented in the *Department of Defense* lesson.

The DoN is separately organized under the Secretary of the Navy. It operates under the authority, direction, and control of the Secretary of Defense. The DoN is composed of the following:

- The Office of the Secretary of the Navy.
- The Office of the Chief of Naval Operations.

- The Headquarters, Marine Corps.
- The entire operating forces, including naval aviation, of the Navy and of the Marine Corps, and the reserve components of those operating forces.
- All field activities, headquarters, forces, bases, installations, activities, and functions under the control or supervision of the Secretary of the Navy.
- The Coast Guard when it is operating as a service in the Navy.

The DoN consists of three elements: the Navy Department, the Operating Forces of the Navy and the Marine Corps, and the Shore Establishment.

a. Navy Department

The Navy Department refers to the central executive offices of the DoN located at the seat of Government. The Navy Department is organizationally composed of the Office of the Secretary of the Navy, the Office of the Chief of Naval Operations, and the Headquarters, Marine Corps. In addition, the Headquarters, Coast Guard, is included when the Coast Guard is operating as a service in the Navy.

b. Operating Forces

The operating forces of the Navy and the Marine Corps comprise the several fleets, seagoing forces, Fleet Marine Forces, other assigned Marine Corps Forces, the Military Sealift Command and other forces and activities that may be assigned thereto by the President or the Secretary of the Navy.

c. Shore Establishment

The shore establishment is composed of shore activities with defined missions approved for establishment by the Secretary of the Navy.

3. The U.S. Navy

a. Functions of the U.S. Navy

In addition to the common military service functions discussed in the *Department of Defense* lesson, the Navy, within the DoN, shall develop concepts, doctrine, tactics, techniques, and procedures and organize, train, equip, and provide forces to perform the following specific functions:

- Conduct offensive and defensive operations associated with the maritime domain such as achieving and maintaining sea control, to include subsurface, surface, land, air, space, and cyberspace.
- Provide power projection through sea-based global strike, to include nuclear and conventional capabilities; interdiction and interception capabilities; maritime and/or littoral fires, to include naval surface fires; and close air support for ground forces.
- Conduct ballistic missile defense.
- Conduct ocean, hydrographic, and river survey and reconstruction.
- Conduct riverine operations.
- Establish, maintain, and defend sea bases in support of naval, amphibious, land, air, or other joint operations as directed.
- Provide naval expeditionary logistics to enhance the deployment, sustainment, and redeployment of naval forces and other forces operating within the maritime domain, to include joint sea bases, and provide sea transport for the Armed Forces other than that which is organic to the individual Military Services and U.S. Special Operations Command.
- Provide support for joint space operations to enhance naval operations, in coordination with the other Military Services, Combatant Commands, and U.S. government departments and agencies.
- Conduct nuclear operations in support of strategic deterrence, to include providing and maintaining nuclear surety and capabilities.

b. General Organization

The structure of the U.S. Navy consists of three components under the Secretary of the Navy: the Office of the Chief of Naval Operations, the operating forces, and the shore establishment. This lesson will focus primarily on the operating forces.

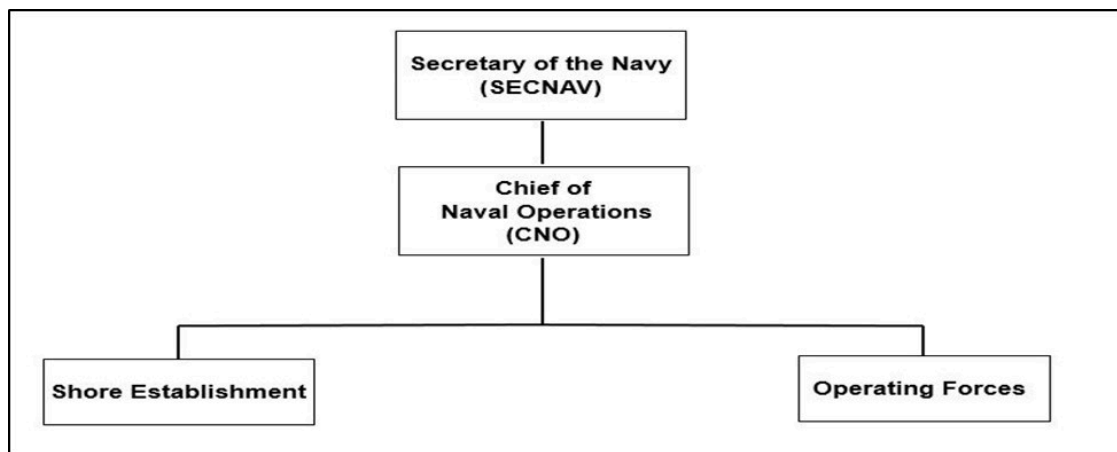


Figure 1: Structure of U.S. Navy.

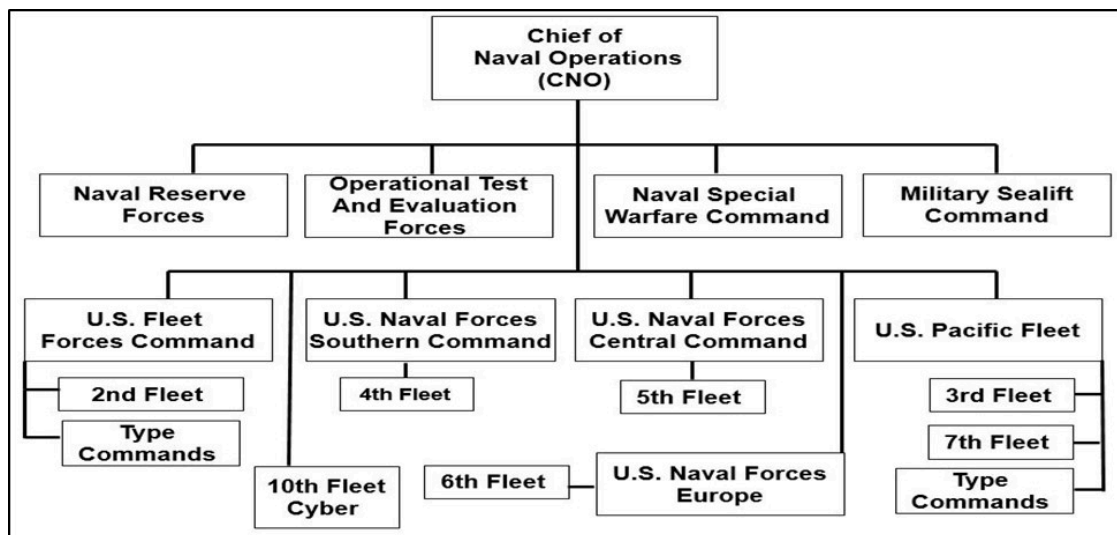


Figure 2: U.S. Navy Operating Forces.

The Navy's operating forces' structure is complex and involves two chains of command, operational and administrative. Typically, units (ships, aircraft, submarines, etc.), will report to their administrative chain of command when they are in a deep maintenance cycle, called a "Chief of Naval Operations Availability" or, subsequently, training and certification "work ups" for deployment. Once the unit is certified for deployment, it will execute a change of operational command and be assigned to one of the fleets for further tasking. The unit's change of administrative and operational commanders is a fluid process, both because of the Navy's expeditionary nature and its global presence, and this change of commanders occurs often.

Note: Do not confuse "operating forces" with "operational chain of command" - operating forces serve within both the administrative and operational chains of command to carry out different functions.

c. Administrative Chain of Command

The administrative chain of command is responsible for organizing, manning, training, and equipping forces. These forces eventually will be assigned to the operational chain of command for use in combat, operations, and exercises. The administrative chain of command runs from the President of the U.S. to the Secretary of the Navy (SECNAV), then through the Chief of Naval Operations (CNO) to the various administrative commands within the operating forces and shore establishment.

The CNO is the senior Naval military officer and a member of the Joint Chiefs of Staff. The CNO is the principal naval advisor to the President and to SECNAV on the conduct of war and is the principal advisor and naval executive to SECNAV on the conduct of the Navy. The CNO is a four-star admiral who is responsible to SECNAV for the command, utilization of resources, and efficiency of the operating forces of the Navy and of the Navy shore activities assigned (several admirals, subordinate commanders, and other staff in the Office of the CNO, referred to as "OPNAV," help oversee specific functions in the Navy).

(1) Administrative Commands

(a) Type Commands. Type Commanders are responsible for coordinating resources and compatibility of procedures across generally like operating forces (i.e., platforms). In common terms, Type Commanders “man, train, equip, and maintain” the operating forces for deployment. The larger type commands may be further broken up into an Atlantic and a Pacific force, and further subdivided into groups, squadrons, or wings. The larger type commands are commanded by a three-star admiral and the smaller ones by a two-star. Geographic delineations, if applicable, are articulated in the bullets below:

- **Naval Surface Forces (SURFOR)** mans, trains, and equips the Surface Force to provide Fleet Commanders with credible naval power to control the sea and project power ashore. Commander (COM) SURFOR is located in San Diego, CA and is dual hatted as COM Surface Forces Pacific (SURFPAC). COM Naval Surface Forces Atlantic (SURFLANT), subordinate to COMSURFOR, is located in Norfolk, VA.
- **Naval Air Forces (AIRFOR)** mans, trains, equips, and maintains a Naval Air Force that is immediately employable, forward deployed, and engaged. COMAIRFOR is located in San Diego, CA and is dual hatted as COM Air Forces Pacific (AIRPAC). COM Naval Air Forces Atlantic (AIRLANT), subordinate to COMAIRFOR, is located in Oceana, VA.
- **U.S. Submarine Forces (SUBFOR)** executes the mission of the U.S. Navy in and from the undersea domain. COMSUBFOR is located in Norfolk, VA and is dual hatted as COM Submarine Forces, U.S. Atlantic Fleet (SUBLANT). COM Submarine Forces, U.S. Pacific Fleet (SUBPAC), subordinate to SUBFOR, is located in Pearl Harbor, HI.
- **Naval Special Warfare Command (SPECWAR)** provides maritime special operations forces to conduct full-spectrum operations, unilaterally or with partners, to support national objectives. COMSPECWAR is located in San Diego, CA.
- **Naval Expeditionary Combat Command (NECC)** organizes, mans, trains, equips, and sustains Navy Expeditionary Combat Forces to execute combat, combat support, and combat service support missions across the full spectrum of naval, joint, and combined operations that enable access from the sea and freedom of action throughout the sea-to-shore and inland operating environments. COMNECC is located in Little Creek, VA. NECC capabilities include Explosive Ordnance Disposal (EOD), Maritime Expeditionary Security, Riverine Operations, Diving Operations, Naval Construction, Maritime Civil Affairs, Expeditionary Training, Expeditionary Logistics, Expeditionary Intelligence, Combat Camera, and Expeditionary Combat Readiness. The Maritime Expeditionary Security Force’s (MESF) primary mission is force protection conducted through fleet support operations worldwide. Two Maritime Expeditionary Security Groups (San Diego, CA, and Portsmouth, VA) supervise the integration of coastal warfare assets trained to operate in high density, multi-threat environments. Coastal and harbor defense of naval assets are under the jurisdiction of two Naval Coastal Warfare Groups, one for the Pacific Fleet and one for the Atlantic Fleet.

(b) Shore Commands. On-shore installations and facilities (i.e., shore establishment) that support the operating forces (ships, submarines, etc.) with repairs, fuel, ammunition, training, and medical support, among other things, are known as shore commands. Current examples of major shore commands are (this list is not all-inclusive):

- Office of the Chief of Naval Operations
- Navy Personnel Command
- Bureau of Medicine and Surgery
- Strategic Systems Programs
- United States Naval Academy
- Naval Education and Training Command
- Naval Meteorology and Oceanography Command
- Office of Naval Intelligence
- Naval Aviation Warfighting Development Center
- Navy Information Operations Command
- Naval Legal Service Command
- United States Naval Observatory
- Naval Safety Center
- Navy Installations Command

(c) System Commands. There are five system commands that oversee the technical requirements of the Navy. These organizations’ roles are principally research, design, acquisition, and sustainment of platforms and systems for the Navy. These organizations work directly with the Navy’s defense industry suppliers to design, build, deliver, and sustain fighting equipment for Sailors and Marines.

d. Operational Chain of Command

You will recall from the *Department of Defense* lesson that the operational chain of command runs from the President of the U.S. through the Secretary of Defense to the various geographic combatant commanders (GCC) and functional combatant commanders (FCC) (with their Service and functional components) and then to the various assigned and allocated operating forces.

U.S. Navy forces are organized, trained, and equipped to carry out naval missions independently or as part of a joint force. The U.S. Navy is operationally organized along geographic lines, with a Navy component commander assigned to each GCC. Under the Navy component commander are numbered fleet commanders who are the primary operational-level Navy commanders within the respective GCCs areas of

responsibility (AORs). All Navy theater component and numbered fleet commanders are configured with a Maritime Operations Center (MOC) to provide command and control at the operational level, to include execution of joint task force (JTF) and joint force maritime component commander (JFMCC) functions. Tactical level operations are conducted by task force (TF) and task group commanders under their respective fleet commander.

(1) Component Commanders

Navy component commanders are the maritime element of the FCC's or GCC's staff and work directly for and in support of their assigned combatant commander (CCDR). CCDRs may delegate operational, tactical, and aspects of administrative control of forces assigned or allocated to the CCDR to the Navy component commander. The Navy component commander may then exercise appropriate command and control of subordinate forces (e.g., numbered fleets) during the planning and execution of joint missions (relationships for each theater are different and change frequently).

Note: Based on the size of the area of responsibility or scope of duties, the Navy component commanders may concurrently serve as the numbered fleet commander

The Navy component commanders are:

(a) Commander, U.S. Fleet Forces Command (COMUSFFC). COMUSFFC, located in Norfolk, VA, is a unique component commander and is central to both the Navy's administrative and operational chains of command. COMUSFFC was established as the Navy component to the now-disestablished Joint Forces Command (the FCC charged with providing forces to the other CCDRs). Within the administrative chain of command, COMUSFFC serves as the Navy's Global Force Provider and has the responsibility to:

- Train, certify, and provide combat-ready Navy forces to CCDR that are capable of conducting prompt, sustained naval, joint, and combined operations in support of U.S. national interests.
- Command and control subordinate Navy forces (e.g., U.S. 2nd Fleet, NECC, Military Sealift Command, and several TYCOMs) and shore activities during the planning and execution of assigned service functions in support of the CNO.

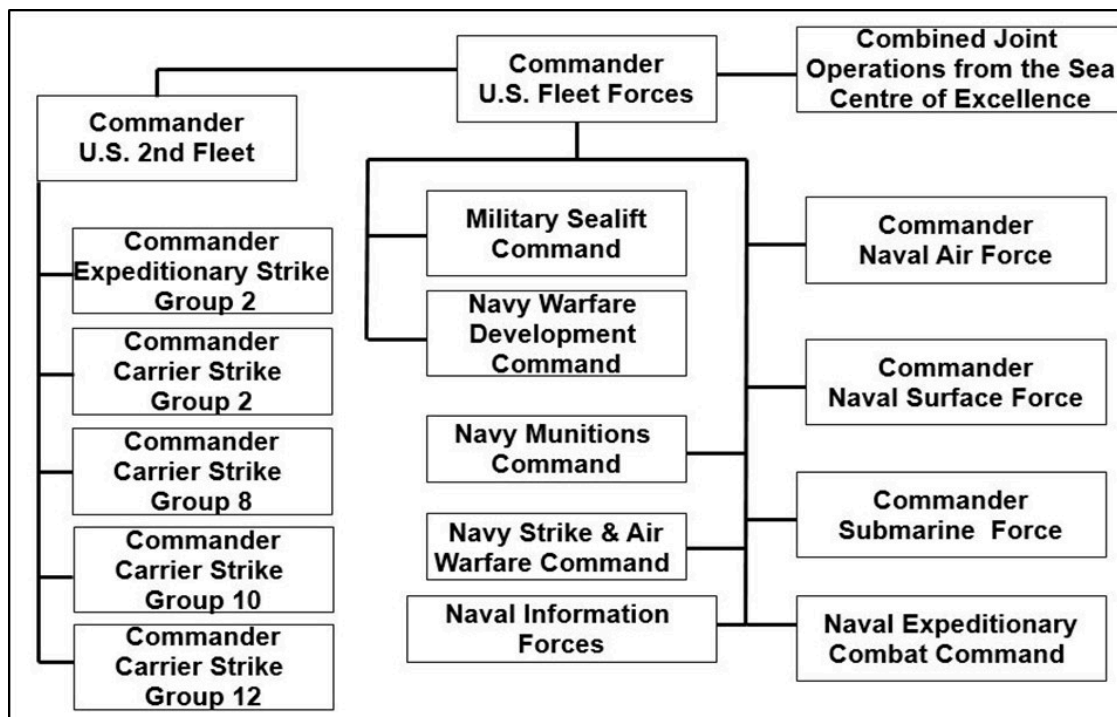


Figure 3: U.S. Fleet Forces Command.

Within the operational chain of command, COMUSFFC also serves as **Commander, Naval Forces Northern Command (COMNAVFORNORTH)**. COMNAVFORNORTH has the responsibility to:

- Provide operational planning and coordination support to Commander, U.S. Northern Command (CDR USNORTHCOM), Commander U.S. Element North American Aerospace Defense Command (CDR USELEMNORAD), and Commander, U.S. Strategic Command (CDR USSTRATCOM).

- Command and control assigned and allocated subordinate forces during the planning and execution of joint missions for CDR USNORTHCOM.

(b) Commander, U.S. Naval Forces Europe (COMNAVEUR)/U.S. Naval Forces Africa (COMNAVAF). This commander is dual hatted to serve as the component commander for both Commander, U.S. European Command (USEUCOM) and Commander, U.S. Africa Command (USAFRICOM).

(c) Commander, U.S. Pacific Fleet (COMPACFLT). COMPACFLT, located in Pearl Harbor Naval Station, HI, and San Diego, CA, serves as the Navy component commander for Commander, U.S. Indo-Pacific Command (USINDOPACOM).

(d) Commander, U.S. Naval Forces Central Command (COMNAVCENT). COMNAVCENT, located in Manama, Bahrain, serves as the Navy component commander and the Joint Force Maritime Component Commander (JFMCC) for Commander, U.S. Central Command (USCENTCOM). This commander also serves as Commander, U.S. 5th Fleet. As the JFMCC, this commander controls a number of task forces that are not a part of 5th Fleet. These include Combined Task Force 150, which conducts maritime surveillance activities in the Gulf of Oman and around the Horn of Africa, and Task Force 152, which has the same mission covering the southern portion of the Persian Gulf.

(e) Commander, U.S. Naval Forces Southern Command (COMNAVSOUTH). COMNAVSOUTH, located in Jacksonville, FL, serves as the Navy component commander for Commander, U.S. Southern Command. This commander is dual hatted as Commander, U.S. 4th Fleet.

(f) Commander, Naval Special Warfare Command (COMNAVSPECWAR). In addition to serving as a TYCOM, COMNAVSPECWAR serves as the Navy component commander for Commander, U.S. Special Operations Command.

(g) Commander, Military Sealift Command (COMMSC). COMMSC, located in Norfolk, VA, serves as the Navy component commander for Commander, U.S. Transportation Command and provides on-time logistics, strategic sealift, and specialized missions anywhere in the world, under any conditions.

(h) Commander, Fleet Cyber Command (COMFLTCYBERCOM). COMFLTCYBERCOM, located in Fort Meade, MD, serves as the Navy component commander for Commander, U.S. Cyber Command, the Navy space component to U.S. Strategic Command, and the Navy's Service Cryptologic Component Commander under the National Security Agency/Central Security Service. Within the administrative chain of command, COMFLTCYBERCOM reports directly to the CNO and has functional responsibility to achieve the integration and innovation necessary for warfighting superiority across the range of military operations in the maritime, information, and cyberspace domains. This commander also serves as Commander, U.S. 10th Fleet.

(2) Numbered Fleets

The numbered fleets are normally commanded by a three-star admiral and are homeported with even-numbered fleets oriented toward the U.S. East Coast and odd-numbered fleets toward the U.S. West Coast. The numbered fleets are described below:

(a) Commander, 2nd Fleet (C2F). 2nd Fleet (2F) is Service retained (not assigned to a CCDR) and is under the command of COMUSFFC. 2F develops and employs maritime forces ready to fight across multiple domains in the Atlantic and Arctic to ensure access, deter aggression, and defend U.S., allied, and partner interests. It is traditionally a force provider and trainer for forces that will be allocated to the GCC. 2F is based in Norfolk, VA.

(b) Commander, 3rd Fleet (C3F). 3rd Fleet (3F) is assigned to USINDOPACOM and is under administrative control of USPACFLT. It provides the realistic, relevant training necessary for an effective global Navy and leads naval forces in the Eastern Pacific. The command fosters joint, interagency, and international relationships, which strengthen our ability to respond to crises and protect the collective maritime interests of the U.S. and its allies and partners. Normally, units assigned to the 3F undergo training cruises prior to deployment with either the 5F or 7F and are not intended for immediate use in combat. 3F is based in San Diego, CA.

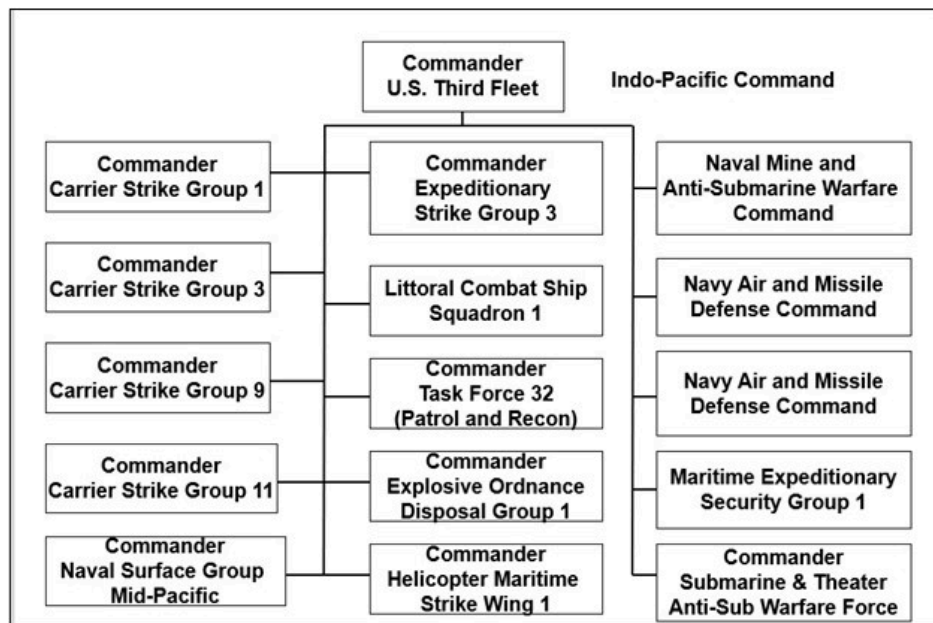


Figure 4: U.S. 3rd Fleet.

(c) Commander, 4th Fleet (C4F). 4th Fleet (4F) is assigned to USSOUTHCOM and has operational control of 2F, 3F, and U.S. Coast Guard assets allocated to operate in the USSOUTHCOM area of responsibility (no ships are permanently assigned to 4F). 4F conducts various missions, including contingency operations, counter-narcoterrorism, and theater security cooperation (TSC) activities. 4F is based in Mayport, FL (but does not have operational or administrative control of 2F assets based there).

(d) Commander, 5th Fleet (C5F). 5th Fleet (5F) is assigned to USCENTCOM and is under operational control of COMNAVCENT. 5F conducts maritime security operations, theater security cooperation efforts, and strengthens partner nations' maritime capabilities to promote security and stability. 5F usually has operational control of an average of 25 ships, including a carrier strike group (CSG), an expeditionary strike group (ESG), and additional forces from 2F and 3F, and administrative control of a few smaller Navy and Coast Guard ships forward deployed to Manama, Bahrain. 5F is based in Manama, Bahrain.

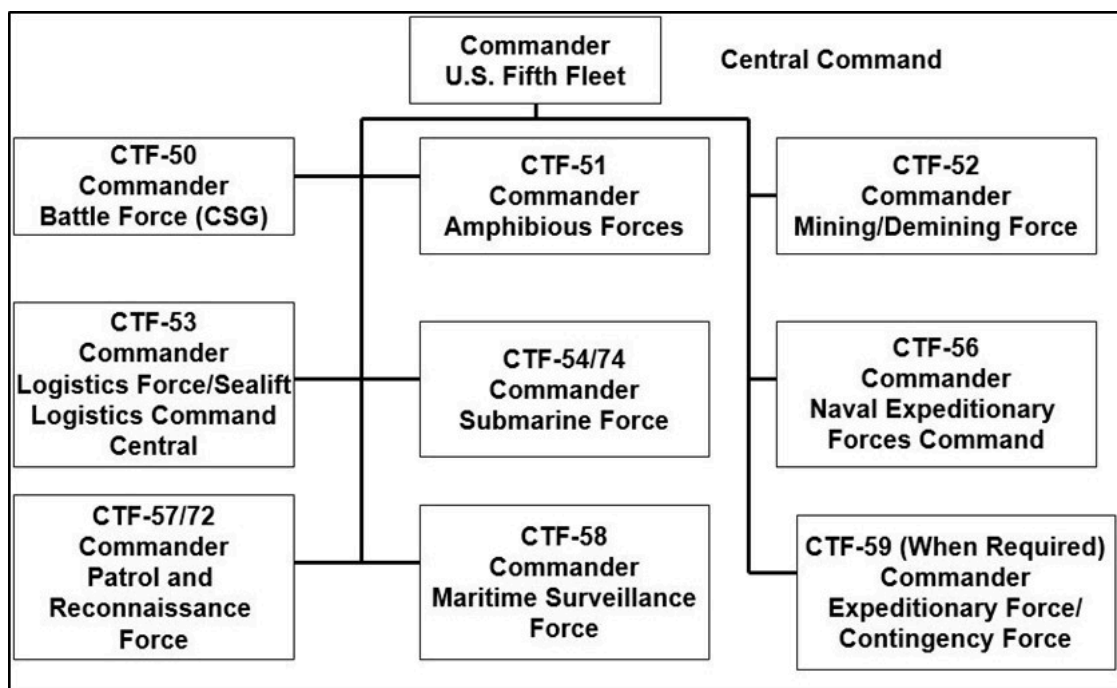


Figure 5: U.S. 5th Fleet.

(e) Commander, 6th Fleet (C6F). 6th Fleet (6F) is assigned to USEUCOM and is under the operational control of COMNAVEUR/COMNAF and conducts the full range of maritime operations and theater security cooperation missions in concert with coalition, joint, interagency, and other parties to advance security and stability in Europe and Africa (primarily in the Mediterranean Sea and Black Sea). In addition to forces provided by 2F and 3F, 6F normally has operational control of one destroyer squadron, one helicopter squadron, and one maritime patrol and reconnaissance aircraft squadron forward deployed to Rota, Spain and Sigonella, Italy. 6F is based in Naples, Italy, and its flagship is the amphibious command ship *USS Mount Whitney* (LCC-20). The *USS Mount Whitney* also serves as an Afloat Command Platform for Naval Striking and Support Forces NATO, a Naples-based maritime headquarters that serves as a deployable Maritime Component Commander as directed by the Supreme Headquarters Allied Powers Europe (SHAPE).

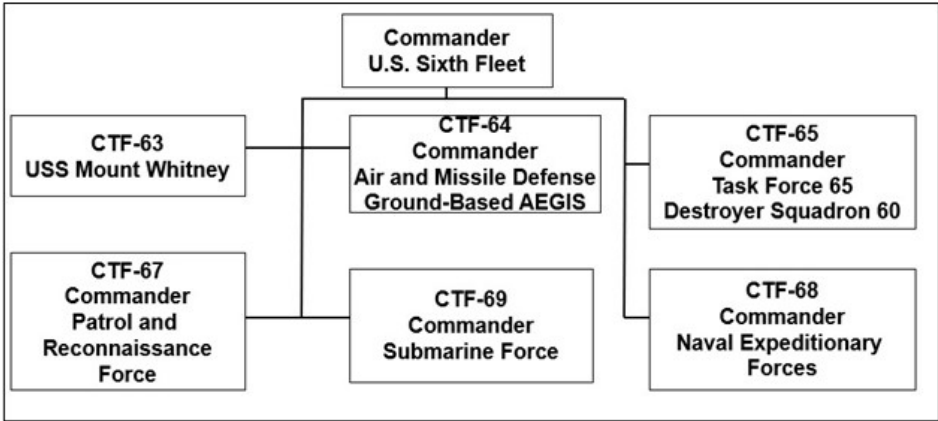


Figure 6: U.S. 6th Fleet.

(f) Commander, 7th Fleet (C7F). 7th Fleet (7F) is assigned to USINDOPACOM and is under the operational control of USPACFLT. It is the largest forward-deployed U.S. fleet and operates in the Western Pacific and the Indian Ocean, stretching to the Persian Gulf and including much of the east coast of Africa. At any given time, 7F controls 40–50 ships operating from bases in South Korea, Japan, and Guam. 7F also has operational control of 3F assets operating in the Pacific or en route to USCENTCOM. The headquarters is located at Yokosuka, Japan, with the *USS Blue Ridge* (LCC-19) as its flagship.

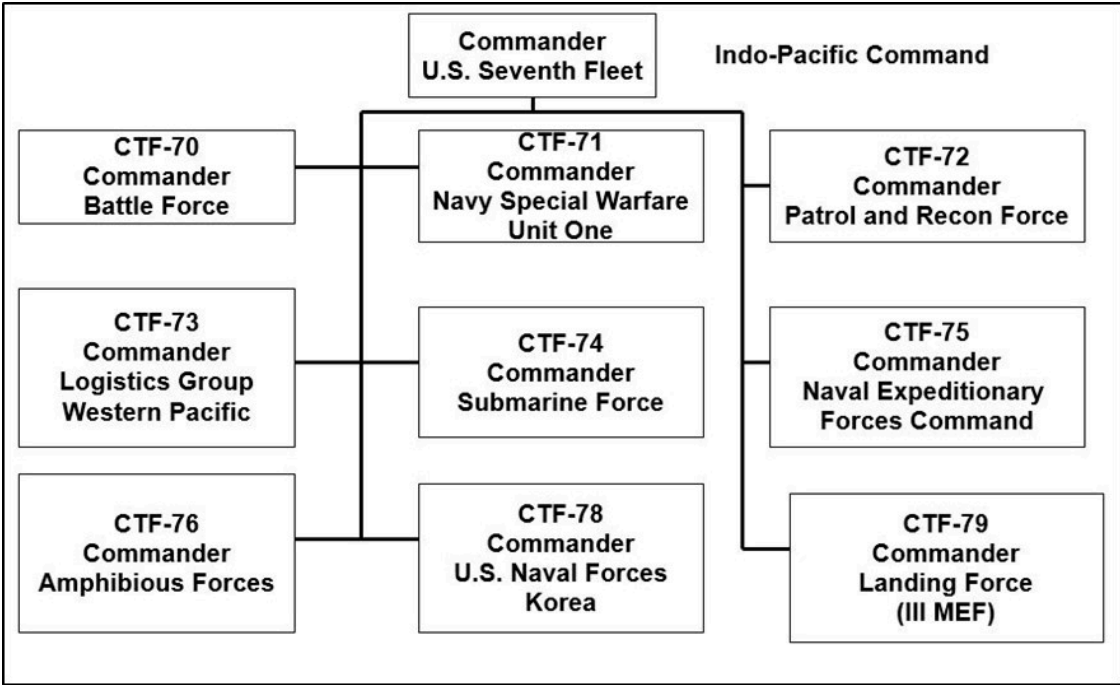


Figure 7: U.S. 7th Fleet.

(g) Commander, U.S. 10th Fleet (C10F). 10th Fleet (10F) is assigned to USCYBERCOM and has operational control of Navy cyber forces to execute information network operations, offensive and defensive cyberspace operations, space operations, and signals intelligence. 10F is based in Fort Meade, MD.

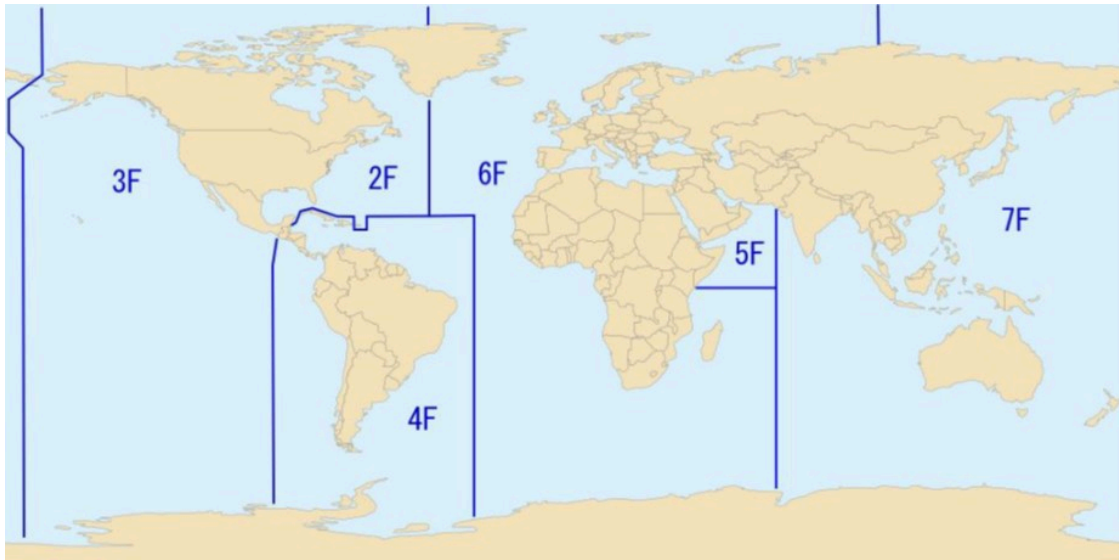


Figure 8: General Areas of Responsibility for Numbered Fleets.

(3) Subordinate Organizations

The fleet commander is in charge of all assigned and allocated naval forces (ships, aircraft, submarines, Sailors, etc.). By nature, fleets are huge and are usually too large for one specific operation. Therefore, fleets will further divide into task- or mission-oriented parts.

(a) Task Force. Each task force is responsible to the fleet commander for certain functions or mission related to the assigned units. This organizational scheme is scalable to meet many operational needs. It may also be conglomerated with other services to form a Joint Task Force (JTF) or with other partner nations to form a Combined Task Force (CTF).

(b) Task Group. A partition of a task force. An individually named group (think regiment/group size organization) that is tasked with a specific mission. A group is usually capable of conducting all composite warfare functions. The most common groups are CSGs, surface action groups (SAGs), and amphibious ready groups (ARGs), the three types of units that form the central organizational structures for the Navy.

(c) Task Units. Units operating together for a specific task, such as the air defense units within a task group, would receive a separate task unit designation.

(d) Task Elements. Individual units within the task unit are assigned as task elements. A unit can be defined as an individual ship, rotary-wing aircraft detachment, fixed-wing aircraft squadron, submarine, security support unit, etc.

The diagram below shows both a numerical and a named depiction of the Navy's operational and tactical task organizations.

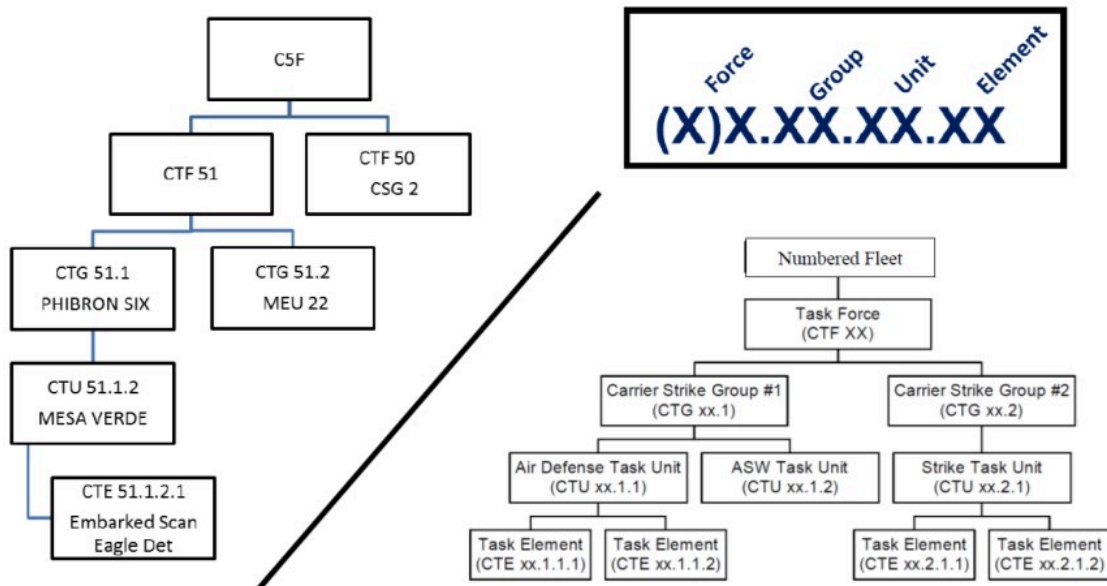


Figure 9: Naming Convention for U.S. Navy Operational Organizations.

e. U.S. Navy Platforms and Capabilities

(1) Overview

The many different types of units that are part of the DoN have their own strengths and weaknesses (as described in the sections below about major elements of the force structure). But as a whole, those units constitute a highly capable force. The Navy's surface combatants, for example, are widely considered to be exceptionally powerful units—generally larger, with bigger and more capable loads of weapons, and with more sophisticated sensors and electronics than the surface combatants of any other navy. Those ships often escort the Navy's aircraft carriers, which are also larger, with a greater complement of aircraft than those of any other navy. The vast majority of other navies in the world resemble the U.S. Coast Guard more than they do the U.S. Navy in that they focus on patrolling their country's coastlines rather than on projecting power overseas.

With the probable exception of China's navy in the western Pacific Ocean, no other nation's navy appears intended to challenge U.S. naval supremacy. Perhaps as a result, the U.S. has not faced any significant naval combat since World War II (although the Soviet Navy was prepared to engage U.S. and NATO naval forces during the Cold War).

In the past, the U.S. has generally had a lower threshold for using air and naval forces in combat than for using ground forces. Naval forces can be stationed in international waters—and thus do not require cooperation from other countries—but are still capable of launching air strikes or cruise missile strikes against potential targets. In addition, they can respond rapidly, provide a relatively visible threat, and are fairly well protected from any reprisals (both by distance from shore and by their own defensive weapons). For those reasons, naval forces have often been the U.S.' preferred first option in crisis situations or in smaller interventions. In such situations, the U.S. has sometimes also employed ARGs, whose ability to land ground combat units on shore can heighten the perceived threat of a U.S. invasion. (However, the relatively small size of the ground combat forces included in an ARG—one combat battalion, with air and logistics support—makes their use as a threat credible only against fairly weak opponents.) Using naval forces (or the Air Force) to conduct air and cruise missile strikes on opposing states, without also committing ground combat forces, has had mixed results in achieving U.S. goals. In some cases—such as operations against Libya in the 1980s and Serbia in the 1990s—air and cruise missile strikes may have been enough to achieve U.S. aims. But in many other cases—including the U.S. bombing of North Vietnam during the 1960s and 1970s and U.S. cruise missile attacks against Afghanistan and Sudan in 1998 (Operation Infinite Reach)—aerial campaigns without the use of ground forces did not prove effective at accomplishing U.S. goals.

By comparison, the U.S. has generally been successful in modern times in using amphibious forces to invade opposing countries. Only small and less capable states are vulnerable to an entirely amphibious invasion; however, in recent decades, the U.S. has taken part in few operationally significant amphibious assaults against major opponents. In major conflicts with such opponents (including the 1991 and 2003 wars with Iraq), the Marine Corps was deployed in essentially the same manner as the Army—as an additional ground force—rather than conducting an amphibious assault. The Marine Corps' amphibious capability has been used most in some of the Corps' least demanding operations, including peacetime missions and operations against opponents such as Grenada or Somalia, which were not capable of presenting concerted resistance.

The U.S. believes that the most likely future scenarios for U.S. naval combat involve operations conducted close to an enemy landmass. Such “littoral” operations pose special challenges for naval forces: They allow an enemy’s land-based forces to affect naval operations (for example, by attacking ships with land-based aircraft or missiles), while making it harder for naval forces to respond (for instance, by limiting their ability to maneuver, making it more difficult for them to find and destroy targets, and exposing them to mines such as those that damaged the USS *Princeton* and USS *Tripoli* during the 1991 war with Iraq). A potential conflict between the U.S. and China over the status of Taiwan, for example, would most likely involve China using land-based aircraft, cruise missiles, and ballistic missiles to try to keep the U.S. Navy out of the immediate area of operations. And a potential conflict in the Strait of Hormuz would most likely see Iran using submarines and land-based cruise missiles to try to deny U.S. Navy and commercial ships safe passage through the narrow waters of the strait.

In addition to their roles during conflicts, naval forces perform a variety of peacetime missions. For example, they are routinely used to evacuate noncombatants from conflict zones, to provide humanitarian and disaster relief, and to conduct antipiracy patrols. Some advocates of naval forces also suggest that the Navy, by being physically present in distant locations around the world, provides a form of visible U.S. presence that is more effective at reassuring friends and allies about U.S. security commitments—and at deterring U.S. opponents—than are Army and Air Force units, which are often farther away. The vast majority of the Navy’s operations today are routine deployments of ships around the globe to provide that presence.

Rather than being deployed at all times, Navy ships progress through an operating cycle of deploying and returning to their home ports, undergoing maintenance, training new crews, and then deploying again. As a result, only a fraction of ships is actually deployed at any one time—typically, about 30 percent to 40 percent (depending on the type of ship, home port, and deployment location), although when necessary, the Navy can increase that number in relatively short order. The Navy generally considers the number of ships deployed—its “forward presence”—to be a more meaningful measure of its contribution to national defense than the total number of ships in its fleet.

The U.S. Navy is expeditionary in nature, and as such has a variety of maritime assets. The following section will briefly describe the major capital assets of the U.S. Navy; not every ship, aircraft, etc. will be covered.

(2) Aircraft Carriers

Aircraft carriers serve as platforms for flight operations by their air wings and also form the nucleus of CSGs. All of the Navy’s current and planned aircraft carriers are nuclear powered, meaning that they can operate for long periods without needing to be refueled. In addition, all of them are large enough and have the necessary design features to allow sustained air operations by fixed-wing aircraft that are not capable of performing short takeoffs and vertical landings. On its own, an aircraft carrier has a limited ability to defend itself from attacks by missiles, aircraft, submarines, or other ships. Its air wing and the other ships in its CSG are responsible for helping to defend the carrier.

(a) Purpose and Limitations. The Navy’s carrier force gives the U.S. the ability to strike a wide variety of targets across the world by air, particularly in places where the U.S. military does not have its own air bases on land or access to other countries’ air bases. The range of Navy fighter aircraft (and the ability to use aerial refueling) means that carrier air wings can strike targets relatively far inland, not just along coasts. In addition, the mobility of aircraft carriers allows the U.S. to reposition them to assist in almost any likely combat scenario. Experience has also shown that carrier-based aircraft are among the most powerful anti-ship weapons and that surface combatants exposed to attack from aircraft are extremely vulnerable (although the U.S. has not had many occasions to use that capability since World War II).

The main limitation of the carrier force is that carrier aviation is a relatively expensive way to employ tactical aircraft in operations in which air bases on land are available to the U.S. The U.S. military has invested heavily in naval aircraft and has used them in every major conflict since World War II (at times, perhaps, because the assets existed rather than because they were the only assets that could perform a particular mission). In many of those conflicts, however, the unique value of aircraft carriers to provide bases in otherwise inaccessible locations was not fully demonstrated because the U.S. had access to air bases on land for at least part of the conflict.

A possible further drawback of aircraft carriers is that during combat operations, they could face a number of threats that might make them vulnerable, despite the defensive capabilities of the other ships in a strike group. However, Navy ships have not faced sustained attacks since World War II, so it is difficult to assess how vulnerable aircraft carriers would be in a conflict in which they came under heavy attack from aircraft, cruise missiles, ballistic missiles, or submarines. Analysts have long debated how well aircraft carriers could survive an attack in a contested naval environment.

Although no adversary has successfully attacked a U.S. carrier since 1945, the importance of aircraft carriers for the U.S.’ ability to project power has created strong incentives for hostile states to develop weapons and tactics to counter those ships and their aircraft. For example, some states are developing high-speed anti-ship cruise missiles and anti-ship ballistic missiles in an effort to penetrate the air defenses of CSGs. In turn, the emergence of those more sophisticated weapons has led the Navy to develop responses, including improvements in air and missile defenses.

(b) Employment. For more than 70 years, the U.S. has used carrier-based aircraft in all of its major combat operations, as well as in a number of smaller operations. In many cases, those aircraft have been the most rapid and flexible form of military response available to the U.S. Aircraft carriers have also been employed, though to a much more limited degree, for some nontraditional missions, such as disaster response. In addition, plans for a U.S. invasion of Haiti in the mid-1990s (called Operation UPHOLD DEMOCRACY) envisioned using two aircraft carriers

as bases for an air assault by an Army division, with the division's helicopters taking the place of the carriers' normal air wings. (The invasion was never carried out because a diplomatic solution to the crisis was found.) The U.S. military seems likely to continue to use aircraft carriers in future conflicts unless a potential adversary proves capable of presenting an unacceptably dangerous threat to carrier strike groups (as some analysts believe China might in a future conflict in the South China Sea).

The Navy's goals for the size of the carrier fleet are based on its analysis of wartime scenarios as well as on its goals for having ships deployed overseas (providing what is commonly called forward presence). In major U.S. military operations since the end of the Cold War—such as the conflicts in Kuwait in 1991, in Afghanistan in 2001, and in Iraq in 2003—the Navy eventually provided five to seven aircraft carriers. Maintaining a fleet of 11 carriers would usually allow 5 of them to be available within 30 days for a crisis or conflict (the rest would be undergoing scheduled maintenance or taking part in training exercises and would be unready for combat). Within 90 days, the Navy would generally have seven carriers available. A larger carrier force would be able to provide more ships for a conflict, and a smaller force would provide fewer.

During peacetime, the carrier fleet conducts routine patrols around the world, providing forward presence to reassure the U.S.'s friends and allies and deter potential aggressors. Given the Navy's normal operating cycles for ships and crews, the current force of 11 carriers—one of which is based in Japan—can provide the equivalent of two carriers deployed year-round and a third carrier deployed for eight months of the year. (At any given time, the other carriers are transiting to or from their deployment areas, engaging in training activities, undergoing routine maintenance, or being overhauled.) Having more carriers, longer deployments, or more carriers based overseas would increase the fleet's capability to provide forward presence, whereas having fewer carriers or shorter deployments, or withdrawing the carrier based in Japan, would decrease that capability.

(c) Nimitz and Ford Class Aircraft Carriers (CVN). There are currently 11 CVNs in the Navy's inventory, with two under construction. One CVN rotates as part of the Forward Deployed Naval Forces–Japan for approximately five years at a time. General characteristics of the *Nimitz* class aircraft carrier are:

- Maximum speed of over 30 kts (35 mph)
- Complement: 3,200 ship's company and 2,480 Carrier Airwing (5,680 total).
- Armament: In addition to the Carrier Airwing, for self-defense the CVN carries a variety of NATO *Sea Sparrow* launchers; the 20mm *Phalanx* Close-In Weapons System (CIWS); Rolling Airframe Missiles (RAM); or the SeaRAM system.
- All CVNs in the U.S. register are nuclear powered.

Nimitz class aircraft carrier construction is complete, and it is being replaced with the *Gerald R. Ford* class aircraft carrier, which is a larger and more technologically advanced ship class. Four *Ford* class aircraft carriers are currently programmed for purchase.



Figure 10: Nimitz Class CVN.



Figure 11: Ford Class CVN.

(d) Carrier Air Wings (CVW). CVWs conduct operations from the deck of aircraft carriers against both land and sea targets. They provide for CSG defense and conduct sustained air operations in support of allied forces. The CVW is nominally composed of eight aircraft squadrons with a total of approx. 64 aircraft across all types:

- A mixture of four U.S. Navy and U.S. Marine Corps squadrons of tactical aircraft (F/A-18 or F-35 variants), approximately 40 aircraft.
 - **F/A-18 Super Hornet Strike Fighter (E and F models).** An all-weather aircraft used as an attack aircraft as well as a fighter. In fighter mode, it serves as a fighter escort and for fleet air defense. In its attack mode, it is used for force projection, interdiction, and close and deep air support.



Figure 12: F/A18 Super Hornet.

- **F-35 Lightning II (B and C models).** A multirole, fifth generation stealth fighter designed for air-to-air combat, air-to-ground strikes, electronic attack, intelligence, surveillance, and reconnaissance. The "B" model is a single-seat aircraft used by the Marine Corps. It has a specially designed lift system that allows for short take-off and vertical landing capabilities. The "C" model is a single-seat aircraft used by the Navy and designed to launch from longer shore-based runways or with assistance from a catapult onboard a CVN.



Figure 13: F-35C *Lightning II*.

- One squadron of command-and-control aircraft (E-2D *Hawkeye*), approximately four aircraft. An all-weather aircraft, the **E-2D *Hawkeye*** is a carrier-based tactical battle management airborne early warning aircraft with a second principal role of airborne command and control. Additional missions include surface surveillance coordination, air interdiction, offensive and defensive counter air control, close air support coordination, time-critical strike coordination, search-and-rescue airborne coordination, and communications relay.



Figure 14: E-2C *Hawkeye*.

- One squadron of multi-mission helicopters (MH-60S and MH-60R), approximately eight aircraft.
 - **MH-60R *Sea Hawk***. Anti-submarine warfare and surface warfare are the MH-60R's primary missions. secondary missions include electromagnetic warfare, search and rescue, vertical replenishment, Naval Surface Fire Support, logistic support, personnel transport, medical evacuation, and radio communications relay.
 - **MH-60S *Sea Hawk***. The MH-60S is a multi-purpose helicopter. Its principal missions are surface warfare, combat support, humanitarian disaster relief, combat search and rescue, aero medical evacuation, special operations forces support, and organic airborne mine countermeasures. This helicopter is normally fitted with an external lift or sling to assist with mission execution.



Figure 15: MH-60S *Seahawk* Outfitted for Surface Warfare.



Figure 16: MH-60S *Seahawk* Outfitted for Anti-Submarine Warfare.

- One squadron of electronic attack aircraft (E/A-18G), approximately eight aircraft. The **E/A-18G Growler** is the electronic attack variant of the F/A-18E/F *Super Hornet*. In addition to the F/A-18E/F capabilities, the Growler incorporates a sophisticated electronic warfare suite and additional radars to perform a wide range of enemy defense suppression missions.



Figure 17: E/A-18G Growler.

- One squadron of logistics aircraft (C-2A), approximately four aircraft. The **C-2A Greyhound** is a multi-purpose aircraft that provides rapid airborne logistics capability to the carrier strike force. It carries both cargo (including mail) and personnel from shore to ship and vice versa. To execute its mission, the Greyhound is equipped not only with a specialized cabin arrangement to allow for prioritization of cargo but also with a cargo cage system that provides restraint for loads during ship launches and landings. Fast turnaround for flight deck operations is enabled by a straight-in rear cargo loading capability, and the cargo ramp can be opened in flight, allowing for airdrops of supplies and personnel.



Figure 18: C-2A Greyhound.

(3) Surface Combat Ships

The Navy divides its fleet of surface combat ships into large surface combatants (destroyers and cruisers) and small surface combatants (littoral combat ships). The larger combatants are powerful ships equipped with the Vertical Launch System (VLS), which allows them to use several different kinds of missiles to attack targets in the air, at sea, or on land. The smaller combatants do not have the VLS but instead carry a variety of smaller and more specialized weapons intended mainly for defensive purposes, particularly antisubmarine warfare. Most of the Navy's surface combatants carry one or two MH-60 *Seahawk* helicopters to assist in various missions.

Since World War II, the Navy's surface combatants have evolved from being vessels distinguished primarily by the size of their main guns—which in turn largely determined the size of the ships—to being versatile platforms for several weapon systems. Since the introduction of the VLS in the early 1980s, the Navy's large surface combatants have been differentiated mainly by their sensors and intended combat specialties rather than by their size or type of weapons. Ships equipped with the VLS can carry an interchangeable set of standard munitions, including *Tomahawk* cruise missiles, ASROC antisubmarine weapons, and Standard air-defense missiles. (Such ships can also carry *Harpoon* anti-ship missiles, which use a launch system other than the VLS.) In addition, the Navy has a limited number of Standard missiles that can intercept short- and medium-range ballistic missiles, although that number is expected to grow. Similarly, the Navy's small surface combatants have

become versatile ships primarily intended to defend larger ships against attack by submarines and small boats and to replace the Navy's mine countermeasures ships. All of the Navy's surface combatants have enough defensive capability that they can operate independently during normal peacetime deployments.

(a) Current Structure. The Navy currently has 115 surface combat ships of various sizes, including destroyers (DDG), cruisers (CG), frigates (FFG), and littoral combat ships (LCSs). Together, surface combatants account for about 14 percent of the Department of the Navy's total operation and support funding.

(b) Purpose and Limitations. A majority of the Navy's surface combatants are used in CSGs to protect aircraft carriers. Although numbers vary at times, a carrier strike group generally includes five or six surface combatants, in addition to the carrier and an attack submarine. Surface combatants could also be used to escort and defend amphibious ready groups in some scenarios, but it is not currently normal peacetime practice for the Navy to deploy surface combatants with those groups. In addition, surface combatants are frequently deployed on their own or in small groups, known as surface action groups (SAGs) either to defend an area against ballistic missiles or to allow the Navy's limited number of ships to provide a greater amount of forward presence in places of interest to the U.S. Missile defense missions and forward presence missions are similar in many respects, though they differ in some ways. In both cases, the essence of the deployment is simply to be available in some area. However, the Navy's ability to carry out missile defense missions depends on the limited number of large surface combatants that have ballistic missile defense capability, and the locations of those missions are determined by the possible flight paths that missiles could travel between an adversary and its potential targets.

The main limitation of surface combatants is that they have less capability than aircraft carriers or amphibious ships to affect ground combat operations, which have dominated the major conflicts in which the U.S. has engaged for the past 70 years. Although large surface combatants can launch *Tomahawk* cruise missiles, the Navy has a significant capability to fire cruise missiles from other vessels (such as attack and guided missile submarines), and most U.S. combat operations rely on tactical aircraft for the vast majority of strikes on ground targets. Surface combatants also have guns that can provide firepower, but those guns have relatively short ranges, which severely limits their ability to affect combat operations on land.

In general, surface ships face a number of potential threats in naval combat operations that might make them vulnerable. However, because the U.S. has engaged in very little naval combat since World War II, it is difficult to gauge how vulnerable the Navy's surface ships would be if they came under heavy attack from aircraft, cruise missiles, ballistic missiles, or submarines. Some events—such as the war between the United Kingdom and Argentina over the Falkland Islands in 1982, and the attack on the USS *Stark* by a missile launched from an Iraqi jet in 1987—suggest that surface ships may be extremely vulnerable to modern weaponry. Moreover, during Operation Desert Shield in the early 1990s, two U.S. surface combatants hit Iraqi mines, which suggests that older naval mines can be effective against Navy ships. Similarly, in 2000, a boat filled with explosives attacked the USS *Cole* in a port in Yemen, indicating that small boats may be capable of inflicting great damage on surface combatants operating close to shore. The Navy, however, has taken a number of steps to respond to those potential threats, and it is difficult to judge how successfully U.S. surface combatants might fare in similar situations in the future.

(c) Employment. In practice, the most common contributions that surface combatants have made to U.S. combat operations in recent decades have been as platforms for launching *Tomahawk* cruise missiles to strike targets on land and as protectors of aircraft carriers and amphibious ships. Those roles reflect the nature of recent conflicts: Iraq and Afghanistan had no significant naval forces to engage. In possible future conflicts, however, the ability of U.S. cruisers and destroyers to provide missile defense and air defense could be significant. For example, scenarios involving possible future conflicts between the U.S. and China over the status of Taiwan would probably require the Navy's large surface combatants to defend Taiwan from attack by ballistic missiles and to defend U.S. carriers from attack by aircraft, ballistic missiles, and cruise missiles. Similarly, scenarios involving attempts by Iran to restrict shipping through the Strait of Hormuz would probably require that large surface combatants defend against aircraft and missiles and that surface combatants of all sizes defend against submarines and small boats.

Analyses of such wartime scenarios have led the Navy to set a goal of having 88 large surface combatants. Although a significant portion of the Navy's cruisers and destroyers are dedicated to protecting aircraft carriers, they also carry out a variety of independent operations and other missions, such as providing regional ballistic missile defense in Europe and Northeast Asia. Major reductions in the force of large surface combatants (without similar reductions in the force of aircraft carriers) might imperil the Navy's ability to provide escorts to carriers, but small or moderate changes to the number of large surface combatants would not, although they might affect the Navy's ability to conduct other missions or to provide forward presence in peacetime.

With a planned force of 88 large surface combatants—including 9 based in Japan and 4 based in Spain—the Navy could have approximately 28 of those ships operating in overseas areas at any one time, given its normal operating cycle. Buying more ships, conducting longer deployments, or basing more ships overseas would increase that number, and the reverse would decrease it.

The Navy's plans for small surface combatants call for having a force of 52 by 2028. This force will consist entirely of littoral combat ships—including, in the future, a new class of improved LCSs that the Navy plans to designate as frigates. LCSs (and the future frigates) use a scheme of rotating crews that would allow the Navy to keep 26 of those 52 small surface combatants forward deployed at any given time.

(d) Ticonderoga Class Guided Missile Cruisers (CG). These are large, multi-mission combat ships that operate independently or can support CSGs or Expeditionary Strike Groups (ESG) (the various groups will be discussed below). Normally, the principal mission of a cruiser is to serve as the Air Defense Commander for a CSG, and the ship is outfitted with additional work and berthing spaces to accommodate the Sailors necessary to execute this mission. Other missions include ballistic missile defense, anti-surface warfare, anti-submarine warfare, and land attack. The CG uses the *Aegis Weapons System*, including the SPY-1D radar, and a variety of missiles to accomplish its role. The CG's general characteristics are:

- Maximum speed of 30+ knots.
- Crew: 24 officers and 340 enlisted (364 total).
- Sensors: In addition to the SPY-1 radar, the CG is equipped with additional surface and air search radars, and an extensive SONAR suite, including both active and passive underwater search capabilities.
- Armament: 124 vertical launching system cells capable of carrying air defense missiles, ballistic missile defense missiles, vertically launched anti-submarine rockets, and *Tomahawk* cruise missiles. The CG is also equipped with two 5-inch deck guns, two 20mm *Phalanx* CIWS systems, torpedoes, and is capable of carrying two lightweight MH-60R or MH-60S helicopters.



Figure 19: USS San Jacinto (CG 56).

(e) Arleigh Burke Class Destroyers (DDGs). These are large surface combatants (warships) that provide multi-mission offensive and defensive capabilities. Destroyers can operate independently or as part of CSGs, SAGs, Amphibious Ready Groups (ARG), and underway replenishment groups. Destroyers are typically known as the “workhorse of the fleet” because they can conduct multi-mission operations in twelve distinct mission areas to include: area air defense, ballistic missile defense, surface warfare, anti-submarine warfare, and land attack. They differ from CGs primarily because they have less vertical launching system cells, and therefore carry fewer missiles, do not have the physical space to embark an additional staff, and the older variants of DDGs do not have a helicopter hangar. Three “flights” of DDGs are currently in the U.S. inventory: the oldest (flight I) can only temporarily service one helicopter at a time; the mid-range DDGs (flight II) have a helicopter hangar and are capable of embarking two helicopters; and finally, the newest variants (flight III), have helicopter hangars, can embark two helicopters, and also have the newest version of the SPY-1 radar system.

- Maximum speed of 31+ knots,
- Crew: 24 officers and 280 enlisted (304 total).
- Sensors: In addition to the SPY-1 radar, the DDG is equipped with additional surface search radars and an extensive SONAR suite, including both active and passive underwater search capabilities.
- Armament: Outfitted with the *Aegis Weapons System*, including the SPY-1 radar; 90-96 usable vertical launching system cells (depending on the flight) capable of carrying air defense missiles, ballistic missile defense missiles, vertically launched anti-submarine rockets, and *Tomahawk* cruise missiles. The DDG is also equipped with a 5-inch deck gun, two 20mm *Phalanx* CIWS systems, torpedoes, and is capable of supporting a maximum of two lightweight MH-60R or MH-60S helicopters (depending on the flight).



Figure 20: Flight II DDG, USS *Porter* (DDG 78).



Figure 21: Flight IIA (Helicopter Hanger) DDG, USS *John Finn* (DDG 113).

(f) Littoral Combat Ships (LCS). The newest ship class in the U.S. Navy's arsenal are fast, agile, focused-mission ships designed for operation in near-shore environments yet capable of open-ocean operation. They are designed to defeat asymmetric anti-access threats such as mines, quiet diesel submarines, and fast surface craft. These ships, with a low draft and fast top speed, were designed to fill the void left by the decommissioning of the *Oliver Hazard Perry* class frigates (FFG) and the *Avenger* Mine Countermeasures Ships. Their employment is intended to concentrate on the littorals and asymmetric warfare areas in support of an ESG or ARG, but they are still able to transit through the blue water environments in company with the strike groups. The manning on these ships is a different construct for the Navy; the ship's crew is "minimally manned" and is assigned duties to take care of the hull, while the combat power comes nearly entirely from the embarked mission module detachment. With a crew of about 75 Sailors to take care of an average 400-foot ship, there is not much tolerance for additional tasking without providing a surge in manpower or equipment. LCS There are two classes of LCS: the *Freedom* and *Independence*, with a total of 32 ships funded between the classes. These ships will be outfitted with reconfigurable payloads, called Mission Packages, which can be changed out quickly. Mission packages are supported by special detachments that will deploy manned and unmanned vehicles and sensors in support of mine, undersea, and surface warfare missions. There are three mission packages planned, and approximately ten LCS will be fitted with each of the mission packages, giving the Navy additional capability in mine warfare, surface warfare, and anti-submarine warfare. Maritime Interdiction Operations and Maritime Security Operations, although not official mission packages, are capabilities any littoral combat ship could execute if provided the additional personnel and equipment. The two variants, which are large enough to be designated as a frigate, are very different from each other.

- **Freedom Class LCS**

- Hull design: steel monohull.
- Draft: 14 feet.
- Speed: 40+ knots.
- Complement: approximately 40 Sailors with berthing for 35 mission module support staff (75 total).
- Armament: one Naval Strike Missile (NSM) system, one RAM system, and one medium caliber deck gun.
- Capable of employing a variety of helicopters, unmanned aerial systems, small boats and unmanned surface systems independently or resident within the mission module complement.



Figure 22: Freedom Class USS Little Rock (LCS 9).

- **Independence Class LCS**

- Hull design: aluminum trimaran.
- Draft: 14 feet.
- Speed: 40+ knots.
- Complement: approximately 50 Sailors with berthing for 25 mission module support staff (75 total).
- Armament: one NSM system, one RAM system, and one medium caliber deck gun.
- Capable of employing a variety of helicopters, unmanned aerial systems, small boats, and unmanned surface systems independently or resident within the mission module complement.



Figure 23: Independence Class USS Oakland (LCS 24).

(4) Combat Logistics Force (CLF) Ships

These logistics-oriented ships are operated by the MSC and provide at-sea sustainment for combat forces. The ships normally assigned to CSGs or ESGs are either **Fleet Replenishment Oilers (T-AOE)** or **Dry Cargo/Ammunition (T-AKE)** ships. Support to the ARG or ESG may also rely on Maritime Pre-positioning Forces (MPF) vessels from the CLF (MPF vessels and capabilities will be discussed in detail in the 8672 *Logistics* sub-course).

Both classes of at-sea replenishment ships carry shipboard and aviation fuels, ammunition, and stores, including spare parts and mail, although the specific capabilities vary between the classes. They deliver their products through either connected replenishment, when ships sail parallel to each other in close proximity and are connected by replenishment rigging, or by vertical replenishment, through the use of rotary-wing logistics assets. The ability to fully replenish at sea is a critical capability of the U.S. Navy and allows combat forces to remain at sea—providing power projection—indefinitely. For instance, on 3 October 2020, the USS *Stout* (DDG 55) set a Navy record of 215 consecutive days at sea throughout the 5th and 6th Fleets. This was made possible through 36 replenishment-at-sea evolutions, supported by both T-AO and T-AKE ships. T-AOEs provide underway replenishment of fuel, fleet cargo, and stores to customer ships at sea. There are currently 15 ships of this class in the MSC inventory. T-AKEs were designed to replenish dry and refrigerated stores as well as ordnance. They have replaced combat stores ships and ammunition ships, incorporating the capabilities of both platforms into one hull. These multi-product ships deliver food, fuel, spare parts, ammunition, and potable water to the U.S. Navy and our allies. There are currently 14 ships of this class in the MSC inventory. The general characteristics of the T-AOE/T-AKE ships are:

- Max speed: 20+ knots.
- Aviation assets: Embark up to two helicopters for vertical replenishment.
- Loads:
 - T-AOE: 170,000+ barrels of aviation and diesel fuel; 7,500+ sq ft of cargo.
 - T-AKE: 18,000 barrels of aviation and diesel fuel; 780,000+ cu ft of cargo.
- Total complement: between 75-140 personnel, depending on class, with berthing for additional passengers.



Figure 24: USNS Supply (T-AOE 6).



Figure 25: USNS Washington Chambers (T-AKE 11).

(5) Amphibious Ships

As their name implies, amphibious ships are designed to conduct operations that involve moving from sea to land—specifically, delivering forces into hostile territory from friendly ships. Amphibious ship employment, capabilities, and limitations will be discussed in detail in the 8673 *Amphibious Foundations* sub-course.

(a) Current and Planned Structure. By law, the Navy must maintain 31 amphibious ships (Those figures do not include 2 command ships, USS *Blue Ridge* and USS *Mount Whitney*, which are considered part of the amphibious fleet in the Department of Defense’s Future Years Defense Program). Amphibious ships account for about 9 percent of the DoN’s total operation and support funding.

(b) Employment. The U.S. has frequently used amphibious ships to deploy Marine Corps forces for small-scale operations, and it seems likely to continue to do so in the future. The U.S. has also deployed amphibious ships for major combat operations, but it has not conducted any large-scale amphibious assaults since the 1950 Inchon landings during the Korean War. Amphibious ships played a fairly minor role in the 1991 and 2003 wars with Iraq. However, during operations against the Taliban in 2002, a small Marine Corps force assaulted Kandahar, Afghanistan, from an amphibious ready group more than 400 miles away in the Indian Ocean. That assault showed the ability of modern amphibious forces to deploy entirely by air over a long range.

Like other surface ships, amphibious ships are used extensively during peacetime for routine patrols to provide forward presence. Their notional operating cycle—one 7-month deployment every 36 months—means that with the current fleet of 33 amphibious ships (4 of which are based in Japan), the Navy can have the equivalent of 8 amphibious ships providing overseas presence year-round and a 9th ship for about 4 months of the year. Acquiring more amphibious ships, lengthening deployments, or basing more amphibious ships overseas would increase the fleet's capacity to provide forward presence, whereas having fewer ships, shortening deployments, or withdrawing ships based in Japan would decrease that capacity. In recent years, high demand for operating amphibious ships overseas has led the Navy to extend deployments for most amphibious ships well beyond the 7 months of their official operating cycle (which is itself an increase from 6 months a decade ago).

(c) Potential New Amphibious Ship Architecture. Statements from the Commandant of the Marine Corps (CMC) suggest strongly that the Integrated Force Structure Assessment (INFSA) might change the Navy's amphibious ship force to an architecture based on a new amphibious lift target and a new mix of amphibious ships.

The 38-ship amphibious ship force-level goal was intended to meet a requirement for having enough amphibious lift to lift the assault echelons of two Marine Expeditionary Brigades (MEBs), a requirement known as the 2.0 MEB lift requirement. The 2.0 MEB lift requirement dates to 2006. The translation of this lift requirement into a Marine Corps-preferred force-level goal of 38 ships dates to 2009, and the Navy's formal incorporation of the 38-ship goal (rather than a more fiscally constrained goal of 33 or 34 ships) into the Navy's overall ship force-structure goal dates to the 2016 Force Structure Assessment (FSA).

In July 2019, General David H. Berger, CMC, released his *Commandant's Planning Guidance* that states that the Marine Corps wants to, among other things, move away from the 38-ship amphibious ship force-level goal and the 2.0 MEB lift force-planning metric, and shift to a new and different mix of amphibious ships. The new mix includes existing types of amphibious ships called for in the 2016 FSA, along with smaller amphibious ships, ships like the Navy's Expeditionary Sea Base (ESB) and Expeditionary Fast Transport (EPF) ships (referred to collectively as E-class ships), ships based on commercial-ship hull designs, and unmanned surface vehicles (USVs). The *Commandant's Planning Guidance*, which effectively announces a once-in-a-generation change in Marine Corps thinking on this and other issues relating to the Marine Corps, states in part (emphasis as in the original document):

We must also explore new options, such as inter-theater connectors and commercially available ships and craft that are smaller and less expensive, thereby increasing the affordability and allowing acquisition in greater quantity. We recognize that we must distribute our forces ashore, given the growth of adversary precision strike capabilities, so it would be illogical to continue to concentrate our forces on a few large ships. The adversary will quickly recognize that striking while concentrated (aboard ship) is the preferred option. We need to change this calculus with a new fleet design of smaller, more lethal, and more risk-worthy platforms. We must be fully integrated with the Navy to develop a vision and a new fleet architecture that can be successful against our peer adversaries while also maintaining affordability. To achieve this difficult task, the Navy and Marine Corps must ensure larger surface combatants possess mission agility across sea control, littoral, and amphibious operations, while we concurrently expand the quantity of more specialized manned and unmanned platforms....

We will no longer use a "2.0 MEB requirement" as the foundation for our arguments regarding amphibious ship building, to determine the requisite capacity of vehicles or other capabilities, or as it pertains to the Maritime Prepositioning Force. We will no longer reference the 38-ship requirement memo from 2009 or the 2016 Force Structure Assessment as the basis for our arguments and force structure justifications. The ongoing 2019 Force Structure Assessment will inform the amphibious requirements based upon this guidance. The global options for amphibs [types of amphibious ships] include many more options than simply LHAs, LPDs, and LSDs. I will work closely with the Secretary of the Navy and Chief of Naval Operations (CNO) to ensure there are adequate numbers of the right types of ships, with the right capabilities, to meet national requirements.

(d) Amphibious Assault Ships. These warships are the largest of all amphibious warfare ships and resemble a smaller aircraft carrier. The U.S. Navy has two types of amphibious assault ships in its inventory: the **Multiple Purpose Amphibious Assault Ship (LHD)**, and the **General Purpose Amphibious Assault Ship (Replacement) (LHA[R])**. These vessels project power and maintain presence by serving as the cornerstone of the ARG and ESG. They transport and land elements of U.S. Marine Corps landing forces with a combination of aircraft and landing craft. A principal secondary mission for these ships is to participate in large-scale humanitarian aid/disaster relief operations, such as post-tsunami efforts in Japan or hurricane recovery efforts in Louisiana and Puerto Rico. Amphibious Assault Ships, in addition to hosting numerous aviation assets and afloat connectors, such as **Landing Craft Air-Cushioned (LCAC)** and **Landing Craft Utility (LCU)**, also serve as command ships for most major operations. The main differences between the LHA and the LHD are the number of aviation assets each can carry and availability of a well deck.

- **Wasp Class LHD.** There are currently eight LHDs in commission. These ships continue to support both aviation operations from their hangar and flight decks and sea-based assets from their well decks for U.S. Marine Corps forces.
 - Speed: 20+ knots.
 - Complement: approx. 1,000 Sailors and berthing for 1,900 embarked troops (2,900 total).
 - Armament: two RAM systems; two or three NATO *Sea Sparrow* systems, three 20mm *Phalanx* CIWS systems.
 - Aircraft: various combinations of F-35s, **MV-22 Osprey** tiltrotor aircraft, **CH-53E Super Sea Stallion** helicopters, **AH-1Z Viper** and **UH-1Y Venom** helicopters, and MH-60S helicopters.
 - Landing/Attack Craft: three LCACs, or two LCUs, or approximately 40 Amphibious Combat Vehicles (ACV).



Figure 26: USS *Makin Island* (LHD 8).

- **America Class LHA(R).** The U.S. Navy currently operates two “flight zero” LHA(R)s, the USS *America* (LHA 6) and USS *Tripoli* (LHA 7). These serve as aviation-oriented ships supporting U.S. Marine Corps forces. The well deck has been replaced with an increased hangar bay, increased aviation maintenance space and capabilities, additional aviation fuel storage capacity, and a reconfigurable command-and-control/intelligence bay. The third LHA(R), the USS *Bougainville* (LHA 8) - a “flight I” LHA(R), will reincorporate a well deck and be able to launch LCACs, LCUs, and ACVs.
 - Speed: 20+ knots.
 - Complement: 1,060 Sailors, with berthing for 1,690 embarked troops (2,750 total).
 - Armament: two RAM systems; two NATO *Sea Sparrow* systems, two 20mm *Phalanx* CIWS systems.
 - Aircraft: various combinations of F-35s, MV-22s, CH-53Es, AH-1Zs, UH-1Ys, and MH-60Ss.
 - Landing/Attack Craft: LHA 6 and 7: none; LHA 8 and future: three LCACs, or two LCUs, or approximately 40 ACVs.



Figure 27: USS *America* (LHA 6).

(e) San Antonio Class Amphibious Transport Dock Ships (LPD). These warships embark, transport, and land elements of a Marine landing force for a variety of expeditionary warfare missions. These ships accomplish this transport by using LCACs, LCUs, ACVs, or through tilt and rotary-wing aircraft. LPDs support expeditionary warfare missions, including amphibious assault and special operations. They serve as secondary aviation platforms for ARGs and ESGs. The U.S. Navy currently operates 14 LPDs with 12 more to be built. The ship characteristics are:

- Speed: 22+ knots.
- Complement: 360 Sailors and berthing for 800 embarked troops (1,160 total).
- Armament: two RAM systems, two MK 46 30mm Close in Guns, and ten .50cal machine guns.
- Aircraft: launch or land two CH-53Es, or two MV-22s, or up to four AH-1Z/UH-1Y/MH-60s.
- Landing/Attack Craft: two LCACs or one LCU, and 14 ACVs.



Figure 28: USS New York (LPD 21).

(f) Dock Landing Ships (LSD). Supporting amphibious operations, including landings through the use of LCACs, conventional landing craft, and helicopters onto hostile shores, these ships are a crucial part of ship-to-objective maneuver and support for embarked Marines. There are two variants of LSDs: the *Whidbey Island* class, which primarily supports LCACs and conventional landing craft, including docking and repair services, and the second variant, the *Harper's Ferry* class, which focuses on cargo capacity by reducing the LCAC and landing craft numbers and services. There are twelve LSDs in commission, and their general characteristics are:

- Speed: 20+ knots.
- Complement: approximately 420 Sailors and berthing for 500 embarked troops (920 total).
- Armament: two RAM systems, two 20mm *Phalanx* CIWS systems, and two 25mm MK 38 chain-guns.
- Landing Craft: two to four LCACs (depending on class).



Figure 29: USS *Fort McHenry* (LSD 43).

(6) Fleet Ballistic Missile Submarines (SSBN)

"Boomers" serve as an undetectable launch platform for intercontinental missiles. A critical part of the U.S. nuclear triad, they are designed specifically for stealth and the precise delivery of nuclear warheads. SSBNs carry the *Trident II* D5 missiles, and under the provisions of the New Strategic Arms Reduction Treaty, can carry a maximum of 20 missiles. The U.S. Navy currently has 14 SSBNs in its register. The *Columbia* class submarine, with the first boat under construction, will replace the current *Ohio* class submarines.



Figure 30: USS *Alaska* (SSBN 732).

(10) Guided Missile Submarines (SSGN)

These undersea vessels are also based on the *Ohio* class submarine platform, with conversion beginning in 2002, focusing on land attack and special operations force support. The four SSGNs in the U.S. inventory are each capable of employing up to 154 *Tomahawk* land-attack cruise missiles or supporting up to 66 SOF personnel, including an optional dry dock shelter attached to the hull. The large missile tubes of the SSGN can also be refitted to support new types of missile or cargo, as necessary to support the Navy's future fighting force.



Figure 31: USS Georgia (SSGN 729) with Optional Drydock Shelter for SOF.

(11) Attack Submarines (SSN)

Attack submarines are designed to seek and destroy enemy submarines and surface ships; project power ashore with *Tomahawk* cruise missiles and special operations forces; carry out intelligence, surveillance and reconnaissance missions; support battle group operations; and engage in mine warfare. The Navy's attack submarines are large vessels powered by nuclear reactors, which allow them to operate underwater for long periods with no practical limits on their range. They are armed with a variety of weapons, such as torpedoes for destroying surface ships and other submarines, and *Tomahawk* cruise missiles for striking targets on land. In addition, some U.S. attack submarines have been fitted with specialized equipment allowing them to deliver teams of special forces ashore. Attack submarines are not capable of performing some naval missions, such as engaging aerial targets or providing missile defense.

(a) Current Structure. The Navy currently has 51 attack submarines (which consist of *Los Angeles*, *Seawolf*, and *Virginia* class submarines). Attack submarines account for about 7 percent of the Department of the Navy's total operation and support funding.

(b) Purpose and Limitations. The Navy's fleet of attack submarines evolved largely to ensure the U.S.'s ability to use sea lanes around the world freely for military and civilian shipping during conflicts. For years, that fleet's main adversary was the Cold War-era Soviet Navy, which built large numbers of submarines in an effort to prevent the U.S. from transporting military forces to Europe by ship in the event of a conflict there. Another major mission for the Navy's attack submarines was to hunt for and destroy Soviet ballistic missile submarines (those carrying strategic nuclear warheads), including submarines operating beneath the Arctic ice pack.

In contrast to the Navy's nuclear-powered submarines, many potential U.S. adversaries have diesel electric submarines. Those submarines use diesel engines to charge batteries, which can then power the submarines for relatively short periods while submerged. Diesel electric submarines are often considered best suited to coastal defense for two reasons. First, the need to carry diesel fuel limits their range, and second, the need for an air supply (generally obtained either by surfacing or by raising an air-intake snorkel periodically) limits their ability to stay underwater. Diesel electric submarines can be more tactically effective than nuclear submarines because battery power is quieter underwater than a nuclear reactor. That quietness gives diesel electric submarines an advantage in detecting or avoiding detection by enemy warships and submarines.

The Navy is generally very secretive about its submarine operations. Nevertheless, it has been asserted that the stealthy nature of attack submarines makes them excellent intelligence-gathering assets, capable of observing foreign nations while undetected. A lack of unclassified information, however, makes it difficult to assess the value of that mission or the number of submarines that it requires. At the same time, the stealthy nature of attack submarines means that they are not useful for providing visible forward presence overseas, except when conducting port visits in other countries.

The main limitation of the attack submarine force is that it has relatively little ability to directly affect ground combat operations, which have dominated the U.S. military conflicts since World War II. Although attack submarines can launch *Tomahawk* cruise missiles, the Navy has an enormous capability to fire cruise missiles from other vessels, such as surface combatants and guided missile submarines. Moreover, most U.S.

combat operations rely on tactical aircraft for the vast majority of strikes on ground targets. Attack submarines can sometimes be used to deploy special forces covertly, but that capability is often more useful in peacetime than during major combat operations, when the U.S. has numerous methods for inserting special forces into a theater (including by fixed-wing aircraft or helicopters).

There is little reason to believe that the Navy's attack submarine fleet is particularly vulnerable to any type of threat in the current military environment. By their nature, submarines are the most difficult types of naval vessels to detect and destroy, and the greatest potential threat to any submarine is generally another submarine.

Some analysts have questioned how U.S. attack submarines might perform against advanced diesel electric submarines in shallow waters, such as those of the Persian Gulf, where diesel electric submarines have some tactical advantages. But the U.S. has various options for attacking and defeating such submarines, including land-based patrol aircraft, ship-based helicopters, and surface combatants.

(c) Employment. In recent decades, the most common roles that attack submarines have played in U.S. combat operations have been as platforms for launching *Tomahawk* cruise missiles at ground targets, for conducting surveillance, or for collecting intelligence. However, those roles reflect the fact that Iraq and Afghanistan had no significant naval forces to engage. In conflicts that the U.S. might face in the future, the ability of attack submarines to intercept an enemy's naval forces and commercial shipping close to the enemy's coastline could be important in the conduct of the conflict. For instance, scenarios involving conflicts between the U.S. and China over the status of Taiwan could easily hinge on the possibility of a Chinese amphibious invasion of Taiwan, in which case the ability of U.S. attack submarines to destroy Chinese vessels would be critical. Similarly, scenarios involving attempts by Iran to restrict shipping through the Strait of Hormuz might require U.S. attack submarines to destroy Iranian submarines (which would most likely be an important part of Iran's strategy to deny the United States access to the Persian Gulf).

On the basis of such wartime scenarios, the Navy's stated goal for the size of the attack submarine force has remained at 48 for the past decade. The Navy's analysis is based on classified information, however, so it is not clear what effects increasing or decreasing the size of that force would have on the Navy's ability to achieve its wartime objectives.

In peacetime, attack submarines' main missions are conducting surveillance, gathering intelligence, and supporting carrier strike groups. The Navy aims to have at least 10 attack submarines deployed overseas at any given time for various peacetime operations, which may also include supporting the activities of special-operations forces. The Navy currently bases three of its attack submarines in Guam, although that number will soon rise to four. The standard operating cycle for attack submarines—one 6-month deployment during a 24-month period—means that a submarine based in the continental U.S. is deployed overseas for an average of about 3 months per year (6 months over two years), whereas a submarine based in Guam is deployed overseas for about twice that amount of time. The Navy could keep more attack submarines overseas at any given time if it had a larger force, deployed submarines for longer periods, or stationed more of them at overseas bases. Conversely, a smaller force, shorter deployments, or fewer submarines based outside the U.S. would reduce the number of attack submarines operating overseas at any one time.

(d) Los Angeles Class. The oldest class of attack submarines, the Navy currently operates over 30 of these submarines. Most are outfitted with a vertical launch system to enable the use of the *Tomahawk* cruise missile system. All of these vessels carry torpedoes, travel in excess of 25 knots, and have a crew of approximately 16 officers and 127 enlisted Sailors. These boats are being replaced by the *Virginia* class attack submarine.



Figure 32: Los Angeles Class USS Newport News (SSN 750).

(e) Virginia Class. The replacement for the *Los Angeles* class submarine, these boats have several innovations that significantly enhance its warfighting capabilities. Improved overall ship design, updated ship control systems, periscopes, sonar suites, and payload tubes increase its lethality, and special features can be added to support special operations forces if desired.



Figure 33: Virginia Class USS Delaware (SSN 791).

(f) Seawolf Class. These boats are exceptionally quiet, fast, well-armed, and equipped with advanced sensors. Though lacking a vertical launching system, the SEAWOLF-Class has eight torpedo tubes and can hold up to 50 weapons.



Figure 34: Seawolf Class USS Seawolf (SSN 21).

(7) Maritime Patrol Aircraft

The Navy's fleet of approximately 90 maritime patrol aircraft consists of land-based, long-range aircraft equipped with a variety of sensors and weapons. They are capable of monitoring large areas of the ocean and improving the Navy's ability to find and track other nations' ships and submarines. They are also capable of conducting limited attacks on ships and submarines. The older P-3 *Orion* model patrol aircraft are currently being replaced by newer P-8 *Poseidon* model aircraft. The Navy is also in the process of fielding an unmanned long-range patrol aircraft, the MQ-4 *Triton*, which is based on the airframe of the Air Force's RQ-4 *Global Hawk*.



Figure 35: P-8 *Poseidon*.

f. U.S. Navy Combat Organizations (How the Navy Fights)

Previously discussed in the Operational Chain of Command section, Navy assets will change operational and tactical commanders often, reorganizing into mission-oriented groups. The Navy will fight from these different groups in the assigned mission areas. Based on the mission, the composition of the groups may vary widely. The descriptions below are the most common groups and their associated compositions.

(1) Carrier Strike Group (CSG) Composition

The CSG, normally commanded by a two-star surface warfare or aviation warfare admiral, is the cornerstone of naval operations and is composed of numerous individual units. Nominally:

$$\text{CSG} = \text{CVN} + \text{CVW} + \text{CG}^* + 4\text{-}6 \text{ DDGs}^* + \text{SSN} + \text{T-AOE}^* \text{ or } \text{T-AKE}^* \\ \text{*embarked helicopters}$$

The CSG is a symbiotic maritime group designed for power projection afloat and ashore, as well as sea control through the use of the carrier air wing and the offensive systems on the CG and DDGs. It is designed to control vast amounts of battlespace in the maritime domain against high-end adversaries. There are 23 different mission areas a carrier strike group can execute, and most of them focus on power projection afloat and ashore, or defense of the carrier strike group. Every unit in this group contributes to the defense of the other units. For instance, the SSN protects all of the ships from submarine attack and can attack enemy submarines. Simultaneously, the CG and DDGs use their sonar systems and embarked helicopters for enemy submarine prosecution, contributing to the mutual defense of the CSG, including the SSN. Additionally, the carrier air wing will conduct strike missions ashore, protected by the radars and offensive missiles of the CG and DDGs, while the CVW also protects the CG and DDGs from enemy aircraft when assigned to defensive counter-air patrol or combat air patrol duties.

Additionally, the CSG may disaggregate (divide or separate) its individual units for other missions and then reconstitute as necessary. Many DDGs are capable of Ballistic Missile Defense, and they are often detached from the CSG to execute that mission. Normally, the CVN and the CVW remain together, and at least two of the *Aegis* ships are tethered in close proximity to support strike group defense. Other units, such as an SSGN, may be added on a temporary basis to augment specific mission taskings of the CSG—in this case, strike missions.

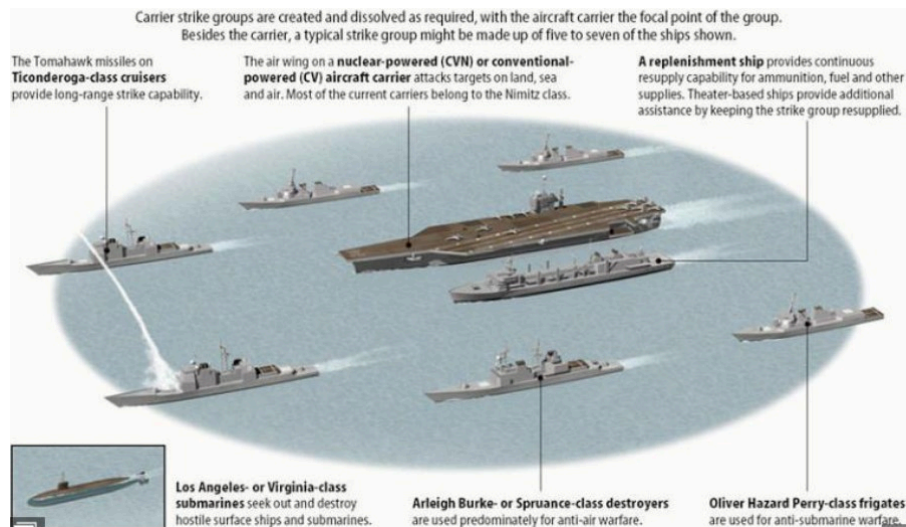


Figure 36: Typical CSG (the Oliver Hazard Perry Frigate Represents the LCS).

Effective command and control of a fighting force of nearly 7,500 Sailors, ten capital platforms and approximately 70 aircraft capable of executing missions in over 25 mission areas across thousands of miles of battlespace requires agility. Therefore, command and control of a CSG is executed through the Composite Warfare Commander construct (this will be later in this reading).

(2) Amphibious Ready Group (ARG) Composition

The ARG, commanded by a Navy captain (normally the commander of the amphibious squadron from which the amphibious ships are drawn), is an agile group of amphibious ships that normally operates in the littoral region. The major elements are:

$$\text{ARG} = \text{LHA or LHD} + \text{LPD} + \text{LSD}$$

The ARG is predominantly assigned amphibious assault missions, other expeditionary warfare missions, and humanitarian aid/disaster response duties. This group has organic capabilities throughout the land, air, and maritime domains, making it extremely versatile and reconfigurable to meet a wide variety of missions. Like the CSG, the ARG may disaggregate and reconstitute its individual units, or in this case, individual aircraft or land forces, for other missions as necessary. In this group, the assigned tiltrotor and rotary-wing assets are in high demand for a variety of missions and are often temporarily reassigned for a few days at a time in support of off-ship operations. The ARG will be discussed in detail in the 8673 *Amphibious Foundations* sub-course.

(3) Surface Action Group (SAG)

A SAG is a temporary or standing organization of combatant ships other than aircraft carriers tailored for a specific tactical mission. Nearly any mission imaginable may be assigned to a SAG: sea denial, chokepoint control, land attack strikes, anti-surface warfare, maritime interdiction operations, etc. For short-term missions, a SAG's assets may be temporarily drawn from a CSG. Command of the SAG normally resides with the most senior ship captain in the group, who is normally a Navy commander. Anti-submarine warfare, however, is a special mission subset formed of different units and given different naming conventions.

(5) Search and Attack Unit (SAU)

A SAU is formed when a group of naval platforms is specifically assigned anti-submarine warfare duties and, like a SAG, may be a temporary or standing organization. A SAU may include any combination of surface combatants, rotary-wing aircraft, fixed-wing aircraft (such as a P-8), and other submarines. This group is specifically oriented to detect, localize, and prosecute enemy submarines and, like a SAG, is usually commanded by the senior ship captain in the group, normally a Navy commander.

(6) Expeditionary Strike Group (ESG) Composition

The ESG, commanded by a Navy admiral (or in some cases a Marine brigadier general), is formed when the mission of the ARG is too great for the amphibious squadron commander to plan and execute. The elements of the ARG form the cornerstone of the group, with other units supplementing its core combat power:

$$\text{ESG} = \text{Staff} + \text{ARG} + 1\text{-}2 \text{ DDGs} + 2\text{-}4 \text{ LCSs} + \text{SSN (maybe)}$$

The ESG staff is a key component of this group, allowing for a greater capacity of both offensive and defensive planning when compared to the ARG alone. The composition of the ESG is extremely fluid, based not only on the mission but also with due regard to the operational area. For instance, if the ESG must transit or operate in an area known to have an enemy submarine threat, an SSN could be assigned to protect the

group, either temporarily or for the duration of the operation. Similar to the other groups, the ESG can and will disaggregate/reconstitute its units as needed. A simple way to think about the composition of an ESG is that it is an ARG combined with another group (e.g., a SAG) under a single commander and staff.

g. Composite Warfare Commander Concept

The U.S. Navy operates its large, distributed force based on a centralized planning structure and decentralized execution through command by negation. Centralized planning ensures a collaborative process with both commanders and subordinates involved in drafting the overarching guidance. The result of centralized planning should be the clear delineation of commander's intent through documents such as Operational Tasking Orders (OPTASKs), OPTASK supplements, Special Instructions Messages (SPINS), Daily Intentions Orders (DIMS), and individual platform Battle Orders. These documents constitute a large library of direction and pre-planned responses, as well as clear delegation of authority and responsibilities, that Sailors must be intimately familiar with and able to instantaneously execute.

Decentralized execution occurs when the commander implicitly trusts that subordinates take action without delay, in accordance with published directions, while continuously keeping the commander informed of the situation. Because of the use of command by negation, the commander will remain silent if the subordinates' actions meet with approval. If not, the commander will intervene and order new or different actions. Efficient execution of commander's intent does not happen instantaneously; it must be trained for and is the main focus of fleet and group integrated exercises.

All of the groups described in the previous section (CSG, ARG, ESG, SAG, SAU, etc.) execute command by negation throughout their missions. The operational term for this combination of centralized guidance and decentralized execution is called the Composite Warfare Commander (CWC) Concept, which is the embodiment of combining centralized planning, decentralized execution, and command by negation:

This concept allows the officer in tactical command to assign some or all of the command functions associated with mission areas to warfare commanders, functional group commanders, and coordinators, thus supporting decentralized execution. The composite warfare organization enables offensive and defensive combat operations against multiple targets and threats simultaneously.

Command by negation acknowledges that, because of the often distributed and dispersed nature of maritime warfare, it is necessary to pre-plan the actions of a force to an assessed threat and delegate some warfare functions to subordinate commanders. Once such functions are delegated, the subordinate commander is to take the required action without delay, always keeping the officer in tactical command informed of the situation."

The U.S. Navy has a model for the command structure inherent in CWC design, and it reaches from the operational level of war down to the tactical level of war. It is important to note that for the Navy, battlespace is fungible, as the operational situation changes frequently because of geography, environmental conditions, and adversary capability and intent. Therefore, commanders, their assigned responsibilities, and their delegated authorities change often. The CWC construct provides the necessary flexibility for decentralized execution in maritime operations. The various commanders and their respective areas of responsibility are covered below. In practice, commanders are activated when the need arises. For instance, if there is neither a mine threat, nor intent to place mines, the mine warfare commander will not be assigned.

(1) Composite Warfare Commander Design

(a) Officer in Tactical Command (OTC). The OTC is the senior officer present eligible to assume command, or the officer to whom the senior officer has delegated tactical command. The OTC and the Composite Warfare Commander (below) conduct planning for the operations with the aid of subordinate and supporting units, and then provide their orders and intentions for execution. This person is normally the numbered Fleet Commander or the Naval Component Commander for a GCC and is a three-star admiral.

(b) Composite Warfare Commander (CWC). The CWC is a command duty subordinate to the OTC. The OTC may choose to retain CWC duties or delegate them to a subordinate commander. The CWC is normally a CSG or ESG commander at the two-star admiral level. An OTC (i.e., a Fleet Commander/Joint Forces Maritime Component Commander) may have more than one CWC (e.g., CSG or ESG commander) under their command. For instance, the Commander, U.S. 7th Fleet, currently has Commander, Task Force 70 (a CSG), and Commander, Task Force 76 (an ESG) as part of his standing maritime forces. These forces continue to execute missions per the CWC concept and are provided authorities commensurate with their mission sets delegated from Commander, U.S. 7th Fleet.

(c) Warfare Commanders. Warfare commanders assist the CWC with planning, and, when assigned, they are given the necessary authority for command, control, direction, and coordination required for the execution of those delegated functions. They may initiate autonomous actions based on the delegations of authority provided in governing documents such as OPTASKs. Traditionally, only certain warfare commanders are assigned, and all remain responsible to the CWC, who may be the Task Force or Task Group commander, for execution of their duties. Warfare Commanders are usually the Commanding Officer of individual ships or submarines, or the Commodore of a destroyer or aircraft squadron, with the rank of a commander or captain. Assigned warfare commanders normally include:

- **Air and Missile Defense Commander (AMDC).** The AMDC is responsible for the air and missile defense of the group, regardless of what platform delivers the threat. Includes both manned and unmanned threats such as tactical aircraft, anti-ship cruise missiles, unmanned

aerial vehicles, air-to-surface missiles, etc. The Commanding Officer of the cruiser is customarily assigned the duty as AMDC, and the cruiser will almost always remain geographically co-located with the aircraft carrier.

- **Surface Warfare Commander (SUWC).** The SUWC is responsible for defense of the group against surface threats. Encompasses operations conducted to destroy or neutralize enemy naval surface forces. These operations typically include the directing of surveillance of the maritime domain, interdiction, and strikes by aircraft and missiles. If activated in lieu of the combined Sea Combat Commander, either the Commodore of the destroyer squadron or one of the senior destroyer Commanding Officers will be assigned this duty.
- **Anti-Submarine Warfare Commander (ASWC).** The ASWC is responsible for the defense of the force against submarine threats. If activated in lieu of the combined Sea Combat commander, usually the Commanding Officer of the most capable anti-submarine warfare destroyer or the Commodore of the P-8 squadron, will be assigned these duties.
- **Sea Combat Commander (SCC).** When the threat environment allows it, the group may combine the duties of the Surface Warfare Commander and the Anti-Submarine Warfare Commander, called the Sea Combat Commander. The SCC is responsible for the defense of the group against both surface and subsurface threats. This duty is customarily assigned to the Commodore of the embarked destroyer squadron.
- **Information Operations Warfare Commander (IWC).** The IWC is responsible for the meteorology and oceanography, cryptology, signals intelligence, cyber, electronic warfare, information operations, intelligence, networks, and space disciplines of the group. A fairly new warfare commander, the IWC is normally the senior Information Warfare officer in the Strike Group Commander's staff.
- **Strike Warfare Commander (STWC).** The STWC plans, directs, monitors and assesses maritime power projection ashore and may strike surface targets at sea at extended ranges from the group as directed by the CWC. Typically, the STWC does not plan or direct *Tomahawk* land-attack missile missions, which are coordinated via the Launch Area Coordinator. STWC is normally the commander of the CVW.
- **Ballistic Missile Defense Commander (BMDC).** The BMDC is responsible for the defense of the group against ballistic missile threats. May be combined with the AMDC if the ships' capabilities allow for it or it may be assigned as a functional group commander. The Commanding Officer of the most capable ballistic missile defense ship, either a cruiser (when not assigned AMDC duties) or destroyer, will be activated as the BMDC.

(d) Functional Group Commanders. These commanders are granted the same authorities as Warfare Commanders, but for a functional group vice a specific warfare area. They may also initiate autonomous actions to execute their mission. Like warfare commanders, functional group commanders are normally the Commanding Officer of individual ships, submarines, or squadrons and can be a commander or captain.

- **BMDC (if not assigned as a Warfare Commander).**
- **Mine Warfare Commander (MIWC).** The MIWC is responsible for the defense of the group against mines and, conversely, planning any offensive mining operations necessary. This is normally the Commodore of the Mine Countermeasures Ship Squadron (MCMRON), a captain, if embarked afloat, or the Commanding Officer of the most capable MCM ship, a lieutenant commander or commander.
- **Maritime Interception (or Interdiction) Operations Commander (MIOC).** The MIOC is responsible for the group efforts to monitor, query, and board merchant vessels in international waters to enforce sanctions against other nations and/or prevent the transport of restricted goods. This duty is normally assigned to a senior Commanding Officer of one of the destroyers or littoral combat ships, a commander.
- **Screen Commander.** The Screen Commander is responsible for the tactical maneuvering of the associated group. This duty normally resides in conjunction with the Sea Combat Commander and, therefore, is most often the Destroyer Squadron Commodore.
- **Underway Replenishment Commander.** The Underway Replenishment Commander plans the underway replenishment schedule (refueling) operations for the group, working closely with the Screen Commander. The Master of the underway replenishment vessel, a Captain in the Military Sealift Command, is normally the Underway Replenishment Commander.

(e) Coordinators. Coordinators serve as asset and resource managers who carry out the policies of the CWC and respond to the specific tasking of either warfare commanders or functional group commanders. Because these are support duties, they can be assigned to a myriad of individuals. Normally, a Navy commander or captain with the requisite warfare skills is assigned the final responsibility for each of these duties.

- **Airspace Control Authority (ACA).** The ACA develops airspace control policies and procedures for all airspace users and airspace control elements.
- **Air Resource Element Coordinator (AREC).** The AREC provides organic carrier air resources as directed and promulgates current information on the availability of aircraft to the group, as well as disseminates information or results (e.g., battle damage assessment) from the organic air resources. Because this duty involves the allocation of tactical aircraft that are multi-mission, the CVW is assigned this duty.
- **Cryptologic Resource Coordinator (CRC).** The CRC coordinates the use of cryptologic resources across the group. This duty is normally assigned to the Intelligence Branch Head (N2) of the most senior staff.
- **Common Tactical Picture Manager (CTPC).** The CTPC is responsible for the accurate and complete display of relevant correlated, associated, and fused tactical data integrated from multiple networks. This duty, while managed by the Operations Branch Head (N3) on the most senior staff, will be delegated to the Commanding Officer of the most data link-capable ships in the group.
- **Force Track Coordinator (FTC).** The FTC manages and collates all track (contact) information. This duty, while managed by the Operations Branch Head (N3) on the most senior staff, will be delegated to the Commanding Officer of the most data link-capable ships

in the group. It can also be co-assigned with the Common Tactical Picture Manager.

- **Helicopter Element Coordinator (HEC).** The HEC coordinates the assignment and use of non-logistics helicopters in the group. This duty shifts based on assigned missions and may be assigned to the Destroyer Squadron Commodore or the Operations Branch Head (N3) of the senior staff.
- **Launch Area Coordinator (LAC).** The LAC coordinates land strikes from the group using *Tomahawk* Land Attack Missiles. Assigned to Commanding Officer of the most proficient *Tomahawk* Land Attack Missile team, this duty can lie with any ship captain carrying *Tomahawks*.
- **Submarine Operations Coordinating Authority (SOCA).** The SOCA ensures safety and prevention of mutual interference, provides water-space management, and controls the submarine broadcast for assigned submarines in a designated operational area. Normally assigned to the embarked Submarine Squadron Commodore, a Navy captain.

The diagram below is a pictorial representation of the CWC construct described above. The OTC is at the top of the chain of command, with coordinators executing well into the tactical level. It is important to remember that various authorities actually lie with the OTC and are deliberately retained or delegated to subordinate commanders. At any time, the OTC or CWC may revoke a delegated authority. In peacetime operations, the Navy normally keeps authorities at a higher level, while in conflict or war, the authorities tend to be delegated to subordinate commanders because of the pace of operations.

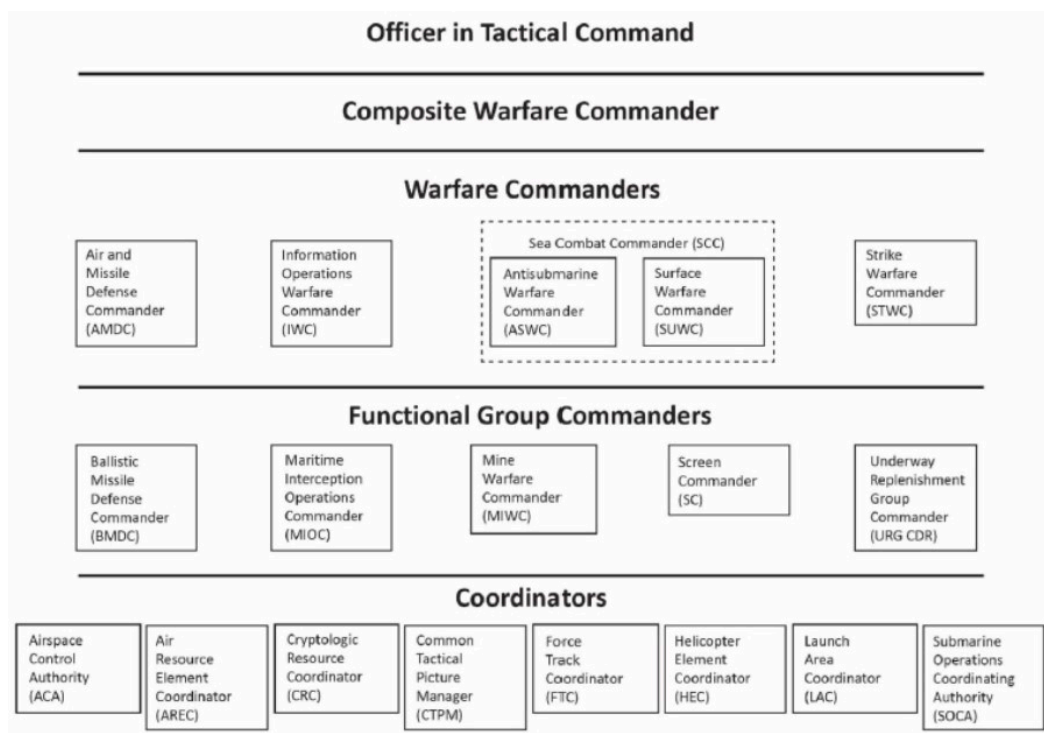


Figure 37: Warfare Commanders, Functional Group Commanders, and Coordinators.

As discussed previously in the Operational Chain of Command section, U.S. Naval Forces change operational and tactical commanders often, and it is no different in the assignments of CWC responsibilities. Within the CWC construct, warfare commanders can change on a daily basis, or even hourly should the tactical situation change. For instance, a DDG may have its embarked staff serving as the SCC, but that ship needs to participate in an Underway Replenishment evolution to receive fuel and stores, which can take up to six hours. If the CWC or SCC determines the assigned mission cannot be accomplished while involved in the underway replenishment, SCC duties will be temporarily shifted to an alternate DDG or CG, and then can be shifted back to the original DDG once the ship has completed replenishment. This happens routinely across all warfare areas, and all ships in a group—if capable of executing the mission—must be trained and prepared to accept warfare commander or coordinator duties on short notice. Assignments for primary and alternate responsibilities are often promulgated throughout group communications, further underscoring the Navy's tenet of centralized planning and decentralized execution.

Operating successfully within the CWC construct relies on centralized planning with decentralized execution using command by negation. It is imperative that the OTC provide subordinates with clear, concise direction and, in turn, that the subordinates precisely execute commander's intent. CWC allows for an agile command structure to meet the constantly changing operational environment the U.S. Navy finds itself.

4. The U.S. Marine Corps

The Marine Corps has long provided the Nation with a force adept at rapidly and effectively solving complex, multifaceted, and seemingly intractable security challenges—so much so that “Send in the Marines” connotes both a demand for action and a presumption of success. While the general public may not be conversant with all the Marine Corps is or does, they display an intuitive understanding that whenever and wherever crises arise, the Marines stand ready to get there fast and do whatever needs to be done.

Like all members of the joint team, the Marine Corps conducts expeditions—military operations by an armed force to accomplish a specific objective in a foreign country. Each Service contributes complementary capabilities to any expedition. The Navy, Air Force, and Army are optimized to dominate the sea, air, and land, respectively. Generally, the greater the capability and capacity required to dominate those portions of the sea, air, or land necessary to accomplish the overall objective of the expedition, the longer it will take to deploy the associated forces. Simply put, there is a tradeoff between size and speed whenever an expedition is put in motion.

While the Marine Corps may operate on and from the sea, in and from the air, and on the land, it is not optimized to dominate any of them. Rather, the Marine Corps is optimized to be expeditionary—a strategically mobile *middleweight force* that can fill the gaps created by the size/speed tradeoff. Usually task-organized and employed as Marine air-ground task forces (MAGTFs), Marine Corps forces are light enough to leverage various combinations of amphibious shipping, maritime prepositioning, and inter-theater airlift to quickly get to the scene of action, yet heavy enough to either accomplish the mission or provide a stopgap pending the arrival of additional forces, whose arrival/employment Marine Corps forces may facilitate.

A key aspect of *being* expeditionary is providing, in partnership with the Navy, the ability to bridge the difficult seam between operations on land and sea. Although Marines are often thought of exclusively as an amphibious *assault* force, World War II was the only time in U.S. history that the Marine Corps was organized, trained, and equipped primarily for that one mission. Both before and after that period, the Marine Corps has focused on being able to respond rapidly and effectively to crises. Within that context, the amphibious capabilities provided by the Navy-Marine Corps team—especially those that are forward postured—have long played a critical role preventing major conflict and deterring the escalation of conflict within the range of military operations (ROMO). They assure littoral access and enable the introduction of capabilities provided by the other Services, other government agencies, nongovernmental organizations, U.S. allies, or other international partners.

Since the end of the Cold War, our amphibious forces have employed with increased frequency to conduct diverse operations, such as foreign humanitarian assistance, noncombatant evacuations, and the suppression of piracy. Additionally, sea-based forces are increasingly prized by the geographic combatant commanders to conduct engagement activities that address mutual security concerns while respecting partner-nation sovereignty.

Many of the crisis response operations that Marine Corps forces conduct involve what have historically been known as “small wars,” from minor interventions to protect U.S. citizens, friends, or interests, to punitive expeditions or longer-duration counterinsurgency operations. Marines have a long track record of success—from recent actions in al Anbar Province back to the suppression of the slave trade and piracy in the Barbary Wars in the early 19th century—in solving complex problems for which purely military solutions would not suffice. The fundamental causes of such conflicts are often a complicated combination of security, economic, political, and social issues. In other cases, Marine Corps forces have responded to contingencies or major operations involving combat against conventional forces, such as Operation Desert Storm. Still others have involved various blends of conventional and irregular conflict, such as Operation Iraqi Freedom. Regardless of the nature of the conflict—conventional, irregular, or a hybrid of the two—expeditions that require larger forces/longer duration may require capabilities or capacities not resident in a middleweight force. The ability to rapidly aggregate additional Marine Corps capabilities—often provided by the United States Marine Corps Forces Reserve—as well as the ability to integrate capabilities provided by the U.S. Army or multinational partners are essential enablers for the conduct of sustained operations ashore.

(1) Maneuver Warfare Philosophy

Marines operate in the spirit of, and are guided by, the philosophy of maneuver warfare articulated in the nine Marine Corps doctrinal publications (MCDPs), which predate this publication—MCDP 1, *Warfighting*; MCDP 1-1, *Strategy*; MCDP 1-2, *Campaigning*; MCDP 1-3, *Tactics*; MCDP 2, *Intelligence*; MCDP 3, *Expeditionary Operations*; MCDP 4, *Logistics*; MCDP 5, *Planning*; and MCDP 6, *Command and Control*. These publications constitute overarching and enduring doctrine that should not be confused with more numerous, frequently revised, subordinate Marine Corps warfighting publications (MCWPs) and Marine Corps reference publications (MCRPs), which address evolving tactics, techniques, and procedures. The MCDPs provide an overarching philosophy of warfighting with the expectation that readers will consider, discuss, and creatively apply the ideas contained therein. They do not prescribe specific techniques or procedures; instead, they provide ideas and values that require a high degree of professional competence and judgment in their application. While these ideas and values are expressed within the context of warfighting, they are pertinent and applicable not only across the range of military operations but also in how the Corps organizes, trains, and equips Marines for conducting operations.

This philosophy recognizes that war is ultimately a violent clash of human wills with an enduring nature characterized by friction, uncertainty, disorder, and complexity. Success in fluid environments demands leaders and organizations that can understand the nature of a given situation and adapt to it faster than their opponents. There are several ways we can increase speed. First, we can emphasize simplicity in all we

do. Second, we can employ mission tactics and commander's intent to decentralize execution of operations. Mission tactics is the assignment of a task to a subordinate without specifying how it must be accomplished, while the accompanying commander's intent provides the overall purpose behind the task. Joint doctrine describes this as mission command. By conveying the higher purpose, seniors give their subordinates the authority—and responsibility—to adapt their methods for achieving the task as the situation unfolds. This approach permits a faster decision cycle than the enemy's, allowing Marines to exploit the advantages of speed, focus, tempo, shock, and surprise. A third way to become faster is through experience, gained through training, planning, or actual operations, which promotes implicit and lateral communication within the organization. Finally, a commander can generate speed by positioning himself or herself at the point of friction.

The human dimension of war is the most critical element; boldness, creativity, intelligence, and the warrior spirit are prime attributes. The warrior spirit includes not only tactical and technical proficiency, but also adherence to the highest standards of personal and professional integrity. In doing so, Marines, individually and collectively, "keep our honor clean."

Marines conduct all military operations in accordance with all applicable laws, regulations, and policies. In particular, Marines comply with the law of war, whether operating alone or with joint or multinational forces. Marines also do their best to prevent violations of the law of war and report all violations to their superiors. The Marine Corps' warfighting philosophy of maneuver warfare is rooted in the long-standing principles of war.

Maneuver Warfare will be discussed in detail in the 8671 *Doctrine* sub-course.

(2) Roles and Functions of the Marine Corps

Roles are the broad and enduring purposes Congress established by law for the Services and for USSOCOM. Functions are specific responsibilities assigned to the Services by the President and Secretary of Defense to enable the Services to fulfill their legally established roles. Various laws, directives, and manuals establish the roles and functions of the Marine Corps and describe its general composition and responsibilities. The key sources are *United States Code*, Title 10; Department of Defense Directive 5100.1, *Functions of the Department of Defense and Its Major Components* (1987, updated in 2002 and 2010); and the *Marine Corps Manual* (1980 with changes in 1982, 1984, and 1986).

(a) United States Code (USC), Title 10, Armed Forces. USC, Title 10, chap. 507, sect. 5063 and chap. 1006, sec. 10173 detail the composition and functions of the Marine Corps. It directs that the Marine Corps:

- Shall be organized to include not less than three combat divisions and three aircraft wings, and other organic land combat forces, aviation, and services.
- Shall be organized, trained, and equipped to provide Fleet Marine Forces of combined arms, together with supporting aviation forces, for service with the fleet in the seizure and defense of advanced naval bases and for the conduct of such land operations as may be essential to the prosecution of a naval campaign.
- Shall provide detachments and organizations for service on armed vessels of the Navy, shall provide security detachments for the protection of naval property at naval stations and bases, and shall perform such other duties as the President may direct. These additional duties may not detract from or interfere with the operations for which the Marine Corps is primarily organized.
- Shall develop, in coordination with the Army and Air Force, those phases of amphibious operations that pertain to the tactics, techniques, and equipment used by landing forces.
- Shall maintain the Marine Corps Forces Reserve for the purpose of providing trained units and qualified individuals to be available for active duty in the Marine Corps in time of war or national emergency and at such other times as the national security may require.
- Is responsible, in accordance with integrated joint mobilization plans, for the expansion of the peacetime components of the Marine Corps to meet the needs of war.

(b) Department of Defense Directive 5100.1, Functions of the Department of Defense and its Major Components. The Marine Corps, within the Department of the Navy, shall develop concepts, doctrine, tactics, techniques, and procedures and organize, train, equip, and provide forces, normally employed as combined arms air ground task forces, to serve as expeditionary force-in-readiness, and perform the following specific functions:

- Seize and defend advanced naval bases or lodgments to facilitate subsequent joint operations.
- Provide close air support for ground forces.
- Conduct land and air operations essential to the prosecution of a naval campaign or as directed.
- Conduct complex expeditionary operations in urban littorals and other challenging environments.
- Conduct amphibious operations, including engagement, crisis response, and power projection operations to assure access. The Marine Corps has primary responsibility for the development of amphibious doctrine, tactics, techniques, and equipment.

- Conduct security and stability operations and assist with the initial establishment of a military government pending transfer of this responsibility to another authority.
- Provide security detachments and units for service on armed vessels of the Navy, provide protection of naval property at naval stations and bases, provide security at designated U.S. embassies and consulates, and perform other such duties as the President or the Secretary of Defense may direct. These additional duties may not detract from or interfere with the operations for which the Marine Corps is primarily organized.

(c) The Marine Corps Manual. The Marine Corps Manual adds two more functions. The Marine Corps shall:

- Provide Marine Corps officer and enlisted personnel in support of the Department of State security program overseas.
- Organize Marine Corps aviation, as a collateral function, to participate as an integral component of naval aviation in the execution of such other Navy functions as the fleet commanders may direct.

(d) Commandant of the Marine Corps. The Commandant has two vital functions: as a member of the Joint Chiefs of Staff and as Marine Corps Service Chief. The Commandant's duties as a member of the Joint Chiefs of Staff take precedence over all other duties.

As a Joint Chiefs of Staff member, the Commandant may submit advice or opinion to the Chairman of the Joint Chiefs of Staff when it disagrees with, or provides additional insight to, the Chairman of the Joint Chiefs of Staff's point of view. When the Chairman of the Joint Chiefs of Staff submits advice or opinion to the President, the National Security Council, and the Secretary of Defense, the Chairman of the Joint Chiefs of Staff is obligated to submit any additional input from the Commandant or other members of the Joint Chiefs of Staff. When acting in this capacity as a military adviser, the Commandant may provide advice to the President, the National Security Council, and the Secretary of Defense when an opinion is requested.

Providing the Secretary of Defense has been informed prior to any meeting with Congress, the Commandant may make recommendations to Congress relating to the Department of Defense. The Commandant will attend the regularly scheduled meetings of the Joint Chiefs of Staff. As long as their independence as a member of the Joint Chiefs of Staff is not impaired, the Commandant will keep the Secretary of the Navy informed of military advice given by the other members of the Joint Chiefs of Staff on matters that impact the Department of the Navy.

As the Service Chief, the Commandant is subject to the authority, direction, and control of the Secretary of the Navy and is directly responsible for the administration, discipline, internal organization, training, requirements, efficiency, and readiness of the Marine Corps. The Commandant is also responsible for the Marine Corps' materiel support system and accountable for the total performance of the Marine Corps.

(3) Organization

The structure of the U.S. Marine Corps consists of four components: Office of the Secretary of the Navy, Commandant of the Marine Corps, the operating forces, and the supporting establishment. This lesson will provide an organizational overview of the operating forces, while the 8672 *Organization of the U.S. Marine Corps* sub-course will provide further detail on the entire U.S. Marine Corps.

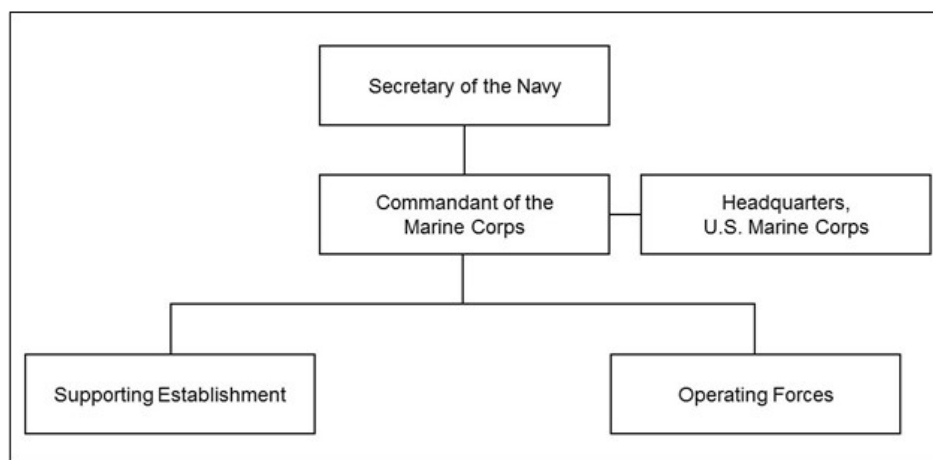


Figure 38: Structure of the U.S. Marine Corps.

The Marine Corps organizes its operational forces as Fleet Marine Forces and as MAGTFs to provide task-organized, self-sustaining, multi-purpose forces to the combatant commanders and U.S. Navy fleet commanders. These uniquely organized Marine Corps forces can respond to a wide range of operational and tactical missions and tasks, providing the President/Sec Def with a combination of deployment and employment options.

(a) Fleet Marine Forces. Development and expansion of the Fleet Marine Forces followed the creation and evolution of first the Advanced Base Force (1902-1920) and then the Expeditionary Forces (1920-1923), the later so named by Major General Commandant Lejeune, who guided the operational orientation of the Marine Corps toward expeditionary service ashore in support of the fleet and the role of amphibious assault. This focus was inspired by a 1921 Headquarters Marine Corps strategic estimate developed by Major Earl E. (Pete) Ellis, who concluded that a Pacific Ocean war would be decided by base seizure and defense and that the Marine Corps should be prepared to “execute opposed landings and attacks...with the greatest rapidity.” In 1933, Major General Commandant Fuller initiated action to integrate the expeditionary forces into the fleet under the operational direction of the fleet commander. These forces were redesignated the Fleet Marine Force and became an organic element of the fleet with the signing of Navy Department Order 241 on 7 December 1933. The first Fleet Marine Force headquarters was established at Quantico, which at the time was also the scene of a major doctrinal development effort resulting in the *Tentative Manual for Landing Operations* (1934)—the foundation for current U.S. amphibious doctrine.

Of the Marine Corps roles and missions, foremost is the provision of Fleet Marine Forces of combined arms, together with supporting air components, for service with the fleet. The National Security Act of 1947 stipulates that the additional duties assigned to the Marine Corps may not detract from or interfere with the operations for which the Marine Corps is primarily organized. Accordingly, the Fleet Marine Forces are the primary Marine Corps contribution to national defense and support of national policy.

The Fleet Marine Force organization of 1933 stands as the first attempt in history to create a major unified air-land-sea-logistics force—the balanced fleet—which achieves unity of effort through centralized direction, decentralized execution, and common doctrine. The balanced fleet concept permits task organization to meet the requirements of the evolving international situation and the progressive nature of a naval campaign. Inherently flexible in form, it achieves unity of effort through common understanding of responsibilities and relationships, task organization, concurrent and parallel planning, and the explicit assignment of authority.

The Goldwater-Nichols DOD Reorganization Act of 1986 removed the preponderance of Fleet Marine Forces from the fleet’s operational control and disrupted the long-standing Navy-Marine Corps relationship by creating separate Navy and Marine Corps components with joint forces. The *38th Commandants’ Planning Guidance* states that the characteristics of the 21st century naval operating environment require a more “integrated approach to operations in the maritime domain.” The reestablishment of the Fleet Marine Forces by the 38th Commandant will help achieve the modern and lethal naval force required to facilitate maritime operations as part of a naval or joint campaign.

(b) Marine Air-Ground Task Forces (MAGTFs). In 1952, Congress directed the integration of Marine air and ground forces. The Marine Corps codified this directive in December 1963 with Marine Corps Order 3120.3, *The Organization of Marine Air-Ground Task Forces*, which formalized the MAGTF construct. The MAGTFs have unique and incomparable warfighting capabilities. They contain organic air, ground, and logistic elements under a single command element, making them integrated and self-sustaining.

MAGTFs provide joint force commanders (JFCs) with the capability to:

- Move forces into crisis areas without revealing their exact destinations or intentions.
- Provide continuous presence in international waters.
- Commence execution of a mission within 6 to 48 hours of receiving a warning order.
- Provide immediate national response in support of humanitarian and natural disaster relief operations.
- Provide credible and over-the-horizon crisis response capability.
- Support diplomatic processes for peaceful crisis resolution before employing immediate response combat forces.
- Respond to crises through the measured projection of combat power ashore—day or night.
- Introduce additional Marine forces sequentially into a theater of operations.
- Operate independent of established airfields, basing agreements, and overflight rights.
- Conduct combat operations ashore by relying on MAGTF-organic combat service support.
- Conduct theater security cooperation (SC) to build partner capacity (BPC).
- Enable the introduction of follow-on forces by securing staging areas ashore or bypassing littoral defenses by going further inland.
- Operate in rural and urban environments, and during chemical, biological, radiological, and nuclear situations.
- Withdraw rapidly at the conclusion of operations.

The Nation’s MAGTFs thus provide a continuum of capabilities to support naval, CCDR, and national requirements. These MAGTFs, joined by special-purpose forces and unique Marine forces, enable the Corps to address the full range of conventional, unconventional, and irregular/hybrid threats as well as other tasking as the President might require.

(c) Elements of a MAGTF. All MAGTFs consist of four core elements—a command element, a ground combat element (GCE), an aviation combat element (ACE), and a logistics combat element (LCE). Although MAGTFs differ in size and capabilities, standard procedures exist for organizing any MAGTF and for planning and executing its operations.

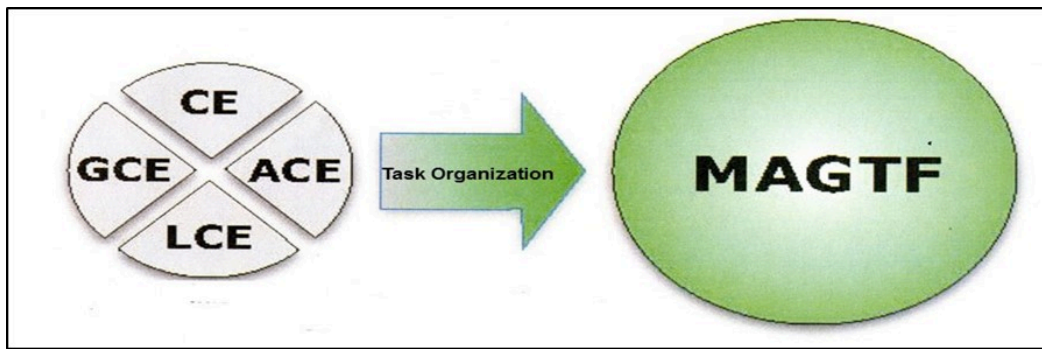


Figure 39: Common Organizational Concept for MAGTFs.

As a modular organization, the MAGTF is tailorable to its mission through task organization. This building block approach also makes reorganization a matter of routine. In addition to its Marine Corps units, a MAGTF may have attached forces from other Services and nations, such as naval construction battalions or infantry/armor brigades.

A key feature of the MAGTF is its expandability. Crisis response may require a larger force than what can initially be brought to bear. Being able to expand the original force—rather than replacing it with a larger one—promotes continuity of operations. The MAGTF's modular structure facilitates rapid expansion into a larger force as a situation demands by simply adding forces as needed to the core units of each existing element.

- **Command Element.** The command element includes the MAGTF headquarters and other enabling attachments. As with all other MAGTF elements, the command element task-organizes to provide the command and control capabilities necessary for effective planning, execution, and assessment of operations. Additionally, the command element can exercise command and control within a joint force from the sea or ashore and act as a joint task force headquarters core element. A command element may include additional command and control and intelligence capabilities from national and theater assets, force reconnaissance assets, signals intelligence capabilities from the radio battalion, and a force fires coordination center. A command element can employ additional major subordinate commands, such as the force artillery headquarters, naval construction regiments, or Army maneuver or engineering units.
- **Ground Combat Element.** The ground combat element provides the ability to conduct ground combined arms maneuver. While it has discrete capabilities to conduct ground combat, the ground combat element does not operate as an independent force. Ground combat element operations are planned and executed as integral components of MAGTF operations. Every ground combat element is task-organized in accordance with mission requirements, ranging from a light, air-transportable force built around a platoon (reinforced) to a relatively heavy mechanized unit that includes one or more Marine, Army, and/or multinational divisions. Marine Corps ground combat units are staffed, organized, trained, and equipped to operate inside the MAGTF construct and cannot be employed separately without a significant loss of capability.
- **Aviation Combat Element.** The aviation combat element task-organizes to conduct air operations, project combat power, and contribute to battlespace dominance in support of the MAGTF's mission by performing some or all of the following six functions of Marine aviation:
 - Anti-air warfare.
 - Assault support.
 - Electronic warfare.
 - Offensive air support.
 - Air reconnaissance.
 - Control of aircraft and missiles.

The aviation combat element consists of an aviation headquarters with air control agencies, aircraft squadrons or groups, and logistic units. It can vary in size and composition from a small aviation detachment of specifically required aircraft to one or more Marine aircraft wings. The ACE may operate from ships or from austere expeditionary locations ashore and can readily transition between them without loss of capability. It exercises command and control throughout the battlespace.

- **Logistics Combat Element.** The logistics combat element task-organizes to provide all functions of tactical logistics necessary to support the continued readiness and sustainability of the MAGTF. The LCE performs some or all six functions of tactical logistics:
 - Supply.
 - Maintenance.
 - Transportation.
 - Health services.
 - General engineering.

- Other services, which include legal, exchange, food, disbursing, postal, billeting, religious, mortuary, and morale and recreation services.

The logistics combat element may vary in size and composition from a support detachment up to one or more logistic groups. The logistics combat element operates from sea bases or from expeditionary bases established ashore. It may be the main effort of the MAGTF during foreign humanitarian assistance missions or selected phases of maritime prepositioning operations.

(d) Types of MAGTFs. There are four types of MAGTFs:

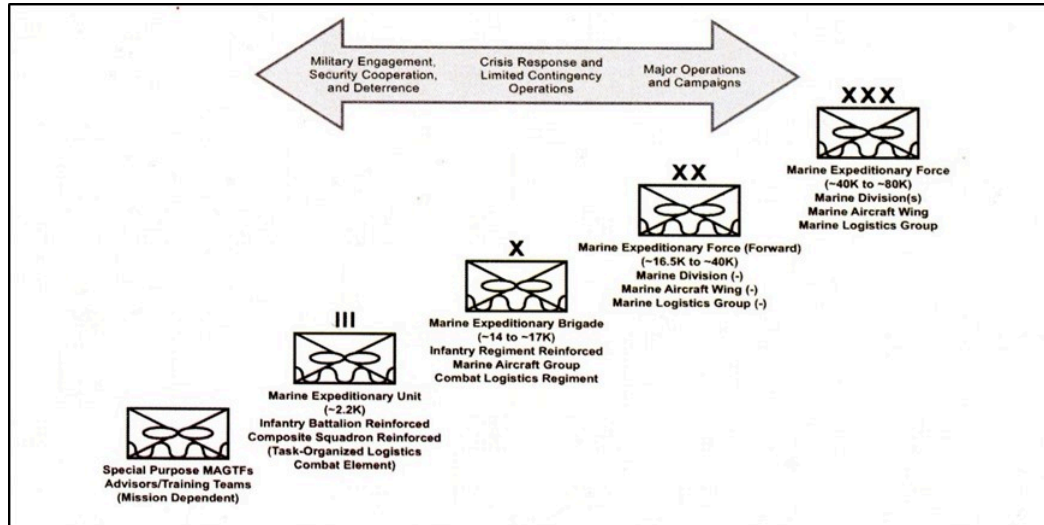


Figure 40: Types of MAGTFs.

(a) Marine Expeditionary Force (MEF). The MEF is the Marine Corps' principal warfighting organization during larger crises or contingencies. Normally commanded by a lieutenant general, a MEF can range in size from one division and aircraft wing to multiple divisions and aircraft wings, together with one or more logistics groups. MEFs are capable of amphibious operations and sustained operations ashore in any geographic environment. With appropriate augmentation, the MEF headquarters is capable of performing as a Joint Task Force (JTF) headquarters. MEFs often deploy in echelon and designate the lead element as the MEF (Forward).

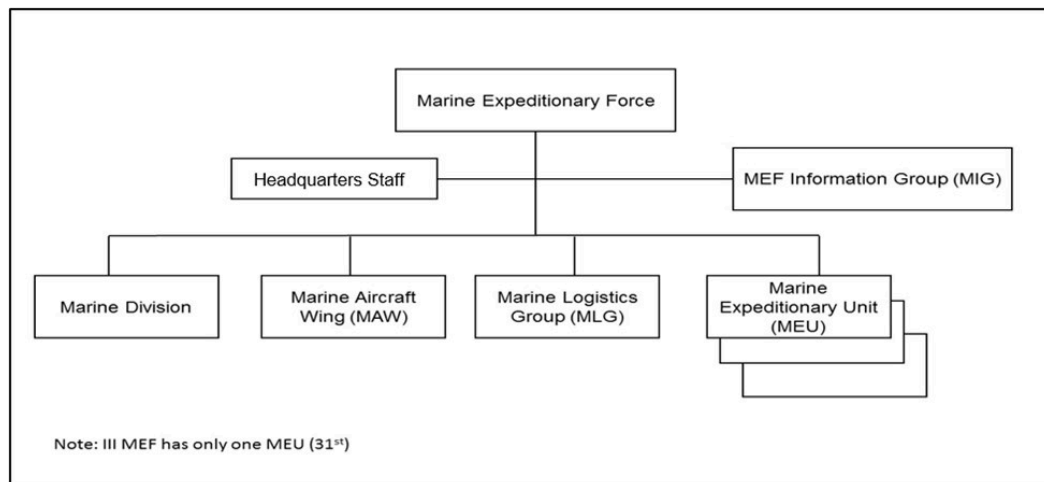


Figure 41: Marine Expeditionary Force.

MEFs are the primary "standing MAGTFs" in peacetime and wartime. The Marine Corps is organized with three standing MEFs, each with a Marine division, aircraft wing, logistics group, and information group.

A MEF (Fwd) is normally the lead echelon of a MEF or, for some contingencies, it can be a stand-alone MAGTF capable of sustained expeditionary operations. A MEF (Fwd) is normally smaller than a MEF and larger than a MEB. For example, a MEF (Fwd) commanded by a Marine major general deployed on a rotational basis to execute combat operations in Operations Iraqi Freedom and Enduring Freedom. The GCE of the MEF (Fwd) normally consists of a division (-) or multiple regiments.

(b) Marine Expeditionary Brigade (MEB). Mid-sized MAGTFs, MEBs conduct major security cooperation operations, respond to larger crises or contingencies, or participate in major operations and campaigns—such as MEB Afghanistan support to Operation Enduring Freedom. They provide the “building blocks” for forcible entry and other power projection operations, providing the landing forces for amphibious assault and the fly-in echelons that “marry-up” with equipment and supplies delivered by maritime prepositioning ships. During Operation DESERT SHIELD, for example, two MEBs deployed via amphibious ships while Marines and Sailors from two other MEBs traveled to Saudi Arabia by inter-theater airlift to fall in on equipment and 30 days of supplies delivered via maritime prepositioning ships. Normally commanded by brigadier generals, MEBs number approximately 16,000 Marines and Sailors once their subordinate units are assigned. A MEB’s elements normally consist of the following:

- A brigade-level headquarters and may include additional assets, such as command and control enablers, reconnaissance, intelligence capabilities, and assets from other Services.
- An infantry regiment reinforced with artillery, reconnaissance, engineer, light armored reconnaissance units, assault amphibian units, and other attachments as required.
- A composite Marine aircraft group with combat assault transport helicopter/tilt-rotor aircraft, utility and attack helicopters, vertical/short takeoff and landing fixed-wing attack aircraft, fighter/attack aircraft, electronic warfare aircraft, unmanned aircraft systems, air refuelers/transport aircraft, and requisite aviation logistic and command and control capabilities.
- A task-organized combat logistics regiment with engineering, supply, services, transportation, medical, and maintenance capabilities, and landing support assets for beach, port, and airfield delivery operations.

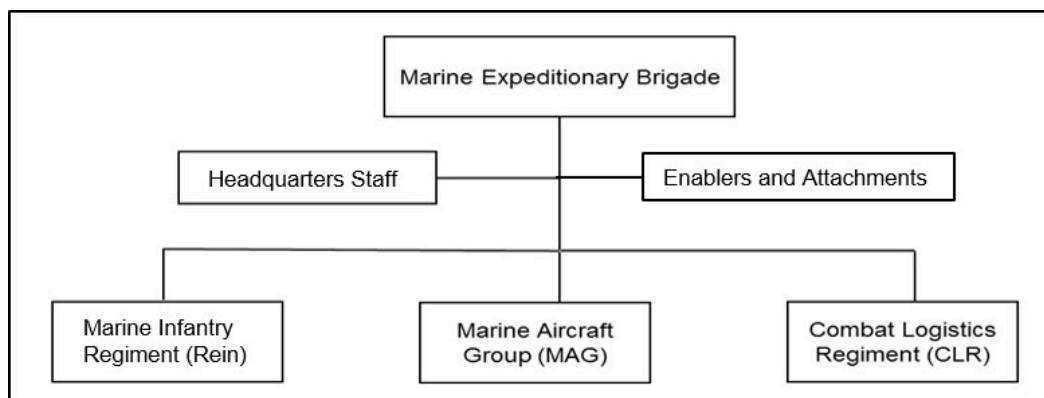


Figure 42: Marine Expeditionary Brigade.

The MEB headquarters maintain close coordination and conduct operational planning with key joint and Service headquarters and are capable of rapidly assuming control of forces for missions across the range of military operations. Like the larger MEFs, MEBs may assimilate units from other Services or nations and grow beyond their notional size. The MEBs are the smallest MAGTFs capable of performing all six functions of Marine aviation. A MEB can conduct the full range of expeditionary operations and may serve as the lead echelon of the MEF. The MEB headquarters can also serve as the nucleus of a joint or multinational task force headquarters.

(c) Marine Expeditionary Unit (MEU). Forward deployed MEUs aboard ARGs operate continuously in the GCCs’ areas of responsibility (AOR). These units provide the President and the GCCs with a forward deployed and flexible, sea-based MAGTF. These units are capable of conducting amphibious operations to respond to crisis, conduct limited contingency operations, introduce follow on forces, and support designated SOF. In effect, they provide an afloat “on-station” force capable of responding to any situation that may arise. MEU’s are characterized by their sea-based forward presence, expeditionary nature, and ability to plan for and respond to crises, combined arms integration, and interoperability with joint, combined, and SOFs. The MEU’s composition is informed by Marine Corps order 3120.13, *Policy for Marine Expeditionary Units (MEU)*.

The MEU is commanded by a colonel and deploys with 15 days of accompanying supplies (for commitments in addition to the supplies used and replenished during transits). Prior to deployment, a MEU undergoes an intensive six-month training program, focusing on its Mission-Essential Task (MET) List and interoperability with Joint and SOF. The training culminates with a thorough evaluation and certification known as “Operationally Ready to Deploy.” The organic capabilities of the MEU are:

- Amphibious operations
 - Amphibious assault
 - Amphibious raid
 - Small boat raid (specific to the 31st MEU)
 - Maritime interception operations
 - Enabling operations
- Crisis response and limited-contingency operations
 - Non-combatant evacuation operations
 - Foreign humanitarian assistance (FHA)
 - Integrating and operating with joint, interagency, intergovernmental, and multinational organizations (JIIM)
 - Tactical recovery of aircraft and personnel
 - Embassy reinforcement
 - Aviation operations from expeditionary sites
 - Theater security cooperation (TSC) activities
 - Airfield and port seizures
 - Expeditionary strike

A MEU's elements normally consist of the following:

- A standing headquarters with enablers and attachments.
- An infantry battalion reinforced with artillery, reconnaissance, engineer, assault amphibian units, and other detachments as required. This reinforced infantry battalion forms a battalion landing team (BLT).
- A Marine medium tiltrotor squadron (VMM) reinforced with detachments from a Marine heavy helicopter squadron (HMH), Marine light attack helicopter squadron (HMLA), Marine fighter attack squadron (VFMA), Marine unmanned aerial vehicle squadron (VMU), Marine air traffic control detachment (MATCD), Marine wing support squadron (MWSS), and Marine aviation logistics squadron (MALS). A notional MEU ACE would contain 10-12 MV-22s, 4 CH-53Es, 4 AH-1Zs, 3 UH-1Ys, and 6 F-35Bs.
- A MEU combat logistics battalion (CLB). The MEU CLB is a permanently structured command that includes a headquarters section for command and control, a communication platoon for internal and external communications, and functional platoons to provide supply, maintenance, distribution, engineering, explosive ordnance disposal (EOD), and health services support to the MEU. Smaller, task-organized detachments may be formed for specific missions.

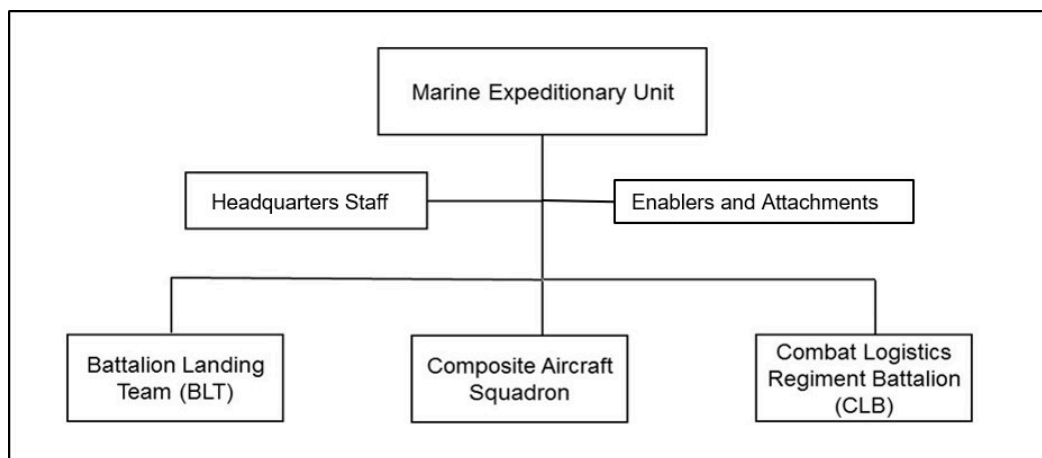


Figure 43: Marine Expeditionary Unit.

There are seven standing MEU headquarters. Six of them are in a rotation cycle with attachments that provide continuous forward presence in key regions. The seventh is permanently forward-deployed in USINDOPACOM's AOR.

(d) Special Purpose MAGTF (SPMAGTF). When situations arise for which a MEU or other unit is either inappropriate or unavailable, a SPMAGTF is formed. A SPMAGTF may be of any size, but normally no larger than a MEU, with tailored capabilities required for accomplishing a particular mission. It may be task-organized from non-deployed Marine Corps forces or formed on a contingency basis from a portion of a deployed MAGTF. Regimental-level headquarters often assume the role as a SPMAGTF headquarters and may conduct training in anticipated mission skills prior to establishment. A SPMAGTF may deploy using amphibious warfare ships, commercial ships, or aircraft, inter-theater airlift,

or organic Marine aviation. Frequently, SPMAGTFs conduct sea and land-based security cooperation activities, while others have been formed to provide sea-based FHA or military support to civil authorities or to participate in freedom of navigation operations. It may embark, for example, on an amphibious warfare ship, such as an LPD or LSD.

A SPMAGTF is task organized to accomplish a specific mission, operation, or regionally focused exercise. They can be organized, trained, and equipped with Marine forces to conduct a wide variety of expeditionary operations ranging from peacetime missions, training exercises, and responses to contingencies and crises. SPMAGTFs can support theater campaign plans, security cooperation activities, and civil-military operations requirements.

Lesson 6: Fundamentals of Joint Operations

Lesson Hours: 1.00

1. Introduction

Joint operations are military actions conducted by joint forces and those Service forces employed in specified command relationships with each other, which of themselves do not establish joint forces. A joint force is one composed of significant elements, assigned or attached, of two or more Military Departments operating under a single joint force commander. This lesson introduces the fundamentals of joint operations to include common operating precepts, joint functions, joint principles of operations, and componency.

2. Educational Objectives

- a. Identify the common operating precepts of joint operations.
- b. Identify the principles of joint operations.
- c. Identify the joint functions.
- d. Identify the joint force options for accomplishing a mission.
- e. Identify the different types of joint task forces.
- f. Identify the different command relationships used by a joint task force.
- g. Identify why a joint force commander would conduct operations using Service components.
- h. Identify the Marine Corps' changing approach to componency.
- i. Identify why a joint force commander would conduct operations using functional components.
- j. Identify the various functional component commands.

3. Key Terms

- Principles of Joint Operations
- Joint Functions
- Joint Task Force
- Command Relationships
- Componency
- Functional Component Commands
- Service Component Commands

4. Preparation

All quizzes, discussion forum posts, and assignments are executed in the Moodle course site.

- a. Lesson Reading: [*Fundamentals of Joint Operations*](#) (24 pages).

b. Other Requirements: None.

c. Supplemental References (Optional):

- [JP 3-0 w/Change 1, Joint Operations](#) (22 Oct 2018).
- [JP 3-33, Joint Task Force Headquarters](#) (31 Jan 2018).
- [MCWP 7-10 w/Change 1, Marine Corps Componentency](#) (4 Apr 2018).
- [Commandant's Planning Guidance, 38th Commandant of the Marine Corps](#) (Jul 2019).

Lesson 6 Reading

FUNDAMENTALS OF JOINT OPERATIONS

References:

- a) JP 3-0, *Joint Operations* (22 Oct 2018)
- b) JP 3-33, *Joint Task Force Headquarters* (31 Jan 2018)
- c) MCWP 7-10 w/Change 1, *Marine Corps Componentency* (4 Apr 2018)
- d) *Commandant's Planning Guidance*, 38th Commandant of the Marine Corps (Jul 2019)

1. Introduction

Joint and combined warfare is an American military heritage, in large part dictated by U.S. geography and strategic circumstances. From Yorktown to Baghdad, these joint and combined experiences have occurred in various patterns. In some, the services and nations fought side by side for a common objective; in others, they fought separately, in either space or time, for a common objective. Since its inception, the Marine Corps has participated in military operations with the forces of other services and nations. Marine forces are seldom employed unilaterally; they are usually part of a larger naval, joint, or combined force. The Marine Corps' size; functional orientation at the confluence of sea, land, and air operations; and worldwide responsibilities will continue to make service with other military forces an operational necessity.

2. Purpose of Joint Doctrine

The purpose of joint doctrine is to govern the activities and performance of the Armed Forces of the U.S. in joint operations, and it provides considerations for military interaction with governmental and nongovernmental agencies, multinational forces, and other interorganizational partners. Joint doctrine provides military guidance for the exercise of authority by combatant commanders and other joint force commanders (JFCs) and prescribes joint doctrine for operations and training. Joint doctrine also provides military guidance for use by the Armed Forces in preparing and executing their plans and orders. It is not the intent of joint doctrine to restrict the authority of the JFC from organizing the force and executing the mission in a manner the JFC deems most appropriate to ensure unity of effort in the accomplishment of objectives. While joint doctrine provides official advice concerning a specific subject matter, *the judgment of the commander is paramount in all situations*.

3. Common Operating Precepts

Ten *common operating precepts* underlie successful joint operations. These precepts flow logically from the broad challenges in the strategic environment to the specific conditions, circumstances, and influences in a JFC's operational environment. These operating precepts are:

- Achieve and maintain unity of effort within the joint force and between the joint force and the U.S. government, international, and other partners.
- Leverage the benefits of operating directly through partners when strategic and operational circumstances dictate or permit.
- Integrate joint capabilities to be complementary rather than merely additive.
- Focus on operational objectives whose achievement suggests the broadest and most enduring result.
- Ensure freedom of action.
- Avoid combining capabilities where doing so adds complexity without compensating advantage.
- Inform domestic audiences and shape the perceptions and attitudes of key foreign audiences as an explicit and continuous operational requirement.
- Maintain operational and organizational flexibility.

- Drive synergy to the lowest echelon at which it can be managed effectively.
- Plan for and manage operational transitions over time and space.

4. Principles of Joint Operations

The principles of joint operations are formed around the traditional principles of war. Three additional principles—restraint, perseverance, and legitimacy—are relevant to how the Armed Forces of the United States use combat power across the range of military operations. These three, added to the original nine, comprise 12 principles of joint operations. The principles do not apply equally in all joint operations. Most principles, if not all, are relevant in combat. Some principles, such as offensive, maneuver, and surprise, may not apply in some crisis response operations like foreign humanitarian assistance (FHA). However, principles such as unity of command, objective, and legitimacy are important in all operations.

a. Objective

The purpose of specifying the objective is to direct every military operation toward a clearly defined, decisive, and achievable goal. The purpose of military operations is to achieve objectives that support attainment of the overall strategic objectives identified to resolve the conflict. This frequently involves the destruction of the adversaries' capabilities and their will to fight. The objective of joint operations not involving this destruction might be more difficult to define; nonetheless, it too must be clear from the beginning. Objectives must directly, quickly, and economically contribute to the purpose of the operation. Each operation must contribute to attaining strategic objectives. JFCs should avoid actions that do not contribute directly to achieving the objective(s). Furthermore, changes to military objectives may occur because national and military leaders gain a better understanding of the situation, or they may occur because the situation itself changes. The JFC should anticipate these shifts in national objectives necessitating changes in military objectives. The changes may be very subtle, but if not made, achievement of the military objectives may no longer support the national objectives, legitimacy may be undermined, and force security may be compromised.

b. Offensive

The purpose of an offensive action is to seize, retain, and exploit the initiative. Offensive action is the most effective and decisive way to achieve a clearly defined objective. Offensive operations are the means by which a military force seizes and holds the initiative while maintaining freedom of action and achieving desired results. The importance of offensive action is fundamentally true across all levels of warfare. Commanders adopt the defensive only as a temporary expedient and must seek every opportunity to seize or regain the initiative. An offensive spirit must be inherent in the conduct of all defensive operations.

c. Mass

The purpose of mass is to concentrate the effects of combat power at the most advantageous place and time to produce decisive results. In order to achieve mass, appropriate joint force capabilities are integrated and synchronized where they will have a decisive effect in a short period of time. Mass must often be sustained to have the desired effect. Massing effects of combat power, rather than concentrating forces, can enable even numerically inferior forces to produce decisive results and minimize human losses and waste of resources.

d. Maneuver

The purpose of maneuver is to place the adversary in a position of disadvantage through the flexible application of combat power. Maneuver is the movement of forces in relation to an adversary to secure or retain positional advantage, usually in order to deliver — or threaten delivery of — the direct and indirect fires of the maneuvering force. Effective maneuver keeps the adversary off balance and thus also protects the friendly force. Maneuver contributes materially to exploiting successes, preserving freedom of action, and reducing vulnerability by continually posing new problems for the adversary.

e. Economy of Force

The purpose of economy of force is to expend minimum essential combat power on secondary efforts in order to allocate the maximum possible combat power on primary efforts. Economy of force is the judicious employment and distribution of forces. It is the measured allocation of available combat power to such tasks as limited attacks, defense, delays, deception, or even retrograde operations to achieve mass elsewhere at the decisive point and time.

f. Unity of Command

The purpose of unity of command is to ensure unity of effort under one responsible commander for every objective. Unity of command means that all forces operate under a single commander with the requisite authority to direct all forces employed in pursuit of a common purpose. Unity of command may not be possible during coordination and operations with multinational and interorganizational partners, but the requirement for unity of effort is paramount. Unity of effort - the coordination and cooperation toward common objectives, even if the participants are not necessarily part of the same command or organization - is the product of successful unified action.

g. Security

The purpose of security is to prevent the adversary from acquiring an unexpected advantage. Security enhances freedom of action by reducing friendly vulnerability to hostile acts, influence, or surprise. Security results from the measures taken by commanders to protect their forces, the population, or other critical priorities. Staff planning and an understanding of adversary strategy, tactics, and doctrine enhance security. Risk is inherent in military operations. Application of this principle includes prudent risk management, not undue caution.

h. Surprise

The purpose of surprise is to strike at a time and place or in a manner for which the adversary is unprepared. Surprise can help the commander shift the balance of combat power, and thus achieve success well out of proportion to the effort expended. Factors contributing to surprise include speed in decision-making, information sharing, and force movement; effective intelligence; deception; application of unexpected combat power; operational security (OPSEC); and variations in tactics and methods of operation.

i. Simplicity

The purpose of simplicity is to increase the probability that plans and operations will be executed as intended by preparing clear, uncomplicated plans and concise orders. Simplicity contributes to successful operations. Simple plans and clear, concise orders minimize misunderstanding and confusion. When other factors are equal, the simplest plan is preferable. Simplicity in plans allows better understanding and execution planning at all echelons. Simplicity and clarity of expression greatly facilitate mission execution in the stress, fatigue, fog of war, and complexities of modern combat, and are especially critical to success in multinational operations.

j. Restraint

The purpose of restraint is to prevent the unnecessary use of force. A single act could cause significant military and political consequences; therefore, judicious use of force is necessary. Restraint requires the careful and disciplined balancing of a need for security, the conduct of military operations, and national objectives. Excessive force antagonizes those parties involved, thereby damaging the legitimacy of the organization that uses it while potentially enhancing the legitimacy of the opposing party. Sufficiently detailed rules of engagement (ROE) that the commander tailors to the specific circumstances of the operation can facilitate appropriate restraint.

k. Perseverance

The purpose of perseverance is to ensure the commitment necessary to achieve national objectives. Perseverance involves preparation for measured, protracted military operations in pursuit of national objectives. Some joint operations may require years to reach the termination criteria. The underlying causes of the crisis may be elusive, making it difficult to achieve decisive resolution. The patient, resolute, and persistent pursuit of national goals and objectives is essential to success. This will frequently involve diplomatic, economic, and informational measures to supplement military efforts.

l. Legitimacy

The purpose of legitimacy is to maintain legal and moral authority in the conduct of operations. Legitimacy, which can be a decisive factor in operations, is based on actual and perceived legality, morality, and rightness of actions from the various perspectives of interested audiences. These audiences will include our national leadership and domestic population, governments, and civilian populations in the area of operations (AO), and the nations and organizations around the world. Committed forces must sustain the legitimacy of the operation and of the host government, where applicable. Security actions must be balanced with legitimacy concerns. All actions must be considered in the light of potentially competing strategic and tactical level requirements and must exhibit fairness in dealing with competing factions where appropriate. Legitimacy may depend on adherence to objectives agreed to by the international community, ensuring the action is appropriate to the situation and to perceptions of fairness in dealing with various factions. Restricting the use of force, restricting the type of forces employed, protecting civilians, and ensuring the disciplined conduct of the forces involved may reinforce legitimacy.

5. Joint Functions

Joint functions are related capabilities and activities grouped together to help JFCs integrate, synchronize, and direct joint operations. Functions common to joint operations at all levels of warfare fall into seven basic groups—command and control (C2), information, intelligence, fires, movement and maneuver, protection, and sustainment. Functions apply to all operations. A number of subordinate tasks, missions, and related capabilities help define each function, and some could apply to more than one joint function.

Note: A similar construct, the *Marine Corps Warfighting Functions*, will be discussed in the 8671 *Doctrine* sub-course.

The joint functions reinforce and complement one another, and integration across the functions is essential to mission accomplishment. For example, joint fires can enhance the protection of a joint security area (JSA) by dispersing or disrupting adversary assets threatening the JSA. In any joint operation, the JFC can choose from a wide variety of joint and Service capabilities and combine them in various ways to perform joint functions and accomplish the mission. Plans describe how the JFC uses military capabilities (i.e., organizations, people, and systems) to perform tasks associated with each joint function. However, forces and other assets are not characterized by the functions for which the JFC is employing them. Individual Service capabilities can often support multiple functions simultaneously or sequentially while the joint force is executing a single task. For example, aviation assets routinely support all seven functions in a single combat operation. Just as component commanders integrate activities across functions to accomplish tasks and missions, the JFC and staff do likewise for the joint force. Various factors complicate the JFC's integration challenge, such as competing demands for high priority capabilities and the fact that joint force

components have different function-oriented approaches, procedures, and perspectives. JFCs and U.S. Government (USG) interorganizational partners synchronize, coordinate, and integrate military operations with the activities of interorganizational participants to achieve unity of effort toward U.S. objectives.

a. Command and Control

C2 encompasses the exercise of authority and direction by a commander over assigned and attached forces to accomplish the mission. The JFC provides operational vision, guidance, and direction to the joint force. The C2 function encompasses a number of tasks, including:

- Establish, organize, and operate a joint force HQ.
- Command subordinate forces.
- Prepare, modify, and publish plans, orders, and guidance.
- Establish command authorities among subordinate commanders.
- Assign tasks, prescribe task performance standards, and designate OAs.
- Prioritize and allocate resources.
- Manage risk.
- Communicate and maintain the status of information across the staff, joint force, and with the public as appropriate.
- Assess progress toward accomplishing tasks, creating conditions, and achieving objectives.
- Coordinate and control the employment of joint lethal and nonlethal capabilities.
- Coordinate, synchronize, and when appropriate, integrate joint operations with the operations and activities of other participants.
- Ensure the flow of information and reports to higher authority.

Command includes both the authority and responsibility to use resources to accomplish assigned missions. Command at all levels is the art of motivating and directing people and organizations to accomplish missions. The C2 function supports an efficient decision-making process. Timely intelligence enables commanders to make decisions and execute those decisions more rapidly and effectively than the adversary. This decreases risk and allows the commander more control over the timing and tempo of operations.

Control is inherent in command. To control is to manage and direct forces and functions consistent with a commander's command authority. Control of forces and functions helps commanders and staffs compute requirements, allocate means, and integrate efforts. Control is necessary to determine the status of organizational effectiveness, identify variance from set standards, and correct deviations from these standards. Control permits commanders to acquire and apply means to support the mission and develop specific instructions from general guidance. Control provides the means for commanders to maintain freedom of action, delegate authority, direct operations from any location, and integrate and synchronize actions throughout the AO. Ultimately, it provides commanders a means to measure, report, and correct performance.

b. Information

The information function encompasses the management and application of information and its deliberate integration with other joint functions to change or maintain perceptions, attitudes, and other elements that drive desired behaviors and to support human and automated decision-making. The information function helps commanders and staffs understand and leverage the pervasive nature of information, its military uses, and its application during all military operations. This function provides JFCs the ability to integrate the generation and preservation of friendly information while leveraging the inherent informational aspects of military activities to achieve the commander's objectives and attain the end state.

All military activities produce information. Informational aspects are the features and details of military activities observers interpret and use to assign meaning and gain understanding. Those aspects affect the perceptions and attitudes that drive behavior and decision-making. The JFC leverages informational aspects of military activities to gain an advantage; failing to leverage those aspects may cede this advantage to others. Leveraging the informational aspects of military activities ultimately affects strategic outcomes.

The information function includes activities that facilitate the JFC's understanding of the role of information in the OE, facilitate the JFC's ability to leverage information to affect behavior, and support human and automated decision-making.

In conjunction with activities under the intelligence joint function, this activity facilitates the JFC's understanding of the pervasive nature of information in the OE, its impact on relevant actors, and its effect on military operations. It includes determining relevant actor perceptions, attitudes, and decision-making processes and requires an appreciation of their culture, history, and narratives, as well as knowledge of the means, context, and established patterns of their communication.

Information affects the perceptions and attitudes that drive the behavior and decision-making of humans and automated systems. In order to affect behavior, the JFC must understand the perceptions, attitudes, and decision-making processes of humans and automated systems. These processes reflect the aggregate of social, cultural, and technical attributes that act upon and impact knowledge, understanding, beliefs, world views, and actions.

The human and automated systems whose behavior the JFC wants to affect are referred to as relevant actors. Relevant actors may include any individuals, groups, and populations, or any automated systems, the behavior of which has the potential to substantially help or hinder the success of a particular campaign, operation, or tactical action. For the purpose of military activities intended to inform audiences, relevant actors may include US audiences; however, US audiences are not considered targets for influence.

c. Intelligence

Understanding the operational environment (OE) is fundamental to joint operations. The intelligence function supports this understanding with analysis of the OE to inform JFCs about adversary capabilities, centers of gravity (COGs), vulnerabilities, and future COAs, and to help commanders and staffs understand and map friendly, neutral, and threat networks. Using the continuous joint intelligence preparation of the operational environment (JIPOE) analysis process, properly tailored JIPOE products can enhance OE understanding and enable the JFC to act inside the adversary's decision cycle. Intelligence activities and assessments also occur while defending the homeland within the guidelines of applicable regulations and laws.

People and organizations other than the enemy may positively or negatively affect the friendly mission. These actors may include the population, HN government, and potential opposition leaders. Other relevant actors may include international organizations, non-state actors, and nongovernmental organizations (NGOs). By first identifying the relevant actors and learning as much as possible about them and their interrelationships, the JFC can develop an approach that will facilitate decision-making and behavior (active or passive) among relevant actors that is consistent with the desired end state of the operation. Sociocultural analysis and identity intelligence (I2) activities enable a better understanding of the relevant actors. Individuals may fit into more than one category of actor. For example, a tribal leader may also work as a district governor, while also working behind the scenes to provide financial and material support to an insurgency.

The intelligence function encompasses the joint intelligence process. The joint intelligence process consists of six interrelated categories of intelligence operations:

1. Planning and direction of intelligence activities.
2. Collection of data.
3. Processing and exploitation of collected data to produce relevant information.
4. Analysis of information and production of intelligence.
5. Dissemination and integration of intelligence with operations.
6. Evaluation and feedback regarding intelligence effectiveness and quality.

JFCs and their component commanders are the key players in planning and conducting intelligence tasks. Commanders are more than just consumers of intelligence. They are responsible for fully integrating intelligence into their plans and operations. They are also responsible for distributing intelligence and information to subordinate commands, and when appropriate, to relevant participants through established protocols and systems. Commanders establish operational and intelligence requirements and continuous feedback to ensure optimum intelligence support to planning and operations. This interface supports the commander and operational planning and execution. It also mitigates surprise, assists friendly deception efforts, and enables joint operation assessment.

d. Fires

To employ fires is to use available weapons and other systems to create a specific effect on a target. Joint fires are those delivered during the employment of forces from two or more components in coordinated action to produce desired results in support of a common objective. Fires typically produce destructive effects, but various other ways and means can be employed with little or no associated physical destruction. This function encompasses the fires associated with a number of tasks, missions, and processes, including:

(1) Conduct Joint Targeting

This is the process of selecting and prioritizing targets and matching the appropriate response to them, taking account of command objectives, operational requirements, and capabilities.

(2) Provide Joint Fire Support

This task includes joint fires that assist joint forces to move, maneuver, and control territory, populations, space, cyberspace, airspace, and key waters.

(3) Countering Air and Missile Threats

This task integrates offensive and defensive operations and capabilities to attain and maintain a desired degree of air superiority and force protection. These operations are planned to destroy or negate adversary manned and unmanned aircraft and missiles, both before and after launch.

(4) Interdict Enemy Capabilities

Interdiction diverts, disrupts, delays, or destroys the enemy's military surface capabilities before they can be used effectively against friendly forces, or to otherwise achieve their objectives.

(5) Conduct Strategic Attack

This task includes offensive action against targets—whether military, political, economic, or other—which are selected specifically in order to achieve national or military strategic objectives.

(6) Assess the Results of Employing Fires

This task includes assessing the effectiveness and performance of fires as well as their contribution to the larger operation or objective.

e. Movement and Maneuver

This function encompasses the disposition of joint forces to conduct operations by securing positional advantages before or during combat operations and by exploiting tactical success to achieve operational and strategic objectives. This function includes moving or deploying forces into an AO and maneuvering them to operational depths for offensive and defensive purposes. It also includes assuring the mobility of friendly forces. The movement and maneuver function encompasses a number of tasks, including:

- Deploy, shift, regroup, or move joint and/or component force formations within the AO by any means or mode (i.e., air, land, or sea).
- Maneuver joint forces to achieve a position of advantage over an adversary.
- Provide mobility for joint forces to facilitate their movement and maneuver without delays caused by terrain or obstacles.
- Delay, channel, or stop movement and maneuver by adversary formations. This includes operations that employ obstacles (i.e., countermobility), enforce sanctions and embargoes, and conduct blockades.
- Control significant areas in the AO whose possession or control provides either side an operational advantage.

JFCs must carefully consider the movement of forces and whether to recommend the formation and/or movement of multinational forces. They must be aware of anti-access/area denial (A2/AD) threats, which may slow or disrupt the deployment of friendly forces. At times, movement of forces can contribute to the escalation of tension, while at other times its deterrent effect can reduce those tensions. Movement of forces may deter adversary aggression or movement.

Maneuver is the employment of forces in the AO through movement in combination with fires to achieve a position of advantage in respect to the adversary. Maneuver of forces relative to adversary COGs can be key to the JFC's mission accomplishment. Through maneuver, the JFC can concentrate forces at decisive points to achieve surprise, psychological effects, and physical momentum. Maneuver may also enable or exploit the effects of massed or precision fires.

The principal purpose of maneuver is to place the enemy at a disadvantage through the flexible application of movement and fires. The goal of maneuver is to render opponents incapable of resisting by shattering their morale and physical cohesion (i.e., their ability to fight as an effective, coordinated whole) by moving to a point of advantage to deliver a decisive blow. This may be achieved by attacking enemy forces and controlling territory, airspace, electromagnetic spectrum (EMS), populations, key waters, and lines of communication (LOC through air, land, and maritime maneuvers).

There are multiple ways to attain positional advantage. A naval expeditionary force with airpower, cruise missiles, and amphibious assault capability, within operational reach of an adversary's COG, has positional advantage. In like manner, land and air expeditionary forces that are within operational reach of an adversary's COG and have the means and opportunity to strike and maneuver on such a COG also have positional advantage. Maintaining full-spectrum superiority contributes to positional advantage by facilitating freedom of action.

f. Protection

The protection function encompasses force protection, force health protection (FHP), and other protection activities.

The function focuses on force protection, which preserves the joint force's fighting potential in four primary ways. One way uses active defensive measures that protect the joint force, its information, its bases, necessary infrastructure, and LOCs from an enemy attack. Another way uses passive defensive measures that make friendly forces, systems, and facilities difficult to locate, strike, and destroy by reducing the probability of, and minimizing the effects of, damage caused by hostile action without the intention of taking the initiative. The application of technology and procedures to reduce the risk of friendly fire incidents is equally important. Finally, emergency management and response reduce the loss of personnel and capabilities due to isolating events, accidents, health threats, and natural disasters.

Force protection does not include actions to defeat the enemy or protect against accidents, weather, or disease. FHP complements force protection efforts by promoting, improving, preserving, or restoring the mental or physical well-being of Service members. Furthermore, as the JFC's mission requires, the protection function also extends beyond force protection to encompass protection of US noncombatants.

Force protection is achieved through the tailored selection and application of multilayered active and passive measures commensurate with the level of risk. Intelligence sources provide information regarding an adversary's capabilities against personnel and resources, as well as information regarding force protection considerations. Foreign and domestic law enforcement agencies can contribute to force protection through the prevention, detection, response, and investigation of crime, and by sharing information on criminal and terrorist organizations. I2 can identify threat networks in the AO, enhancing the commander's ability to establish effective antiterrorism programs, screening and vetting activities, and lasting relationships with the HN and local population.

g. Sustainment

Sustainment is the provision of logistics and personnel services to maintain operations through mission accomplishment and redeployment of the force. Sustainment provides the JFC the means to enable freedom of action and endurance and to extend operational reach. Sustainment determines the depth to which the joint force can conduct decisive operations, allowing the JFC to seize, retain, and exploit the initiative. The sustainment function includes tasks to:

- Coordinate the supply of food, operational energy (fuel and other energy requirements), arms, munitions, and equipment.
- Provide for maintenance of equipment.
- Coordinate and provide support for forces, including field services; personnel services support; health services; mortuary affairs; religious support; postal support; morale, welfare, and recreational support; financial support; and legal services.
- Build and maintain contingency bases.
- Assess, repair, and maintain infrastructure.
- Acquire, manage, and distribute funds.
- Provide common-user logistics support to other government organizations, international organizations, NGOs, and other nations.
- Establish and coordinate movement services.
- Establish large-scale detention compounds and sustain enduring detainee operations.

JFCs should identify sustainment capabilities early in planning. Sustainment should be a priority consideration when the timed-phased force and deployment data is built. Sustainment provides JFCs with flexibility to develop branches and sequels and to refocus joint force efforts. Given mission objectives and adversary threats, the ultimate goal is for planners to develop a feasible, supportable, and efficient CONOPS that takes into account the threat and defense of logistical forces. Prior to the development of contingency plans, combatant commands (CCMDs) develop a theater logistics analysis, a theater logistics overview (TLO), and a distribution plan to provide detailed mobility and distribution analysis to ensure sufficient capacity or planned enhanced capability is available to support the CCMD's theater cooperation plan (TCP).

In the context of joint operations, having a common understanding of the role and interaction of the joint functions is critically important for proper planning, preparation, employment, and support of one's own forces in the attainment of assigned objectives. Over or under resourcing any single joint function occurs at the expense of the force's aggregate available combat power. The best operational leaders recognize the close interrelationship between time and space and the joint functions early in the development of campaigns or major operations. A developed understanding of military history, theory, and doctrine enables savvy operational commanders and planners to meld their available assets to the environment in order to achieve a marked advantage in combat power.

6. Joint Task Forces

Joint task forces (JTFs) operate in a complex strategic environment that consists of a variety of national, international, and global factors that affect the decisions of senior civilian and military leaders with respect to the employment of U.S. instruments of national power in peace and periods of conflict. This environment is uncertain and complex and can change rapidly, requiring JTF commanders and staff to maintain persistent military engagement with multinational partners. The JTF commander will encounter threats that are increasingly *transregional*, *multi-domain*, and *multi-functional* in nature. Joint forces today operate across multiple combatant commands (CCMDs), employ capabilities across all domains, and involve conventional forces (CF) and special operations forces (SOF). The strategic environment is fluid, with continually changing alliances, partnerships, and threats that rapidly emerge, disaggregate, and reemerge. While it is impossible to predict precisely how challenges will emerge and what form they might take, we can expect that uncertainty, ambiguity, and surprise will persist.

A JTF is one of several C2 options for conducting joint operations. A JTF may be established when the scope, complexity, or other factors of the operation require capabilities of Services from at least two Military Departments operating under a single joint force commander (JFC). A combined JTF (CJTF) may be formed when the operation requires capabilities from more than one country. The JTF establishing authority designates the JTF, assigns the mission, designates forces from currently assigned or allocated forces, or, in the case of CCMDs without or with

not enough assigned or allocated forces, submits a request for initial or additional forces through the global force management (GFM) process to stand up a JTF. Additionally, the JTF establishing authority delegates command authorities and relationships and provides other C2 guidance necessary for the JTF to form the joint force and begin operations. The appropriate authority may establish a JTF on a geographic or functional basis or a combination of the two. In either case, the establishing authority typically assigns a joint operations area (JOA) to the JTF. There may also be situations in which a JTF has a logistics-focused mission. In these situations, the CDR should delegate directive authority for common support capabilities over specific logistic forces, facilities, and supplies to the JTF.

A JTF has authority to assign missions, redirect efforts, and require coordination among subordinate commanders. Unity of command, centralized planning and direction, and decentralized execution are key considerations. *Generally, a JTF should allow Service tactical and operational groupings to function as they were designed.* The intent is to meet the JTF's mission requirements while maintaining the functional integrity of Service components. A JTF typically organizes the joint force with a combination of Service and functional components and additional attachments as required. A functional component is a command normally composed of forces of two or more Military Departments, such as a joint force air component command. A JTF may elect to centralize selected functions within the joint force, but should avoid reducing the versatility, responsiveness, and initiative of subordinate forces. Figure 1 shows the basic JTF organizational options.

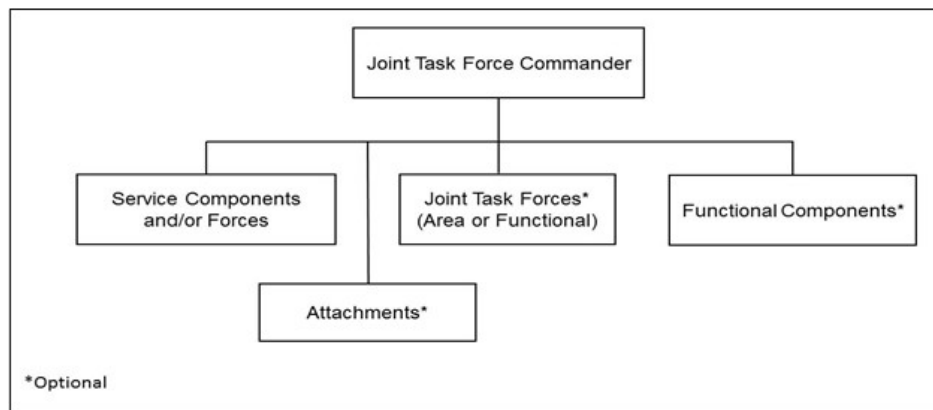


Figure 1: Joint Task Force Organizational Options.

The JTF and staff have a fundamental role in ensuring unified action—the synchronization, coordination, and/or integration of the activities of governmental and nongovernmental entities with military operations to achieve unity of effort. The interface between a JTF and these entities is considered interorganizational cooperation—the interaction that occurs among elements of the Department of Defense (DOD); participating U.S. Government (USG) departments and agencies; state, territorial, local, and tribal agencies; foreign military forces and government agencies; international organizations; nongovernmental organizations (NGOs); and the private sector. JTF commanders are challenged to achieve and maintain operational coherence when operating in conjunction with such a potentially diverse group of participants. However, JTF HQ staff can often act as facilitators for unified action under challenging conditions through inclusive planning processes, coordination mechanisms, and the identification of (and focus on) common interests in accomplishing the broader mission.

The JFC has full authority to assign missions, redirect efforts, and direct coordination among subordinate commanders. JFCs should allow Service tactical and operational assets and groupings to function generally as they were designed and organized (e.g., carrier strike group; Marine air-ground task force [MAGTF]; U.S. Air Force air and space expeditionary task forces; U.S. Army corps, divisions, brigade combat teams; and SO JTF). The intent is to meet the needs of the JFC while maintaining the tactical and operational integrity of the Service organizations. The following policy for C2 of U.S. Marine Corps tactical air (TACAIR) recognizes this and deals with MAGTF aviation during sustained operations ashore:

The MAGTF commander will retain OPCON of organic air assets. The primary mission of the MAGTF aviation combat element is the support of the MAGTF ground combat element. During joint operations, the MAGTF air assets normally will be in support of the MAGTF mission. The MAGTF commander will make sorties available to the JFC, for tasking through the joint force air component commander (JFACC), for air defense, long-range interdiction, and long-range reconnaissance. Sorties in excess of MAGTF direct support requirements will be provided to the JFC for tasking through the JFACC for the support of other components of the joint force or the joint force as a whole.¹

a. Types of Joint Task Forces

A JTF may be established as operational, contingency, or standing.

(1) Operational JTF

This is the most common type of JTF and is established in response to a SecDef-approved military operation or crisis. An establishing authority normally disestablishes an operational JTF when the mission is complete, but can also transition it into a standing JTF. Examples include JTF Katrina (USNORTHCOM), JTF Lebanon (United States European Command [USEUCOM]), JTF-515 (United States Pacific Command [USINDOPACOM]), and Combined JTF-101 (United States Central Command [USCENTCOM]).

(2) Contingency JTF

This is a JTF identified and designated to support OPLAN execution or specific on-call missions, such as national special security events, on a contingency basis. These JTFs normally remain in a cadre or reduced manning status for planning purposes within a Service component HQ or one of its subordinate HQs until required to conduct operations. A contingency JTF becomes an operational JTF when SecDef issues the execution order (EXORD). Examples include JTF National Capital Region (USNORTHCOM), Joint Task Force-Israel (USEUCOM), and Joint Task Force-Port Opening (JTF-PO) (USTRANSCOM).

(3) Standing JTF

This is a JTF originally established as an operational JTF but that has an enduring mission that is projected to continue indefinitely. Some standing JTFs will have transitioned to permanently assigned personnel, while others will rely on continued rotation of core HQs’ elements and joint individual augmentees (JIAs). Examples include JTF Bravo (USSOUTHCOM), JTF Global Network Operations (United States Strategic Command [USSTRATCOM]), JTF-N (USNORTHCOM), and JIATF-West (USINDOPACOM).

b. Command Authorities

JFCs exercise command authorities delegated to them by law or senior leaders and commanders over assigned and attached forces. Command relationships is another term used for these authorities. Unity of command in joint operations is maintained through the application of the various command relationships as follows. Figure 2 summarizes their relationships.

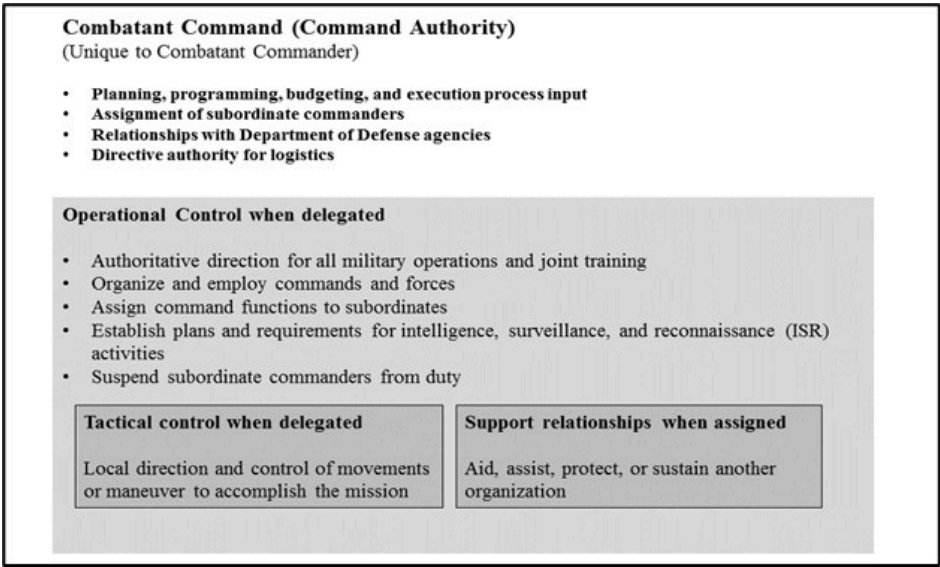


Figure 2: Command Relationships.

(1) Combatant Command (COCOM)

COCOM is the nontransferable command authority established by Title 10, United States Code (USC), Section 164, exercised only by commanders of unified or specified CCMDs unless otherwise directed by the President or SecDef. COCOM, which cannot be delegated, is the authority of a CDR to perform those functions of command over assigned forces involving organizing and employing commands and forces, assigning tasks, designating objectives, and giving authoritative direction over all aspects of military operations, joint training, and logistics necessary to accomplish the missions assigned to the command. COCOM should be exercised through the commanders of subordinate organizations. Normally, this authority is exercised through subordinate JFCs and Service and/or functional component commanders. COCOM provides full authority to organize and employ commands and forces as the CDR considers necessary to accomplish assigned missions. In crisis response and combat, or where critical situations require changing the normal logistic process, the CDRs’ logistic authority enables them to use all logistic capabilities of all forces assigned and attached to their commands as necessary to accomplish their mission. Under peacetime conditions, the CDR will exercise logistic authority consistent with the peacetime limitations imposed by legislation, DOD policy or regulations, budgetary considerations, local conditions, and other specific conditions prescribed by SecDef or CJCS.

(2) Operational control (OPCON)

OPCON is inherent in COCOM and may be delegated within the command. OPCON is command authority that may be exercised by commanders at any echelon at or below the level of CCMD to perform those functions of command over subordinate forces involving organizing and employing commands and forces, assigning tasks, designating objectives, and giving authoritative direction necessary to accomplish the mission. OPCON includes authoritative direction over all aspects of military operations and joint training necessary to accomplish missions assigned to the command. This authority should be exercised through the commanders of subordinate organizations, normally through subordinate JFCs and Service and/or functional component commanders. OPCON normally provides full authority to organize commands and forces and to employ those forces as the commander in operational control considers necessary to accomplish assigned missions; it does not, in and of itself, include authoritative direction for logistics or matters of administration, discipline, internal organization, or unit training.

(3) Tactical control (TACON)

TACON is inherent in OPCON. TACON is command authority over assigned or attached forces or commands, or military capability or forces made available for tasking, which is limited to the detailed direction and control of movements or maneuvers within the operational area necessary to accomplish missions or tasks assigned. TACON may be delegated to and exercised at any level at or below the level of CCMD. TACON provides sufficient authority for controlling and directing the application of force or tactical use of combat support assets within the assigned mission or task. Commanders may delegate TACON to subordinate commanders at any echelon at or below CCMD and may exercise TACON over assigned or attached forces or military capabilities or forces made available for tasking. TACON does not provide organizational authority or authoritative direction for administrative and logistic support or discipline (Uniform Code of Military Justice authority); the commander of the parent unit continues to exercise those responsibilities unless the establishing directive specifies otherwise. Except for special operations forces (SOF), functional component commanders typically exercise TACON over military capabilities or forces made available for tasking.

(4) Administrative control (ADCON)

ADCON is the direction or exercise of authority over subordinate or other organizations in respect to administration and support, including organization of Service forces, control of resources and equipment, personnel management, unit logistics, individual and unit training, readiness, mobilization, demobilization, discipline, and other matters not included in the operational missions of the subordinate or other organizations.

c. Componentcy

Before the *Goldwater-Nichols Department of Defense Reorganization Act of 1986*, the Fleet Marine Force (FMF) provided Marine Air-Ground Task Forces (MAGTF) to the combatant commanders (at the time, called "commanders in chief", or "CINCs") directly or through the U.S. fleet commanders. An example of a successful command arrangement involving Marine Corps support of a joint force occurred in the early days of the Korean War. In August-September 1950, the 1st Marine Brigade supported the 8th Army in the defense of the Pusan perimeter. Later that September, the 1st Marine Division, with supporting Marine aviation, served as part of X Corps, the land combat component of Joint Task Force 7, during the Inchon landing. The joint force commanders for these operations effectively integrated Marine forces into the joint force. The MAGTF operated using Marine Corps service doctrine and its optimal task organization, including organic Marine aviation. The concept of MAGTF operations was proven during these operations. The combined arms capability and well-established support relationships between the two combat arms—ground and aviation—were the envy of the other services.

The *Goldwater-Nichols Act* created a new model for joint operations by clearly placing responsibility for the accomplishment of a combatant command's assigned mission on the combatant commander and by ensuring that authority is fully commensurate with responsibility. The act also established closer, more formal command linkages from the services to the supported combatant commander, thereby creating the requirement for service components to the combatant commands.

General David H. Berger, U.S. Marine Corps, Commandant of the Marine Corps, has provided guidance on the Marine Corps' component command relationships. The following is quoted from The Commandant's Planning Guidance (CPG):

In 1933, the establishment of the FMF under the operational control of the Fleet Commander generated great unity of effort, operational flexibility, and the integrated application of Navy and Marine capabilities throughout the maritime domain. The 1986 Goldwater-Nichols Act, however, removed the preponderance of the FMF from fleet operational control and disrupted the long-standing Navy-Marine Corps relationship by creating separate Navy and Marine Corps components within joint forces. Furthermore, Navy and Marine Corps officers developed a tendency to view their operational responsibilities as separate and distinct, rather than intertwined. With the rise of both land and sea-based threats to the global commons, there is a need to reestablish a more integrated approach to operations in the maritime domain. Reinvigorating the FMF can be accomplished by assigning more Marine Corps forces to the Fleet, putting Marine Corps experts in the fleet Maritime Operations Centers, and also by shifting emphasis in our training, education, and supporting establishment activities. Refining the component relationship, within the framework of Goldwater-Nichols, is a more complicated issue that must be explored in partnership with the Navy. With one exception, our MARFORs are not operational headquarters, nor will they be resourced as such. Our MARFORs are intended as administrative headquarters that advise their respective commands on the Marine Corps. In a functional component construct, we will complement and augment the JFMCC [joint force maritime component commander].

The Marine Expeditionary Force (MEF) will remain our principal warfighting organization; however, our MEFs need not be identical. III MEF will become our main focus-of-effort, designed to provide U.S. Indo-Pacific Command (U.S. INDOPACOM) and the Commander, 7th Fleet with a fight-tonight, stand-in force capability to persist inside an adversary's weapon systems threat range, create a mutually contested space, and facilitate the larger naval campaign. When modernized in a manner consistent with the vision above, III MEF will be a credible deterrent to adversary aggression in the Pacific. I MEF will also be focused on supporting the Commander, USINDOPACOM and the Commander, 3rd Fleet. I MEF will continue to provide forces to USINDOPACOM to build partner capacity and reinforce deterrence efforts, and must be prepared to impose costs on a potential adversary, globally. We will increasingly accept risk with I MEF's habitual relationship with CENTCOM; however, 7th Marines is at present purpose-built to support CENTCOM requirements; thus, I MEF will continue to support CENTCOM requirements within the capacity of 7th Marines. II MEF will undergo substantial changes to better align with the needs of Commanders of 2d and 6th Fleets. During a major contingency operation or sustained campaign ashore, necessary combat power will be provided to the committed MEF through global sourcing by the Total Force.

We will continue to recommend Marine operating forces be employed as combined-arms teams; however, we must be flexible enough to satisfy Fleet and Combatant Commander needs whether they require a MEF, a single LHA with Marine complement in support of an Expeditionary Strike Group (ESG), or an aviation detachment in support of U.S. Special Operations Command (USSOCOM). First and foremost, we must be prepared to be employed as Fleet Marine Forces. The Service will provide ready forces, and our component headquarters will advise their respective commanders on the best employment of those forces; however, the ultimate decision on tactical employment resides with the Combatant Commanders.

All Service forces (except as noted in United States Code, Title 10, *Armed Forces*, subtitle A, part I, chapter 6, section 164, *Commanders of Combatant Commands: Assignment, Powers and Duties*) are assigned to combatant commands by the Secretary of Defense through the annual *Forces for Unified Commands* memorandum. Once assigned, a force may be transferred from that command only at the direction of the Secretary of Defense. CDRs coordinate with their Service component commanders to organize forces to accomplish their assigned mission.

While the Services' roles and functions do not determine the theater command structure, the CDR should consider the traditional roles of the Services in the organization of the theater. The CDR seeks to organize their forces in the most effective manner to accomplish the mission. The CDR attempts to preserve normal Service responsibilities and organizational integrity to fully exploit their inherent capabilities. The options for organizing the forces available to the CDR are by Service components, functional components, or a combination of the two.

(1) Service Components

Any joint force, by definition, will include forces belonging to two or more Military Departments. The JFC may conduct operations through their Service components. This is appropriate when stability, continuity, economy, ease of long-range planning, and the scope of operations dictate that the organizational integrity of Service forces be maintained. Component commands may exercise operational control (OPCON) of forces assigned or attached to their CDR, or they may be limited to tactical control (TACON) of these forces. Marine Corps Component commands (MARFOR) are shifting to exercise administrative control (ADCON) only as described in the Commandant's Planning Guidance. Since the individual Services are organized, trained, equipped, and employed using each Services' doctrine, this arrangement fully exploits the capabilities and experience the individual Services can bring to a joint command (Figure 3).

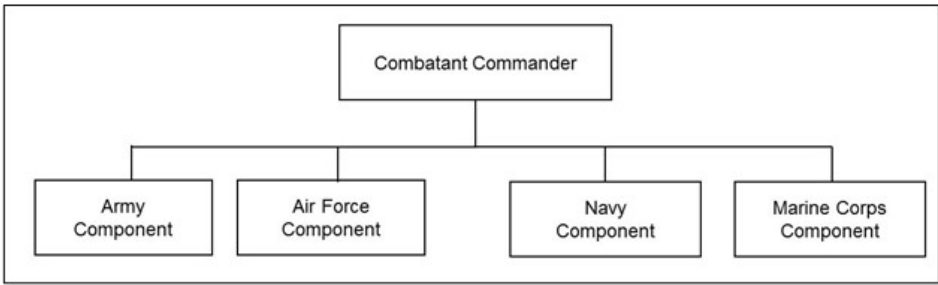


Figure 3: Combatant Command Organized by Service Components.

Exercising ADCON of Marine Corps forces allows the Marine Corps component commander to the CDR to take full advantage of the Marines' established lines of command and standing operating procedures. It enables Marine Corps forces to function as they were designed; i.e., as a MAGTF.

MARFOR commanders have responsibilities that derive from their roles in fulfilling Service functions. Their primary responsibility is that of a force provider and sustainer. Their specific responsibilities are as follows:

- Recommend the allocation and coordinate provision of Marine Corps forces or individuals to support the CDR's operations.

- Make recommendations to the CCDR on the proper deployment and employment of Marine Corps forces.
- Conduct deployment/redeployment planning and execution of assigned/attached Marine Corps forces.
- Select and nominate specific units or individuals of the Service component for attachment to the CCDR's subordinate forces and recommend command relationships.
- Conduct joint and combined training, including the training, as directed, of components of other Services in joint operations for which the Service component commander has or may be assigned primary responsibility or for which the Service components facilities and capabilities are suitable.
- Conduct joint, component, and combined contingency, crisis action, and exercise planning to support CCDR-assigned missions.
- Ensure internal Service functions; e.g., general and personnel administration, finance, discipline, training, logistics, processing of urgent universal needs requests, force protection, safety, and Service intelligence in support of assigned or attached forces and individuals, are performed as directed.
- Ensure commanders, staffs, and forces are trained to conduct joint, combined, and Service exercises and operations.
- Retain ADCON of forces attached to a Service component or force commander of a subordinate joint force command or another unified CCDR. However, the United States Marine Corps Forces, Special Operations Command (MARSOC), which is the Marine Corps component to the U.S. Special Operations Command (USSOCOM), performs only those ADCON functions that USSOCOM does not perform.
- Develop plans and procedures for the effective and efficient utilization of reserve and active forces.
- Provide and/or coordinate logistic support and inform the CCDR of plans or changes in logistic support that would significantly affect operational capability or sustainability.
- Develop and submit the Marine Corps forces input to the CCDR's integrated priority list.
- Develop program and budget requests that comply with the CCDR guidance on warfighting requirements and priorities.
- Inform the CCDR of program and budget decisions that may affect joint operation planning.
- Coordinate, execute, and report external funding for Marine Corps operations and exercises.
- Establish and maintain a resource evaluation and analysis function to ensure effective and accurate control and use of funds and resources provided for CCDR mission accomplishment.
- Ensure compliance with force protection, force health protection, and personnel recovery requirements for assigned and attached forces, civilian personnel, and families.
- Ensure assigned and attached forces have C2 systems that are interoperable in joint and combined environments.
- Maintain information management programs.

(2) Functional Component Commands

The CCDR may establish functional component commands to centralize selected functions and reduce the span of control by placing forces with similar capabilities under a single commander. Conducting operations through functional component commands requires the CCDR to accomplish the following:

- Assign the authority and responsibilities of functional component commands based on the concept of operations.
- Designate the forces or capabilities to be made available for tasking by the functional component commander.
- Establish the command relationship of the functional component commander over the forces or assigned capabilities. The functional component command must be aware of the organization, capabilities, and limitations of assigned or attached forces and the responsibilities retained by the Service component command. The functional component commander's assigned authority and responsibilities will not affect the command relationships between the Service component commander and the CCDR (Figure 4).

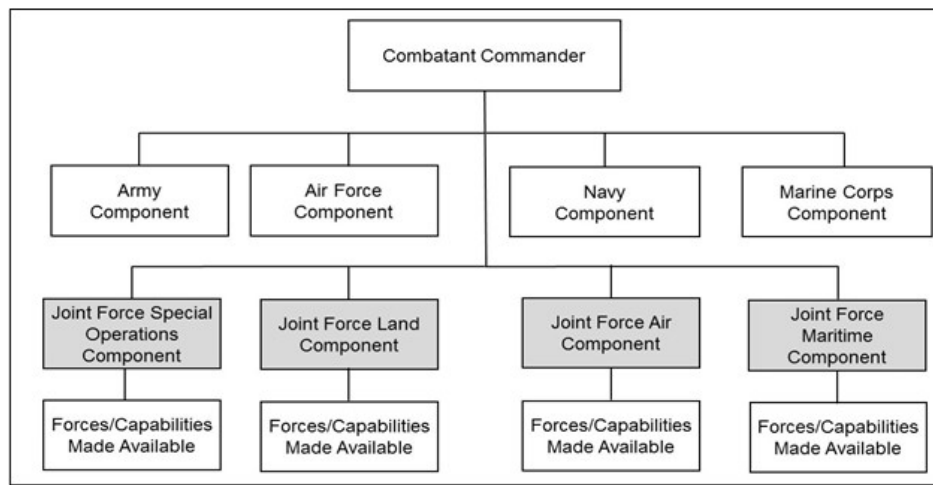


Figure 4: Combatant Command Organized by Functional Components.

(3) Combination of Service and Functional Components

Normally, joint forces are organized with a combination of Service and functional component commands with operational responsibilities. Figure 5, is an example of a combatant command organized with the four Service forces commanded by Service component commands and a special operations component.

The combination of Service and functional components takes advantage of the benefits of Service componentcy while allowing the CCDR to centralize certain functions to achieve the strategic or operational objective. In these cases, the Service component commanders retain ADCON of their forces while providing forces and capabilities through the CCDR to the functional component command. Regardless of how the CCDR organizes assigned or attached forces, if they have any Marine Corps forces, they will have a Marine Corps Service component.

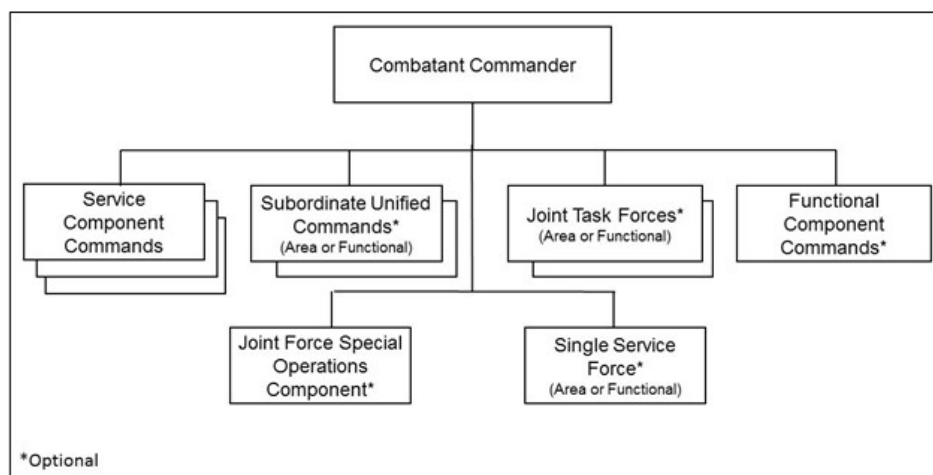


Figure 5: Combatant Command Organized by Combination of Service and Functional Components.

d. Subordinate Joint Command Structure

Subordinate joint commands may be either a subordinate unified command or a JTF. Differences in authority exist between the combatant command and the subordinate joint command. Forces, not command relationships, are transferred between commands. The COCOM is a non-transferrable command authority and remains with the assigned CCDR. The subordinate JFC normally exercises OPCON or TACON of assigned or attached forces. Both the combatant command and subordinate joint force may be organized to conduct operations through Service components, functional components, or a combination of the two. Both may be established in a geographic area or on a functional basis. Subordinate unified commands, such as United States Forces, Korea, are established to conduct operations on a continuing basis in accordance with criteria set by the unified commander, while a JTF has a specific, limited objective and does not require overall centralized control of logistics. Like combatant commands, JTFs contain components and are organized to conduct operations by Service, function, or a combination of the two. When the JTF is small and largely composed of Marine Corps units, the MAGTF commander and their staff can simultaneously perform the roles of JTF nucleus, JTF Marine component headquarters, and MAGTF command element.

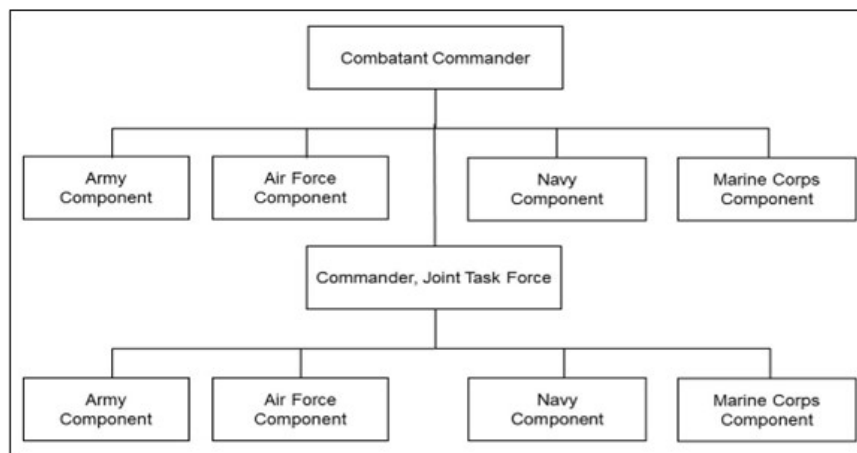


Figure 6: Subordinate JTF Organized by Service Components.

e. Designation as a Functional Component Command

The JFC can designate the MARFOR command as a functional component command. Examples include, but are not limited to:

- Joint force maritime component command
- Joint force land component command
- Joint force air component command
- Joint force special operations component command

While one commander may have two designations (i.e., Marine Corps component command and joint force functional component command), the responsibilities are separate, distinct, and not interchangeable. Because command functions are separate, so are staff functions. The Marine Corps component command's staff performs Service component functions and is staffed by the Marine Corps component's normal staff members. The functional component command staff performs functional component activities and should be staffed to reflect the composition of the functional component command. Designation as a functional component command brings additional responsibilities but does not replace Service component responsibilities for assigned Marine Corps forces. Regardless of the joint command structure, the Marine Corps component command must still provide administrative and logistic support to assigned forces. In addition to functional component duties, the JFC can assign the Marine Corps component command other joint duties, such as the area air defense commander or airspace control authority. These functions are normally accomplished by the assigned MAGTF.

(1) Joint Force Maritime Component Commander

This commander is responsible for planning, coordinating, and executing joint maritime operations. Joint maritime operations are performed with maritime capabilities or forces made available by components to support the joint force commander's operation or campaign objectives or to support other components of the joint force. The maritime environment includes oceans, littorals, riverine areas, and amphibious objective areas, and the coordinated airspace above them as defined by the joint force commander. The joint force commander will designate the component commander best suited to accomplish the mission as the joint force maritime component commander. When maritime operations are focused on littoral operations—and Marine Corps forces have the preponderance of the mission or capabilities to accomplish the mission—the Marine Corps component commander may be designated the JFMCC.

(2) Joint Force Land Component Commander

This commander is responsible for planning, coordinating, and executing joint land operations. Joint land operations are performed with land capabilities or forces made available by components to support the joint force commander's operation or campaign objectives or to support other components of the joint force. Marine Corps component commanders normally have the preponderance of land forces and the necessary command and control capability to direct their activities during expeditionary operations in a smaller-scale contingency. In the early stages of a major theater war, the Marine Corps component commander may serve as the JFLCC, but as forces continue to build up in theater, the JFC will normally designate the Army Service component commander as the JFLCC.

(3) Joint Force Air Component Commander

This commander is responsible for planning, coordinating, and executing joint air operations. Joint air operations are performed with air capabilities or forces made available by components to support the joint force commander's operation or campaign objectives or to support other components of the joint force. The expeditionary nature of Marine aviation and its associated command and control capability allow the Marine Corps component commander to function as the joint force air component commander in a smaller-scale contingency. In the early

stages of a major theater war, the Marine Corps component commander may serve as the joint force air component commander, but as forces continue to buildup in theater, the joint force commander will normally designate another component commander as the joint force air component commander (JFACC).

(4) Joint Force Special Operations Commander

The JFSOCC is the commander within a unified command, subordinate unified command, or JTF responsible to the establishing commander for making recommendations on the proper employment of assigned, attached, and/or made available for tasking SOF and assets, including planning and coordinating special operations, or accomplishing such operational missions as may be assigned. The JFSOCC acts as the senior SOF representative in the operation or within the theater. The JFSOCC supports the JFC with unique and effective SOF capability while ensuring the SOF components are organized for optimum contribution to the JTF. The JFSOCC provides a CF/SOF synchronizing process with the JTF through the joint forces land component commander, if designated. SOF is inherently organized as a joint force, providing a timely capability to coordinate across multiple services with communications connectivity that provides accurate and timely information to the JFSOCC. The JFSOCC's capability to assist with C2 functions is critical during the initial phases of crisis response.

¹ Joint Publication 1, *Doctrine for the Armed Forces of the United States*, 12 July 2017, p. IV-4 – IV-5.

Lesson 7: Future Operating Environment - Russia

Lesson Hours: 2.00

1. Introduction

21st-century military operations are increasingly complex. They cross multiple domains and involve myriad state and non-state actors. The Marine Corps' role is to deter adversaries through competition below the threshold of armed conflict and, when necessary, to engage in armed conflict and force a return to competition on favorable terms. Adversaries, such as China and Russia, are adept at competition and do not make the same distinction between competition and conflict the way U.S. joint forces do. These competitors are constantly seeking advantage over other actors, and especially the U.S., across all domains, and with all instruments of national power. By understanding how, where, and why adversaries are competing, the Marine Corps can better focus how it trains, equips, organizes, and employs operational forces to succeed in these competitions in the future operational environment.

This lesson presents the Commandant of the Marine Corps' article, "The Case for Change," that articulates why the Marine Corps must evolve to meet the challenges of the present and future. The lesson continues with a short discussion of a model that we may use to describe the interactions of people in the operating environment, and then takes a closer look at the military goals, strategy, organization, and capabilities of the Russian Federation.

Note: The next lesson will similarly present the military goals, strategy, organization, and capabilities of the People's Republic of China.

2. Educational Objectives

- a. Identify the broad themes associated with the future operating environment.
- b. Identify the characteristics of the future operating environment.
- c. Identify the conditions used to categorize strategic relationships.
- d. Identify the perceived goals of the Russian Federation.
- e. Identify the key aspects of Russian military strategy.
- f. Identify Russia's core military capabilities.
- g. Identify the general organization of the Russian armed forces.

3. Key Terms

- Regional competition
- Instability
- Cooperation
- Competition below armed conflict
- Armed conflict
- Russian Strategic Deterrence
- Russian Strategic Stability

4. Preparation

All quizzes, discussion forum posts, and assignments are executed in the Moodle course site.

- a. Lesson Reading: [Future Operating Environment - Russia](#) (47 pages).
- b. Other Requirements: None.
- c. Supplemental References (optional):
 - [MARSOF 2030: A Strategic Vision of the Future](#) (2018).
 - ["Strategic Competition: Beyond Peace and War," Joint Forces Quarterly](#) (2017).
 - [Russian Armed Forces: Capabilities](#), Congressional Research Service (Jun 2020).
 - [Russia Military Power: Building a Military to Support Great Power Aspirations](#), Defense Intelligence Agency (2017).
 - [Russia's War in Ukraine: Military and Intelligence Aspects](#), Congressional Research Service (14 Sep 2023).
 - [2025 Worldwide Threat Assessment](#), Defense Intelligence Agency (11 May 2025).

Lesson 7 Reading

FUTURE OPERATING ENVIRONMENT - RUSSIA

References:

- a) "The Case for Change," General David H. Berger, Commandant of the Marine Corps, *Marine Corps Gazette* (Jun 2020)
- b) MARSOF 2030: A Strategic Vision of the Future (2018)
- c) "Strategic Competition: Beyond Peace and War," *Joint Forces Quarterly* (2017)
- d) *Russian Armed Forces: Capabilities*, Congressional Research Service (Jun 2020)
- e) *Russia Military Power: Building a Military to Support Great Power Aspirations*, Defense Intelligence Agency (2017)
- f) *2025 Worldwide Threat Assessment*, Defense Intelligence Agency (11 May 2025)

1. The Case for Change - General David H. Berger

a. Introduction

The U.S. Marine Corps I lead in 2020 finds itself, like the rest of the U.S. defense establishment, at a crossroads. The passing of our Nation's "unipolar moment" and the emergence of revisionist great power competitors in China and Russia, coinciding with a sea change in the character of warfare driven by social and technological change, demands that we move rapidly to adapt to the circumstances of a new era.

This article lays out the case, as I see it, for the sweeping changes the Marine Corps needs to make to meet the principal challenges facing the institution: effectively playing our role as the Nation's naval expeditionary force-in-readiness while simultaneously modernizing the force to play its necessary roles in the operating environment described in the National Defense Strategy (NDS)—and doing both within the fiscal resources we are provided. Deep institutional change is inevitable when confronting modernization on this scale, and that type of change is hard. The urgency of change and the institutional reform and innovation necessary to achieve it has not diminished in the two years that have passed since the publication of the NDS. The ideas expressed below are not unique or original to me—forward thinkers across the defense establishment, academia, and industry have given voice to them for years. But the time to act is now.

b. Today's Marine Corps

Today's Marine Corps, despite many surface adaptations to the demands of the past two decades of counterinsurgency operations, is at its core optimized for amphibious forcible entry and sustained operations ashore. This essential design has endured since the 1950s, though it has changed in details of equipment and doctrine in response to the secular trend, extending back to the dawn of modern warfare, toward greater range and lethality of weapon systems. My predecessors made significant advances in keeping pace with this trend in pursuit of capabilities they deemed essential to the Nation's defense, based on the operating environment and the resources available at that time. Despite those advances, however, in light of the unrelenting increases in the reach, effectiveness, and lethality of modern weapons, the rise of revisionist powers with the technical acumen and economic heft to integrate those weapons and other technologies for direct or indirect confrontation with the U.S., and the persistence of rogue regimes possessing enough of those attributes to threaten U.S. interests, I am convinced that the defining attributes of our current force are no longer what the Nation requires of the Marine Corps. The rest of this article will review the reasons why.

c. Threat Technology - Secular Trends and the Rise of the Precision Strike Regime

The secular trend toward the increasing range and lethal effect of military technology is a commonplace of the history of modern warfare. Accompanying the development of range and lethality at every stage, albeit sometimes unevenly, has been the advance of the ability to apply that lethality effectively to military ends through the necessary command, control, communications, computer, intelligence, surveillance, and reconnaissance (C4ISR) organizations and technologies. These trends are of very long standing. Leaving aside the sweep of military history before 1945, the coming of the Atomic Age provided the clearest possible signal of their ultimate expression. The means to deliver lethality by very long-range unmanned means followed swiftly; Bernard Brodie noted in 1959, in strategic nuclear context, the advent of the "Missile Age."

As technology continued to develop, the outlines in maritime warfare of what the influential defense analyst Andrew Krepinevich identified as the "Mature Precision Strike Regime" (MPSR) began to become evident at the tactical and operational levels of warfare. Although the advanced military establishments of the Cold War superpowers thankfully never met in open combat, indications of the evolution and proliferation of long-range precision strike and accompanying C4ISR technologies appeared as early as 1967 with the sinking of the INS *Eilat* by an Egyptian operated, Soviet-manufactured SS-N-2 *Styx* anti-ship cruise missile (ASCM). Examples recur with regularity in the decades since, through the Tanker Wars and the South Atlantic War of the 1980s, to the crippling of the INS *Ah-Hanit* by a Hezbollah ASCM in 2006, and the attempted engagement of the USS *Mason* by similarly armed Yemeni rebels in 2016. Of critical note is the fact that these capabilities are now widely proliferated, to a limited degree of sophistication and integration, to regional powers and their non-state proxies, with the revisionist (and nuclear-armed) great powers possessing capabilities that increasingly mirror our own. The world we live in today, much less tomorrow's, displays most of the attributes of a truly mature precision strike regime.

Unsurprisingly, the trends driving the maritime precision strike regime also define the state of the art in joint warfare more broadly. As many observers have noted, the U.S. awakened the world to this reality with its one-sided annihilation of Iraqi forces in Kuwait in 1991. The revisionist powers have taken some time to close the U.S. lead, but evidence that they have done so is clear in their fielded forces and in the steady drumbeat of real-world incidents drawn from the recent history of military action below the threshold of great power conflict. The revolutionary impact of early ATGMs in the 1973 Arab-Israeli war prefigures the trend in the same manner as the extensive employment of naval ASCMs in that same conflict. More recent conflicts, including Israel's 2006 conflict with Hezbollah in Lebanon showcase the increasing range and lethality of modern precision-guided ground ordnance, while Russia's devastating employment of massed long-range artillery, directed and enabled by advanced C4ISR and electronic warfare capabilities, against Ukrainian forces in 2014 provide the most recent example of the proliferation on land of something approximating the maritime MPSR. Accompanying these indicators is the clear lesson from the U.S.' own experience in Iraq and Afghanistan of the vulnerability to even improvised explosive devices of the light armor systems (Stryker, AAVP7, LAV-25) that form so large an element of the current force design of our Nation's expeditionary land forces.

Corresponding trends are visible in the aviation component of joint warfare, with the steady advance and proliferation of ever more sophisticated aircraft, surface-to-air missiles, electronic warfare capabilities, and associated C4ISR technologies to integrate and control these capabilities. Real-world evidence of the "live" employment of these capabilities (outside of the horrific and regrettable incidence of their misemployment against defenseless civilian airliners) is sparser than in the maritime and land domains. This is likely because the greater difficulty of integrating these advances still renders them largely the province of advanced state militaries, and because the U.S. still maintains a substantial qualitative lead in this domain. Nevertheless, two points bear emphasis. First, the state of the art in threat capabilities, especially sensors and both surface-to-air and air-to-air missiles, has already forced enormous and potentially prohibitively costly adaptation upon U.S. forces in response. The current emphasis upon low-observable stealth technology in aircraft design, for example, represents a large element of our technological response to the development and proliferation of the precision-strike regime in the air domain. Second, while advances in range, precision, and lethality drive high-end competition in the air domain, related technologies offer increasing risks and opportunities at much lower levels of conflict, potentially blurring the lines between air, maritime, and land domains and giving less-sophisticated actors the ability to contest great-power air supremacy in previously unavailable ways.

Real-world incidence of the employment of unmanned aerial systems and loitering munitions, from the crude efforts of the Islamic State and its non-state competitors in Iraq and Syria from 2014-2016, to the more sophisticated employment by Armenian separatists of an Israeli-manufactured HAROP loitering munition in 2016, to the swarming drone attack on the Saudi Aramco oil processing facilities that evaded air defenses, points to the expression of the secular trend at levels far below the realm of great power competition. Why am I devoting space to a review of such well-established and documented trends? Because, despite the available evidence and near-consensus in many defense circles

as to the implications of these changes, we have been slow to adapt as a Service. Specific implications for our force design are addressed in greater detail below. In the meantime, I must consider the impact of a more recent (though far from historically unprecedented) development in warfare—the emergence of so-called “gray zone” strategies by an array of real and potential adversaries, most notably the two revisionist powers identified in the NDS.

d. Gray Zone Strategies-Multi-domain Competition within the MPSR

With the advance and proliferation of the precision strike regime, our adversaries have already proven they can deter us, to a degree, from employing our existing force design to counter their malign activities and defend the interests of our Nation, as well as those our allies and partners. Recognizing that the U.S. must at a minimum, and to a degree that varies by threat and theater, employ greater caution in the employment of its existing military capabilities, these actors use the degree of deterrence thus achieved to advance their respective agendas by means of “gray zone,” “hybrid warfare,” proxy warfare, and related strategies. There is little profit, for my purposes, in a debate over the intellectual merits of these various terms. The connecting file, from the perspective of force design, is the combination of deterrent effect with asymmetry of interest. Our adversaries, confronting the U.S.’ long-standing lead in the technologies and capabilities of the precision-strike regime, have chosen to employ “salami slicing” strategies that confront us with the alternatives of waging or threatening war over comparatively minor stakes, or accepting *faits accompli* in the form of local encroachments, annexations, or other violations of the rules of the established international order. Facing an adversary that has credibly fielded elements of a long-range reconnaissance-strike complex or possesses other capabilities (such as Iran’s well-established capacity for irregular warfare, augmented by increasing capability for long-range precision strike), the U.S. is in greater or lesser degree deterred. If the objective the adversary seeks appears relatively insignificant, the U.S. incentive to overcome the deterrent effect is correspondingly reduced.

Recent history offers a number of examples, exhaustively analyzed in the national security literature of the past decade. China’s “cabbage strategy” with respect to the disputed features of the South and East China Seas is commonly described as the classic example of a gray zone strategy, while Russia’s destabilization of Ukraine and illegal annexation of Crimea epitomize the so-far successful implementation of something more closely approximating “hybrid warfare.” Meanwhile, Iran’s pursuit of regional hegemony manifests as a more traditional program of political and religious subversion and proxy warfare in Iraq, Lebanon, Syria, and Yemen, all backed by an increasingly capable long-range reconnaissance-strike complex that displays, in local context, many of the attributes of the mature precision-strike regime. These strategies are designed to avoid obvious counters by the U.S. and its allies and partners. The idea, again, is to present us with what Michael O’Hanlon describes as the “Senkaku Paradox”: *faits accompli* on matters of such relative insignificance, in areas at the margins of our current ability to project and logistically support significant forces, that we perceive a lethal response as simply “not worth it.”

e. Imperative for Maritime Campaigning

The principal area where these trends play out today are in maritime theaters. Thus, it is no surprise that the NDS has directed our attention seaward, where the threats posed by both revisionist powers and rogue states are most significant. Our “need to refocus on how we will fulfill our mandate to support the fleet” is clear enough in the Planning Guidance I issued in July 2019. Still, it is worth restating the arguments that underlay that contention in the context of the argument for significant change in our present force design. Of the four state adversaries specifically described in the NSS and NDS, two—the “revisionist power” of the People’s Republic of China and the “rogue state” of Iran—present actual and potential threats that are either principally or partially of a maritime character. Russia, the other revisionist power, and North Korea, the second rogue state, present a variety of threats and challenges to the U.S., but the majority of these manifest outside the maritime domain. These adversaries are more accurately categorized principally as land powers.

It follows that, although the Naval Services will play certain roles as elements of Joint forces engaged in any principally land-oriented campaign that may take place involving Russia or North Korea, it is likely that these roles will be of a supporting nature, including (especially in the case of Russia) the provision of capabilities to support the deterrence or defeat of malign activities outside of areas in the “near abroad,” close to the borders of their sovereign territory. The Marine Corps will contribute to such campaigns in accordance with relevant plans and orders but will not use them as principal determinants of its force design or force structure.

The predominantly maritime threat posed by China globally, against which the Naval services will need to operate in close concert to execute missions involving sea control and denial, long-range strike, and limited operations to provide assured access for elements of the Joint force, does represent the primary pacing threat against which our force design and force structure will be measured. Any fight against China, in particular, and for the present most critically the deterrence of any such fight, is an inherently joint endeavor to which the Marine Corps can sensibly contribute only as an integral part of the Naval force in the prosecution of a naval campaign. We will optimize our design for this threat, though, as in the case of Russia, we will not consider exclusively the threat that China may pose in its immediate vicinity within the first island chain. Both China and the U.S. enjoy a range of options for confrontation and competition in a wider regional and global arena, though few of these involve credible scenarios featuring sustained land operations, and most of them are essentially founded upon the capabilities of the Naval Services.

f. Tomorrow’s Marine Corps-Implications for Force Design

The preceding review of the imperatives for change explains why I concur with the 37th Commandant’s assessment that “The Marine Corps is not organized, trained, equipped, or postured to meet the demands of the rapidly evolving future operating environment.” The imperatives of maritime competition, deterrence, and conflict in an era of warfare dominated by the emergence of a mature precision-strike regime demand

change. The NDS offers clear guidance at the strategic level as to the general nature of the change required; at my level, as a service chief, appear the institutional challenges and tradeoffs of recruiting, training, educating, and equipping Marines to give the combatant commanders the tools they need to execute the strategy.

So, what are the specific changes required? I have recently released a force design report describing in detail my conclusions thus far, and I will not repeat the whole of that here. It is also important to remember that “answers” are elusive when the task is preparation for an unknowable future. I keep constantly in mind the words of the great British historian Michael Howard, who was “tempted,” he once said, “to declare dogmatically that whatever doctrine the Armed Forces are working on now, they have got it wrong.” Sympathetic to the challenge of preparing forces for a test that can only truly be administered in battle, Howard went on to allow that it is not too much to hope that we will not “get it too badly wrong,” and that is most certainly my intention.

Clearer to me than the specifics of what we must do in future are a few things we need to stop doing now. As I noted above, the Marine Corps we have today is weighted too heavily toward amphibious forcible entry and sustained land operations. The fact that these design imperatives are not necessarily complementary does not help us—much of our present equipment, for example, is larger and heavier than we might wish it to be for amphibious operations of any kind. Its development was shaped, practically speaking, more by the demands of sustained operations ashore (from Desert Storm forward) than of amphibious operations per se. If we take the three considerations outlined above—rise of the precision strike regime, gray zone strategies, and the imperative of maritime campaigning—and the NDS’s guidance regarding pacing for inter-state strategic competition as yardsticks by which to measure the adequacy of what we have today, the basic outlines of the necessary change become clear enough, at least at a fairly high level of abstraction.

First, a focus on a pacing threat that is both a maritime power and a nuclear power eliminates entirely the salience of large-scale forcible entry operations followed by sustained operations ashore. Such operations are problematic even in the case of the lesser rogue regime threats, as both of those identified in the NDS are also either nuclear or near-nuclear powers. As I noted in my Planning Guidance last year, this does not mean that forcible entry is no longer a capability the Nation might require at some level—merely that the requirement will be, for the foreseeable future, limited in scale, and focused specifically on the need to provide assured access for elements of the Naval or Joint force rather than as a precursor to sustained Marine Corps operations ashore.

Second, even if there were a strong and credible requirement for large-scale forcible entry operations, such operations could not be carried out in the face of an adversary that has integrated the technologies and disciplines of the mature precision strike regime. As I noted in my Planning Guidance, the days of massed naval armadas nine miles offshore from some contested feature are long over. It has been traditional in the Marine Corps to note that “naysayers” have taken this position since the failure of the Gallipoli campaign in 1915, and to point to the U.S. Naval Services’ success in the interwar period in developing techniques of amphibious warfare that would prove the naysayers wrong. It is essential to note that the true lesson of this story is that the innovators of the 1930s created a complex of then-revolutionary ideas and technologies to solve the then-salient problem of the strongly opposed amphibious assault. The force we have today, with the notable but operationally insufficient exception of rotary-wing vertical envelopment, is an incrementally advanced, higher-tech version of that same 1930s solution. We now must recognize that time has flowed on. Our problems today, in terms of threat, geography, and technology (among other considerations), are not those of the 1930s. With respect to the effects of land-based precision fires, especially those launched from the homeland of a nuclear-armed great power, the naysayers of the 1930s are now simply the realists of the 2020s. Our job is to come up with doctrine and technology appropriate for the challenges of today (and tomorrow).

Finally, given the geopolitical realities of today and the nature of China’s society and strategic culture, it is highly likely that even if we did have an answer for the challenges of amphibious power projection in a mature precision strike regime, this capability would not be sufficient to deter or prevent our pacing threat from accomplishing its objectives in regions we judge important to our national security. The threat today accepts the (present) reality of U.S. conventional force superiority, and he has an answer for it in the form of the complex of aggressive behaviors “short of war” that we have come to characterize as gray zone operations. Credible “lethality” is only a part of the answer to this challenge—the ability to compete directly, daily, and globally, by means acceptable to the American people and the rule of law, is the missing piece.

2. Preparing for the Future Operating Environment

Preparing the Marine Corps for service in a volatile and uncertain future requires a degree of prediction about the drivers of conflict, threats, and the contexts of the environment in which Marines will serve. Predictions are inherently risky and imprecise. We fully acknowledge the inexact nature of future projection. Nonetheless, some prediction is required to plan. The view of the future is grounded in a wide survey of the thinking across the Joint Force, the intelligence community, our allies, academia, and nongovernmental business and technology sectors. The prevalence of certain key trends is undeniable, though the exact timing and impact of specific trends within the broader sweep of time may be debatable. Barring a “Black Swan” event, we are confident that our vision accommodates key features and the general context of the future operating environment.

The world is changing rapidly. Failing to adapt to those changes is to condemn our forces to unacceptable risk, as our opponents innovate and new threats surface to challenge the nation. Across a wide range of megatrends, forecasts, and developing technologies, our wargaming and analysis consistently pointed to several key trends. These, in turn, drove conditions in the future operational environment that we can view

as either threats or opportunities.

Among the large set of trends that will drive future conflict, the convergence of two broad themes will likely shape the specific problems against which the U.S. will employ military forces. These two factors are regional competition and instability.

a. Regional Competition

Regional competition describes a condition brought about by a number of trends already visible today. Those trends include, but are not limited to, competition as a result of shifting strategic relationships and changes in the relative power of certain countries. In particular, the economic and military rise of China and India, increasing rivalries in the Middle East, and Russia's struggle to remain a world power will fuel competition in an increasingly connected global landscape. In some cases, regional powers will attain global reach by acquiring nuclear or advanced technological armaments, a fact that will challenge U.S. freedom of action and perhaps diminish its influence with less committed partners. The very foundations of current U.S. military strength may be difficult to bring to bear in scenarios where a regional adversary possesses a local advantage that precludes the unilateral application of airpower, intelligence, surveillance, and reconnaissance (ISR), or space-based communications/guidance. In certain key capability areas, competitors may effectively close technological gaps with the U.S. military or use asymmetric strategies to blunt our strengths. This military advance will likely coincide with economic growth that fuels global expansion into areas of traditional U.S. influence and gives rise to sharp, multi-faceted competition short of open military conflict. Regional powers are likely to attempt to challenge the U.S. role in areas they increasingly view as their own. The U.S. response to these revisionist bids will, in many cases, be the employment of military forces to define the problem, achieve ends, and demonstrate resolve without unnecessarily escalating them into open conflict. As previously mentioned in Lesson 1, this is known as the competition continuum, which includes three categories: cooperation, adversarial competition below armed conflict, and armed conflict/war, in order to distinguish relationships between actors and clarify options for the use of military force within each context.

b. Instability

Meanwhile, more of the globe will find the conditions of daily life increasingly unstable. Explosive population growth in areas of traditionally poor governance and limited resources will fuel instability across entire regions. This will, in turn, increase the emerging patterns of migration, destabilizing areas perhaps quite removed from the initial crisis source. Much of the migration will be to urban areas and ever-growing megacities. These key hubs will become dense, disordered 'knots' of competing power structures. They will comprise both licit and illicit activity, exert outsized influence on international affairs, and involve overlapping and competing interests for the U.S. and an array of global and regional actors.

The war of ideas will not be the exclusive domain of nation states. Individuals and groups will rise to prominence to challenge traditional power structures and norms. Both multinational corporations and powerful transportation hubs that straddle air, land, and water routes are likely to seek larger influence as issues bring national interest into friction with their own. As resource disparities increase and personal outlooks for success dim, an increased population of disaffected or marginalized youth, ever more connected by the internet and social media, will be increasingly susceptible to virtual and non-state groups promoting radical or destabilizing ideologies.

This environment will challenge U.S. policy during the best of times, but with the addition of a regional crisis, a natural disaster, a pandemic, or internal conflict, the problem multiplies dramatically. Such circumstances challenge the U.S. government's ability to respond and make the role of the U.S. military simultaneously more urgent and more difficult.

The future operating environment can be characterized by the following:

- Erosion of U.S. military advantage
 - Peers/near peers close technological gaps and find asymmetric offsets; we often fail to comprehend these asymmetries.
 - The military instrument alone is insufficient to solve the complex problems for which it will be used.
 - Signature management severely limits U.S. freedom of action and maneuver.
- Multi-polar arenas featuring sharp competition across DIME
 - Other instruments of power's "contactless action" are employed by adversaries to achieve strategic objectives.
 - Rising powers/non-state actors take advantage of the way in which the U.S. conceptualizes war; adversaries operate within the category of competition below armed conflict.
- Dramatic expansion of global connectiveness and interactions
 - Trend toward transregional and multi-domain actions increases and accelerates.
- A "Tangle" of capabilities spread across agencies/levels limiting responsiveness and efficacy
 - Finding an appropriate and durable solution challenges traditional U.S. "means" and modes of organization.
- Attractiveness of Irregular and Information Warfare as a tool for adversaries
 - At a relatively low cost of entry, the "Gray Zone" offers our adversaries a means to mitigate traditional U.S. strengths, precisely because it operates below the norms of international response.

- The gray zone is a form of competition accompanied by ambiguity concerning the actors involved, the nature of the conflict, and the relevant policy and legal frameworks.
- Increased complexity and uncertainty in the operating environment
 - Speed of change and action only increases the imperative to understand complex environments and inform decisions.

3. Conditions-Based Model

The international system is a vastly complex and densely populated network comprised of actors with interests and relationships that are overlapping to various degrees and, at times, conflicting. To understand the dynamics within this intricate system, one must necessarily simplify or generalize aspects of it.

The conditions-based model attempts to reflect dynamics that already exist in the operating environment while providing a framework for thinking about and organizing relationships in the international system. Rather than being predictive, this model is a guide to understanding interactions between actors of strategic importance. Additionally, it assumes rational actors, defined as states having situational awareness of their external environment and behaving logically to achieve their own goals.

In the conditions-based model, the term *condition* describes the way in which two strategic actors are associated in the international system. The three conditions used to categorize relationships are cooperation, competition below armed conflict, and armed conflict. (These conditions were introduced in Lesson 1.) The model pertains to the state or nonstate actor as a whole and concerns all its instruments of strategic power: diplomacy, information, military, and economics (DIME). The three possible conditions result from the interaction of interests, the importance of those interests, and the capabilities available to advance them. Since rational actors behave according to their interests, the activities they employ are indicative of the condition at hand.

While the model does not imply that the three conditions follow a linear progression, there is an implied hierarchy of coercive measures employed. The lowest level condition is cooperation since actors primarily use cooperative activities to facilitate mutually beneficial relationships. Coercion is more central to the condition of competition below armed conflict, while armed conflict involves the highest intensity of coercive force. Since a rational actor will not engage in activities that reflect a higher level of coercive intensity than their interests dictate, the highest-level activity is indicative of the current condition between two actors.

Thinking about strategic relationships in terms of these three conditions provides several advantages. Compared to a limited peace/war model, the conditions-based model is more descriptive in its portrayal of reality through its accounting of activities below the threshold of armed conflict. It addresses perception biases, identifies conditions resulting from interests, and outlines possibilities for influencing conditions. The following sections describe the three conditions in more depth, provide examples reflecting each condition, and briefly illustrate the role of the joint force in the context of this model.

Note: This model, also known as the “competition continuum,” will be discussed further in the 8671 *Warfighting* course.

a. Cooperation

The peace/war paradigm lends a passive connotation to actors not in conflict, even though various instruments of national power are required to actively maintain and strengthen peace. Mutually beneficial relationships between actors with similar or compatible high-priority interests are the basis for the condition of cooperation. In the global context, cooperation occurs in a variety of forms and across a range of issues, including, for example, security, nuclear nonproliferation, environmental issues, and economics. Actors may cooperate over the long term, or they can cooperate on a specific issue in an isolated instance. Activities within a condition use various instruments of power. Cooperative activities across DIME instruments could include friendly diplomatic actions, training exercises to increase interoperability, security cooperation, and economic partnerships.

One example of bilateral cooperation is the U.S.-Japan Security Treaty. According to this agreement, Japan acts as a host nation to U.S. forces, and the United States is committed to defending Japan and is essentially responsible for Japanese security. Since 1960, the United States and Japan have perpetuated a symbiotic security relationship, and the longevity of this alliance is evidence that the two countries’ interests are more compatible than incompatible. In the context of the Cold War, the alliance allowed Japan to concentrate its efforts on rebuilding its economy while the U.S. was able to maintain a forward presence in East Asia and extend its nuclear umbrella. This forward strategy allowed the U.S. to observe Soviet maritime movement in the region. Today, both countries are invested in maintaining the status quo power balance in the Far East. Over time, the U.S. and Japan have negotiated the terms of the alliance and adjusted them to meet the changing needs of both actors. Security cooperation has been enhanced by an increase in military- to-military engagement, benefiting force and intelligence interoperability.

Cooperation is strategically important for the U.S. It underpins the current international order, enhances collective security, helps to ensure access to global commons, enables burden-sharing, and deters conflict. Military power supports and enables cooperation in many ways. Joint force participation in military engagement builds trust and enables information-sharing with U.S. partners. Joint actions such as nation

assistance and foreign humanitarian assistance bolster friendly relations and cooperation efforts. Show of force and enforcement of sanction missions augment deterrence and assure partners of U.S. resolve. Assurance is also vital for enabling nations to maintain military forces at levels unlikely to trigger arms races.

b. Competition Below Armed Conflict

The condition of competition below armed conflict exists when two actors in the international system have incompatible high-priority interests and one or both actors engage in or intend to engage in behavior that will be detrimental to the other's interests. The incompatible interest is either too low a priority or too difficult to attain given actor capabilities to rise to the level of open armed conflict. To be an act of competition, the behavior must negatively affect another actor's vital interests or suggest that future activities are likely to do so.

Regarding the instruments of power, diplomatic acts of competition could include espionage and sabotage. Information operations range from deception and disinformation techniques to propaganda. The military aspect of power can be employed through proxy warfare, guerrilla tactics, covert operations, or a mix of covert and overt operations. Economic activities in competition can take the form of sanctions, trade barriers, or tariffs. Competitive behavior is often asymmetric and can include criminal action employed for political gain, terrorism, and annexation of foreign territory. Competitive behavior is normally covert, ambiguous, gradual, indirect, or some mixture thereof.

Conditions are perspective dependent. For instance, the Ukrainian government perceived Russia's annexation of Crimea and the ensuing disorder in 2014 as armed conflict, but from the point of view of the United States, it reflected a condition of competition below armed conflict. Russia's behavior was consistent with the Kremlin's interest in geostrategic expansion to former Soviet territories. By preparing the "battlespace," President Vladimir Putin was able to create an opportunity to accomplish his goals without engaging the West in armed conflict. Preparations included a robust information operations offensive, consisting of a heavy barrage of propaganda targeting Russian-speaking viewers of state-run media in the near abroad. As the expansion unfolded, Russian tactics included espionage and both covert and overt military action. Even though Putin engaged the military instrument of power, he did not consider the behavior as constituting war, and he neither declared war nor stated an intention to seize Crimea. Repeated denials of Russian involvement from the Kremlin also contributed to widespread confusion about the actors involved in the crisis. The international community did not take military action in the conflict, most likely because analysis revealed the cost of intervention would outweigh any resultant benefits. As the risk of a North Atlantic Treaty Organization (NATO) military backlash subsided, Russian forces gradually transitioned to more overt uses of force.

U.S. activities in the condition of competition below armed conflict can aim at either directly accomplishing U.S. goals or countering the advancement of adversaries. U.S. military activities are a critical component of achieving and maintaining national security interests within the condition of competition. Joint actions to counter rival actors include security force assistance, building partner capacity to improve collective deterrence, show of force, counterterrorism, and foreign internal defense.

c. Armed Conflict

When one or both actors have extremely incompatible high-priority interests and sufficient capabilities to pursue these interests, they are likely to enter into a condition of armed conflict. Armed conflict is not ubiquitous, and the intensity ranges from limited warfare to traditional great power warfare and even to total war with nuclear weapons. An actor's use of force can target civilian or military citizens, infrastructure, or resources, and may result in adversary retaliation. Whether the activity triggers a military response depends on a variety of factors, including the value of the object in view, the scale of the attack, the actor's available capabilities, and the desired condition from the point of view of the target actor.

Activities reflecting the condition of armed conflict involve coercive use of DIME instruments of power. One role of diplomacy in this condition is to communicate the conditions of war termination directly or through the cessation of diplomatic interaction. Information operations can include cyber-attacks to impede or destroy the opponent's capabilities. Military action in the condition of armed conflict can aim to either contain, defeat, or destroy an enemy. Economic activities reflecting the condition of armed conflict can include embargo, sanctions more severe than those used in competition, and the use of naval, air, and/or ground forces to cut the adversary off from resources.

The Japanese attack on Pearl Harbor and the ensuing conflict between Japan and the U.S. is an example of two great powers crossing the threshold of armed conflict and engaging in traditional warfare. Given the magnitude of U.S. power at the time, many have contemplated and studied Japan's reasons for what appeared to be "national suicide." While it is beyond the scope of this reading to dissect the substantial literature surrounding causes of war, this model asserts that armed conflict occurs when high-priority interests are not reconcilable through measures short of coercive force. Analyses of the attack on Pearl Harbor present varying arguments about the direct causes of war, but it is evident that Japan and the United States had directly incompatible interests at the time: Japan sought expansion into Southeast Asia, and U.S. interests prioritized the prevention of this expansion. Although neither government desired war with the other, a series of events resulted in misperception and miscalculation and led the Japanese government to conclude that it had no acceptable alternative.

Although the destructive nature of war makes it an undesirable option, strategic actors may view armed conflict as the best available means to achieve their political ends. The high level of coercion implicit in armed conflict aims to affect another actor's cost-benefit analysis so that the other believes the costs to his own entity will outweigh the benefits of pursuing whatever interest is in question. As Clausewitz stated, "War is thus an act of force to compel our enemy to do our will." The military is the instrument of power most capable of incurring costs on the

adversary. The joint force must be prepared to prevail in open armed conflict. The military instrument of power has utility both for offensive coercive purposes and for defending against the threat of external coercion. Thus, the primary purpose of the U.S. military is to fight and win the Nation's wars.

4. Implications

The uncertainties of the future and the realities of the present require a paradigm shift in the way the joint force views the operating environment. Even as the joint force must be prepared to prevail in war, it has significant utility for conditions outside of armed conflict. The conditions-based model is a comprehensive approach to understanding strategic relationships in an increasingly complex world. Categorizing relationships in terms of cooperation, competition below armed conflict, and armed conflict equips joint leaders with an improved lexicon for providing the best military advice and conveying intent. Furthermore, the model has utility beyond the joint force, offering a basis for all instruments of national power to achieve policy aims with a consistent view of U.S. strategic relationships.

With the widening complex operational requirements, the U.S. and its partners and allies may have to rethink existing concepts of operations, doctrinal command-and-control methodologies, organizational force structures, training programs, and ultimately, military-technological acquisition priorities. Indeed, major changes in the direction and character of conflicts will have implications on defense planning, resource allocation, training, organization, and the use of force—propelling the need for a sustained conceptual, organizational, and technological innovation intended to enhance the military's ability to prepare for, fight, and win new types of wars. In this context, U.S. strategic partners and allies must enhance their ability to change military posture rapidly based on the changes in geostrategic environment, while having the flexibility and robustness to employ novel strategies, tactics, and technologies in different ways and scenarios. As new strategic realities create new powers, new types of future conflicts will emerge. Select military technologies and capabilities will diffuse to other major and many minor military powers, reshaping the paths and patterns of regional military modernization.

5. Russian Military Power

a. Introduction

Since returning to power in 2012, Russian President Putin has sought to reassert Russia as a great power on the global stage and to restructure an international order that the Kremlin believes is tilted too heavily in favor of the United States at Russia's expense. Moscow seeks to promote a multi-polar world predicated on the principles of respect for state sovereignty and non-interference in other states' internal affairs, the primacy of the United Nations, and a careful balance of power preventing one state or group of states from dominating the international order. To support these great power ambitions, Moscow has sought to build a robust military able to project power, add credibility to Russian diplomacy, and ensure that Russian interests can no longer be summarily dismissed without consequence.

Russia's assertive promotion of its national interests, punctuated by its military actions in Ukraine and Syria, demonstrates a more confident and somewhat less risk averse Kremlin, but it also has revived international concerns about the re-emergence of a more militaristic Russia. Russian military forces played a key role in the seizure of Crimea and fomenting an artificial separatist revolt in eastern Ukraine, blunting Kyiv's aspirations to join NATO, at least for the foreseeable future. Additionally, Russia's military intervention in Syria changed the entire dynamic of the conflict by ensuring that no resolution to the conflict could be possible without Moscow's agreement. Nevertheless, these actions also belie a deeply entrenched sense of insecurity regarding a United States that Moscow believes is intent on undermining Russia at home and abroad.

Moscow undoubtedly views the U.S. and its NATO partners as the principal threat to Russian security, its geo-political ambitions, and most importantly, the Kremlin's continued hold on power. This perception of vulnerability vis-à-vis the U.S. is most clearly evident in the latest Russian National Security Strategy published in December 2015. The document identifies the U.S. and its NATO allies as Russia's main threat and accuses the West of pursuing a deliberate policy of containment against Russia to sustain its domination of the post-Cold War international order and deprive Moscow of its rightful place on the world stage. It explicitly states, "the Russian Federation's implementation of an independent foreign and domestic policy is giving rise to opposition from the U.S. and its allies, who are seeking to retain their dominance in world affairs." The security strategy also cites the buildup of NATO military capabilities closer to the Russian border, the deployment of U.S. missile defense capabilities in Europe, and the ongoing U.S. pursuit of strategic non-nuclear precision weapon systems as a serious threat to Russian security.

Russia also has a deep and abiding distrust of U.S. efforts to promote democracy around the world and what it perceives as a U.S. campaign to impose a single set of global values. Moscow worries that U.S. attempts to dictate a set of acceptable international norms threaten the foundations of Kremlin power by giving license for foreign meddling in Russia's internal affairs. The December 2015 National Security Strategy warns of the importance of preserving traditional Russian spiritual and cultural values against foreign Western ideas and influences aimed at undermining Russia from within. The Kremlin is convinced the U.S. is laying the groundwork for regime change in Russia, a conviction further reinforced by the events in Ukraine. Moscow views the United States as the critical driver behind the crisis in Ukraine and the Arab Spring and believes that the overthrow of former Ukrainian President Yanukovich was consistent with a long-established pattern of U.S.-orchestrated regime change efforts, including the Kosovo campaign, Iraq, Libya, and the 2003–05 "color revolutions" in Georgia, Ukraine, and Kyrgyzstan.

Russian threat perceptions are not limited to the U.S. Moscow views the danger posed by Islamic militants and terrorists with grave concern. The Kremlin is particularly sensitive to the growth and spread of these ideologies and their potential to further radicalize Russian Muslims in the turbulent North Caucasus and other Muslim areas of central Russia. Russian military operations in Syria were also intended to eliminate jihadist elements operating there that originated in the territory of the former Soviet Union, to prevent them from returning home and posing a threat to Russia. At the same time, Moscow remains anxious about the deteriorating situation in Afghanistan and the potential for Afghan-based Islamic extremists to spill over into the Central Asian states of the former Soviet Union and ultimately into Russia.

Russian threat perceptions with regard to China are more divided and nuanced. Russian officials regularly praise the cooperative nature of the bilateral relationship, and Putin himself has declared that the current Russian Chinese relationship is the best it has been in decades. In fact, the Russian National Security Strategy lists developing a strategic partnership with China as one of Russia's most important goals. Moscow and Beijing share a common interest in weakening U.S. global influence and are actively cooperating in that regard. Military cooperation between the two countries is slowly expanding, as are economic ties. Nevertheless, some Russians are keenly aware of the growing power disparity between Russia and an ascendant China and worry that Moscow is at risk of becoming Beijing's junior partner. Others continue to harbor suspicions that China, over the longer term, will once again become a military threat to Russia.

President Putin regards the war in Ukraine as an existential struggle against the West that will determine Russia's place in the world, Putin's hold on power, and his historical legacy. Putin remains confident in Russia's ultimate victory in Ukraine and is prepared to continue the use military force. Putin also remains steadfast in his demand that Ukraine be permanently prohibited from joining the North Atlantic Treaty Organization (NATO) while insisting Kyiv withdraw all its military forces from Donetsk, Luhansk, Zaporizhzhia, and Kherson Oblasts. Despite Western provision of lethal aid to Ukraine, Russia almost certainly seeks to avoid direct conflict with NATO because it assesses it cannot win a conventional military confrontation with the alliance. However, Moscow remains fully capable of employing asymmetric capabilities against the U.S. and its allies, including cyber and information campaigns, and ultimately possesses an existential threat capability with its strategic nuclear forces that can range the U.S. Homeland. The war in Ukraine is part of Putin's broader objectives to recoup the prestige and global influence that he perceives Russia lost when the Soviet Union collapsed in 1991 and to exercise control over former Soviet states. Russia aims to wield strong influence over these states' foreign, domestic, and economic policies, while limiting inroads by rival foreign powers.

Putin almost certainly is committed to victory in Ukraine, and his objectives remain mostly unchanged since the beginning of the war: Ukrainian neutrality and a further partition of the Ukrainian state. In the absence of a negotiated settlement or, alternatively, robust Western aid, the battlefield outlook probably will continue to slowly trend in Russia's favor through 2025, though Russian battlefield gains are slowing and continue to come at the expense of high personnel and equipment losses. Since the war's start, Russia has lost at least 10,000 ground combat vehicles, including more than 3,000 tanks, as well as nearly 250 aircraft and helicopters, and more than 10 naval vessels. Russia has experienced more than 700,000 personnel casualties during the war. Russian forces maintain small, incremental progress toward capturing portions of Donetsk and Luhansk Oblasts and expelling Ukrainian forces from Kursk Oblast. To this end, Russia introduced more than 10,000 North Korean soldiers along the Kursk salient to support Russian offensive operations. Simultaneously, Russia is targeting critical Ukrainian infrastructure with a combination of one-way attack UAVs and long-range missiles to degrade its national will to fight. While affirming an openness to a negotiated settlement, Russia has remained largely steadfast in its demands for post-conflict Ukrainian neutrality, authority over the size of Ukraine's armed forces, and a further partitioning of the Ukrainian state. Russia is likely to continue its strategy of attrition, focused on degrading Ukraine's ability and will to resist through 2025, and allowing Russia to impose its preferred terms in any future negotiated settlement. Although Putin and his military leaders probably would prefer more rapid battlefield gains, Moscow seems comfortable with the current cost of its slow advances, calculating that they can steadily drain Ukraine's resources and will, and outlast the West's support for Ukraine. The war against Ukraine has proven costly to the Russian military, especially to Russia's ground forces, which probably have suffered more than 170,000 fatalities since the war began.

Since 1991, the Russian perception of the nature of modern conflict has evolved. Russia views wars as often undeclared, fought for relatively limited political objectives, and occurring across all domains, including outer space and the information space. Russian leaders have noted the tendency for crises to arise quickly and develop impetuously, and to potentially escalate from local wars into global ones. In addition, Moscow judges that modern conflicts are characterized by a destructive and rapid "initial period of war"—a subject on which Russian military leaders and theorists have written extensively since the 1920s—which is becoming more decisive than ever before. In modern cyber-enabled information and battlefield spaces, this destructive non-kinetic initial period can be reduced to milliseconds, and kinetically to hours.

Moscow fears that the speed, accuracy, and quantity of non-nuclear strategic precision-guided weapons can achieve strategic effects on par with nuclear weapons, one of the primary reasons that, since at least 1993 (and most recently codified in the 2014 Military Doctrine), Russia has reserved the right to a nuclear response to a non-nuclear attack that threatens the existence of the state. In addition to rejecting no-first-use, Moscow has discussed using nuclear weapons to de-escalate a conflict. While most military theorists and leaders believe great-power conflict is unlikely, they nevertheless express concern about the usability of the information space to achieve state goals. Russia has tied this decisive and shortened initial period to the idea that only more proactive or even preemptive action is required to counter it. Russian developments in precision-guided munitions indicate a desire for "deep strike" capability to preempt attacks from an adversary.

Russia's Military Doctrine, last updated in December 2014, contained several new elements not in the 2010 Doctrine, which reflect Moscow's military focus and threat perceptions. First codified in the doctrine was the concept of non-nuclear deterrence, an idea that has been evolving since the Soviet period. The doctrine also underscored perceived threats to Russia's domestic security and described the military's

requirement to inflict unacceptable damage on any adversary at any time. This requires the military to calculate or understand what level of damage would constitute unacceptable damage to an adversary. Mobilization readiness of the state was stressed, as were measures to unify state, societal, and individual efforts to protect Russia and increase the effectiveness of military-patriotic indoctrination of citizens and their preparation for military service.

The concepts of readiness, non-nuclear deterrence, and unacceptable damage are closely linked in Russian thinking; Russian military leaders judge that a highly ready non-nuclear force, able to inflict unacceptable damage on an aggressor—including against its economy—at any moment, is its own deterrent. For Moscow, the word translated as “deterrence” is more closely linked to a concept of active restraint, or literally to hold back something moving with force. In the West, deterrence is often seen as an established condition, whereas in Moscow it is an active, flexible process that continues throughout the conflict spectrum.

The Russians define *strategic deterrence* as a package of coordinated political, diplomatic, economic, ideological, moral, spiritual, informational, scientific, technological, military, and other actions taken by a country to demonstrate the decisiveness of the political leadership to tap all instruments of state power consecutively or simultaneously—to stabilize the military, political, and strategic environment, to anticipate aggression, and to deescalate military conflict. Some Russian theorists break deterrence down further into non-forceful and forceful means and even into deterrence by “type” (economic, military, nuclear, non-nuclear, etc.).

Closely linked to strategic deterrence is the concept of strategic stability. At its basic level, Russia’s concept of deterrence, appropriately applied in its view, assures strategic stability. Strategic stability is the sum of political, economic, military, and other measures (e.g., force) retained by states in a stable balance whereby neither side has the opportunity, interest, or intent to carry out military aggression.

Russia has observed modern conflicts and incorporated aspects of these observations into its deterrence and warfighting strategies. Russia seeks to shape the environment in peacetime to avoid or deter conflict and, if war does occur, will use its military force to establish a favorable outcome for Moscow. Moscow’s warfighting strategy includes the use of indirect action and asymmetric responses, including using technical and psychological operations to disrupt technical systems, influence public opinion, and “erode the opponent’s resolve.” The modernization of its nuclear and conventional forces to include precision-guided strike weapons provides it with a major military force to shape the outcome of war along the entire spectrum of modern conflict.



Figure 1: Dolgorukiy Class Nuclear Powered Ballistic Missile Submarine.

c. Core Russian Military Capabilities

(1) Nuclear Forces and Weapons

Russia is one of the oldest nuclear powers, first detonating a nuclear device in 1949. As heir to the former Soviet Union's nuclear arsenal, Russia has one of the world's two largest inventories of strategic weapons. While participating in strategic arms reduction treaties (START) with the United States, Russia is also committed to maintaining and modernizing its nuclear forces. Land-based intercontinental ballistic missiles are controlled by the Strategic Rocket Forces (SRF), and the sea-based and air strategic systems are managed by the Navy and Aerospace Force, respectively. Moscow plans to spend about \$28 billion by 2020 to upgrade the capacity of its strategic nuclear triad.

In the first leg of the triad, the SRF operates three older ICBM systems for more than half of their land-based nuclear delivery vehicles. As these aging missiles reach the end of their operational lives, they will be withdrawn from service by 2019–2021 and replaced with newer, modern road-mobile and silo-based ICBMs by 2020. The SRF's missile inventories will be equally split between road-mobile and silo-based ICBMs.



Figure 2: Russian SS-27 Mod 2 road-mobile ICBM.

The second element of the nuclear triad is a fleet of at least 10 nuclear-powered ballistic missile submarines (SSBN) under administrative control of the Naval High Command. The Russian strategic Navy is modernizing, mainly by building and deploying new SSBN platforms and sea-launched ballistic missiles (SLBM).

The third element of the nuclear triad is the Russian Aerospace Force's fleet of strategic bombers, which forms the core of the Long-Range Aviation (LRA) Command. Like other components of the triad, the LRA is modernizing, and Russia has announced that it will resume production of existing strategic bomber designs and complete development of a new generation bomber within a decade. The new bomber design is expected to have some stealth and short- or rough-runway capabilities and employ both conventional and nuclear armament.



Figure 3: Russian Tu-160 heavy bomber and Il-78M tanker.

The main function of strategic forces is effective, reliable deterrence. Scenarios for the use of strategic nuclear forces fall into three main categories: preemptive strike (first strike), counterstrike (launch on warning, prior to impact in-country), and retaliatory strike (response to impacts in-country). Because the retaliation option imposes the most difficult situation on the strategic forces—which must respond even after an enemy's strategic strike has impacted and disabled elements of the force—strategic forces, weapons, and battle management systems are designed and built to be hardened, stealthy, redundant, and reliable—and trained to function in a WMD-degraded environment.

Russia continues to retain a sizable nuclear stockpile even after several decades of arms reduction treaties. Russia has a large nuclear weapons infrastructure and a production base capable of producing large numbers of new nuclear weapons annually.

(2) Anti-Access/Area Denial (A2/AD)

Russia repeatedly cites in open-source literature the need to repel or defend against a Western aerospace attack. Russia would seek to deter any Western use of aerospace power against Russia using its conventional, non-strategic nuclear, and, in extreme circumstances, its strategic nuclear forces. Russian military theorists have examined the likelihood of a great power war arising out of a local conflict, similar to the events leading up to World War I and escalating to combat with U.S./NATO or another peer. Based on insight gleaned from studies of warfare since 1991, Russia would seek to limit the capability of an adversary to conduct aerospace strikes on its territory. Russian strategy for A2/AD would focus on a combination of various elements that military planners and theoreticians have identified as critical to the development of a comprehensive approach to A2/AD. These involve the incorporation of the following elements.

(a) Information Operations. Information operations are seen as a critical capability to achieve decisive results in the initial period of conflict, with a focus on control of the information spectrum in all dimensions of the modern battle space. Authors often cite the need in modern warfare to control information—sometimes termed “information blockade” or “information dominance”—and to seize the initiative early and deny an adversary use of the information space in a campaign so as to set the conditions needed for “decisive success.” Russia continues to emphasize electronic warfare and other information warfare capabilities, including denial and deception, as part of its approach to all aspects of warfare, including A2/AD.

(b) Strategic Air Operations. Russian military theorists continue to emphasize the key importance of strategic air operations in modern war. This concept originated in the 1920s, where Soviet planners viewed the initial period of war as the time that aviation would strike deep in enemy rear areas to destroy mobilization and concentration areas. At the same time, air forces would also prioritize the defense of the country against enemy air attack and conduct close air support of ground operations, achieving air supremacy in the first days of the war using all means. This concept was underscored in 1993, when Defense Minister Grachev indicated that “war will begin with an offensive aerospace operation on both sides.” Russian planners have indicated that in such a war there will be no front and no rear, with space emerging as an independent theater of military operations. Russian doctrine, down to the present day, continues to emphasize that strategic objectives can be achieved with mass aerospace strikes early in a conflict, with victory achieved without the seizure and occupation of territory by forces.

Russian planners have analyzed U.S. operations such as DESERT STORM, NOBLE ANVIL, and IRAQI FREEDOM for insight, observing military art at the strategic, operational, and tactical levels in campaigns that displayed U.S. aerospace capabilities and underscored the importance of developing comparable indigenous capabilities that can be employed defensively. This emphasis on strategic air operations is reflected in long-term procurement goals of platforms and weapons focused on space, aerospace defense, and precision-guided munitions.

(c) Integrated Air Defense System. Russian doctrine places a great deal of emphasis on aerospace defense as a key component in its overall A2/AD strategy. Though still in development, Russia’s 21st-century integrated air defense system will be designed to integrate future and existing systems around a central command structure that is designed to promote the interaction of all air defense forces and weapons. Capabilities optimized against cruise missiles are key to this defense component, not just those optimized to target aircraft.



Figure 4: Russian S-400 Surface-to-Air Missile System.

(d) Modern Precision Strike Capabilities: Air and Sea Systems in Combination with Older Technologies. Russia continues to develop a variety of sea and aerospace-based programs that offer a variety of offensive and defensive capabilities that could enable the implementation of its integrated A2/AD strategy. These include the continued production and deployment of coastal defense cruise missiles, air/surface/sub-

surface-launched anti-ship cruise missiles (ASCMs), submarine-launched torpedoes, and naval mines, along with Russian fighter, bomber, and surface-to-air missile capability. These are intended to provide Russia with the ability to limit access to its territory and extend its strategic depth by providing long range kinetic strike capability.

(3) Precision Strike

Russian doctrine on Precision Strike is essentially a 21st-century extension of the Russian doctrine of “deep battle,” initially codified during the 1920s and 1930s by Chief of the General Staff Marshal Mikhail Tukhachevskiy, and represents an attempt to incorporate new technology into traditional Russian strategic, operational, and tactical strategy. Deep battle was a strategic concept that focused on terminating, overwhelming, or dislocating enemy forces not only at the line of contact, but throughout the depth of the battlefield. Deep battle encompassed maneuvers by multiple Soviet Army frontsize formations simultaneously. It was not meant to deliver a victory in a single operation; instead, multiple operations, which might be conducted in parallel or successively, would induce a catastrophic failure in the enemy's defensive system. Initially, deep battle focused on improved ground and air forces and was influential in Soviet operations in World War II from 1943 onward. Chief of the General Staff Marshal Nikolay Ogarkov, writing in the 1970s to 1980s, updated the deep battle concept to develop a more aerospace-centric approach in an attempt to incorporate traditional Russian doctrine with precision technology.

Ogarkov theorized throughout his tenure as chief of the Soviet General Staff that conventional precision-guided munitions were part of a revolution in military affairs. In an influential 1983 *Krasnaya Zvezda* article, Ogarkov took notice of the impact of new types of precision weapons and micro-circuitry on the development of conventional capabilities. For Ogarkov, the development of new conventional forms of non-nuclear weapons would enable the sorts of multi-front operations that were envisioned in the original deep battle concept. On a theoretical basis, Ogarkov forecast that precision strike could exercise a direct and decisive outcome of a future war.

Despite enthusiasm by the Soviet General Staff, very little progress was made in the development of precision-guided munitions except at the theoretical level for the remainder of the Soviet period. In 1991, DESERT STORM provided the Soviet military with proof of concept regarding the use of precision-guided munitions. Former Soviet officials and Russian authorities argued that the DESERT STORM campaign demonstrated the capability of precision-guided airstrikes in the land attack role to paralyze the rear area and an adversary's economy. Targets could include vulnerable areas of the economy, command and control centers, and transportation centers. The introduction of precision-guided munitions changed the nature of modern war by reinforcing traditional concepts that emphasized decisive action during the initial stage of warfare and at the same time undermined the traditional Russian reliance on large ground force groupings to achieve tactical and strategic objectives.

Russia was unable to achieve real progress in the development of precision strike until the first decade of the 21st century, when it was able to create a viable state armaments program that allowed prioritization of certain key components of 21st-century warfare. Between 2010 and 2015, Russia's strategic forces, space and aerospace defense platforms, and precision-guided munitions²⁵⁴, such as ISKANDER, KALIBR, or KH-101 were defined as priorities, and system development, production, and testing occurred. The effectiveness of precision-guided munitions is being tested in a variety of settings, including Command Staff exercises KAVKAZ-2012, VOSTOK-2014, and KAVKAZ-2016, as well as operationally against targets in Syria beginning in 2015.



Figure 5: Kh-101/102 Air-Launched Cruise Missiles on a Tu-95MS Heavy Bomber.

(4) Space/Counterspace

The Russian General Staff postulates that modern warfare is increasingly reliant on information, particularly from space, because of the expansion of the geographic scope of military action and the information needs of high-precision weapons. Russia has a significant constellation of satellites in orbit. According to Colonel Sergey Marchuk, chief of the Main Test Space Center, Russia has more than 130 spacecraft, civilian and military, performing communications, navigation, geodetic survey support, meteorological, reconnaissance, and intelligence gathering missions.

Russia's space program is both formidable and in a state of rebuilding. Moscow seeks to maintain the health of its current constellations while deploying a next-generation architecture on par with Western space systems. Over the next several years, Russia will prioritize the modernization of its existing communications, navigation, and earth observation systems, while continuing to rebuild its electronic intelligence and early warning system constellations.

Russia's current systems provide an array of capabilities, including high-resolution imagery, terrestrial and space weather, communications, navigation, missile warning, electronic intelligence, and scientific observations. With a long-standing heritage in space, Russia gains a sense of national pride from its space program, which has included manned missions and led the world in space launches. Russia is currently ranked third in total number of satellites in orbit behind the U.S. and China.

(5) Cyber

Russia views the information sphere as a key domain for modern military conflict. Moscow perceives the information domain as strategically decisive and critically important to control its domestic populace and influence adversary states. Information warfare is a key means of achieving its ambitions of becoming a dominant player on the world stage.

Since at least 2010, the Russian military has prioritized the development of forces and means for what it terms "information confrontation," which is a holistic concept for ensuring information superiority, during peacetime and wartime. This concept includes control of the information content as well as the technical means for disseminating that content. Cyber operations are part of Russia's attempts to control the information environment.

The weaponization of information is a key aspect of Russia's strategy and is employed in times of peace, crisis, and war. In practice, information battles draw upon psychological warfare tactics and techniques from the Soviet Era for influencing Western societies. Moscow views information and psychological warfare as a measure to neutralize adversary actions in peace to prevent escalation to crisis or war.

Chief of the General Staff Gerasimov announced that "information operations troops" were involved for the first time in the Kavkaz-2016 strategic command staff exercise in September 2016, demonstrating Russian military commitment to controlling the information domain.

(a) Propaganda Helps Shape the Information Environment. Russian propaganda strives to influence, confuse, and demoralize its intended audience, often containing a mixture of true and false information to seem plausible and fit into the preexisting worldview of the intended audience. Russian propaganda targets a wide variety of audiences, including its own population, selected populations of other countries, domestic and foreign political elites, and the West writ large. The variety of techniques for disseminating Russian propaganda includes pro-Kremlin "news" websites and TV and radio channels such as Russia Today and Sputnik News, bots and trolls on social media, search engine optimization, and paid journalists in Western and other foreign media.

(b) Cyber-Enabled Psychological Operations. One of the newest tools in Russia's information toolkit is the use of cyber-enabled psychological operations that support its strategic and tactical information warfare objectives. These new techniques involve compromising networks for intelligence information that could be used to embarrass, discredit, or falsify information. Compromised material can then be leaked to the media at inopportune times.

- **Hacktivists.** Russian intelligence services have been known to co-opt or masquerade as other hacktivist groups. These groups appeal to Russia due to the difficulty of attribution and the level of anonymity provided. It is widely accepted that Russia, via patriotic hackers, conducted a cyber attack on Estonia in 2007. Under the guise of hacktivism, a group called "CyberCaliphate," seemingly ISIS associated, conducted a hack against French station TV5 Monde in January 2015. The CyberCaliphate group was later linked to Russian military hackers. The same group hijacked the Twitter feed of the U.S. Central Command.
- **CyberBerkut – A False Persona.** Russian hackers also use false personas. *CyberBerkut* is a front organization for Russian state-sponsored cyber activity, supporting Russia's military operations and strategic objectives in Ukraine. *CyberBerkut* employs a range of both technical and propaganda attacks, consistent with the Russian concept of "information confrontation." Since emerging in March 2014, *CyberBerkut* has been implicated in multiple incidents of cyber espionage and attack, including distributed denial of service attacks against NATO, Ukraine, and German government websites. More recently, it has focused on the online publication of hacked documents, ostensibly obtained from the Ukrainian government and political figures' computers. *CyberBerkut* uses information gained through these hacks to discredit the Ukrainian government. The intent is to demoralize, embarrass, and create distrust of elected officials.



Figure 6: CyberBerkut Arm Patch.

- **Trolls.** Russia employs a troll army of paid online commentators who manipulate or try to change the narrative of a given story in Russia's favor. Russia's Troll Army, also known as the Internet Research Agency, is a state-funded organization that blogs and tweets on behalf of the Kremlin. Trolls typically post pro-Kremlin content and facilitate heated discussions in the comments sections of news articles. Their goal is to counter negative media and "Western influence." While the goal of some trolls is to simply disrupt negative content, other trolls promote completely false content.
- **Bots.** Another way Russia manipulates the information space is through the use of bots. Bots are automated pushers of content on social media. These bots vary in sophistication and can continuously push content or imitate real life patterns. Bots can drown out unwanted content or push a specific message. Bots have the ability to overwhelm the information space and discourage readers from looking for real content.

(c) Information Defense. The Russian Federation Security Council's 2016 Information Security Doctrine mandates protecting Russian citizens from outside threats to the information sphere. The doctrine aims to secure Russian information freedom and protect information technologies from foreign influence, cyberattacks, intelligence collection, and terrorism. The doctrine emphasizes the need to develop a national system for government control of the Russian Internet, information warfare forces, and cyber weapons.

Since at least 1999, Russia has attempted to gain consensus on international governance of the Internet and international norms and rules guiding the behavior of states in the information space. A major component of the proposal pertains to a state's ability to govern its information space as a means of maintaining state sovereignty and preventing an arms race in cyberspace. Although state sovereignty traditionally refers to domestic enforcement law, Russia commonly uses this term to denounce other nations' meddling in their internal affairs. Russia also proposed a code of conduct for cyberspace with specific dictums regarding non-state cyber-actors, such as criminal hackers involved in cyber activities.

(d) Media Laws - A Hedge Against Instability. In the past decade, Russia has implemented numerous laws curbing domestic media in broadcast, print, and cyber media, taking an abrupt turn from the post-Soviet glasnost policies of media "openness" and its own constitutional guarantees of freedom of speech. The use of social media to organize opposition street protests in 2011 and 2012 prompted a reappraisal of official internet policy. Since then, the authorities have treated the Internet as a serious threat, pushing through laws increasing government controls over technology and content, giving the state powers to block content, ban websites, monitor online activity, and limit media ownership. The ultimate goal of this policy appears to be to create what some have called a "sovereign internet."

The Kremlin's strategy of reducing foreign influence on the media has not been confined to the internet. Numerous other pieces of legislation have been passed restricting the level of foreign ownership of the media, impeding the work of the foreign NGOs supporting independent media in Russia and forcing Russian media to account for any foreign funding they receive. A recent law has even banned foreign companies from conducting TV audience research in Russia.

(6) Indirect Action

Indirect action is a component of Russia's strategic deterrence policy developed by Moscow in recent years. Its primary aim is to achieve Russia's national objectives through a combination of military and non-military means while avoiding escalation into a full blown, direct, state-to-state conflict. Drawing on a combination of facets from Russia's whole-of-government or interdepartmental strategy and overt or covert military means, indirect action seeks to exploit weaknesses and fissures in target countries in order to fulfill Moscow's desired national goals.

In Ukraine, indirect action manifested itself in non-military measures first, with less visible efforts taken to exert pressure on Kiev, like restricting food imports to Russia, but then broadening to wider actions involving financial, economic, and information warfare. Later, this was followed by unconventional military action involving Russian Spetsnaz and other non-attributable military units in Crimea and eastern Ukraine. This phase involved the actual seizure of facilities and infrastructure by these covered units, along with the use of local agents, sympathizers, and irregular forces in the vicinity to cause unrest and subversion, all of which are distinct hallmarks or evolutions of Soviet-era Spetsnaz wartime operations.

(7) Electronic Warfare

Based on authoritative military academic writings, the Russian military views electronic warfare as an essential tool for gaining and maintaining information superiority over its adversaries. Russia's world-class electronic warfare forces support denial and deception operations and allow identification, interception, disruption, and, in combination with traditional fires, destruction of adversary command, control, communications, and intelligence capabilities.

In addition to technical disruption, effective use of electronic warfare can confuse adversary commanders and decision-making at any or all levels, demoralize opposing troops, and allow Russian forces to seize the operational initiative. Russia has fielded a wide range of ground-based electronic warfare systems to counter GPS, tactical communications, satellite communications, and radars. Further, military academics have suggested that electronic warfare fuse with cyber operations, allowing electronic warfare forces to corrupt and disable computers and networked systems as well as disrupt the use of the electromagnetic spectrum. Russia has aspirations to develop and field a full spectrum of electronic warfare capabilities to counter Western C4ISR and weapons guidance systems.



Figure 7: Russian jammer on display at Kubinka in 2016.

(8) Power Projection

Moscow continues to prioritize modernizing its military forces, viewing military power as critical to achieving key strategic objectives and global influence. Russian acquisition plans for its ground, air, naval, and missile forces are designed to enable the ability to conduct out of area operations during peacetime and to contest U.S./NATO military superiority in the event of a regional conflict. The rebuilt Russian military includes modernized, agile general-purpose forces, vital to limited out-of-area power projection. While the objectives of the Russian military do not suggest a return to the Cold War posture, Moscow intends to use its military to promote stability on its own terms and to assert its great power status.

Russia's State Armaments Program will continue to emphasize priority programs related to the development of a viable 21st-century military, prioritizing strategic forces, space, precision-guided munitions, and aerospace defense capabilities. Russia's strategic triad, along with the increasing capability of its conventional forces, remains a critical deterrent in preventing an attack. Russian long-range aviation remains a priority for Russian leadership as a key part of its strategic deterrent capability, while also providing an advanced conventional option to rapidly project power well beyond Russian borders. Russia is also modernizing its naval forces, which conduct operations globally in order to "show the flag" and contribute to Moscow's narrative of Russia's re-emergence as a global power. Russia is also focused on enhancing its C4ISR capabilities, which will enable improved targeting and timely responses to perceived threats.

(a) Long-Range Aviation (LRA). Russia periodically deploys assets of its LRA bomber force to conduct limited out-of-area operations as a power projection tool. LRA operations have included activity in the Pacific, the Arctic, and even as far south as Venezuela in 2008. The capabilities of LRA aircraft allow for missions as far as 5,000–10,000 kilometers away.

(b) Naval Forces. The Russian Navy will continue to conduct operations in parts of the world that are deemed important to national objectives. In recent times, these have included operations in the Mediterranean, the Arctic, and periodic deployments to the western hemisphere and the Indian Ocean. Russia's naval recapitalization program will focus on the development of modern general-purpose submarines and surface combatants to enable continued out-of-area operations.

(c) Expeditionary Operations. Along with more conventional power projection missions, Russia has displayed a new capability to field an expeditionary force capable of intervening in a foreign conflict. In Syria, Russia used a mix of maritime and air assets to forward deploy its forces, and Russia will almost certainly be able to logistically support its current level of operations in Syria via a mix of those means for the foreseeable future.

After politically supporting the Syrian regime throughout the Syrian civil war, Moscow began to deploy military forces to Syria in September 2015, likely both to shore up the regime and assert Russia's status as a military player and powerbroker in the Middle East. The majority of Russian air strikes and artillery operations have supported regime ground offensives and focused on opposition targets, with an increased focus against Islamic State forces at certain points in their campaign.

Russia has also sought to use the Syrian intervention as a showcase for its military modernization program and advanced conventional weapons systems, including employing systems from outside of Syrian territory to demonstrate its power projection capacity. Moscow has launched *Kalibr* land-attack cruise missiles from naval units in the Caspian Sea and the Mediterranean Sea, demonstrated new capabilities with air-launched cruise missiles from its Tu-160M1 BLACKJACK and Tu-95MS BEAR H heavy bombers, forward-staged long-range Tu-22M3 BACKFIRE bombers for strikes from Iranian territory, and deployed some of its most advanced air and air defense systems to Syria. These operations are meant to demonstrate strategic capabilities and message the West about the manner in which the Russian military could operate in a major conventional conflict, while also providing combat experience for the personnel and allowing the systems to be field tested.



Figure 8: A Russian naval task force centered on its only aircraft carrier, the *Admiral Kuznetsov*, conducted a 5-month deployment to the Mediterranean to support strike operations in Syria.

(9) Underground Facilities

Russia inherited a vast underground facilities (UGFs) program from the Soviet Union, primarily designed to ensure the survival of the leadership and military command and control in wartime. This program involved the construction of underground bunkers, tunnels, secret subway lines, and other facilities beneath Moscow, other major Russian cities, and the sites of major military commands. Although the majority of these hardened facilities are near-surface bunkers, many critical sites are built deep underground and, in some cases, are hundreds of meters deep.

Deep underground command posts both within and outside of Moscow are interconnected by a network of special deep subway lines that provide leadership a quick and secure means of evacuation. The leadership can move from their peacetime offices through concealed entryways to protective quarters beneath the city. A deep underground facility at the Kremlin and an enormous underground leadership bunker adjacent to Moscow State University are intended for the National Command Authority in wartime. They are estimated to be 200–300 meters deep and can accommodate an estimated 10,000 people.

The leadership can remain beneath Moscow or travel along the special subway lines that connect these urban facilities to their preferred deep underground command posts outside the city, and possibly to the VIP terminal at Vnukovo Airfield, 27 kilometers southwest of the Kremlin. Two of the most important underground complexes for the National Command Authority and General Staff are located some 60 kilometers south of the city.

The support infrastructure for the UGFs in and around Moscow is substantial. A highly redundant communications system, consisting of both on-site and remote elements, allows the leadership to send orders and receive reports. Highly effective life support systems may permit independent operations for many months following a nuclear attack.

Russian military officials suggest the UGF program has been retained. In October 2014, the chief of the General Staff's Main Operations Directorate, General-Lieutenant Andrey Kartapolov, told a Rossiyskaya Gazeta correspondent that the new National Defense Management Center in Moscow is safe from a nuclear strike. The National Defense Management Center became operational in December 2014 and is at the apex of the national command structure. General Kartapolov noted that protection against nuclear strike is always considered in building the most important facilities.

(10) Denial and Deception

The Russian military relies on extensive use of denial and deception (*maskirovka*) to obscure intentions and conceal military movement. The family of capabilities that composed traditional *maskirovka* includes camouflage, deception, denial, subversion, sabotage, espionage, propaganda, and psychological operations. Russian operational and tactical *maskirovka* is a form of operational combat support. It encompasses a set of interrelated organizational and technical measures and practical actions of staffs, troops, and facilities intended to deceive foreign intelligence. *Maskirovka* promotes surprise, maintenance of combat capability, and survivability. For example, *maskirovka* in rocket units and subunits is organized and carried out for the purpose of ensuring that the enemy experiences maximum difficulty in collecting intelligence data to reduce the effectiveness of strikes, but is also carried out to create the false appearance of a combined unit in support of deception at the operational level of war.

Moscow employed *maskirovka* at the beginning of the 2014 conflict in Ukraine, when media reported on the presence of “little green men” in Crimea who strongly resembled Russian soldiers, although they wore uniforms without insignia identifying their origins. President Putin insisted they were “self-defense groups” or “volunteers.” By the time Moscow admitted to the presence of Russian troops in Crimea, this deception had created enough confusion to forestall significant international intervention in the conflict, and the ground reality was irreversibly tipped in Russia's favor.



Figure 9: Moscow used troops without insignia – the “little green men” – to seize the Crimean Peninsula in early 2014, claiming these forces were local militia.

d. Outlook

The Russian military has built on the military doctrine, structure, and capabilities of the former Soviet Union, and although still dependent on many of the older Soviet platforms, the Russians have modernized their military strategy, doctrine, and tactics to include the use of asymmetric weapons like cyber and indirect action such as was observed in Ukraine.

One of Russia's biggest hurdles since the dissolution of the former Soviet Union has been its need to rely heavily on its nuclear forces to deter aggression, resulting in its stated willingness for first use of nuclear weapons. Russia has been building its conventional force capability along with modernizing its nuclear forces to create a more balanced military. Moscow has stressed development of conventional precision-strike weapons, a critical gap in its inventory, and recently has tested them in combat in Syria, providing it with an advanced non-nuclear capability to impact the battlefield.

In 2009, after almost two decades of deterioration and neglect of the Russian military, Moscow began developing a more modern military force capable of power projection outside Russia's borders. The New Look reforms instituted structural and organizational reforms, and the State Armaments Program emphasized development of modernized platforms and weapons systems. In 2013, readiness became an additional area of emphasis with institution of no-notice "snap" exercises and accompanying mobilization and deployments. Moscow's long-term goal is building a military prepared to conduct the range of conflicts from local war through regional conflict to a strategic conflict that could result in massive nuclear exchange.

Russia's conventional capabilities to deter, fight, or militarily compete with NATO are likely to be degraded for at least the next three years as the majority of all combat-capable Russian units are committed to the war in Ukraine. Russia faces tradeoffs between modernizing weapons and resupplying its forces in Ukraine, and Russia's defense industry is giving priority to refurbishing its equipment and existing platforms ahead of producing new or modernized weapons systems. The exception is submarines, where planned modernization and fielding have continued mostly without interruption. Russia is experiencing labor shortages in the economy, which, along with sanctions, are driving up costs for its defense industry. These rising defense production costs will probably hamper Russia's long-term modernization efforts, particularly for producing advanced equipment. The Russian Navy almost certainly will serve as Russia's primary tool of global power projection for the near future. The Russian Navy conducted deployments to Latin America and deployed a nuclear-powered submarine in the region in 2024 for the first time since the fall of the Soviet Union. The Navy also conducted a multi-fleet, global exercise in September—OKEAN-2024—to showcase its ability to defend its national security interests despite attrition in Ukraine. The Russian navy also fielded new, advanced ballistic missile submarines, cruise missile submarines, and attack submarines in its Pacific Fleet.

Russia seeks to enhance its force projection by pursuing basing or pre-approved port call agreements worldwide. Russia continues hosting bilateral and multilateral military exercises, particularly with Indo-Pacific countries, and probably will expand bilateral defense ties with additional Global South countries over the next year. Russia envisions the development of additional basing in the Mediterranean Sea, Red Sea, and Indian Ocean regions to enable inter-fleet transfers.

Russia's desire to be a leader in a multipolar world and recapture the "great power" status it had in Tsarist times and the latter days of the Soviet Union requires a force capable of deterring aggression, fighting the range of conflicts from local crises to nuclear war, projecting power, and employing force if necessary to intervene in conflicts across the globe.

For more information on Russia's current military capabilities and outlook, see *Russia's War in Ukraine: Military and Intelligence Aspects* provided in the Supplemental References to this lesson.

e. Russian Strategic Rocket Forces

The Strategic Rocket Forces (SRF) (Russian name: *Raketniye Voyska Strategicheskovo Naznacheniya [RVSN]*) is one of the most potent missile forces in the world. The SRF was established as a separate military service in December 1959 to operate the first nuclear-armed intercontinental-range land-based ballistic missile (SS-6), as the third element of Russia's growing strategic nuclear force deterrent triad.

The Russian SRF headquarters is in Moscow. The SRF's three missile armies—the 27th, 31st, and 33rd—have a total of 12 subordinate missile divisions. Eight of the divisions operate road-mobile ICBMs, with the other four armed with silo-based missiles. The Russian SRF have approximately 60,000 personnel.



Figure 10: Locations of Strategic Rocket Forces Missile Divisions.

In 2016, the SRF had deployed 299 operational missiles, with half that number equipped with multiple independently targetable reentry vehicles (MIRV) payloads. The SRF arsenal includes three older ICBM types—46 SS-18s and 30 SS-19s in silos, and 72 road-mobile SS-25s—and two newer ICBM types—60 silo-based and 18 road-mobile SS-27 Mod 1s, and 73 of the most modernized SS-27 Mod 2s.

The development of new ballistic missile systems is a high priority for Russia. The Russian military has outlined that the SRF should be completely re-armed with modern (post-Soviet) missile systems by 2022. Russia has stated that it will soon begin testing a developmental, heavy, liquid-propellant ICBM called the Sarmat to replace the aging SS-18. Russia's goal is to begin Sarmat deployment in the 2018–2020 timeframe.

Russia has announced a new missile called the Rubezh (Border) or RS-26, which is smaller than the SS-27 Mod 2 ICBM and will be deployed in 2017. According to the SRF commander, the RS-26 is envisioned as a mobile system and has been referred to by Russian Vice-Premier Rogozin as a “missile defense killer.” Russian industry officials also claim development of the Barguzin rail-mobile ICBM is continuing. A decision on full development, production, and deployment will occur in the coming months.

The currently deployed SS-18, which Russia plans to replace with the Sarmat, is a silo-based, 10-MIRV heavy ICBM first deployed in 1988; it needs to be replaced by 2018–2020, when the SS-18s' 27- to 30-year service lives expire. The SS-19 is a silo-based, six-MIRV ICBM that entered service in 1980, which the SRF will replace with silo-based SS-27 Mod 2 by 2019, as the SS-19s retire.

The SS-25 solid-propellant, single-warhead, road-mobile ICBM was first deployed in 1985 and will retire by 2019–2021, to be replaced by regiments of new production SS-27 Mod 2s, and possibly the two-stage, road-mobile RS-26 Rubezh.

In addition, Russian leadership claims a new class of hypersonic glide vehicle is being developed to allow Russian strategic missiles to penetrate missile defense systems. Hypersonic glide vehicles (HGVs) are maneuverable vehicles that travel at hypersonic (typically greater than Mach 5) speed and spend most of their flight at much lower altitudes than a typical ballistic missile. The combination of high speed, maneuverability, and relatively low altitude makes them challenging targets for missile defense systems.

Russia's overall number of strategic systems is constrained by the New Strategic Arms Reduction Treaty (START), which entered into force on 5 February 2011. This treaty limits the U.S. and Russia to no more than 1,550 deployed warheads each (including warheads on ICBMs and SLBMs and counting each heavy bomber as one warhead) 7 years after entry into force.

Russia retains about 1,200 nuclear warheads for ICBMs. Most of these missiles are maintained on alert, capable of being launched within minutes of receiving a launch order. Although the number of missiles in the Russian ICBM force will continue to decrease because of arms control agreements, aging missiles, and resource constraints, Russia intends to retain the largest ICBM force outside the U.S.

Despite Russia's modernization efforts, the size of the SRF may drop below 300 deployed ICBMs by the early 2020s, but most of those missiles will be equipped with multiple warheads. The composition of the force is changing significantly to meet the deployed strategic warhead total limit of 1,550. Notably, prior to 2010, no SRF road-mobile ICBMs carried MIRVs; by the early 2020s, all will do so.

f. Russian Ground Forces

The Russian ground forces are composed of the Ground Troops, Airborne Troops (VDV), Naval Infantry, Coastal Troops, Coastal Missile Artillery Troops, and National Guard. The Russian Federation armed forces are geographically distributed across four military districts: Western, Southern, Central, and Eastern. The Western Military District has three numbered combined-arms armies (CAAs), the Southern and Central Military Districts each have two, and the Eastern Military District has four.



Figure 11: Russian Armored Fighting Vehicles Parade in Red Square.

(1) The Ground Troops

The Ground Troops, or *Sukhoputniye Voyska*, are the land warfighting component of the Russian Ministry of Defense. The Ground Troops constitute the largest component of the Russian Federation armed forces. The Ground Troops are currently organized into approximately 40 active and reserve maneuver brigades and eight maneuver divisions. There are about 350,000 military personnel in the ground troops. According to Russia's Ministry of Defense, the roles of its Ground Troops include repelling enemy aggression and the protection of Russia's territorial integrity and Russian national interests. Its main peacetime missions include maintaining adequate combat readiness, participating in international peacekeeping operations, participating in disaster recovery efforts, and assisting in the maintenance of internal security, if needed. Examples of what Moscow designates as peacekeeping operations include ongoing efforts in breakaway enclaves in Georgia and Moldova.

In times of heightened tension, the Ground Troops will mobilize forces, operationally deploy to threatened areas, call up and train reservists, and prepare for defensive operations. Finally, in a time of war, Russia's Ground Troops are charged to suppress military conflicts if possible, repulse enemy aggression, conduct defensive and counter-offensive operations to defeat the aggressor, and defend critical infrastructure.

Organizationally, the Ground Troops are composed of main combat components—motorized rifle, tank, missile and artillery, and air defense units. Support elements for these units include reconnaissance, engineer, nuclear, biological and chemical defense, and signal troops.

- **Motorized Rifle Troops** units are the most abundant formations in Russia's Ground Troops. Essentially mounted infantry, these are highly mobile forces tasked with holding territory, repulsing enemy attacks, breaking through enemy defenses, capturing important areas, and defeating the enemy.
- **Tank Troops** are the main strike component of the Ground Troops. They support Motorized Rifle Troop missions with direct fires during meeting engagements.
- **Missile Troops** and Artillery are the main means of indirect fires for Russian combined arms operations. Missile Troops and Artillery forces are organized into missile, rocket-artillery, and combined artillery units. Missile units operate close/shortrange ballistic missiles. Rocket-artillery units operate multiple rocket launchers (MLRs), and combined artillery units operate composite towed or self-propelled artillery and MLRs.

- **Air Defense Troops** provide air defense for the Ground Troops. These units are equipped with anti-aircraft missiles, anti-aircraft artillery, anti-aircraft gun-and-missile systems, and portable anti-aircraft missile systems.
- **Reconnaissance Troops** perform a wide range of tasks in order to provide decision makers with information about enemy strength, disposition, terrain, and weather conditions.
- **Engineer Troops** perform a variety of specialized tasks, including the construction of fortifications, installation of obstacles (mine fields, etc.), the preparation of field deployment locations, the preparation and maintenance of deployment routes, the construction of bridges and ferry crossings, and water purification.
- **Nuclear Biological Chemical Defense Troops** are specialized forces tasked with mitigating the effects of nuclear, biological, or chemical contamination.
- **Signal Troops** are specialized forces designed for the deployment and maintenance of mobile redundant command, control, and communications systems.

(a) The New Look Reforms and the Ground Troops. The centerpiece of the 2008–2009 New Look reforms was the elimination of the divisional/regimental structure and its replacement by the brigade. The Russian Ground Troops currently have about 40 combined arms brigades. In the winter of 2013, one motorized rifle brigade and one tank brigade were reformed as divisions, and in the spring of 2016, it was announced that four new divisions would be formed in the Western and Southern Military Districts and one in the Central Military District.

The transition to the brigade structure was intended to optimize Russia's ground forces to fight in what the Russians call "local wars and armed conflicts," limited wars along Russia's periphery, which the Russian General Staff believes to be very likely under modern conditions. In November 2011, then-Chief of the General Staff Nikolai Makarov said: "The possibility of local armed conflicts virtually along the entire perimeter of the border has grown dramatically." The Russian ground forces fielded brigades of this type that had been field tested in Afghanistan (1979–1989) and had proved to be quite effective in combat.

Another development that had received great impetus in the Afghanistan war was the reinforced battalion, or battalion tactical group (BTG), a motorized rifle or tank battalion, strengthened by other assets, such as artillery, reconnaissance, and air defense resources. BTGs are similar to NATO battalion task forces and are ad-hoc organizations, individually created and optimized to fulfill a particular mission.

The new Russian divisions are much smaller than their Soviet predecessors. While a Soviet motorized rifle division numbered around 13,000 officers and soldiers, Russia's new motorized rifle divisions number around 9,000.

A proponent of the mixed division-brigade ground forces, then-acting chief of the ground forces, General Lieutenant Vladimir Popov, stated that Russia's combined arms brigades "in terms of structure are intended for fighting in local wars," but that they also "can be successfully employed in large-scale wars. They differ from divisions by lesser numbers of personnel and military equipment and are capable of executing missions with the very same high effectiveness as divisions, but in a smaller zone of responsibility." The reintroduction of some smaller divisions may be based more on their potential intimidation value than they are on their potential value in combat.

These positive developments have led some analysts to claim that Russia is developing entirely new military concepts. Modern Russian tactics show a strong continuity with past practices. Recent Russian activity in eastern Ukraine, for example, demonstrates a creative use of their traditional combined arms and reconnaissance-strike tactics combined with a more aggressive application of information warfare concepts that date back to the Soviet period. Russian ground forces troops have cooperated with non-traditional semi-military forces such as partisans and Cossacks for centuries. Regardless, the contemporary Russian ground forces pose a serious challenge to U.S. military planners, and they should be seen as neither a simple continuation of past Soviet practices, nor an entirely new force employing entirely new military concepts, but a highly nuanced and adaptive combination of both.

The main combat power of the Ground Troops is centered in tank and motorized rifle divisions and separate tank and motorized rifle brigades that are normally subordinate to combined arms armies. Although Russia's military strategy is officially defensive, the Russian Ground Troops, basic principle of land warfare is violent, sustained, and deep offensive action, just as it was during the Soviet era. Mechanized and armored formations supported by aviation and artillery are to seize the initiative at the outset of hostilities, penetrate the enemy's defenses, and drive deeply and decisively into the enemy's rear area.

(b) Combined Arms Armies. The combined arms army is an operational and administrative organization that forms the basis of the Russian field army. A typical combined arms army includes two to four combined arms brigades, usually motorized rifle brigades and, in a few cases, a tank brigade, plus artillery, missile, air defense, engineer, chemical defense, communications, intelligence and reconnaissance, and rear support units. By altering the mix of motorized rifle and tank formations and artillery and missile support, the army can operate in either offensive or defensive roles in different geographical areas and under various operational constraints.

(c) The Tank Army. The Russian armed forces currently only have one tank army, the First Guards Tank Army (1st GTA). It, like the combined arms army, is both an operational and administrative unit. Currently, the 1st GTA includes a tank division, a motorized rifle division, and a tank brigade, plus artillery, missile, air defense, engineer, chemical defense, communications, intelligence and reconnaissance, and rear support units. The traditional role of a tank army is to exploit penetrations deep into the enemy's rear areas.

(d) The Separate Combined-Arms Brigade. The primary combat formation of the Ground Troops is the separate combined-arms brigade, either motorized rifle (MR) or tank. There are three basic tables of organization and equipment (TO&Es) for separate MR brigades and one for separate tank brigades. All Russian combined-arms brigades, however, regardless of specific primary combat vehicle, are organized in essentially the same manner.

Tank and MR brigades differ in organization in that the MR brigade has three MR battalions, one tank battalion, and an antitank (AT) battalion; the independent tank brigade reverses this basic structure with three tank battalions, one motorized tank battalion, and no AT battalion. The three MR TO&Es differ from one another in their primary combat vehicle, either wheeled armored personnel carrier (APC) or tracked infantry fighting vehicle (IFV). The organizational structure of a typical tracked APC-equipped independent MR brigade is shown in the accompanying illustration. In addition, the primary equipment of a separate motorized rifle brigade is shown in the table below.

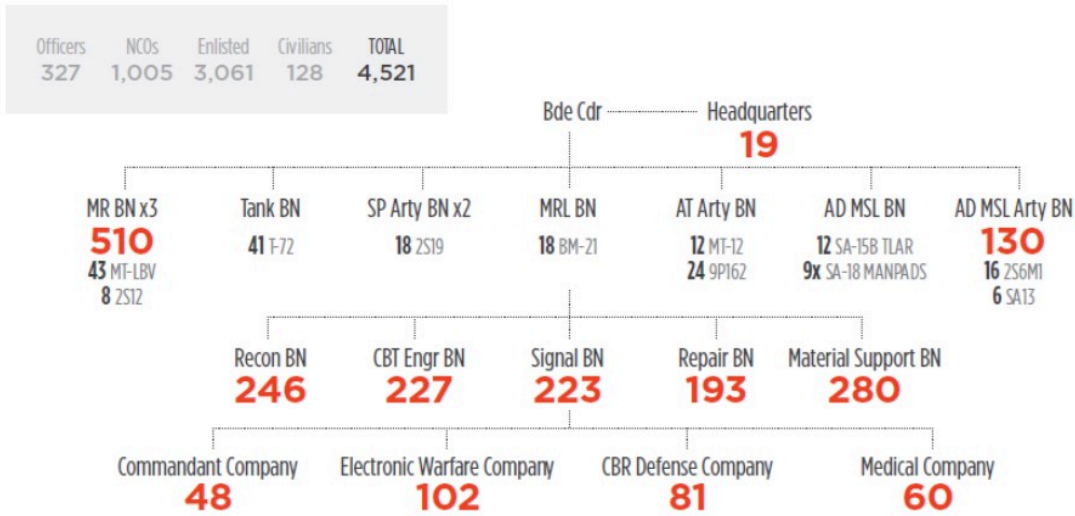


Figure 12: New Look Motorized Rifle Brigade Table of Organization and Equipment: Primary and Supporting Subunits.

Nomenclature	Quantity
Personnel	4521
T-72B3 Main Battle Tank	41
BMP-3 Infantry Fighting Vehicle or	129
BMP-2 Infantry Fighting Vehicle or	129
MT-LBV Tracked Armored Personnel Carrier	129
BTR-82A Wheeled Armored Personnel Carrier	129
2S19 152-MM SP Howitzer	18
BM-21 Multiple Rocket Launcher	18

Figure 13: Russian Separate Motorized Rifle Brigade Personnel and Primary Offensive Equipment.

(e) Battalion Tactical Groups. BTGs are task-organized battalion-plus-sized tactical combat entities that are capable of performing independent combined-arms combat missions. They are similar in purpose, structure, and tactical use to U.S. Army battalion task forces. Most, if not all, New Look maneuver brigades have one BTG, manned entirely or mostly with contract soldiers, which is used to perform the most difficult or complicated combat tasks assigned to the brigade. An order issued on 19 September 2012 required all maneuver brigade commanders to create a contract-manned BTG within the brigade if they had not already done so.

The need to have effective BTGs is a primary driver of the New Look structural reforms. BTGs have their theoretical origins in the late Soviet period, where they were envisioned to fight against NATO on both a nuclear and non-nuclear battlefield in a nonlinear, large-scale environment. BTGs have been used in every local war or armed conflict in which Soviet and Russian forces have been involved since the Afghanistan War (1979–1989). The tactical use of BTGs in combat has impacted the tactical principles that govern their construction and use. BTGs currently serve—and will continue to serve—as Russia's primary tactical fighting unit in all tactical circumstances, both in large-scale and small-scale conflicts, well into the future.

(2) The Airborne Troops

Russia's Airborne Troops, or *Vozdushno-Desantniye Voyska* (VDV), is an independent arm of service within the Russian Federation armed forces. It is composed of four maneuver divisions, four maneuver brigades, and a separate special purpose (*Spetsnaz*) reconnaissance brigade. The VDV serves as Russia's high-mobility initial invasion and rapid response force. In its role as rapid response or initial assault forces, the VDV may be used to achieve specific objectives that shape the battlespace for follow-on ground forces. These may include:

- Seizing key terrain (i.e., bridges, airports, and seaports).
- Establishing blocking positions and vertical envelopment of a retreating enemy.
- Disrupting enemy logistical supplies, communications, and command centers.
- Destroying high-value targets.

Maneuver formations within the VDV are designated as either parachute or air assault, the primary difference being in whether they arrive at their objective via airdrop or overland means. In all cases, VDV personnel are trained to operate both ways. Major VDV formations are:

- Two parachute divisions – the 98th Guards and 106th Guards.
- Two air assault divisions – the 7th Guards Mountain and 76th Guards.
- Four air assault brigades – the 11th, 31st, 56th, and 83rd.
- One special-purpose reconnaissance brigade – the 45th *Spetsnaz*.

In line with its highly mobile function, the VDV is equipped with a large number of amphibious air droppable combat vehicles, the BMD-series IFVs and BTR-D series APCs. The VDV's increased mobility comes at the expense of armor and firepower; its primary combat vehicles are generally lighter than their Ground Troops counterparts. In late 2016, however, the air assault divisions and brigades received up to a company (10 to 13) of T-72-series main battle tanks. The tank companies within these air assault units will very likely increase to tank battalions (30 to 42) by the end of 2018. The MBTs are not intended for air drops but will accompany VDV ground maneuver formations to increase firepower and lethality.

(3) Naval Infantry

Russian Naval Infantry is organized into units that are operationally subordinate to fleet commanders. Naval Infantry is focused on amphibious assaults, coastal defense, counterterrorism, anti-piracy, and ship security missions. The organization and equipment of Naval Infantry units are generally similar to those of motorized rifle units in the Ground Troops.

The Naval Infantry consists of four independent brigades, one separate brigade, and three separate battalions.

(4) Coastal Troops

The Russian Coastal Troops consist of Coastal Missile Artillery Forces (CMAF) and Coastal Troops. CMAF consists of three independent brigades, two independent regiments, and one independent battalion.

The Coastal Troops are organized as ground forces but are subordinate to the Navy. The Coastal Troops consist mainly of motorized rifle brigades and artillery brigades. Their primary mission is coastal and regional defense. The Navy Ground and Coastal Troop Headquarters, a command unit based in Moscow, heads the Coastal Troop force, but coastal missile units likely take operational orders from their respective fleets.

Russia's coastal missile and artillery forces provide anti-ship defenses for Russia's coastline and littoral regions. Coastal missile defense in Russia is primarily centered on anti-ship missile systems. Most units are still dependent on two systems that entered production in the late 1970s to early 1980s—the STYX and SEPAL. Efforts are underway to rearm the coastal missile force with new, longer-range missile systems. These systems include the BAL and BASTION, and they are slowly being introduced to the force.



Figure 14: BASTION Coastal Defense Missile Launcher.

g. Russian Aerospace Forces

The former Russian Federation Air Forces and Aerospace Defense Troops merged to create the Russian Federation Aerospace Forces in August 2015. The merger places former space and aerospace defense assets vital to strategic aerospace operations under one organizational structure. The Russian Aerospace Forces include four tactical air armies, which are aligned with the military districts. They also contain the Long-Range Aviation and Military Transport Aviation, as well as the Space Troops, which are not subordinate to the military districts but to Aerospace Command in Moscow. Overall manpower for the Russian Aerospace Forces is listed at 148,000, including conscripts.

Aircraft Type	Total Number	Most Capable
Bomber	141	Tu-160
Fighter	420	MiG-29
Fighter Ground Attack	345	Su-35S
Attack	215	Su-25SM
ELINT	32	Il-22M
Airborne Warning and Control	22	A-50
C2	6	Il-86VKP
Tanker	15	Il-78M
Heavy Transport	122	An-124
Training	198	Yak-130

Figure 15: Russian Air Forces Order-of-Battle.

The 6th Air Force and Air Defense Army (AFADA) is subordinate to the Western Military District, the 14th Air Force and Air Defense Army to the Central Military District, the 11th Air Force and Air Defense Army to the Eastern Military District, and the 4th Air Force and Air Defense Army to the Southern Military District.



Figure 16: Russian Air Forces Air Bases.

(1) Long Range Aviation (LRA)

The LRA is the bomber force of the Russian Aerospace Forces and operationally subordinate to the Supreme High Command of the Russian armed forces. The LRA is tasked with long-range bombardment of strategic targets with conventional or nuclear weapons. Currently, TU-95MS aircraft are being modernized to include the Kh-101/102 missile system. The LRA has an inventory of 16 Tu-160, 60 Tu-95MS, and more than 50 Tu-22M3 bombers.

(2) Military Transport Aviation (VTA)

The VTA is subordinate operationally to the Supreme High Command of the Russian armed forces and is the main provider of the air lift for Russian troops and equipment. The recent Ukraine and Syria conflicts have resulted in heavy use of the VTA forces, allowing pilots to gain significant flight hours. The primary aircraft operated by the VTA include the Il-76, An-124, An-22, An-26, An-72, and An-12. The various sizes of aircraft allow the VTA to support many different missions, to include VIP flights, small cargo, and transporting tanks and aircraft. Additionally, these aircraft tend to have larger fuel tanks, allowing for extended missions without refueling to increase efficiency.

(3) Space Troops

These forces within the Aerospace Forces have the mission of conducting space launches and maintaining the ballistic missile early warning system, the satellite control network, and the space object surveillance and identification network.

(4) Integrated Air Defense System

Russia employs what is considered to be among the very best of modern military integrated air defense systems. Historically, Russia has been a leader in developing technologically advanced detection and engagement elements. During the 1990s, Russia largely maintained its research and development programs for air defense equipment. During this period, Russia purchased very few of these systems for domestic use. However, the State Armaments Program of 2015 and the subsequent 2020 plan significantly enhanced support for the purchase and employment of the newest and most capable air defense equipment, including radar, surface to air missiles, command and control, and electronic warfare equipment. Concurrent to the acquisition plan, Russia continues to support research and development efforts in the air defense realm.

The military integrated air defense system kill chain provides the framework for the Russian design, deployment, and command hierarchy of deployed air defense assets. The kill chain contains the seven elements that a fully functional Russian military integrated air defense system would employ in an air defense scenario. To support the kill chain (outlined below), Russia employs redundant and overlapping systems.

Kill Chain Element		Associated Equipment/Process
	Indications and warning	Human intelligence, signals intelligence, open-source information, over-the-horizon radar
	Detection	Air surveillance radar, airborne early warning and control radar, passive Detection system, ship-based radar, visual observation, secondary surveillance radar
	Identification	Secondary surveillance radar, visual identification, non-conformity with approved routes
	Tracking	Integration of detection data into data processing and command & control elements to maintain positional data on an airborne threat
	Assignment	Command & control (variable echelon, based on threat condition and other factors) takes track data and assigns target tracks to weapons platforms
	Engagement	Surface to Air Missiles, Air to Air Missiles, Air Defense Artillery, Electronic Warfare
	Assessment	Verification of engagement: air surveillance radar, airborne early warning and control radar, passive detection system, ship-based radar, visual observer, secondary surveillance radar in conjunction with command & control elements

Figure 17: Russian Military Integrated Air Defense System Kill Chain.

Russia employs its military integrated air defense system at home and abroad. The military integrated air defense system also plays a significant role in Russia's domestic defense and expeditionary operations. Russia's initial deployments to Crimea and Syria included the *Pantsir* and S-300 air defense systems. These systems allowed Russia to build an echelon-based air defense system in the Crimean Peninsula. At a later point in the Crimean and Syrian operations, Russia deployed long-range strategic surface to air missile systems in both countries.

(5) Unmanned Aerial Systems

In 2008, Russia lagged behind the world in development of UAVs. However, the 2008 Georgia conflict accelerated efforts with initial requirements focused on tactical reconnaissance systems that currently dominate inventory. Russia has introduced a class of mini-UAVs for use by the military, but the most significant defense developments are occurring with larger, more capable systems for tactical and strategic use. Russia is also working on unmanned combat aerial vehicles (UCAVs).

(6) Outlook

Russia's fleet of aircraft is aging, but they are rapidly modernizing their air force as well as their air defense systems. When the Soviet Union collapsed in 1991, Russia's air force entered a decline as industry and operational units languished. Since 2008, however, the Russian Air Force and Navy have invested unprecedented financial resources toward airpower to include the upgrade and/or new build of approximately 700 combat fighter/bomber aircraft through 2020 to replace legacy systems.

Newly modified aerodynamic systems in Syria demonstrate that Russian airpower has returned in limited numbers. Similar efforts with newly upgraded air combat systems have been used in the Crimea/Black Sea and Baltic regions, with Russian expectations that provocative moves by the United States and/or NATO will be met with more capable Russian air power. Finally, a slowly improved strategic bomber force (i.e., Tu-95MS BEAR and Tu-160 BLACKJACK) is again using Cold War concepts to conduct international flights that impinge upon the sovereign borders of foreign nations.

Overusage of aircraft and pilots in combat operations in Ukraine, as well as poor force implementation, will probably strain Russia's Aerospace Forces' operations for the near future but will not disrupt the pace of Aerospace Forces employment in the conflict. These forces have demonstrated varying degrees of proficiency and capability in Ukraine, leading to both substantial losses of equipment and veteran personnel,

as well as the development of new tactics for weapons employments, such as glide bombs.

h. Russian Navy

The disintegration of the Soviet Union in 1991, as well as the economic stagnation that followed, led to a severe downsizing for the Navy. Naval construction ground to a halt, and the fleet fell into disrepair and obsolescence. Under Vladimir Putin, however, the Russian military’s capabilities have undergone significant improvement, and the Navy is no exception.

The Russian Navy has approximately 130,000 personnel. The combined major forces of the current Russian Navy number about one-sixth to one-quarter of what the Soviet Navy was in its heyday. That legacy force today has an average age exceeding 20–25 years. With the economic stabilization of the Russian Federation in the early 2000s, the past 10 years have seen a steady increase in the maintenance, training, and deployment activity of the Navy and, more importantly, the activation of a broad submarine and ship construction program to recapitalize the fleet. The Navy’s missions remain focused on strategic deterrence and homeland defense. Periodic distant deployments support the Russian Federation’s global foreign policy interests.

The Navy operates nuclear-powered ballistic missile submarines, which are an essential arm of Russia’s nuclear triad and capable of delivering nuclear warheads from thousands of kilometers away. This strategic capability puts the Russian Navy in the top tier of foreign navies.

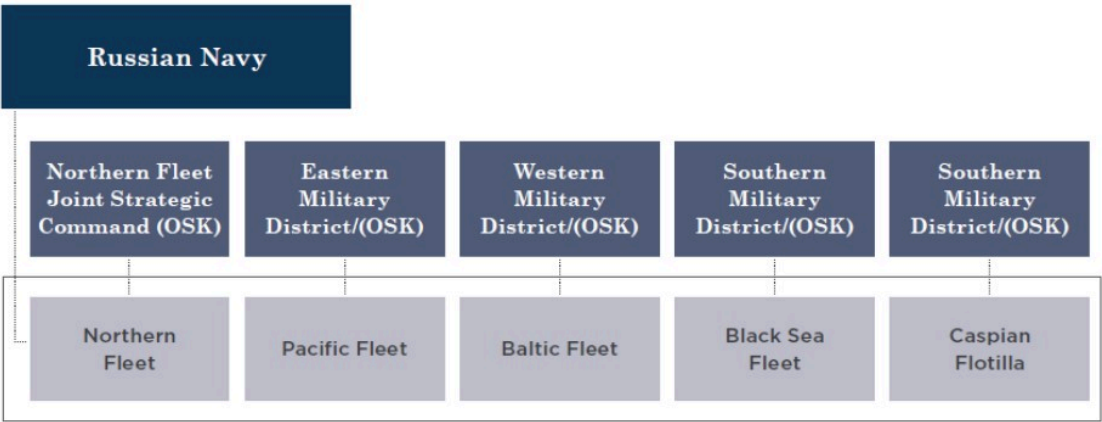


Figure 18: Russian Navy Organization.

(1) Structure

The headquarters of the Russian Navy is located in St. Petersburg. The Russian Federation Navy consists of four fleets (Baltic, Black Sea, Northern, and Pacific) and a flotilla in the Caspian Sea. The fleets receive administrative orders and guidance from the Navy Staff in St. Petersburg, whereas operational orders are issued from the various Joint Strategic Commands (OSKs). Each fleet and the Caspian Flotilla are operationally subordinate to one of these OSKs.

(2) Northern Fleet

The Northern Fleet is Russia’s most capable naval force. Based in Severomorsk, located in the Kola Gulf (the only ice-free direct access to the North Atlantic), its seven operational ballistic missile submarines provide the bulk of the firepower for the Navy’s arm of the strategic nuclear triad. Russia’s only operational aircraft carrier is also based in the Northern Fleet, along with the Navy’s only nuclear-powered heavy cruiser. Surface combatants and submarines deploy worldwide from the Kola Gulf, playing an active role in the ongoing Syria crisis, conducting counter-piracy patrols off the Horn of Africa, along with power projection in the North Atlantic and Caribbean. The Northern Fleet’s two primary missions are to provide strategic deterrence with its ballistic missile submarines and to defend the maritime approaches to northwest Russia.

(3) Pacific Fleet

The Pacific Fleet lags behind the Northern Fleet in terms of maintenance and overall capability; however, it is still able to conduct strategic nuclear strikes against the U.S. mainland, and its surface units are active from the Pacific region to the Horn of Africa. The Pacific Fleet has its headquarters in Vladivostok, but its forces are split between two main locations, with the majority of surface ships and diesel-powered submarines in the Vladivostok region and the nuclear-powered submarines, including the SSBNs, located in Petropavlovsk-Kamchatskiy. The workhorses of the PACFLT are four UDALOY-class destroyers, which are regularly deployed throughout the region.

(4) Black Sea Fleet

The Black Sea Fleet, for years, has been a fleet in decline, forced to operate with a handful of Soviet-era vessels. Beginning in 2014, after the occupation of Crimea, new units began to enter the order of battle, including modern coastal missiles and naval infantry. Then in 2015, new submarines and surface combatants began to arrive to bolster the fleet. Now armed with the KALIBR missile system, the Black Sea Fleet is a significant force in the region and, over the next few years, could have as many as six new attack submarines and six new surface ships, which can not only exert control on the Black Sea but can also operate in the Mediterranean to counter NATO forces and support operations in Syria.

(5) Baltic Fleet

The majority of Baltic Fleet vessels are located at Baltiysk in the Kaliningrad Oblast, with a handful further north near St. Petersburg. Headquartered in Kaliningrad, the fleet's mission focuses on specifically ensuring sea-lines of communication and trade are open between Kaliningrad and St. Petersburg, and in countering NATO forces in the region. The Baltic Fleet has also been a key player in support of Russian interests in the Eastern Mediterranean Sea and Horn of Africa. With the arrival of two KALIBR-equipped vessels in 2016, the Baltic fleet presents a significant long-range precision conventional and theater nuclear strike threat to Western Europe.

(6) Caspian Sea Flotilla

The Caspian Sea Flotilla is the dominant naval force on the Caspian Sea and was the first Russian surface force operationally equipped with the KALIBR missile system. Russia's naval superiority ensures Moscow has leverage in regional economic disputes. The KALIBR land attack cruise missile gives Moscow a precision strike weapon that can range targets in Central Asia, the Middle East, and parts of Europe, as evidenced by strikes into Syria in October 2015. Most of the flotilla's combat power (all of the KALIBR shooters) is based at Makhachkala, possibly to be closer to regional threats and also to avoid having to navigate the Volga River Delta to reach the sea, as is the case with ships based at Astrakhan.

(7) Naval Aviation

Naval aviation assets are spread through four fleet air forces, each with composite regiments under their command. The main missions of naval aviation are to track and destroy enemy submarines and warships and also help achieve air superiority where the fleet is operating. Most naval aviation aircraft are land-based; the only aircraft carrier, *Admiral Kuznetsov*, can accommodate 22 strike aircraft and 17 attack helicopters.

(8) Submarine Forces

Russia's sea-based strategic deterrent is deployed in the Northern and Pacific Fleets. There are six DELTA IV SSBNs, one DOLGORUKIY SSBN, and one remaining TYPHOON SSBN used as a test platform in the north. Three DELTA III and two DOLGORUKIY SSBNs are in the Pacific. All sea-launched ballistic missiles (SLBMs) carried by these submarines—SS-N-18 (DELTA III), SS-N-23 (DELTA IV), and SS-N-32 (DOLGORUKIY)—can reach U.S. targets from their home base piers and, if required, could be launched with the submarines on the surface.

These SSBNs are protected by nuclear-powered cruise missile and torpedo attack submarines, which also engage enemy surface and submarine forces and pose a land attack cruise missile threat against an enemy homeland. In the Northern Fleet, these attack submarines include three OSCAR II and one SEVERODVINSK SSGNs, and three VICTOR III, six AKULA I/II, and four SIERRA SSNs. The SEVERODVINSK class is new, extremely quiet, and is armed with a wide range of advanced cruise missiles to destroy enemy ships and targets ashore. The Pacific Fleet has five OSCAR II SSGNs and four AKULA I SSNs. It will eventually receive SEVERODVINSK SSGNs. A new fifth-generation general purpose nuclear-powered submarine is under development.

Non-nuclear diesel-electric submarines round out the Russian submarine forces. These units are assigned to all fleets for close-in area defense missions in adjacent seas. Older and newer versions of the KILO class compose most of this force: six in the Northern Fleet, two in the Baltic, three new KALIBR-equipped units in the Black Sea, and eight older KILO class in the Pacific. The newest KILO version continues in construction with three more units destined for the Black Sea Fleet and eventually another six for the Pacific Fleet. A single PETERSBURG-class improved design experimental unit is in the Northern Fleet with two additional units to be completed. A future non-nuclear, KALINA design, likely having an air independent propulsion plant, is in development with construction projected after 2020.

(9) Surface Forces

The Russian Navy's major combatant surface ships, frigates and larger, comprise some 32 units assigned across all 4 fleets.

- The **Northern Fleet** has Russia's only aircraft carrier (KUZNETSOV), one nuclear-powered KIROV-class cruiser, one conventionally powered SLAVA-class cruiser, and four UDALOY-class destroyers. The first new GORSHKOV-class (KALIBR) guided missile frigate was recently commissioned, with more expected. This fleet also has 12 minor anti-ship and anti-submarine combatant ships, as well as 4 ROPUCHA-class amphibious assault ships.
- The **Baltic Fleet** has nine major ships—two older SOVREMENNY-class destroyers and seven frigates: one KRIVAK-class, two NEUSTRASHIMYY-class, and four new STEREGUSHCHIY-class units. It recently received two SVIYAZHISK-class (KALIBR) guided missile patrol ships. These are supplemented by 18 minor combatants and 4 amphibious assault ships.

- The **Black Sea Fleet** has one SLAVA-class cruiser, one 47-year-old KASHIN-class destroyer, two older KRIVAK-class frigates, and the first of a planned six new GRIGOROVICH-class (KALIBR) frigates. More new construction units are expected for the Black Sea Fleet. The fleet is supplemented by 15 minor combatants and 7 amphibious assault ships.
- The **Caspian Flotilla** has two GEPARD-class frigates (one with KALIBR) and recently received two new ASTRAKHAN-class patrol ships and three SVIYAZHSK-class (KALIBR-capable) guided missile patrol ships.
- Finally, the **Pacific Fleet** has seven major ships: one SLAVA-class cruiser, four UDALOY class and two SOVREMENNY-class destroyers. These are supplemented by 24 minor anti-ship and anti-submarine combatants and 4 amphibious assault ships.

(10) Outlook

The Russian Navy has several weapons upgrade programs in progress. The new SS-N-32 BULAVA submarine launched ballistic missile is being produced for the DOLGORUKIY-class SSBNs. The most consequential development is that Russia plans to deploy KALIBR capability on all new design construction nuclear and non-nuclear submarines, corvettes, frigates, and larger surface ships. KALIBR provides even modest platforms, such as corvettes, with significant offensive capability and, with the use of land attack missiles, all platforms have a significant ability to hold distant fixed ground targets at risk using conventional warheads. The proliferation of this capability within the new Russian Navy is profoundly changing its ability to deter, threaten, or destroy adversary targets.

Although the Navy is mainly made up of Soviet-era surface ships and submarines, an extensive modernization program is underway, focusing first on the submarine force. Progress in submarine modernization is underway; however, the majority of the naval inventory still consists of aging units from the 1980s and 1990s. While more new classes of ships are planned, the Navy will have to maintain its older fleet for several years until these new vessels come online. Despite this, Russia is still capable of deploying its assets worldwide, best evidenced by continuous support to Russian operations in Syria since 2012 and recurring counter-piracy deployments to the Gulf of Aden since 2008.

i. Russian Special Operations Forces

Russia's special operations forces are popularly known by the abbreviation "*Spetsnaz*," short for *Spetsial'noye naznachenie*, a term meaning "special purpose." *Spetsnaz*, although similar to special operations forces in Western countries, retains a slightly different set of roles and missions than those commonly found elsewhere. The current incarnation of *Spetsnaz* traces its origin back to naval and ground units created back in the mid-1950s to establish a dedicated special purpose force to operate with the armed forces, although its wider lineage can be attributed to counterrevolutionary and partisan units formed during the Russian Revolution and World War II, respectively.

These predecessor units became pivotal in defining the hallmark missions of the modern *Spetsnaz* force, from diversionary acts conducted by partisan units behind German lines in World War II to deep reconnaissance and intelligence collection, reflecting the postwar perception of a potential conflict with the West during the Cold War. These traditional missions—with slight modifications and variations—still manifest themselves in *Spetsnaz* doctrine and are associated with Moscow's recent strategy of using indirect action, albeit with the wider aim of achieving goals while avoiding a large-scale conflict. Within Russia, the term *Spetsnaz* is often misappropriated and misattributed. Moscow's true *Spetsnaz* force is a relatively small, select group of mission-dedicated special purpose forces, primarily belonging to the military and its Main Intelligence Directorate (GRU) (e.g., the Defense Ministry's ground and naval *Spetsnaz* units), and to a lesser extent, the security services of the National Guard, Federal Security Service (FSB), Foreign Intelligence Service (SVR), and Justice and Emergency Situations Ministries. Estimated to number 20,000–30,000 personnel, *Spetsnaz* units in these organizations all retain distinct and separate missions from one another. The single largest contingent with which recent and visible exploits of *Spetsnaz* are attributed resides with the GRU.

j. Russian Intelligence Services

Russia has three primary intelligence services: the Federal Security Service (FSB), the Foreign Intelligence Service (SVR), and the Main Intelligence Directorate of the General Staff (GRU). The FSB and SVR trace their lineage to the old Soviet Committee of State Security (KGB). Although the FSB and SVR are considered military services under Russian federal law, they are more akin to civilian intelligence agencies.

(1) The Federal Security Service

The FSB has three primary missions: countering foreign intelligence services, combating organized crime, and ensuring economic and financial security. It is also the Russian lead counterterrorism organization. President Vladimir Putin launched a major reorganization of the FSB during his first term, placing the organization under the president's direct control. The FSB has continued to grow, integrating the Border Guard Service of Russia and the Federal Agency of Government Communication and Information (FAPSI).

(2) The Foreign Intelligence Service

The SVR is Russia's main external intelligence agency; it focuses on civilian affairs, whereas the GRU focuses on military affairs. According to Russian law, the SVR is authorized to carry out the following missions:

- Conduct intelligence.
- Implement active measures (disinformation, propaganda, etc.) to ensure Russia's security.
- Conduct military, strategic, economic, scientific, and technological espionage.
- Protect employees of Russian institutions overseas and their families.

- Provide personal security for Russian government officials and their families.
- Conduct joint operations with foreign security services.
- Conduct electronic surveillance in foreign countries.

(3) The Main Intelligence Directorate of the General Staff

The GRU is the Ministry of Defense's foreign intelligence organization that provides military intelligence for the General Staff, Ministry of Defense, and senior government officials. The GRU's responsibilities include providing senior political and military leadership with all-source intelligence, including indications and warning of strategic threats and information to assist leadership in making decisions regarding Russia's armaments program. GRU operations abroad involve human intelligence (HUMINT) collection on potential enemies' military-industrial capabilities, troop movements, and weapon systems. The GRU manages military attachés, intelligence analysis, cryptanalysis, space-based assets, telecommunications intercept capabilities, and radio-electronics and telecommunications-based offensive capabilities. It also oversees *Spetsnaz* units and special operations forces. GRU operational combat and advisory roles have expanded dramatically in the past decade, involving military operations in Ukraine's Crimea region, eastern Ukraine, and Syria.

Lesson 8: Future Operating Environment - China

Lesson Hours: 1.00

1. Introduction

The purpose of this lesson is to build upon the previous lesson to increase the understanding of the future operating environment by presenting an overview of the People's Republic of China, specifically the national goals and the strategy, organization, and capabilities of its military.

2. Educational Objectives

- Identify the perceived goals of the People's Republic of China.
- Identify the key aspects of Chinese military strategy.
- Identify China's core military capabilities.
- Identify the general organization of China's armed forces.

3. Key Terms

- Chinese Communist Party (CCP)
- Central Military Commission (CMC)
- People's Liberation Army (PLA)
- People's Liberation Army Army (PLAA)
- People's Liberation Army Navy (PLAN)
- People's Liberation Army Air Force (PLAAF)
- People's Liberation Army Rocket Force (PLARF)
- People's Liberation Army Strategic Support Force (PLASSF)
- People's Liberation Army Joint Logistics Support Force (PLAJLSF)
- First and Second Island Chains

4. Preparation

All quizzes, discussion forum posts, and assignments are executed in the Moodle course site.

- Lesson Reading: Preface and Executive Summary from [Military and Security Developments Involving the People's Republic of China](#) (18 Dec 2024) (8 pages).

b. Other Requirements: Review (i.e., skim to become generally familiar with) the entire *Military and Security Developments Involving the People's Republic of China* document (Chapters 1-6, Special Topics, and Appendix I) (182 pages).

c. Supplemental References (optional):

- [*China Military Power: Modernizing a Force to Fight and Win*](#), Defense Intelligence Agency (2019).
- [*ATP 7-100.3, Chinese Tactics*](#), Headquarters, Department of the Army (9 Aug 2021).
- [*U.S.-China Relations*](#), Congressional Research Service (3 Sep 2019).
- [*U.S.-China Strategic Competition in South and East China Seas: Background and Issues for Congress*](#), Congressional Research Service (6 Feb 2020).
- [*China Naval Modernization: Implications for U.S. Navy Capabilities—Background and Issues for Congress*](#), Congressional Research Service (18 Mar 2020).
- [*U.S. Strategic Competition with China*](#), RAND Corporation (Jun 2021). This material is protected by copyright; further distribution is prohibited.
- [*China's Grand Strategy*](#), RAND Corporation (2020). This material is protected by copyright; further distribution is prohibited.
- [*2021 Recommended Reading List*](#), U.S. Marine Corps China Research Group (2021).
- [*Chairman Xi Remakes the PLA, Assessing Chinese Military Reforms*](#), National Defense University Press (2019).
- [*Military and Security Developments Involving the People's Republic of China*](#), U.S. Department of Defense (18 Dec 2024).

Lesson 9: Operational Terms and Graphics

Lesson Hours: 1.00

1. Introduction

Tactical terms and the related graphics associated with military operations are tools for communicating and directing the activities of subordinates. These words and symbols control resources and time and incur different levels of risk. As leaders and planners, we need to know what effects we want to have upon an adversary, the number of resources we are willing to expend, and the risk we are willing to accept. We must then communicate this vision and direct subordinates during execution. To accomplish this, we must properly articulate our vision by using doctrinally correct military terms and supporting graphics.

2. Educational Objectives

- Identify the purpose of a military symbol.
- Identify the composition of icon-based symbols.
- Identify the purpose of a control measure.
- Identify the purpose of a control measure symbol.
- Identify the composition of control measures symbols.
- Identify the purpose of fire support coordination measures.
- Identify the different types of fire support coordination measures.
- Identify the purpose of tactical tasks.
- Identify the different types of tactical tasks.

3. Key Terms

- Military symbols
- Control measure symbols
- Fire support coordination measures

- Tactical tasks

4. Preparation

All quizzes, discussion forum posts, and assignments are executed in the Moodle course site.

a. Lesson Reading:

- [Operational Terms and Graphics](#) (27 pages).
- [Operational Terms and Graphics Reference Sheet](#) (4 pages).

b. Other Requirements: None.

c. Supplemental References (optional):

- [MIL-STD-2525D, Department of Defense Interface Standard: Military Symbology](#) (10 June 2014).
- [FM 1-02.1, Operational Terms](#), Department of the Army (28 Feb 2024).
- [FM 1-02.2, Military Symbols](#), Department of the Army (13 Aug 2025).
- [MCDP 1-0 w/changes 1, 2, and 3, Marine Corps Operations](#) (29 March 2019).
- [MCTP 3-10F, Fire Support Coordination in the Ground Combat Element](#) (4 April 2018).

Lesson 9 Reading

OPERATIONAL TERMS AND GRAPHICS

References:

- MIL-STD-2525D, *Department of Defense Interface Standard: Military Symbology* (10 June 2014)
- FM 1-02.1, *Operational Terms*, Department of the Army (28 Feb 2024)
- FM 1-02.2, *Military Symbols*, Department of the Army (13 Aug 2025)
- MCDP 1-0 w/changes 1, 2, and 3, *Marine Corps Operations* (29 Mar 2019)
- MCTP 3-10F, *Fire Support Coordination in the Ground Combat Element* (4 Apr 2018)

1. Military Symbols

The main purpose of military symbology is to allow commanders and staffs to graphically display certain elements of the operational picture. Situation maps, overlays, and annotated aerial photographs are used to express an operation plan or order, concept, or friendly or hostile (enemy) situation. The combination of unit and equipment symbols and control measures, as well as other military symbols, creates an indispensable tool for quickly portraying military operations.

A military symbol is a graphic representation of units, equipment, installations, control measures, and other elements relevant to military operations. As a part of doctrine, these symbols provide a common visual language for all users. Standardization of military symbols is essential if operational information is to be passed among military units without misunderstanding. This reading defines the single standard used to develop and depict hand-drawn and computer-generated military symbols for situation maps, overlays, and annotated aerial photographs for all types of military operations; it also provides rules for building specific sets of military symbols. These rules provide details of construction but allow enough flexibility for the user to build symbols to meet operational needs.

a. Composition of Military Symbols

A military symbol is a graphic representation of a unit, equipment, installation, activity, control measure, or tactical task relevant to military operations that is used for planning or to represent the common operation picture (COP) on a map, display, or overlay. Military symbols are governed by the rules in the *Military Standard (MIL-STD) 2525D*. Military symbols fall into two categories: framed, which includes unit, equipment, installation, and activity symbols; and unframed, which includes control measures and tactical symbols.

(1) Frame

The frame is the border of a symbol. It does not include associated information inside or outside of the border. The frame serves as the base to which other symbol components are added. The frame indicates the standard identity, physical domains, and status of the object being represented.

(a) Standard Identity. Standard identity reflects the relationship between the viewer and the operational object being monitored. The standard identity categories are unknown, pending, assumed friend, friend, neutral, suspect, and hostile. In the realm of surface operation symbols, a circle or rectangle frame denotes friend or assumed friend standard identity, a diamond frame denotes hostile or suspect standard identity, a square frame denotes neutral standard identity, and a quatrefoil frame denotes unknown and pending standard identity. Table 1 shows frame shapes for standard identities for land symbols.

DIMENSION STANDARD IDENTITY	UNKNOWN	SPACE	AIR	LAND UNIT	LAND EQUIPMENT AND SEA SURFACE	LAND INSTALLATION	SUBSURFACE	ACTIVITY/ EVENT
PENDING (YELLOW)								
UNKNOWN (YELLOW)								
FRIEND (CYAN)								
NEUTRAL (GREEN)								
HOSTILE (RED)								
ASSUMED FRIEND (CYAN)								
SUSPECT (RED)								

Table 1: Frame Shapes for Standard Identities for Land Symbols.

(b) Physical Domain. The physical domain defines the primary mission area for the object within the operational environment. An object can have a mission area above the earth's surface (in the air domain or space domain), on the earth's surface, or below the earth's surface (that is, in the land domain or maritime domain). The land domain includes those mission areas on the land surface or close to the surface (such as caves, mines, and underground shelters). Maritime surface units are depicted in the sea surface dimension. Aircraft, regardless of Service ownership, can be depicted in either the air dimension (in-flight) or land dimension (on the ground), while air units are depicted as a land unit and facilities as a land installation. Land equipment is depicted in the land dimension. Likewise, landing craft whose primary mission is ferrying personnel or equipment to and from shore are represented in the sea surface dimension. However, a landing craft whose primary mission is to fight on land is a ground asset and is represented in the land dimension.

(c) Status. Status (Table 2) depicts whether an object exists at the location identified (status is "present"), will in the future reside at that location (status is "planned" or "anticipated"), or is thought to reside at that location ("suspected"). The symbol frame will be a solid or black and white dotted line when indicating a present status and a dashed line when indicating anticipated, planned, or suspected status. When the frame is assumed friend, suspect, or pending, the status will not be displayed.

<i>Dimension</i>	<i>Space</i>	<i>Air</i>	<i>Land Unit</i>	<i>Land Equipment and Sea Surface</i>	<i>Land Installation</i>	<i>Sea Subsurface</i>	<i>Activity or Event</i>
<i>Status</i>							
Present or confirmed position							
Anticipated, planned or suspected position							

Table 2: Friendly Frames in Present, Planned, or Suspected Status.

(2) Color (Fill)

In framed symbols, color provides a redundant clue with regard to standard identity. The fill is the interior area in a symbol. If color is not used, the fill is transparent. In unframed symbols (equipment), color is the sole indicator of standard identity, excluding text amplifiers. The default colors used to designate standard identity are blue for friendly or assumed friend, red for hostile or suspect, green for neutral, and yellow for unknown or pending. Affiliation color without the fill may also be used for the frame, main icon, and modifiers.

(3) The Octagon

The octagon serves as a three-sector spatial reference for placement of main icons and modifiers in the frame of a symbol. The three sectors specify where main icons and modifiers are positioned and how much space is available for sizing of main icons and modifiers. The vertical bounding octagon allows for effective use of the space when dealing with vertical icons. Table 3 provides examples showing the horizontal and vertical bounding octagons examples for all frame shapes.

Horizontal Bounding Octagon				
Vertical Bounding Octagon				

Table 3: Examples of Horizontal and Vertical Bounding Octagons.

b. Main Icons for Framed Symbols

The main icon is the innermost part of a symbol. The icon provides an abstract pictorial or alphanumeric representation of units, equipment, installations, or activities. This reading distinguishes between icons that must be framed and icons for which framing is optional. In general, main icons should not be so large as to exceed the dimensions of the main sector of the octagon or touch the interior border of the frame. However, there are exceptions to this size rule. In those cases, the main icons will occupy the entire frame and must, therefore, exceed the dimensions of the main sector of the octagon and touch the interior border of the frame. These are called full-frame main icons and occur only in land domain symbols. Figure 1 shows an example of a full-frame main icon for all frame shapes.

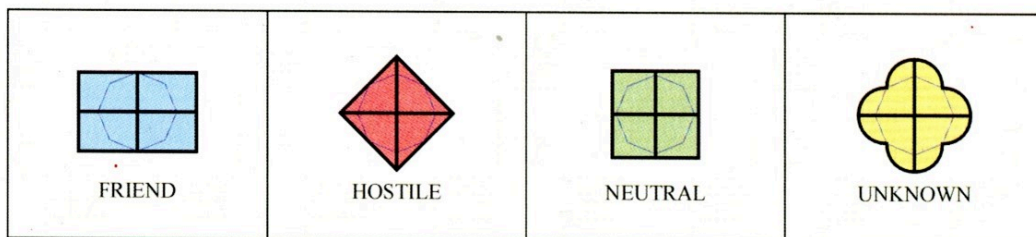


Figure 1: Examples of Full Frame Icons (Medical Unit).

c. Modifiers for Framed Symbols

A text or graphic modifier provides additional information about a symbol. This information is displayed on the outside of the frame.

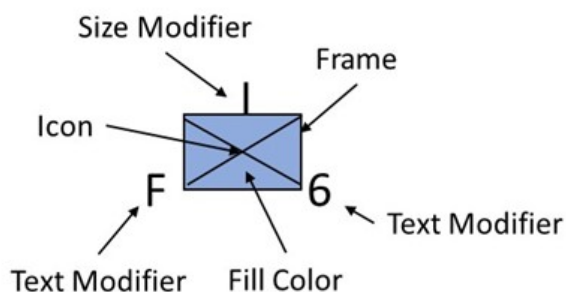


Figure 2: Components of a Military Symbol (Fox Company, 2nd Battalion, 6th Marines).

Note: In the Marine Corps, the second battalion always contains "E," "F," and "G" companies, so it is not necessary to show the "2" for the second battalion

d. Building an Icon

The following shows the steps for the building process of framed symbols.

(1) Frame Affiliation

Choose the frame that matches the affiliation of the unit (friendly, hostile, neutral).



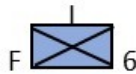
(2) Main Icon

Choose the appropriate main icon and combine it with the frame. For this example, the infantry icon will be used.



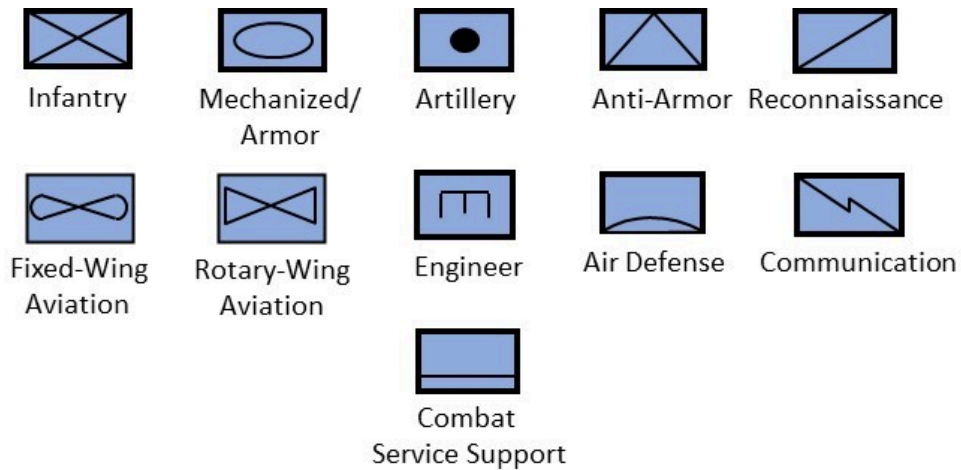
(3) Text or Graphic Modifiers

Choose the text or graphic modifiers as necessary to provide further amplifying information. In this example, there is a graphic modifier and a text modifier that would identify this infantry unit as Company F, 2nd Battalion, 6th Marines.



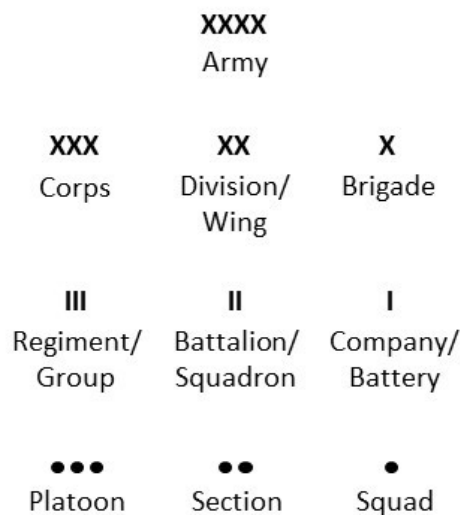
The final product above consists of both role indicators and size indicators. Role indicators describe the basic role or mission of the unit. Combined with size indicators, a detailed and informative symbol is created.

(a) Role Indicators



Note: This lesson provides an extensive number of icons and modifiers for building a wide variety of framed symbols. No attempt has been made to depict all possible combinations. Instead, a standard method for constructing symbols is presented. Once the user is familiar with the prescribed system, any desired symbol can be developed using the logical sequence provided.

(b) Size Indicators

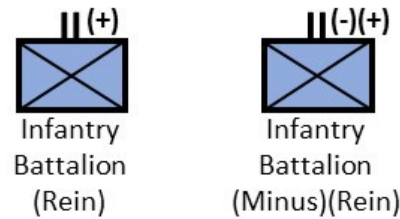


Examples:



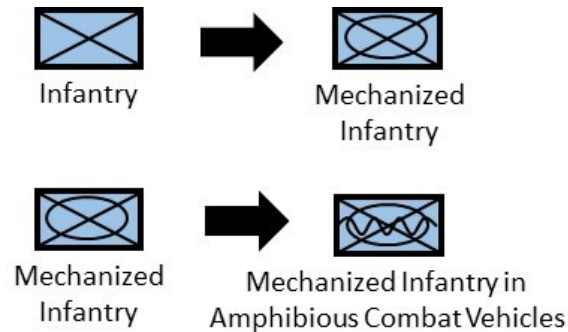
(c) Reinforced and Minus Amplifiers. The reinforced amplifier + (plus sign) indicates that the capability of one unit has been reinforced by the capability of another unit. The minus amplifier - (minus sign) indicates that the capability of a unit has been reduced by the detachment of one or more of its units. If the unit has been both reduced and reinforced, then both signs (-) (+) are used.

Examples:



(d) Unit Symbol Modifiers. Unit symbol modifiers are combined with role indicators to create symbols representing a unique or specific type of unit.

Examples:



(e) Special Size Indicator. Special size indicators are used if a unit is temporarily organized or “task organized” for a specific operation into a task force or company team. A “hood” is placed over the size indicator. Note that if the hood is used, the (+) (-) symbols are not used. A battalion task force consists of, at a minimum, the battalion’s headquarters, one organic element, and one attached unit of comparable size of an unlike element. If the unit is of company size, it is referred to as a “company team” instead of a task force.

Example:



2. Tactical Maneuver Control Measures

A control measure is a means of regulating forces or warfighting functions. Control measures may be boundaries, special area designations, or other unique markings related to an operational environment’s geometry, and are necessary for planning and managing operations. Control measure symbols represent control measures that can be graphically portrayed, and they provide operational information. They can be displayed as points, lines, and areas. Control measure symbols can be combined with other military symbols, main icons, and amplifiers to display operational information. Control measure symbols follow the same fundamental building rules as framed symbols, but they are built in accordance with their template. Control measure symbols can be black or white, depending on their display background. Display backgrounds can be blue (for friendly), red (for hostile), green (for obstacles), or yellow (for a chemical, biological, radiological, and nuclear contaminated area fill). Control measure symbols can be combined with other symbols, icons, and amplifiers to display operational information (Figure 3). They do not follow the same building rules as the icon-based symbols, but shall be built in accordance with the draw rules specified in the symbol tables.

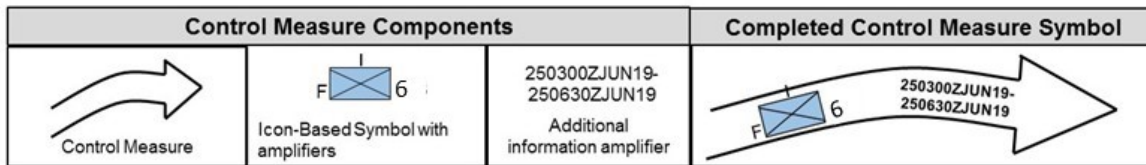


Figure 3: Composition of Control Measure Symbols.

a. Standard Identity Coloring Control Measures

Friendly graphic control measures will be shown in black or blue when drawn manually or on a color computer-generated display. Hostile graphic control measures will be shown in red. If red is not available, they will be drawn in black with the abbreviation "ENY" placed on the graphic in at least two places. Obstacles, as shown in this chapter (friendly, hostile, neutral, unknown, or factional) will be drawn using the color green. If the color green is not available, obstacles should be drawn using black. The color yellow will be used for the hatching for chemical, biological, radiological, and nuclear (CBRN) contaminated areas.

b. Labeling Control Measures

Make all text labeling in upper case letters. The reader should be able to read the labels for all text labels of modifier or amplifier fields for control measures symbols when the bottom of the overlay is closest to the reader. Labeling written on an angle should be readable to viewers, so they do not have to turn their heads.

c. Status

Status refers to whether a control measure exists at the location identified (status is "present") or will in the future reside at that location (status is "planned," "anticipated," "suspected," or "on order"). If a control measure is on order, the status code shall be specified "A – anticipated or planned" and field amplifier "W" shall be present and specified "O/O." In general, linear control measures (including boundary lines) and area control measures shall be a solid line when indicating present status and a dashed line when indicating anticipated or planned status, as depicted in Table 4. There are certain control measures, such as counterattack, which are drawn in the "present" status with dashed lines.

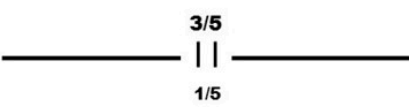
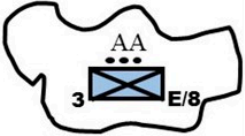
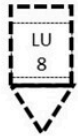
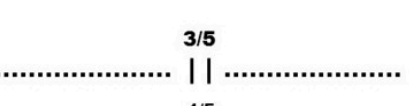
	Point Graphics	Boundary Line Graphics	Area Graphics
Present Position (P)			
Anticipated, planned, suspected, or On Order(O/O)			

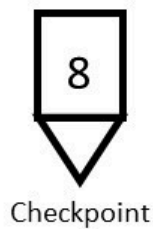
Table 4: Present and Planned Status for Control Measure Symbols.

d. Point, Line, Boundary, and Area Symbols

(1) Point Symbols

A point is a control measure symbol that only has one set of coordinates. Most Marine Corps point symbols follow a standard format. Point symbols cannot be rotated; therefore, text must only be written horizontally.

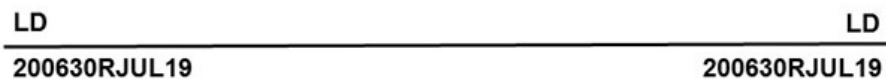
Examples:



(2) Lines

A line is a control measure symbol with multiple sets of coordinates. Most lines are also labeled as phase lines (PL) for easy reference in orders and during transmissions. When lines representing other purposes are marked as phase lines, they should show their primary purpose in the icon field (such as NFL for no fire line). The purpose of the line is labeled on top of the line at both ends inside the lateral boundaries or as often as necessary for clarity.

(a) Line of Departure (LD). An LD is a line designated to coordinate the departure of attacking elements. The LD should generally be perpendicular to the direction of attack and on easily identifiable terrain. The time of attack should be written below the LD.



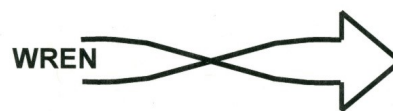
(b) Phase Line. A phase line is used for control of forward movement and coordination of units. It should be extended across the entire zone or sector, preferably along an easily identifiable terrain feature or man-made object. It is labeled with a code name, letter, or number for identification.



(c) Axis of Advance. The axis of advance indicates a general direction of movement for a unit. A commander assigned an axis of advance may deviate from the axis; however, a major deviation must be reported. It is labeled with a code name, letter, or unit designation.



(d) Airborne Axis of Advance. The airborne axis of advance is the same as a ground axis of advance except that it has a twist in the shaft of the open arrow.

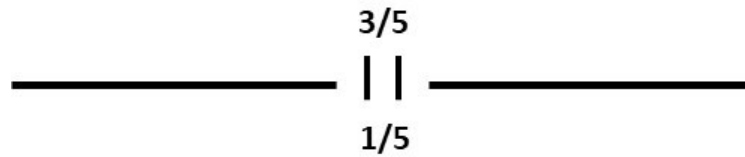


(3) Boundaries

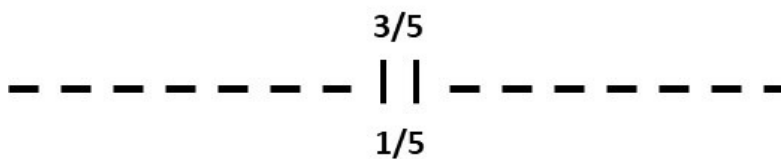
A boundary is a line that delineates surface areas for the purposes of facilitating coordination and deconfliction of operations between adjacent units, formations, or areas. There are three types of boundary lines: lateral, rear, and forward. Boundaries are:

- Drawn along easily recognizable terrain features
- Should be marked perpendicular to the boundary line
- Use abbreviation for unit size indicators
- National abbreviation is used if forces are from different countries
- Enemy indicates equipment, line areas, and boundaries

Examples:



Friendly present



Planned



Suspected Enemy

(4) Area Symbols

An area is a control measure symbol with multiple sets of coordinates that start and finish at the same point. Areas are marked with an abbreviation for the type of area followed by the name.

Examples:



Landing Zone/Drop Zone



Objective

An objective is usually a key terrain feature, area, or an enemy force to be seized or engaged in an attack. Objectives are enclosed with a solid line and identified by the abbreviation "OBJ" and number, letter, code name, or unit designation.

3. Fire Support Coordination Measures

Commanders use control measures to assign responsibilities, coordinate fire and maneuver, and control combat operations. These measures delineate areas of operations or other areas in which commands will conduct their operations or coordinate maneuver or fires between adjacent units. Each of these measures has specific purposes. Therefore, it is critical Marines use these measures as they were designed and clearly articulate to all affected higher, adjacent, and subordinate units any modifications to the doctrinally based measures.

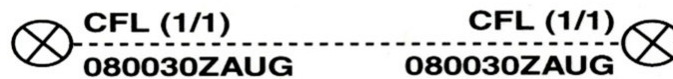
Fire support coordinating measures (FSCMs) enhance the expeditious attack of targets and provide safeguards for friendly forces. Their use is governed by the tactical situation. FSCMs fall into two broad categories: permissive and restrictive. Some maneuver control measures also affect fire support coordination. FSCMs are recommended by the Fire Support Coordinator and established by the commander. FSCMs are graphically shown as a black line or lines (solid or dashed). All measures are labeled with the title or abbreviated title of the measure, followed by the establishing headquarters (in parentheses). The significance of identifying the establishing headquarters is that it designates the area where the FSCM applies. For example, an FSCM established by a battalion is effective within that battalion's zone of action. The effects do not extend beyond the battalion's zone. When a date-time group is used, the time zone must be shown. FSCMs can be effective immediately, at a scheduled time or on-call.

Note: FSCMs will be discussed in detail in the 8671 *Fires* sub-course.

a. Permissive Fire Support Control Measures

(1) Coordinated Fire Line (CFL)

The CFL is a line beyond which conventional surface fire support means (e.g., mortars, field artillery, and naval surface fire support ships) may fire at any time within the zone of the establishing headquarters without additional coordination. The CFL expedites the attack of targets beyond the CFL without coordination with the ground commander in whose zone of action targets are located. It also provides the ground commander with an area within their zone.



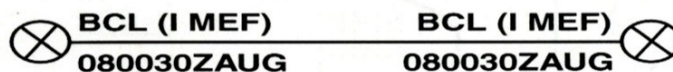
(2) Fire Support Coordination Line (FSCL)

The FSCL is an FSCM that delineates the coordination requirements for the attack of surface targets. It is established by the appropriate land or amphibious commander within his/her boundaries in consultation with superior, subordinate, supporting, and affected commanders. An FSCL is not a boundary and does not divide an area of operations. The FSCL applies to all fires of air, land, and sea-based weapon systems using any type of ammunition against surface targets. Forces attacking targets beyond an FSCL must inform all affected commanders in sufficient time to allow necessary reaction to avoid fratricide, both in the air and on the ground.



(3) Battlefield Coordination Line (BCL)

A BCL is a fire support coordinating measure, established based on METT-T, which facilitates the expeditious attack of surface targets of opportunity between the measure and the FSCL. When established, the primary purpose is to allow Marine Air-Ground Task Force (MAGTF) aviation to attack surface targets without approval of a Ground Combat Element commander in whose area the targets may be located. The BCL is an exclusive Marine Corps FSCM, similar to an FSCL, which facilitates the expeditious attack of targets with surface indirect fires and aviation fires between this measure and the FSCL.

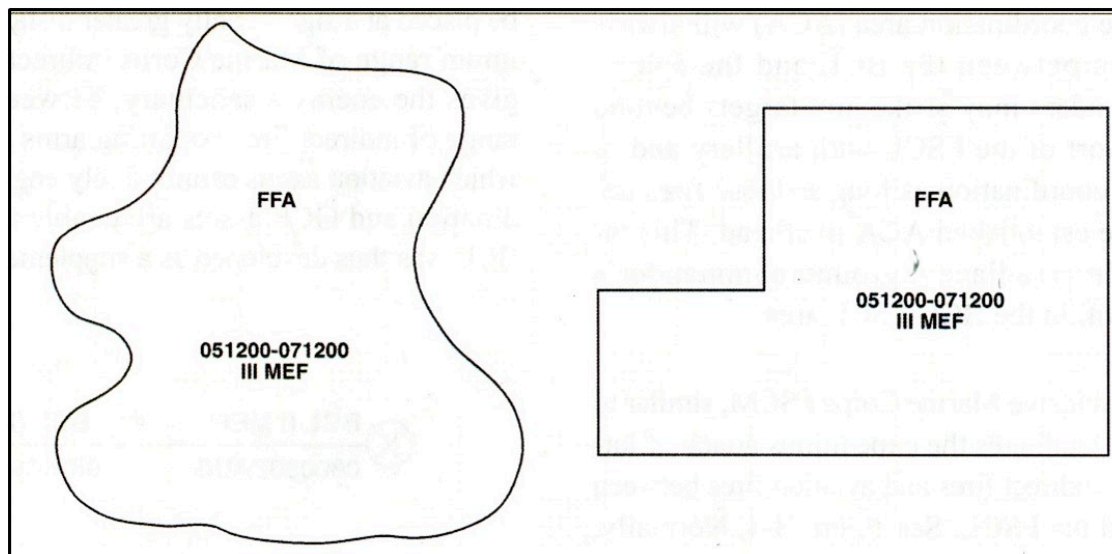


(4) Free-Fire Area (FFA)

The FFA is a specifically designated area into which any weapon system may fire without additional coordination with the establishing headquarters. FFAs expedite fire and facilitate jettisoning of aviation munitions if an aircraft is unable to drop on a target. FFAs may be established only by the military or civilian commander with jurisdiction over the area (usually division or higher). An FFA request must be

forwarded to the appropriate approving official through the MAGTF commander.

Preferably, the FFA should be easily identifiable from the air, but it may be designated by grid coordinates, GPS coordinates, or latitude and longitude. The force commander disseminates the FFA, including the effective dates and times of the FFA to all subordinate units. The FFA is enclosed by a black line with the words Free-Fire Area or the letters FFA written inside the circumscribed area. The establishing headquarters may be identified as part of the graphic portrayal.



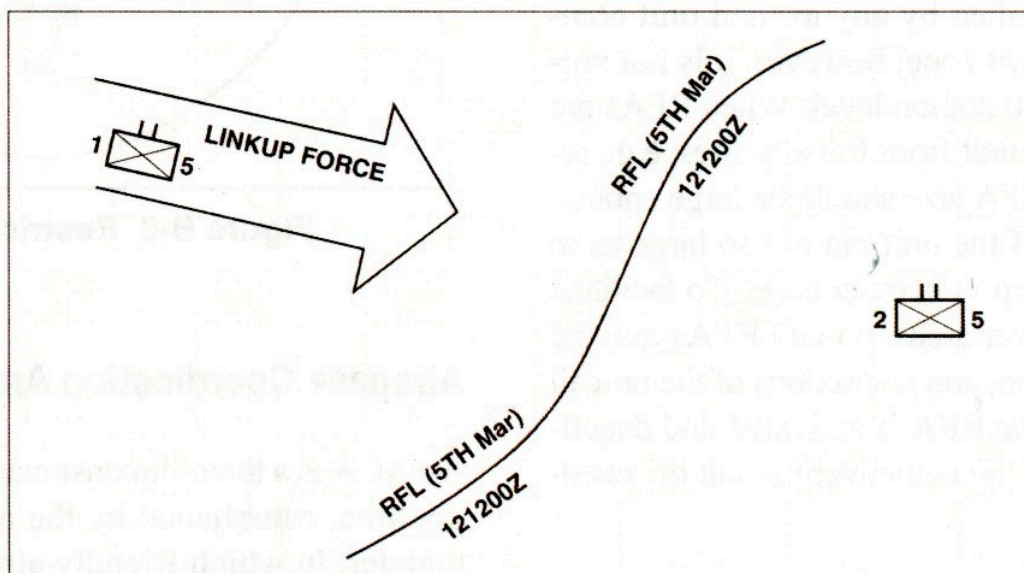
b. Restrictive Fire Support Control Measures

(1) Restrictive Fire Line (RFL)

A restrictive fire line (RFL) is a line established between converging friendly forces (one or both may be moving) that prohibits fires, or effects from fires, across the line without coordination with the affected force. It prevents fratricide between converging friendly forces.

The common commander of the converging forces establishes the RFL. They may delegate establishing authority to the senior commander of the two converging forces or to the commander of the maneuvering force in a linkup operation between a moving and a stationary force. To provide for recognition, the RFL should be located on identifiable terrain. In linkup operations, the RFL is moved as close as possible to the stationary force to allow maximum freedom of action for the maneuver and fire support of the linkup force.

The location of the RFL is graphically portrayed by a solid black line with the letters RFL followed by the name of the establishing headquarters above the line. The effective date-time group is written below the line.



(2) No-Fire Area (NFA)

An NFA is an area where no fires or effects of fires are allowed. Two exceptions are when establishing headquarters approves fires temporarily within the NFA on a mission basis, and when the enemy force within the NFA engages a friendly force, which the commander may engage the enemy to defend friendly forces.

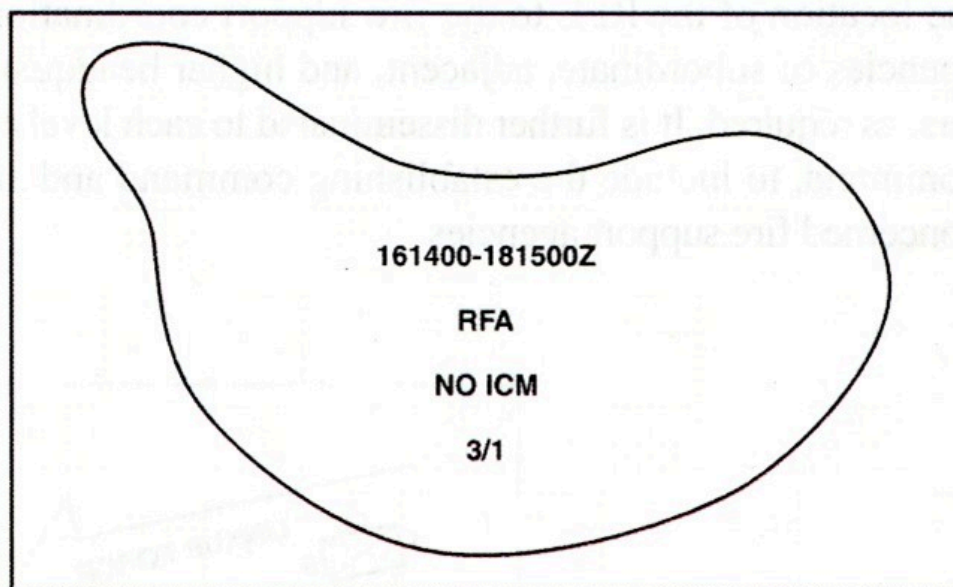
The NFA prohibits fires or their effects in the area, normally to protect civilians or cultural areas. Typically, the Navy amphibious commander/MAGTF commander or MEF establishes an NFA. On arrival of military forces, the force commander coordinates the location of an NFA with local authorities. Normally, an NFA is on recognizable terrain, but its location may also be expressed by grid coordinates or by radius in meters from a center point.



(3) Restrictive Fire Area (RFA)

A restrictive fire area (RFA) is an area in which specific restrictions are imposed and into which fires that exceed those restrictions will not be delivered without prior coordination with the establishing headquarters. The RFA regulates fires into an area according to the stated restrictions. This means that fires or certain types of ordnance (e.g., improved conventional weapon or white phosphorus) can be controlled where friendly forces are or will be located.

An RFA may be established by any ground unit commander within their own zone; however, it is not normally established below battalion level. An RFA may be on recognizable terrain, expressed by grid coordinates or by radius from a point. An RFA is portrayed by a solid black line defining the area with the letters RFA, the designation of the unit establishing the area, any special instructions, and the effective date-time group written inside the area.



(4) Airspace Coordination Area (ACA)

An ACA is a three-dimensional block of airspace in a target area, established by the appropriate ground commander, in which friendly aircraft are reasonably safe from friendly surface fires. The airspace coordination area may be formal or informal. Formal ACAs require detailed planning. More often, informal ACAs are established using time or lateral or altitude separation between surface- and air-delivered weapon

effects. ACAs can vary from physically defined areas to the various methods of separation. Physical areas can include routes (control point to initial point) or areas (over the target, initial point).



c. Boundaries as Fire Support Control Measures

Boundaries are the basic maneuver control measures used by commanders to designate the geographical area for which a particular unit is tactically responsible. They affect fire support in two ways:

- They are *restrictive* in that no fire support means may deliver fires across a boundary unless those fires are coordinated with the force having responsibility for the area within that boundary.
- They are *permissive* in that the maneuver commander has complete freedom of fire and maneuver within his/her boundaries, unless otherwise restricted by higher headquarters. Many times, boundaries negate the need for fire support coordination measures.

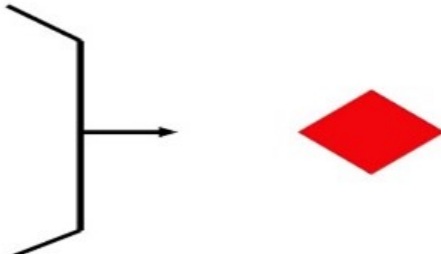
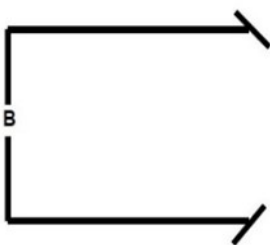
4. Tactical Tasks

Tactical tasks may be specified, implied, or essential. They define actions commanders may take to accomplish their missions. In special circumstances, commanders may modify tasks to meet situational requirements. They must clearly state that they are departing from the standard meaning of these tasks. One way this can be done is by prefacing the modified task with the statement, “What I mean by [modified task] is...”

a. Enemy-Oriented Tactical Tasks

The following tactical tasks focus friendly efforts on generating effects against enemy forces.

Task	Definition	Graphic
Ambush	A surprise attack from concealed positions on a moving or temporarily halted enemy.	

Attack by fire	Fires (direct or indirect) in the physical domains and/or through the information environment to engage the enemy from a distance to destroy, fix, neutralize, or suppress.	
Block	As a tactical task, to deny the enemy access to an area or prevent the enemy advance in a direction along an avenue of approach. It may be for a specified time. As an obstacle effect, to integrate fire planning and obstacle effort to stop an attacker along a specific avenue of approach or to prevent an attacker from passing through an engagement area.	
Breach	To break through or secure a passage through an obstacle.	 Orientation: The opening defines the span of the breach and typically faces enemy forces.
Bypass	To maneuver around an obstacle, position, or enemy force to maintain the momentum of the operation while deliberately avoiding combat with the enemy force.	
Canalize	To restrict enemy movement to a narrow zone by the use of existing or reinforcing obstacles, fires, or friendly maneuver.	
Contain	To stop, hold, or surround enemy forces or cause the enemy to center activity on a given front and prevent the withdrawal of any part of the enemy's forces for use elsewhere.	
Corrupt	To change, debase, or otherwise alter information from its original or correct form or version by intentionally introducing errors or altercations, thereby rendering it useless.	No Graphic
Deceive	To manipulate an enemy into believing and acting upon something that is not true for a selected period of time and/or at a particular location to create a friendly advantage.	No Graphic

Defeat	To disrupt or nullify the enemy commander's plan and overcome the will to fight, thus making the enemy commander unwilling or unable to pursue the adopted course of action and yield to the friendly commander's will.	No Graphic
Degrade	To diminish the effectiveness or efficiency of an enemy's C2 systems, and/or information collection efforts or means; lower the morale of an enemy unit; reduce a target's worth or value; and/or impair an enemy's decision-making capability.	No Graphic
Deny	To hinder or prevent the enemy from using terrain, space, personnel, supplies, facilities, and/or specific capabilities.	No Graphic
Destroy	To physically render an enemy force combat ineffective unless it can be reconstituted, or render a target or capability so damaged that it can neither function as intended nor be restored to a usable condition. Note: Defeat and destroy are not the same. Destruction of the enemy force normally leads to their defeat, but defeat does not necessarily require destruction.	
Disrupt	As a tactical task or effect, to employ or integrate fires and obstacles to break apart an enemy's formation and tempo, interrupt the enemy's timetable, or cause premature commitment or the piecemealing of enemy forces. In information operations, to prevent the efficient interaction of enemy combat and combat support systems by inflicting damage over the short term to specific facets of the system's operations.	
Exploit	To employ, to the greatest possible advantage, the success achieved in a military operation or enemy information that has come into friendly hands. Note: Exploitation is an offensive operation following a successful attack that is designed to disorganize the enemy in depth. It extends the initial success of the attack by preventing the enemy from disengaging, withdrawing, and reestablishing an effective defense.	No Graphic
Feint	Contact with the enemy to deceive the enemy about the locations or time of the actual main offensive action. Note: Feint is the counterpart to the type of attack.	

Fix	As a tactical task, to prevent the enemy from moving any part of its forces, either from a specific location or for a specific period of time, by holding or surrounding them to prevent their withdrawal for use elsewhere. As a tactical obstacle effect, to integrate fire planning and obstacle effort to slow an attacker within a specified area — normally an engagement area. Note: Fixing an enemy does not mean destroying it. However, the friendly force has to prevent the enemy from moving in any direction, which can be resource intensive.	
Influence	To cause an enemy to behave in a manner favorable to friendly forces.	No Graphic
Interdict	To divert, disrupt, delay, or destroy the enemy's surface military potential before it can be used effectively against friendly forces.	
Isolate	To seal off—both physically and psychologically - an enemy from sources of support, deny the enemy freedom of movement, and prevent that enemy force from having contact with other enemy forces.	
Neutralize	As a tactical task, to render the enemy or enemy resources ineffective or unstable. As an effect of fires delivered, to render a target ineffective or unusable, thereby degrading the enemy's capability of accomplishing its mission.	
Penetrate	To break through the enemy's defense and disrupt the enemy's defensive system. Note: The tactical task "penetrate" is the counterpart to the form of offensive maneuver that is known as "penetrate."	
Reconnoiter	To obtain, by visual observation or other methods, information about the activities and resources of an enemy or adversary.	No Graphic

Support by Fire	Movement to a position where the maneuver force can engage the enemy by direct fire in support of another maneuvering force. Note: Support by fire closely resembles the task of attack by fire. The difference is a unit conducting attack by fire only uses direct and indirect fires. A unit conducting support by fire uses direct and indirect fires to support the maneuver of another friendly force.	
<u>Note:</u> Support by fire closely resembles the task of attack by fire. The difference is a unit conducting attack by fire only uses direct and indirect fires. A unit conducting support by fire uses direct and indirect fires to support the maneuver of another friendly force.		
Suppress	The transient or temporary degradation of an opposing force or the performance of a weapons system below the level needed to fulfill its mission objectives.	No Graphic

b. Terrain-Oriented Tactical Tasks

The following tactical tasks focus friendly efforts on achieving some sort of condition as it relates to terrain.

<u>Task</u>	<u>Definition</u>	<u>Graphic</u>
Breach	To break through or secure a passage through an obstacle.	
Clear	To remove enemy forces and eliminate organized resistance in an assigned zone, area, or location by destroying, capturing, or forcing the withdrawal of enemy forces that could interfere with the unit's ability to accomplish the mission.	
Control	To maintain physical influence by occupation or range of weapon systems over the activities or access in a defined area. Note: Control differs the tactical task "secure" in that control prevents the movement of enemy ground forces through an area, but does not require the complete clearance of enemy forces or the prevention of enemy fires into that specified area.	No Graphic
Cordon	To prevent an enemy unit's withdrawal from or reinforcement to a position.	No Graphic

Occupy	To move to an objective, key terrain, or other manmade or natural area without opposition and control the entire area.	
Reconnoiter	To secure data, by visual observation or other methods, about the meteorological, hydrographic, or geographic characteristic of a particular area.	No Graphic
Retain	To occupy and hold a terrain feature to ensure it is free of enemy occupation or use.	
Secure	To gain possession of a position, terrain feature, piece of infrastructure, or civil asset with or without force, and prevent its destruction or loss by enemy action. The attacking force may or may not have to physically occupy the area.	
Seize	To clear a designated area and gain control of it. Note: The tactical task "seize" differs from occupy in that "seizure" occurs in the face of enemy opposition.	

c. Friendly-Oriented Tactical Tasks

The following tactical tasks focus friendly efforts on supporting the actions of other friendly forces.

<u>Task</u>	<u>Definition</u>	<u>Graphic</u>
Cover	To conduct offensive and defensive actions independent of the main body to protect the covered force and develop the situation. Note: It is the tactical task associated with the security operation "cover."	

Disengage	To break contact with the enemy and move to a point where the enemy cannot observe or engage friendly forces by direct fire.	No Graphic
Displace	To leave one position to take another while remaining in contact with the enemy. Note: Displace differs from the tactical task "disengage" in that units disengage to break contact with the enemy, while units displace to continue the mission or execute alternate missions.	No Graphic
Exfiltrate	To remove personnel or units from areas under enemy control by stealth.	No Graphic
Follow and Assume	A second committed force follows a force conducting an offensive operation and is prepared to continue the mission if the lead force is fixed, attrited, or unable to continue. Note: The follow-and-assume force is a not a reserve but is prepared to execute all missions of the followed unit.	
Follow and Support	A committed force follows and supports a lead force conducting an offensive operation. Note: The follow-and-support force is not a reserve but is a force committed to supporting the followed unit. The difference between follow and assume and follow and support is that the follow and assume force is prepared to take over the lead element's mission, whereas the follow and support force acts to create the conditions necessary to allow the lead element to continue its success (such as destroying bypassed elements, blocking enemy movement of reinforcements, clearing obstacles, or controlling dislocated civilians).	
Guard	To protect the main force by fighting to gain time while also observing and reporting information. Note: It is the tactical task associated with the security operation "guard."	
Protect	To prevent observation by engagement with or interference from an adversarial or enemy force, system, capability, or location.	No Graphic
Screen	To observe, identify, and report information, and only fight in self-protection.	

d. Population-Oriented Tactical Tasks

The following tactical tasks focus friendly efforts on achieving some sort of condition as it relates to the population within the area of operations.

<u>Task</u>	<u>Definition</u>	<u>Graphic</u>
Advise	To improve the individual and unit capabilities and capacities of host nation security forces through the development of personal and professional relationships between the U.S. host nation forces.	No Graphic
Assess the Population	To evaluate the nature, situation, and attitudes of a designated population or elements of a population inhabiting the area of operations.	No Graphic
Assist	To provide designated support or sustainment capabilities to host nation security forces to enable them to accomplish their objectives.	No Graphic
Build/Restore Infrastructure	To construct, rebuild, or repair local infrastructure to support the host nation and gain or maintain the cooperation of the local population.	No Graphic
Contain	To prevent or halt elements of a population or designated party from departing or projecting physical influence beyond a defined area. <i>See also enemy-oriented tactical tasks.</i>	No Graphic
Control	To use physical control measures and information operations to influence elements of the population or designated actors to respond as desired. <i>See also terrain-oriented tasks.</i>	No Graphic
Coordinate with Civil Authorities	To interact with, maintain communication, and harmonize friendly military activities with those of other interorganizational agencies and coalition partners to achieve unity of effort.	No Graphic
Cordon	To temporarily prevent movement to or from a prescribed area such as a neighborhood, city block, series of buildings, or other features. <i>See also terrain-oriented tasks.</i>	No Graphic
Enable Civil Authorities	To support or assist the host nation government and designated interorganizational agencies in providing effective governance.	No Graphic
Exclude	To prevent or halt elements of a population or designated party from entering or projecting physical influence into a defined area.	No Graphic
Influence	To persuade the local population, including potential and known adversaries, within the operational area to support, cooperate with, or at least accept the friendly force presence, and to dissuade the local population from interfering with operations. <i>See also enemy-oriented tactical tasks.</i>	No Graphic
Occupy	To move to an objective, key terrain, or other manmade or natural area without opposition and control the entire area. <i>See also terrain-oriented tactical tasks.</i>	

Reconnoiter	To obtain, by visual observation or other methods, information about civil considerations. <i>See also enemy- and terrain-oriented tactical tasks.</i>	No Graphic
Secure	To gain possession of a position, terrain feature, piece of infrastructure, or civil asset with or without force, and prevent its destruction or loss by enemy action. The attacking force may or may not have to physically occupy the area.	
Train	To teach designated skills or behaviors. To improve the individual and unit capabilities and capacities of host nation security forces.	No Graphic
Transition to Civil Control	To hand over civil government and security responsibilities from friendly force military authorities to legitimate civil authorities.	No Graphic

Lesson 10: Cognitive Bias

Lesson Hours: 0.50

1. Introduction

The complexity and ambiguity of the contemporary operational environment demands that Marine Corps leaders develop increased mental agility and embrace a refined variety of decision-making skills. These skills include an understanding of cognitive biases. Cognitive biases are mental errors caused by simplified information processing strategies. It is important to distinguish cognitive biases from other forms of bias, such as cultural bias, organizational bias, or bias that results from one's own self-interest. In other words, a cognitive bias does not result from any emotional or intellectual predisposition toward a certain judgment, but rather from subconscious mental procedures for processing information. A cognitive bias is a mental error that is consistent and predictable.

2. Educational Objectives

- Identify the definition of cognitive bias.
- Identify the differences between the two main cognitive systems.
- Identify the most common types of cognitive bias.
- Identify the most common causes of bias.

3. Key Terms

- Cognitive bias
- Areas of cognition
- Heuristics
- Confirmation bias
- Availability bias

4. Preparation

All quizzes, discussion forum posts, and assignments are executed in the Moodle course site.

a. Lesson Reading: [Cognitive Bias](#) (8 pages).

b. Supplemental References (optional):

- [*Measuring Human Capabilities: An Agenda for Basic Research on the Assessment of Individual and Group Performance Potential for Military Accession*](#). Washington, DC: The National Academies Press, (2015). This material is protected by copyright; further distribution is prohibited.
- [*Cognitive Biases: What They Are and How They Affect You*](#), <https://effectiviology.com/cognitive-biases/>, (2019). This material is protected by copyright; further distribution is prohibited.
- [*The Applied Critical Thinking Handbook, v7.0*](#), U.S. Army Training and Doctrine Command (2015).

Lesson 10 Reading

COGNITIVE BIAS

References:

- a) *Thinking Fast and Slow* (2011). This material is protected by copyright; further distribution is prohibited.
- b) *Measuring Human Capabilities: An Agenda for Basic Research on the Assessment of Individual and Group Performance Potential for Military Accession* (2015). This material is protected by copyright; further distribution is prohibited.
- c) *Cognitive Biases: What They Are and How They Affect You* (2019). This material is protected by copyright; further distribution is prohibited.
- d) *The Applied Critical Thinking Handbook* (2015).

1. Introduction

Decision-making at all levels is critical to the success of the Marine Corps. As the strategic and operational environment is increasingly characterized by ambiguity and complexity, military professionals are faced with an increasing variety of responsibilities. This emerging situation presents new challenges to decision-makers at all levels within the Marine Corps. As the quantity and variety of available information become more abundant, military professionals will be expected to make critical decisions under conditions of uncertainty. Research from the fields of cognitive psychology and behavioral economics tells us that under constrained conditions—time limitations, over-abundance of information, unfamiliar dynamics—people are more prone to make errors in judgment caused by cognitive biases. As such, it will be critical for the Marine Corps to train, educate, and develop its personnel to understand and cope with these cognitive biases appropriately. The consequences of not doing anything—that is, of making poor decisions—in the current and future operating environment are amplified due to expanding media and communication networks and increasingly sophisticated use by more and more people. In order for the military to retain overmatch and remain adaptive amid the unpredictable conditions of the future, its leaders will need to be able to negotiate complex circumstances and make effective and meaningful decisions. Just as the military commits substantial resources to develop appropriate new technologies to counteract emerging threats, it will be critical over the coming years for the Marine Corps to more clearly understand how individuals make decisions, what factors and dynamics influence these decisions, and how the military can foster better and more agile decision-making so that leaders are able to operate effectively in this new environment.

2. Cognitive Biases

A *cognitive bias* is a systematic pattern of deviation from rationality, which occurs due to the way our cognitive system works. Accordingly, cognitive biases cause us to be irrational in the way we search for, evaluate, interpret, judge, use, and remember information, as well as in the way we make decisions. For example, one well-known cognitive bias is the illusion of transparency, which causes us to believe that our thoughts and emotions are more apparent to others than is actually the case. We experience this bias because when we try to imagine how others see us, we struggle to properly shift away from our own internal perspective.

Cognitive biases affect our thought process in every area of life, from how we form our memories, to how we shape our beliefs, and to how we connect with other people. The patterns of irrationality that represent the various cognitive biases vary in terms of how they affect us and in terms of their scope. Accordingly, cognitive biases can lead to minor issues, such as forgetting a small detail from a past event, or they can lead to more serious issues, such as choosing to reject life-saving treatment due to a misguided belief in a pseudoscientific course of treatment.

Because cognitive biases have such a pervasive influence, it's important to understand them. In this reading, you will learn more about cognitive biases, understand why they affect us, see what types of them exist, and find out what you can do to mitigate them successfully.

3. Why We Have Cognitive Biases

In short, cognitive biases are caused by the imperfect way in which we process information and make decisions, and primarily as a result of our tendency to use mental shortcuts that allow us to think faster but also make us more likely to make mistakes.

a. Bounded Rationality

To understand why we have cognitive biases, it's important to first understand the concept of bounded rationality. Bounded rationality is the idea that our decision-making ability is constrained by the limitations of our cognitive systems and depends on the type of information that we have to process, as well as on the amount of time that we have to process it. This means that when we try to solve a problem, we often end up reaching a solution that is different from the one we would reach if our cognitive systems were perfect, or from the one that we would reach if we had more time to consider the problem.

Under this framework, people are seen as being bounded rationally and as satisficing. This means that we make decisions that are not necessarily optimal, since instead of looking for the best solution that is available for a certain problem, we generally tend to look for a solution that is perceived as good enough given the circumstances.

b. Heuristics

A common way in which we satisfice is by using heuristics, which are mental shortcuts that help us quickly reach a 'good enough' solution to our problems. For example, if we are asked how many apples there are in a certain basket, then instead of counting each apple individually, we might decide to estimate this amount by counting the number of apples in one section of the basket, and then multiplying that amount by the number of sections. Overall, this means that cognitive biases occur when we select a sub-optimal solution to a problem, due to the way that our cognitive systems work.

c. Cognitive Systems

Daniel Kahneman, who wrote *Thinking Fast and Slow*,¹ identifies two main cognitive systems at play here: System 1 and System 2.

(1) System 1

This cognitive system is responsible for our *intuitive processing*. It is *fast, automatic, associative, emotional, effortless*, and learns through repeated experiences and gradually over time. Processes in this system run in *parallel*, meaning that it is possible to engage this system on multiple fronts simultaneously. Strongly influenced by emotional considerations, System 1 generates involuntary expressions that do not need to be expressed in words. This type of thinking requires little effort but is often error prone. Most everyday activities (driving a vehicle, talking, cleaning, etc.) make heavy use of System 1 thinking. It is System 1 thinking that is responsible for many everyday decisions, judgments, and the purchases we make, and explains many of the heuristics (shortcuts or rules of thumb) that we use when making decisions. An example of a situation where this system is engaged is when we feel pleased because someone laughed at a joke that we told.

(2) System 2

This cognitive system is responsible for our *conscious reasoning*. It is *slow, logical, and effortful*. Processes in this system run in a *serial* way, meaning that this system can only focus on one thing at a time. This system is emotionally neutral, and therefore is not influenced by emotional considerations. System 2 thinking allocates attention to the effortful mental activities that demand it, including complex computations. We use System 2 thinking to make rational decisions. It is this slower system that retrieves mental data and weighs the pros and cons for us. An example of a situation where this system is engaged is when we try to solve a complex math problem in our heads. (See Figure 1.)



Figure 1: Daniel Kahneman's Systems of Thinking

d. Cognitive System Failures

One way in which the use of our cognitive systems can cause us to experience a cognitive bias is when we rely on our intuition (System 1) to make a decision that normally requires complex reasoning skills (System 2).

For example, in one experiment, students at Princeton University were asked a simple brain teaser:

"A bat and a ball cost \$1.10 in total. The bat costs \$1 more than the ball. How much does the ball cost?"

Almost anyone who hears this question feels an initial tendency to answer, "10 cents," because the total sum of \$1.10 separates naturally into \$1 and 10 cents, so that the answer "10 cents" sounds about right in terms of magnitude. The problem is that over half the people in the experiment ended up sticking with this initial estimate, leading them to answer this question incorrectly, since the right answer is that the bat costs \$1.05, while the ball costs \$0.05.

Essentially, what happens here is that most people make an intuitive assessment of the situation using System 1, which gives them a close but incorrect answer to the question. Some people then use System 2 to reassess this initial solution, which allows them to realize that it's incorrect, and which consequently leads them to calculate the correct solution to this question. However, other people fail to engage System 2 properly or at all, which means that they end up relying on the incorrect initial assessment that they got from System 1.

Of course, this does not mean that the intuitive assessments provided by System 1 are inherently faulty. Rather, our intuition can be a powerful and accurate tool, and there are many situations where our intuition provides us with the right solution to problems that we encounter. Instead, the issue here is that our intuitive system (System 1) usually requires monitoring by our conscious reasoning system (System 2) to identify and correct any errors that occur due to faulty intuitions.

Overall, based on this framework, cognitive biases occur primarily due to a failure of three cognitive mechanisms:

- Failure of System 1 to generate a correct intuitive impression. This means that System 1 gives us a quick but incorrect solution to our problem.
- Failure of System 2 to monitor impressions generated by System 1. This means that System 2 fails to notice and correct faulty impressions which are generated by System 1.
- Failure of System 2 to reason correctly. This means that System 2 gives us an incorrect solution to our problem, due to a failure of our conscious reasoning process.

4. Areas of Cognition

Different biases affect different areas of our cognition. Based on this criterion, some of the most common types of biases include:

a. Information Biases

These are biases that affect the way in which we acquire and process information. For example, the overkill effect is a cognitive bias that causes people to reject explanations that they perceive as too complex. This occurs because people generally prefer information that is easy for them to process from a cognitive perspective.

b. Belief Biases

These are biases that affect the way in which we form our beliefs. For example, the backfire effect is a cognitive bias that causes people to strengthen their support of their preexisting beliefs when they encounter evidence that shows that those beliefs are wrong. This occurs because when people argue strongly against unwelcome information, they end up with more arguments that support their original stance.

c. Decision-Making Biases

These are biases that affect the way we make decisions. For example, the bandwagon effect is a cognitive bias that causes people to do something because they believe that other people are doing the same thing. This occurs because people feel a need to conform and act in accordance with others, and because people often rely on other people's judgment when deciding how to act.

d. Calculation Biases

These are biases that affect the way in which we calculate things such as probabilities or values. For example, the gambler's fallacy is a cognitive bias that causes people to mistakenly believe that if something happens more frequently than normal during a given time period, then it will happen less frequently in the future. This occurs because people believe that a short sequence of random independent events should be representative of a longer sequence of such events.

e. Memory Biases

These are biases that affect the way our memory works. For example, rosy retrospection is a cognitive bias that causes people to remember past events as being more positive than they were in reality. This occurs because when people experience a certain event, they tend to have both positive and negative thoughts, but as time progresses, they are more likely to forget their negative thoughts than their positive ones.

f. Social Biases

These are biases that affect our social perception and behavior. For example, the spotlight effect is a cognitive bias that causes people to think that they are being observed and noticed by others more than they actually are. This occurs because people naturally see everything from their own point of view, so they struggle to accurately judge how they look through other people's eyes.

Note that many of these areas of cognition are interrelated, meaning that some biases can affect our cognition in several domains. For example, the bandwagon effect, which as we saw above causes people to conform to the attitudes of others, affects not only the way we make decisions, but also the way in which we form our beliefs, and the way in which we shape our social behavior. Another bias that can affect several areas of our cognition is the confirmation bias, which is a cognitive bias that causes people to search for, favor, interpret, and recall information in a way that confirms their preexisting beliefs. This bias can, for example, affect the way we acquire new information, as well as the way we remember old information, and the way we evaluate different choices.

5. Causes of Bias

Biases can occur due to various reasons, with the most common causes being:

a. Heuristics

These are biases that are caused by our use of mental shortcuts. For example, the representativeness heuristic is a heuristic that causes people to focus on how representative someone or something is of a particular class when assessing the likelihood that they belong to that class. This occurs because focusing on representativeness allows people to quickly assess the situation and reach a roughly correct conclusion in many cases, without them having to deal with complex background information.

b. Limited Cognitive Capacity

These are biases that are caused by our limited cognitive capacity. For example, the is a cognitive bias that causes people to forget information that they believe can be easily found online. This occurs because people sometimes prefer to rely on their ability to remember how to find information, rather than on their ability to remember the information itself.

c. Noisy Information Processing

These are biases that are caused by the effects of various background factors on the way we process information. For example, the humor effect is a cognitive bias that causes people to remember information better when that information is perceived as humorous. This occurs because humorous information benefits from increased attention and improved encoding, compared to non-humorous information.

d. Emotional Motivation

These are biases that are caused by various emotional considerations. For example, the Ben Franklin effect is a cognitive bias that causes people to like someone more after they do that person a favor. This occurs because people want to avoid cognitive dissonance, which could arise as a result of behaving in a favorable way towards someone that they either dislike or do not like enough.

e. Social Influence

These are biases that are caused by the influence of various social factors. For example, the outgroup homogeneity bias is a cognitive bias that causes people to view members of outside groups as being more similar to each other than members of groups that they are a part of. This occurs because people tend to allocate more of their attention to members of their own group, since interactions with those people are generally perceived as more important.

Note that some biases can occur due to a combination of reasons. For example, many heuristics-based biases occur due to our limited cognitive capacity, and many biases that have an emotional motivation are also affected by social factors.

6. Military Planning and Decision-Making

There are several cognitive biases that frequently affect military planning and decision-making.

a. Anchoring

Oftentimes, humans are unable to compute an item's true value. Instead, they resort to an overreliance upon an initial "anchor" value provided by someone else, and thereafter bias all subsequent value decisions relative to that initial "anchor."

b. Status Quo Bias

Many humans find the status quo comfortable and avoid changing it.

c. Confirmation Bias

As touched on above, a trap that humans often fall into is a tendency to look for evidence that supports the conclusion we have made prematurely, not realizing that evidence can often support several hypotheses. "No matter what we humans think about, we tend to pay more attention to stuff that fits in with our beliefs than stuff that might challenge them..." In the process, inclined to "see what we expect to see," we actively dismiss evidence that contradicts our conclusion. It is this phenomenon that lends importance to the active search for disconfirming evidence: evidence that would disprove the conclusion we have formed.

d. Sunk-Cost Bias ("Loss Aversion")

A bias in which humans increasingly persist in deciding and acting illogically, based upon decisions they made previously. This occurs despite the fact that the present context dictates deciding otherwise. Sunk-cost bias occurs because it relieves one of the necessity to admit that preceding decisions might have been made in error. It may also avoid incurring a permanent loss, at least in the short term. It is a failure to cut bait, to use a fishing metaphor.

e. Framing Trap

The way we frame an issue affects the way we perceive it, and hence, affects a solution's potential options.

f. Halo Effect

We tend to select that which we see as more attractive, regardless of actual capabilities or qualities. The opposite of the halo effect is the Pitchfork Effect: we tend to disregard (or put at the bottom of a list) that which we find unappealing.

g. Narrative Fallacy

The human compulsion to turn a series of connected or disconnected facts into story or pattern, inventing reality. This is a human tendency to construe meaning in a completely random situation, where no meaning actually exists.

h. Self-Fulfilling Prophecy

A term coined by Robert Merton in 1948. "The self-fulfilling prophecy is, in the beginning, a false definition of the situation evoking a new behavior which makes the originally false conception come true... the prophet will cite the actual course of events as proof that he was right from the beginning..."

7. Historical Examples of Cognitive Bias

The following examples illustrate how cognitive biases are at least partly responsible for errors in judgment that may have led to the incident. First, consider the Iran Air Flight 655 incident, July 3, 1988, in which an Iran Air civilian passenger flight was shot down by surface-to-air missiles fired from the *USS Vincennes* over the Persian Gulf, killing all 290 passengers on board. The *USS Vincennes* had previously (same day) been attacked by Iranian gunboats, of which two were sunk. The *USS Vincennes* was participating in *Operation Earnest Will*, in which U.S. Navy ships were protecting oil tankers in the Persian Gulf during the Iran-Iraq war (1980-1988).² The commanding officer had incorrectly acted upon the belief that the Iranian Airbus was actually an F-14 fighter from the Iranian Air Force; a belief developed in the context of a high-pressure situation with complicated, confusing, and contradictory information to be interpreted and reconciled within minutes. As tragedies like this often go, there were many factors that contributed to the mistake. A possible contributing cause, however, is certainly confirmation

bias, in which the context of high tensions and prior incidents in the area (including the 1987 incident in which an Iraqi jet determined to be nonhostile shot upon the *USS Stark*, killing 37 sailors and injuring 21 more) contributed to confirmatory thinking such that the evidence of a military aircraft was over weighted compared to the nonconfirmatory evidence of a civilian aircraft.³

Confirmation biases negatively affect decisions when individuals interpret information, including conflicting evidence, as confirmation of previously held beliefs. This is a tendency of special concern in situations where information is incomplete or unclear and critical decisions must be made under high levels of uncertainty, such as decisions that must be made in combat or by intelligence analysts.

Another example is provided by the 1973 Yom Kippur War between Israel and Egypt. Then Director of Military Intelligence for Israel, Major General Eli Zeira, was instrumental in developing *The Concept*, which captured assumptions about Egypt's military strategy. *The Concept* articulated that Egypt would only attack Israel if it had the air power generation capability to strike Israel's rear operating air bases, so as to reduce their air superiority. As Egypt would only have achieved this capability at least two years later, Zeira assessed that Egypt would not attack Israel. Little did Zeira know that Anwar Sadat, the Egyptian leader, sought to retake the Suez Canal, rather than defeat the Israeli forces. Therefore, the Egyptian forces' lack of air power was not a showstopper. On October 6, 1973, 100,000 Egyptian troops commenced their attack on the 450 Israeli soldiers along the Suez Canal. The Israelis were fundamentally surprised.

Along the way, Zeira committed several disconfirmation biases. Disconfirmation bias is the tendency for individuals to set more stringent standards of evidence for hypotheses other than their own. Before the attack, the Israelis had clear evidence of the Egyptians' attack plans and military deployment. However, Zeira discounted the possibility that Egyptian forces were massing for an attack and chose to situate the evidence in the context of the Egyptian forces conducting a training exercise, buying into the Egyptian forces' deception plan. This was despite the lack of evidence for actual training being conducted on the ground. Even when Jordan's King Hussein undertook a clandestine flight to Tel Aviv to warn the Israeli Prime Minister about Syria and Egypt's intention to attack Israel, Zeira was unmoved. Disturbingly, even when an operative at the highest levels of Egypt's government sent a warning to Israel just hours before the offensive, Zeira chose not to believe the report. It was proposed that his insistence on a flawed theory and playing down of contrary indicators contributed to his failed assessment of Egypt's intention to attack.³

In September 2009, a German commander ordered an airstrike on North Atlantic Treaty Organization fuel trucks that had been hijacked by Taliban forces. The hijacked trucks had gotten stuck while crossing the Kunduz River, and dozens of civilians had gathered around the trucks to siphon free fuel from the tankers. As a result, the airstrike killed up to 142 people, including over 100 Afghan civilians. Despite the fact that the trucks had not moved for more than four hours, the German commander had perceived an 'imminent threat' (this being a necessary condition for ordering an airstrike), noting that "my feeling was that if we let them get away with these tankers, they will prepare them to attack police stations or even the Provincial Reconstruction Team."⁵ The German commander seemed to have based his decision on recent and salient events. Indeed, Taliban fighters had detonated a tanker truck in Kandahar not too long ago. Moreover, there had been several other hijackings of the German Provincial Reconstruction Team's tanker trucks by the Taliban in Kunduz. In addition, just one month before the airstrike, intelligence had suggested that the Taliban was planning to overrun the German camp, using explosive-laden trucks. All these easily retrievable events could have contributed to the **availability bias** that affected the German commander's assessment of the seriousness and imminence of the threat that the hijacked trucks had posed, ultimately leading to his ill-fated decision.⁶

8. Summary

Cognitive biases are systematic patterns of deviation from rationality, that cause us to be irrational in the way that we search for, evaluate, interpret, judge, use, and remember information. We experience cognitive biases when we fail to correct faulty intuitive impressions, or when we fail to conduct a valid reasoning process. Both these issues occur as a result of the way that our cognitive systems work.

Cognitive biases can be categorized based on the area of cognition in which they affect us (e.g., decision-making or memory) and based on their cause (e.g., social influence or limited cognitive capacity). Biases can also be categorized based on a hot/cold distinction, with hot biases being motivated by emotional considerations and cold biases being driven by emotionally neutral processes.

Cognitive biases can affect us negatively in many areas of life when they cause us to find sub-optimal solutions to our problems. However, biases can sometimes be beneficial, such as when they help us find quick solutions to our problems. It is possible to debias yourself and others successfully through the use of metacognitive strategies, which can help you conduct a valid reasoning process. Furthermore, you can reduce the number of biases that you experience by training yourself to form better intuitive impressions.

Notes

¹ Kahneman, D. (2011). *Thinking Fast and Slow*. Macmillan.

² Marvin, Andrew R. (2020) "Operation Earnest Will—The U.S. Foreign Policy behind U.S. Naval Operations in the Persian Gulf 1987–89; A Curious Case," *Naval War College Review*: Vol. 73: No. 2, Article 8. Available at: <https://digital-commons.usnwc.edu/nwc-review/vol73/iss2/8>.

³ "Cognitive Biases." National Research Council. 2015. *Measuring Human Capabilities: An Agenda for Basic Research on the Assessment of Individual Group Performance Potential for Military Accession*. Washington, DC: The National Academies Press.

⁴ CPT Chen Jing Kai "Cognitive Biases – The Root of Irrationality in Military Decision-Making," CPT Chen Jing Kai, POINTER, *Journal of the Singapore Armed Forces*, Vol 42., No. 2, 2016.

⁵ *Washington Post*. Chandrasekaran, Rajiv. "Decision on Airstrike in Afghanistan Was Based Largely on Sole Informant's Assessment." 2009. <http://www.washingtonpost.com/wpdyn/content/article/2009/09/05/AR2009090502832.html?hpid=topnews>. (Permission for Use is Authorized: Only staff-produced materials may be used. Rajiv Chandrasekaran was a staff writer for the Washington Post in 2009. He is currently a senior editor and associate editor for the Washington Post.)

⁶ CPT Chen Jing Kai "Cognitive Biases – The Root of Irrationality in Military Decision-Making," CPT Chen Jing Kai, POINTER, *Journal of the Singapore Armed Forces*, Vol 42., No. 2, 2016.

Lesson 11: Accessing Doctrine

Lesson Hours: 0.50

1. Introduction

This lesson provides an introduction to Joint and Service doctrinal publications and provides the links to access the respective doctrine. When using and referencing doctrinal publications, it is critical to have the current version; this lesson provides information that will help you determine whether or not you have the current version of a doctrinal publication.

2. Educational Objectives

- a. Identify the purpose of joint doctrine.
- b. Identify the organization of joint doctrine.
- c. Identify the structure of U.S. Army doctrine.
- d. Identify the structure of U.S. Air Force doctrine.
- e. Identify the structure of U.S. Navy doctrine.
- f. Identify the structure of U.S. Marine Corps doctrine.

3. Key Terms

- Joint doctrine
- Service doctrine
- Multinational doctrine
- Marine Corps Warfighting Publication
- Marine Corps Reference Publication
- Marine Corps Tactical Publication
- Tactics
- Techniques
- Procedures

4. Preparation

All quizzes, discussion forum posts, and assignments are executed in the Moodle course site.

- a. Lesson Reading: [Accessing Doctrine](#) (7 pages).
- b. Supplemental References (optional):
 - [Chairman Joint Chiefs of Staff Instruction 5120.02G, Joint Doctrine Development System](#) (13 Nov 2025).
 - [Marine Corps Order 5600.20S w/ Change 1, Marine Corps Doctrinal Publications System](#) (28 May 2025).

- [Marine Corps Order 5600.48D, United States Marine Corps Procedures for Participation in the Development of Joint Doctrine and NATO Allied Doctrine](#) (14 Sep 2023).
- [Naval Doctrine Publication 1, Naval Warfare](#) (Apr 2020).
- [Army Doctrinal Publication 1-01, Doctrine Primer](#) (Jul 2019).
- [Air Force Doctrinal Publication 1, The Air Force](#) (10 Mar 2021).
- [MARADMIN 452/20, Implementing the Vision for Marine Corps Doctrine and Training Publications](#) (6 Aug 2020).

Lesson 11 Reading

ACCESSING DOCTRINE

References:

- a) Chairman Joint Chiefs of Staff Instruction 5120.02G, *Joint Doctrine Development System* (13 Nov 2025)
- b) Marine Corps Order 5600.20S w/ Change 1, *Marine Corps Doctrinal Publications System* (28 May 2025)
- c) Marine Corps Order 5600.48D, *United States Marine Corps Procedures for Participation in the Development of Joint Doctrine and NATO Allied Doctrine* (14 Sep 2023)
- d) Naval Doctrine Publication 1, *Naval Warfare* (Apr 2020)
- e) Army Doctrinal Publication 1-01, *Doctrine Primer* (Jul 2014)
- f) Air Force Doctrinal Publication 1, *The Air Force* (10 Mar 2021)
- g) MARADMIN 452/20, "Implementing the Vision for Marine Corps Doctrine and Training Publications" (6 Aug 2020)

1. Introduction

Doctrine serves as a starting point for thinking about and conducting operations. When military officers allow it to fulfill this role, doctrine makes five basic contributions to the conduct of operations and the development of military professionals. Each is vitally and equally important, and each contributes directly to the conduct of operations and mission effectiveness. Doctrine's contributions are as follows:

- Provides a coherent vision of warfare
- Enhances operational effectiveness
- Provides a common frame of reference and cultural perspective
- Provides a common professional language
- Allows for study and reflection

2. Currency

One of the challenges of using military doctrine is currency: Do you have the most current version of the doctrine? It is important to know that doctrine has never been static, nor should it be. It must change to account for evolving circumstances. While some publications have had longer "shelf lives" than others (for example, MCDP 1), all are subject to change and almost all have been changed over their lifespan. Doctrine also changes when practice changes. Changes in practice are driven by changes in the operational environment. Changes occur for any number of reasons: changing threats, technology, organizational design, and new and effective ways to perform tasks. For the Marine Corps, a significant change came in May 2016 when Marine Corps doctrine was reorganized with a new hierarchy to better reflect the Marine Corps' warfighting focus on the 21st century. This new doctrine hierarchy is included in this lesson. It is important to understand that most Marine Corps doctrine was renamed. For example, MCWP 2-1, *Intelligence Operations*, was redesignated MCWP 2-10. A view of the cover indicates the change.



Furthermore, the name change was identified through the use of an Erratum page with the date of the name change (2 May 2016).



This publication also has one change, so the correct title of the publication would be MCWP 2-10 w/change 1, *Intelligence Operations*. The date of change was 4 April 2018.





Finally, we can see when the document was originally published by looking at the date on the Forward page. *Intelligence Operations* was originally published on 10 Sept 2003. So, MCWP 2-10 w/change 1, *Intelligence Operations*, is current as of 4 April 2018.

The next doctrinal publication to be used as an example is MCDP 1-0 w/changes 1, 2, and 3, *Marine Corps Operations*. MCDP 1-0 was published in September 2001. Since then, this publication has had three changes. Let us take a look at these. Notice that the changes are identified on the cover.





In order to have the most current version, you must have the version with all three changes. This publication is a little confusing because Change 2 is dated 29 March 2019 and Change 3 is dated 4 April 2018.

It is also important to check the date of the original publication. The current version of MCDP 1-0, *Marine Corps Operations* was signed on 9 August 2011 and superseded the first version dated September 2001. This date is listed on the Forward of the publication.



3. Types of Doctrine

Marine Corps doctrine exists within a larger set of doctrine publications. In theory, this is also a hierarchy in which the higher-level doctrine should drive the lower-level doctrine. Lower-level doctrine should, in turn, be consistent with the higher-level doctrine. There are, however, exceptions to this. Sometimes, lower-level doctrine drives higher-level doctrine.

The Marine Corps recognizes four general types of doctrine:

- Joint doctrine
- Multinational doctrine
- Multi-Service Doctrine
- Service Doctrine

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a. Joint Doctrine

Joint doctrine contains fundamental principles that guide the employment of U.S. military forces in coordinated action toward a common objective and may include terms, tactics, techniques, and procedures. By Federal Law, the Chairman, Joint Chiefs of Staff (CJCS) is responsible for developing doctrine for the joint employment of the armed forces. Joint doctrine broadly covers the strategic and operational levels of war and is (only by exception) tactical in nature. U.S. joint doctrine provides the overarching joint context for multi-Service doctrine.

Joint doctrine is written to reflect existing capabilities and roles, missions, and functions as delineated in Title 10, United States Code, and DOD Directive 5100.01, *Functions of the Department of Defense and Its Major Components*. The DOD Directive tasks the Marine Corps to assist the CJCS in the development of joint doctrine. The Marine Corps has been designated the lead agent (LA) for two joint publications (JPs). These JPs are:

- JP 3-25, *Countering Threat Networks* (28 Feb 2023)
- JP 3-68, *Joint Noncombatant Evacuation Operations* (26 May 2022)

The Marine Corps is also the technical review authority (TRA) for the following JPs:

- JP 3-02, *Amphibious Operations* (29 Apr 2025)
- JP 3-20, *Security Cooperation* (5 Jul 2024)
- JP 3-24, *Counterinsurgency* (30 Apr 2021)
- JP 3-32, *Joint Maritime Operations* (4 Dec 2023)
- JP 3-42, *Joint Explosive Ordnance Disposal* (14 Sep 2022)

Joint doctrine is organized into series based on the predominant content:

- Capstone Series
- Reference Series
- 1-0 Personnel Series
- 2-0 Intelligence Series

- 3-0 Operations Series
- 4-0 Logistics Series
- 5-0 Planning Series
- 6-0 Communication Series

Joint doctrine can be accessed at the following link: <https://www.jcs.mil/Doctrine>

b. Multi-National Doctrine

The Marine Corps also recognizes doctrine developed in coordination with other countries.

(1) Allied Joint Publications (AJP)

AJPs address the doctrine of military forces of allied nations in the North Atlantic Treaty Organization (NATO) and are targeted at headquarters at a joint level. Their development is closely monitored by the Joint Staff, staffed to all Services and deviations or conflicts with US joint doctrine are identified. Where appropriate, reservations specifying a restricted use of an AJP are included in the US ratification process. AJPs provide the doctrinal basis for NATO operations and will be widely used in NATO operations. Most NATO nations use AJPs as their own national joint doctrine.

(2) Allied Tactical Publications (ATP)

These address doctrine and TTP for land, air, and maritime forces and consist of a set of over 300 publications. They are developed in working groups by subject matter experts from NATO nations (including the U.S.) and may be used when there is not a Service doctrinal publication on the topic. Certain ATPs are directly used by U.S. forces, especially when allied interoperability is a major factor. Rather than having each U.S. Service maintain its own publication, all efforts are devoted to ensuring the ATP is accurate and current. Do not confuse these publications with Army Techniques Publications discussed below.

(3) American, British, Canadian, Australian, and New Zealand (ABCANZ) Armies' Program Publications

A select number of ABCANZ publications have been identified for inclusion on the Marine Corps doctrine hierarchy. They generally provide guidance on operating with allied or coalition forces.

NATO publications may be accessed at: <https://nso.nato.int/nso/nsdd/main/list-promulg>

c. Dual-Designated; Multi-Service; and Air, Land, and Sea Application (ALSA) Center Publications

Each Service is authorized to use other Services' doctrinal publications and to effect necessary modifications to align each with their own mission, roles, and functions. These publications, identified as dual-designated or multi-Service publications, contain tactics, techniques, and procedures to which two or more of the Services have subscribed. They are ratified by and bear the designation of each participating military Service. Dual designated or multi-Service publications ratified by the Marine Corps are included in the Marine Corps doctrine hierarchy, given a Marine Corps designation, and have the same authority as other Marine Corps warfighting or reference publications. An example is ATP 2-01.3/MCRP 2-10B.1, *Intelligence Preparation of the Battlefield/Battlespace*, 1 Jul 2023.

The ALSA Center is chartered by the Services' doctrine development agencies and tasked to rapidly develop publications that fill interoperability voids between Services, joint forces, and staffs. In most cases, these publications take on the form of multi-Service tactics, techniques, and procedures (MTTP) and address operational details often not included in our joint publications. Once signed by the Deputy Commandant for Combat Development and Integration (DC C&I) or designated representative (Director, Capabilities Development Directorate), ALSA publications are given an appropriate Marine Corps designator and placed within the Marine Corps' publication hierarchy. As with non-ALSA dual-designated publications, these MTTPs have the same authority and status as our other Marine Corps doctrinal publications. Although developed by the ALSA Center, ALSA publications must be approved and signed by the individual Service doctrine commands. ALSA publications can be accessed at: <https://www.alssa.mil/>. An example of a Marine Corps approved ALSA publication is MCTP 3-03B, *Peace Operations*.

d. U.S. Navy Doctrine

The U.S. Navy is charged with developing and publishing doctrine for naval, air, surface, and subsurface warfare. Naval Doctrinal Publications (NDPs) and Naval Warfare Publications (NWP) are the designators for Navy doctrine. Navy doctrine also includes Navy Tactical Techniques and Procedures (NTTPs). An example of an NTTP is NTTP 3-13.3M, *Operational Security*, which is dual designated as MCTP 3-32B. Navy Warfare Development Command (NWDC) is the executive agent for the Navy Doctrine Library System (NDLS) that is used to distribute doctrine to fleet users and to manage the content of the entire doctrine library. There are six NDPs:

- NDP 1: *Naval Doctrine*
- NDP 2: *Naval Intelligence*
- NDP 3: *Naval Operations*
- NDP 4: *Naval Logistics*
- NDP 5: *Naval Planning*

- NDP 6: *Naval Command and Control*

U.S. Navy doctrine may be accessed at: <https://www.nwdc.navy.mil/site/doctrine.html> (.mil and .gov only)

e. U.S. Army Doctrine

U.S. Army doctrine's current structure places all Army doctrine into four types of publications, each with a different purpose:

(1) Army doctrinal publications (ADPs)

ADPs contain the fundamental principles by which operating forces and elements of the generating force that directly support operations guide their actions in support of national objectives. An Army doctrine publication provides the intellectual underpinnings of how the Army operates as a force.

(2) Army doctrinal reference publications (ADRP)

When necessary, an Army doctrine publication has a corresponding ADRP that provides a more detailed explanation of the principles contained in the related ADP.

(3) Field manuals (FMs)

FMs contain principles, tactics, procedures, and other doctrinal information. They describe how the Army and its organizations conduct and train for operations. FMs describe how the Army executes operations described in the ADPs. They fully integrate and comply with the doctrine in the ADPs and ADRPs.

(4) Army technical publications (ATPs)

ATPs contain techniques. These publications fully integrate, nest, and comply with the doctrine contained in ADPs, ADRPs, and FMs.

U.S. Army doctrine can be accessed at: <https://armypubs.army.mil/ProductMaps/PubForm/ADP.aspx>

f. U.S. Air Force Doctrine

As implemented in the Air Force, doctrine affects operations at three levels: basic, operational, and tactical. These levels speak to the intellectual content of the doctrinal concepts, not to the architectural structure of doctrine publications.

(1) Basic Doctrine

Basic doctrine states the most fundamental and enduring beliefs that describe and guide the proper use, presentation, and organization of forces in military action. It describes the "elemental properties" of airpower and provides the Airman's perspective. Because of its fundamental and enduring character, basic doctrine provides broad and continuing guidance on how Air Force forces are organized, employed, equipped, and sustained. Because it expresses broad, enduring fundamentals, basic doctrine changes relatively slowly compared to the other levels of doctrine. As the foundation of all doctrine, basic doctrine also sets the tone and vision for doctrine development for the future. *Air Force Doctrine* Volume 1 is the Air Force's basic doctrine publication. Volume 2 is *Leadership*, and Volume 3 is *Command*.

(2) Operational Doctrine

Operational doctrine contained in doctrine annexes describes a more detailed organization of forces and applies the principles of basic doctrine to military actions. Operational doctrine guides the proper organization and employment of air, space, and cyberspace forces in the context of distinct objectives, force capabilities, broad functional areas, and operational environments. Operational doctrine provides the focus for developing the missions and tasks to be executed through tactical doctrine. Doctrine at this level changes a bit more rapidly than basic doctrine, but usually only after deliberate internal Service debate.

(3) Tactical Doctrine

Tactical doctrine describes the proper employment of specific Air Force assets, individually or in concert with other assets, to accomplish detailed objectives. Tactical doctrine considers particular objectives (stopping the advance of an armored column) and conditions (threats, weather, and terrain) and describes how Air Force assets are employed to accomplish the tactical objective (B-1 bombers dropping anti-armor cluster munitions). Air Force tactical doctrine is codified as tactics, techniques, and procedures (TTP) in Air Force TTP (AFTTP) -3 series manuals. Because tactical doctrine is closely associated with the employment of technology and emerging tactics, change will likely occur more rapidly than other levels of doctrine. Also, due to their sensitive nature, many TTPs are classified.

U.S. Air Force doctrine can be accessed at: <https://www.doctrine.af.mil/>

g. U.S. Marine Corps Doctrine

Marine Corps Service doctrine consists of publications created to describe the warfighting doctrine, tactics, techniques, and procedures of the Marine Corps. Doctrine provides the fundamental principles by which our forces guide their actions in support of national objectives. Tactics, techniques, and procedures provide the detailed description of how our forces perform assigned tasks.

(1) Tactics, Techniques, and Procedures

Although frequently (and usually inappropriately) lumped together into the acronym "TTPs," tactics, techniques, and procedures are distinct with different levels of prescription.

(a) Tactics. Tactics are the employment and ordered arrangement of forces in relation to each other (DOD Dictionary of Military and Associated Terms). It includes the ordered arrangement and maneuver of units in relation to each other, the terrain, and the enemy in order to translate potential combat power into winning battles and mission success. Effective tactics translate combat power into decisive results. Tactics vary with terrain and other circumstances; they change frequently as the enemy reacts and friendly forces explore new approaches. Applying tactics usually entails acting under time constraints with incomplete information. Tactics always require judgment in application; they are always descriptive, not prescriptive. In a general sense, tactics concern the application of the tasks associated with offensive, defensive, stability, or defense support of civil authorities' operations. Employing a tactic may require using and integrating several techniques and procedures. Tactics are primarily contained in Marine Corps warfighting publications (MCWPs) and Marine Corps tactical publications (MCTPs).

(b) Techniques. Techniques are non-prescriptive ways or methods used to perform missions, functions, or tasks. They are based on accepted doctrine principles and tactics. They are primarily contained in MCRPs.

(c) Procedures. Procedures are standardized, detailed steps that prescribe how to perform specific tasks (DOD Dictionary of Military and Associated Terms). They include formats for orders and reports as well as control measures. They are prescriptive. Procedures consist of a series of steps in a set order that are executed the same way, at all times, regardless of circumstances. Procedures require stringent adherence to steps without variance. Failure to follow procedures may result in serious injury, loss of life, or loss of property. Procedures are usually contained in the appendices of MCWPs, MCTPs, and MCRPs. Publications that contain a preponderance of procedures are not doctrine and are considered training, technical-related, or safety publications.

(2) Designation of Publications

Marine Corps doctrinal publications are designated as one of the following (as directed by Marine Corps Order 5600.20):

(a) Marine Corps Doctrinal Publication (MCDP). MCDPs are promulgated publications that are the highest order of Marine Corps Service doctrine. They contain the fundamental principles and institutional thinking regarding warfighting and the conduct of major warfighting activities. Collectively, MCDPs are the philosophical underpinning of the Marine Corps and the Marine ethos. MCDPs cannot be partnered with another Service. The CMC is the signatory authority for all MCDPs, and all MCDPs are integrated (managed) by the Deputy Commandant for Combat Development and Integration (DC CD&I). MCDPs are also written in a way that can be understood by all Marines. MCDPs are to be assessed for information currency and accuracy by the respective proponent lead a minimum of every ten years.

(b) Marine Corps Warfighting Publication (MCWP). MCWPs are publications that are the overarching and fundamental operational doctrine and TTP utilized by the Marine Corps in the prosecution of war or other assigned missions. Collectively, MCWPs represent the operational foundation of how the Marine Corps fights and prosecutes its missions in support of national strategy and objectives. MCWPs inform and influence the content of their aligned MCTPs and MCRPs. MCWPs may or may not be partnered with another Service. The DC CD&I or CMC is the signatory authority for all MCWPs. All MCWPs must have proponent leads within CDD. MCWPs are to be assessed for information currency and accuracy by the respective proponent lead a minimum of every six years.

(c) Marine Corps Tactical Publication (MCTP). MCTPs are publications that contain focused or specific TTPs that relate to a specific function, area, or subject. Each MCTP falls under an MCWP and must follow, align, amplify, and further detail the doctrine discussed in its respective MCWP. MCTPs may or may not be partnered with another Service. DC, CD&I is the signatory authority for all MCTPs. MCTPs are to be assessed for information currency and accuracy by the respective proponent lead a minimum of every six years. MCTPs are given the same number as the parent MCWP, followed by a letter. Example: MCWP 3-03, *Stability Operations*; MCTP 3-03A, *MAGTF Civil Military Operations*.

(d) Marine Corps Reference Publications (MCRP). MCRPs are publications that primarily focus on TTP used by small unit leaders or individual Marines to ensure mission success. Some MCRPs have application across the entire Marine Corps doctrine hierarchy, providing harmony and understanding of organizations, and terms and techniques to be understood and used by all Marines. They may or may not have a respective MCTP or MCWP that they align to, and MCRPs may or may not be partnered with another Service. MCRPs do not contain unit or individual training standards. Training standards are properly located in training and readiness manuals but are based on accepted doctrine. MCRPs are also not procedural publications that are prescriptive in nature. Non-doctrinal manuals that include procedures throughout the text such as construction procedures, rigging, physics, or explosives procedures will not be considered for inclusion in the Marine Corps doctrine hierarchy. These are prescriptive training and operational materials. The DC CD&I is the signatory authority for all

MCRPs. MCRPs are to be assessed for information currency and accuracy by the respective proponent lead a minimum of every four years. MCRPs are given the same number as the parent MCWP/MCTP, followed by a decimal and a single digit (or double if more than nine in grouping). Example: MCWP 3-03, *Stability Operations*; MCRP 3-03.2, *MTTP For Integrated Money Shaping Operations*.

(e) Marine Corps Interim Publications (MCIPs). MCIPs are publications that introduce new or emerging doctrine. The overall purpose of an MCIP is to disseminate new TTP, based on findings from lessons learned, training, and experimentation. MCIPs are not to be used for concepts. Concepts and concepts of operations are used to inform the Marine Corps of an envisioned set of capabilities for future operations, and not current best practices and lessons learned. The DC CD&I is the signatory authority for all MCIPs. MCIPs usually contain TTP but may contain discussions of principles. Proponents request authorization to develop an MCIP using the same process by which they request authorization for an MCWP or MCRP. Information disseminated via an MCIP should be incorporated into a new or revised MCWP or MCRP that supersedes or cancels the MCIP. An MCIP will expire three years after its publication date unless it is extended, superseded, or cancelled earlier. Proponents may prepare an MCIP to meet an immediate doctrine need if issuing a change to an existing MCWP or MCRP is inappropriate. Once promulgated, the doctrine in the MCIP is approved for interim use in training and operations. The numbering of an MCIP will end in an "i" when it is within the initial three-year period.

Marine Corps doctrinal publications are further organized by series, as described below:

- 1-0 Series: Organization and Standard
- 2-0 Series: Intelligence
- 3-0 Series: MAGTF
- 4-0 Series: Logistics
- 5-0 Series: Planning
- 6-0 Series: Leadership
- 7-0 Series: Training and Education
- 8-0 Series: Information
- 10-0 Series: Protection
- 11-0 Series: Legal
- 12-0 Series: Environments
- 13-0 Series: Naval
- MCDPs: Capstone
- FMFMs & FMFMRPs: Historical

U.S. Marine Corps doctrine can be accessed at: <https://www.marines.mil/News/Publications/MCPEL/> and https://usmc.sharepoint-mil.us/sites/MCEN_Support_MCDoctrine.

h. U.S. Coast Guard Doctrine

U.S. Coast Guard doctrine may be accessed at: <https://www.dcms.uscg.mil/Our-Organization/Assistant-Commandant-for-C4IT-CG-6/The-Office-of-Information-Management-CG-61/aboutCGDS/>

i. U.S. Space Force Doctrine

U.S. Space Force doctrine may be accessed at: <https://www.starcom.spaceforce.mil/About-Us/STARCOM-Deltas/Space-Delta-10-Doctrine-Wargaming/>

4. Summary

The principles discussed within doctrine are generally enduring, yet they may evolve based on policy and strategy, new technology, and/or organizations, and from lessons gained from experience and from insights derived from operational assessment. The focus of doctrine is on *how* to think about operations, not *what* to think about operations. Doctrine provides a basis for analyzing the mission and its objectives and tasks and for developing the commander's intent and associated planning guidance. It provides a foundation for training and education. The success of a military force is directly associated with how well its doctrine—

- Captures and addresses lessons learned and vetted conceptual thinking.
- Addresses current challenges.
- Addresses current capabilities.
- Is understood and inculcated into the thought processes of the forces.

Thus, doctrine is a shared way of thinking that is authoritative, but not directive in nature. It is a starting point from which we develop solutions and options to address specific demands and challenges. Adherence to doctrine provides the basic vernacular with which the Services can communicate. By providing the how in general terms, we gain a degree of standardization without relinquishing freedom of judgment and the commander's requirement to exercise initiative.

Lesson 12: Human Performance

Lesson Hours: 0.50

1. Introduction

There are numerous components that contribute to optimal human performance. Marine Corps human performance requires a holistic approach to physical and mental fitness. Physical fitness reflects a Marine's ability to perform the tasks assigned individually and as a unit, both in garrison and while deployed. Physical training is part of and contributes to physical fitness and the ability to perform tasks, but it also includes compliance with body composition standards, managing fatigue, maintenance of good health, and making lifestyle decisions that support health and physical ability to perform work. Mental fitness incorporates a Marine's self-confidence, strong coping, and decision-making skills. A fit Marine engages in healthy behaviors that help maintain focus and perseverance, which enables them to successfully perform their duties in pursuit of organizational mission accomplishment.

The lesson introduces some of the components required to optimize a Marine's level of human performance. These components include physical fitness, nutrition, and injury prevention. A balanced approach requires consideration of these components for optimum performance enhancements and combat conditioning.

2. Educational Objectives

- a. Identify the four components of physical fitness.
- b. Identify the principles of the Strength and Conditioning Program.
- c. Identify the components of the Dynamic Warm-Up.
- d. Identify the four phases of Strength Training.
- e. Identify the methods that can be used to enhance speed and agility.
- f. Identify the various factors associated with flexibility and mobility.
- g. Identify the best practices for injury prevention.
- h. Identify the nutritional requirements for optimal human performance.

3. Key Terms

- Endurance
- Mobility
- Strength
- Flexibility
- Dynamic Warm-Up
- Energy Balance

4. Preparation

All quizzes, discussion forum posts, and assignments are executed in the Moodle course site.

- a. Lesson Reading: [Human Performance](#) (12 pages).
- b. Other Requirements: None.
- c. Supplemental References (optional):
 - [Total Force Fitness for the 21st Century: A New Paradigm](#), *Military Medicine*, Vol. 75 No. 8, Supplement (2010). This material is protected by copyright; further distribution is prohibited.

- [*FMF Physical Training Playbook Template*](#) (13 pages).
- [*USMC Warfighter Guide to Performance Nutrition*](#) (55 pages).

Lesson 12 Reading

HUMAN PERFORMANCE

References:

- a) Force Fitness Instructor Course, *Fitness Methodologies, Performance Nutrition, and Injury Prevention* (2019)
- b) *Total Force Fitness for the 21st Century: A New Paradigm* (2010)

1. Physical Fitness Methodologies

Military fitness programs need to be based on the mission task-oriented fitness philosophy. In fact, the most recent physical fitness guidance from the Army and Marine Corps places an emphasis on this philosophy. It isolates individual tasks performed by the service member during a mission and breaks them down into their physical components. An example of this task breakdown is when a squad must patrol through a city, sprint through the market, kick down a door, and steady their weapons to fire. Many aspects of fitness are being utilized to accomplish this mission. The service member must possess the aerobic endurance to walk the distance of the patrol while wearing combat gear. Sprinting through the marketplace and jumping through the window require not only the anaerobic ability to sprint, but also the ability to change directions quickly to negotiate obstacles. Kicking down the door requires strength. The service member's overall endurance must be high enough to allow him or her to accomplish all these tasks without fatiguing. Failure to accomplish any task in the sequence puts the service member in danger and could result in death. Therefore, sub-optimal or poor physical fitness in service members is much more serious than in an athlete, and training should reflect this.

a. Components of Physical Fitness

Physical fitness training can be split into four major components, allowing for the diversity of training necessary to accomplish mission tasks: endurance, mobility, strength, and flexibility. Endurance is the body's ability to continually accomplish the same task in a repetitive fashion (e.g., loading supplies onto a ship). Mobility is the ability to move the body in space with the precision necessary to negotiate an obstacle (e.g., climbing a wall). Strength is the ability to generate force attempting to overcome resistance (e.g., lifting an ammunition can). Finally, flexibility is possessing the optimal joint range of motion (e.g., reaching overhead). All four aspects of physical fitness are important and essential in creating a physically well-balanced, injury-resistant, and mission-fit service member.

(1) Endurance

Endurance is the body's ability to continually accomplish the same task in a repetitive fashion. Proper endurance training will increase the service member's ability to perform repetitive tasks at a lower relative workload; patrolling 10 miles will become easier. Endurance is aerobic training, which is dependent on the body's ability to take in and utilize oxygen. The American College of Sports Medicine and the American Heart Association recommend a minimum of 30 minutes of moderately vigorous aerobic activity for a minimum of 5 days a week for adults under the age of 65. However, participation in physical activity above the minimum and within the guidelines will increase the health benefits. Since the military requires service members to perform challenging physical tasks, they should perform more than the minimum duration and frequency of aerobic training. Examples of endurance training are long-distance running; swimming; foot marching; cycling; low-weight, high-repetition strength training; and stair master. Distance, duration, repetition number, and speed need to be increased gradually to reduce the risk of injury and increase performance. A very general example would be increasing the distance run by only 10% per week.

(2) Mobility

Mobility is the ability to move the body in space with the precision necessary to negotiate an obstacle, such as entering a window. Mobility training is designed to increase the service member's speed, balance, jumping, and ability to change direction. Anaerobic training can improve mobility. This involves training with tasks that take no more than 5 minutes. Examples of mobility training include jumping, sprinting, and climbing. Plyometrics (jumping exercises) have been shown to be effective in increasing vertical jump height more than strength training alone. Short repetitive sprints, slowly increased in duration and number over time have been shown to increase jump height, jump power, jump length, squat strength, sprint speed, and agility.

It is important that speed training incorporates changes in direction as well. If a service member is training to accomplish running tasks that require quick directional changes, the principle of specificity of training should be applied; namely, train by sprinting in a straight line as well as with multiple directional changes. Using only straight sprint training results in increases in straight speed but not change-of-direction

speed, and vice versa. Therefore, a combination of the two training techniques is needed.

(3) Strength

The purpose of strength training is to increase the service member's ability to generate force and power, thus lowering the relative work required to complete mission tasks. Muscle strength is the maximum force created by a muscle; muscle endurance is the ability to create a force repetitively or sustain it, and power is the ability to create force quickly across distance. Hence, strength is required to lift a 40 lb. box, while power is the ability to move the box 4 feet in 1 second.

Many weight training regimes are based on a one-repetition maximum (1 RM). This is simply the most weight that the service member can lift once. Novice individuals should begin with 60–70% of their 1 RM for 8–12 repetitions and 2–3 sets. However, once a service member has been strength training for 6 months or more, 80% of 1 RM is needed to continue to increase one's strength, and 85% has been shown to be most effective in athletes. It is important to remember that even though 80% is stated to be the most successful, this effectiveness is maximized when incorporated into a periodized weight training program. Periodization is a phased training program using a variety of sets, repetitions, weights, and exercises.

The strength program goal always needs to be considered. For maximum strength to increase, one of three things must happen: (1) the load must increase, (2) repetitions must increase, or (3) rest periods must be lengthened. For muscle endurance to increase, repetition speed should be increased and rest time reduced. If the goal is to increase power, light loads (0–60% 1 RM for lower body and 30–60% 1 RM for upper body) are used with fast repetitions. The amount of rest time between sets varies, depending on the goal of the weight-lifting program. If the goal is for maximum one-time strength, such as lifting a heavy casualty, then the rest periods between sets need to be longer, at least 2–3 minutes, and often up to five. If the goal of the program is to increase endurance, such as lifting more boxes, then the rest period should be shorter, less than 1 minute for sets of 10–15 repetitions and 1–2 minutes for 15–20 repetition sets. For power, such as pushing up out of the prone, rest periods should be 3–5 minutes between sets. Ballistic exercises use extremely fast, explosive movements and are most effective when training power. It is important to remember that maximum strength, muscle endurance, and power are all inter-related and a general program should include aspects of each one. The American College of Sports Medicine recommends that weightlifting be performed 3–4 days a week.

It is important to remember that this does not mean that each muscle group is stressed every day. When performing a whole body routine, the service member should lift a maximum of three times a week or, when using an upper and lower body split workout, the service member should lift four days a week: upper body two days and lower body two days. Core strength is the endurance strength of the abdominal and back muscles and should be included as the foundation of any physical fitness program. Low muscular forces applied at specific areas along the spine during core strengthening will drastically increase the load carriage capability of the spine (combat gear or lifting loads). Since this submaximal coactivation must be maintained regardless of which component is being trained (endurance, mobility, strength, or flexibility), endurance is much more important than overall strength for maintaining and increasing load carriage capability.

Core stability is the biomechanical building block upon which all other training rests. Core stability is also important because low core strength is associated with a high prevalence of low back pain (LBP), and higher core strength is associated with less reoccurrence of LBP; patients were 12 times less likely to have recurrence in LBP for 1 year after receiving core stability training. The core muscles must be conditioned to increase the load carriage capability of the spine, reduce the risk of injury, and keep the body biomechanically efficient.

(4) Flexibility

Flexibility is the fourth physical fitness component. The purpose of flexibility training is to maintain or achieve optimal muscle length–tension relationships across joints that enable a service member to perform military-specific tasks with less likelihood of injury or undue difficulty. Hypo- or hyper-flexibility can be detrimental. Poor flexibility is related to injury in the Achilles tendon, hamstrings, patellar tendon, plantar fascia, and iliotibial band. For individuals who require increased flexibility, static stretching (motionless stretches held for 15–30 seconds) should be done at the end of the fitness session. If static stretching is performed before the main exercises, there is often a decrease in performance. In fact, one study shows that maximum voluntary muscle contraction was still reduced 9% an hour after stretching. Hence, to avoid performance decrements, stretching is best done at the end of a workout. It is important to note that it is possible to be too flexible. Those who are hyperflexible do not need to stretch these muscles, as it will not reduce their likelihood of injury.

b. Principles of the Strength and Conditioning Program

A properly planned and executed strength and conditioning program will prevent potential for injury, increase performance, and account for all seven foundational movements.

(1) Prevent Potential for Injury

A well-designed, balanced, and progressive program will help eliminate muscle imbalances, strengthen tendons and ligaments, and increase maximum muscle strength, power, and endurance. A sound program will also address flexibility/mobility and proper nutrition and will aid in recovery, regeneration, and injury prevention. The program should:

- Include proper dynamic warm-ups and cool-downs.

- Address key deficits identified in initial human performance assessments.
- Incorporate rest and recovery periods.
- Integrate medical personnel to track and mitigate injury trends.
- Adhere to an individual's limitations imposed by medical professionals to safely rehabilitate an injury.

(2) Increase Performance

A sound strength and conditioning program should be designed based on a comprehensive approach to balance all facets of performance enhancement and exercises that will transfer the Marine's increased performance in the field. Any athlete or Marine will tend to gravitate or prefer exercises or a program they are "good" at performing. The program should also address their deficiencies and weaknesses.

(3) Seven Foundational Movements

The seven foundational movements were put into place to help make sure there is a balance of movement patterns. The foundational movements will ensure the Marine is strong in all aspects and able to apply force in a 360-degree environment. The Seven Foundational Movements are Push, Pull, Squat, Hinge, Lunge, Plank, and Rotation. Despite the load, implement, task, or location, the seven foundational movements are the basis of programming.

c. Dynamic Warm-Up

The dynamic warmup is paramount when it comes to physical fitness. This portion of the workout will prime the body for maximum effective performance.

(1) Thermoregulation

Elevated core body temperature lowers the tissue viscosity of the muscles, tendons, and ligaments. A lower viscosity, or resistance, in these tissues facilitates an increased range of motion. An increased range of motion in the muscles, tendons, and ligaments allows for improvements in movement mechanics as well as force production.

(2) Injury Prevention

As the viscosity is lowered and the range of motion is increased, athletes experience a decrease in muscle and joint stiffness. This reduces the likelihood of non-contact injuries caused by the stresses of sudden and unexpected movements.

(3) Mental Preparation and Coaching

While warming up, this is an opportunity to mentally prepare the Marine and gain more information about the unit or individual Marines. The FFI needs to know how they are both physically feeling and if they are mentally prepared to put maximum effort into that day's training, or if an adjustment needs to be made to the day's plan. This is also a time to emphasize and work on both effective movement patterns and body mechanics.

(4) Components

(a) Multi-joint, multi-muscle functional movements. These are like mission specific movements and extend the dynamic range of joints. These movements require balance and coordination, and they increase concentration levels.

(b) Dynamic versus Static ROM. The dynamic warm-up addresses the differences in movement expressions seen when comparing dynamic versus static ranges of motion.

(c) Tempo. As a warm-up, becomes more physically and mentally demanding; the muscular and neuromuscular systems are provided the opportunity to gradually reach training intensity without underworking or over-fatiguing the Marine.

(d) Effects. Through low intensity activities, the dynamic warm-up should:

- Elevate body temperature, heart rate, respiration rate, and blood flow.
- Lower joint fluid viscosity.
- Enhance overall movement patterns that will be needed in the following workout.
- Increase the speed of an impulse from the brain to working muscle to either contract or relax (i.e., potentiation).

(5) Phases of a Dynamic Warm-Up

A dynamic warm-up is usually executed in four phases.

(a) General Mobility. Activities used to increase blood flow, take joints through ranges of motion, and prepare the body for movement generally executed at a low exertion level at the beginning of the dynamic warm-up.

(b) Muscle Activation. These are isolated movements used to stimulate specific muscles and to positively affect both the central and peripheral nervous system.

(c) Transit Mobility. This phase involves taking joints through a specific range of motion while traveling over a prescribed distance. These movements are designed to reinforce athletic movement, increase dynamic flexibility, and increase the intensity of physical exertion.

(d) Dynamic Mobility. This is an activity that takes joints through a rapid range of motion and change of direction. This will be the phase where the Marine is approaching 100% efforts.

d. Phases of Strength Training

There are four phases of strength training that provide a foundational baseline for developing combat efficiency. These phases are not exclusive or individual. The Marine will work through all phases or cycle through phases in practicality.

(1) Foundational/Hypertrophy

The foundational/hypertrophy training phase will adapt the body for more strenuous resistance training later, by increasing both muscle fiber size and work capacity, along with other adaptations such as preparing the ligaments and connective tissue for more strenuous training. The Marine will also develop a base of strength and coordination that will give them the foundation to progress to a Strength Phase.

(2) Strength

Strength is defined as the max load the Marine can move one time. A strength training phase will lead to a high neuromuscular adaptation that will aid in speed development and an increase in joint stability. The body will continue to adapt and be able to recruit more muscle fibers on demand. In this phase, the Marine will be working with loads closer to their 1 rep max (1RM) with the goal of increasing their maximal amount of strength. They will be working on the lower end of the force-velocity curve. Example: Deadlift 4x4 @ 90% 1RM.

(3) Power

During this phase, the Marine will be training with higher velocities of movement being the focus. They will be moving less than maximal loads, but at a higher rate of speed. The Marine will be working on the higher end of the force-velocity curve. There is a direct relationship between force and velocity. As the speed of the movement increases, force decreases. Force production is extremely low at extremely high speeds. Between these two extremes is an optimal point for power development. In fact, maximal power occurs at intermediate velocities when lifting moderate loads.

Assuming a Marine has maximized their ability to apply force through strength training, it would be beneficial if they could train to increase the rate of force production. Increasing the rate at which strength can be generated positively alters the time aspect of the power equation (Figure 1). When the Marine can competently lift relatively heavy loads, target repetitions should be in the 3-5 per set with loads of 75-85% 1-RM. Normally, with exercises such as the bench press and squat, loads in the region of 87-93% 1 RM can be lifted for 3-5 repetitions. Peak power, however, is achieved with moderate, not maximal loads that are executed with a fast tempo. Proper form and speed of execution can be maintained by using a load of 75% 1-RM and completing a maximum of five repetitions.



Figure 1: Force-Velocity-Power-Curve Training Methods.¹

(4) Endurance

Muscular endurance is required to overcome a relatively low resistance for a prolonged period. Aerobic endurance is training the cardiovascular system to be more efficient. Examples include Deadlift three repetitions per minute for 10 minutes at 70% 1-RM or running three miles at a sub-maximum pace.

e. Developing Speed and Agility

Speed is the ability to apply force into the ground and move with as much velocity as a Marine can in a specific direction. Agility is the ability to decelerate and change directions and accelerate again while being balanced and under control. There are three methods that can be used to develop speed and agility.

(1) Primary Method

The primary method for speed and agility development is execution of sound movement and techniques in a specific task. Marines should perform tasks at sub-maximal learning speeds to establish proper mechanics. To some extent, technique training can focus on perfecting form and correcting faults more than on teaching novel mechanics.

(2) Secondary Method

One of the secondary methods is sprint resistance. This method includes gravity-resisted running (e.g., incline sprints) or other means of achieving an overload effect (e.g., harness, parachute, sled, or weighted vest). The objective is to provide resistance without arresting the athlete's movement mechanics, primarily as a means of improving explosive strength and stride length.

The other secondary method is sprint assistance. Sprint assistance includes gravity-assisted running (e.g., downgrade sprinting on a shallow slope), high-speed towing (e.g., harness and stretch cord), or other means of achieving an over speed effect. The objective is to aid without significantly altering the athlete's movement mechanics, primarily as a means of improving stride rate.

(3) Tertiary Methods

During running, the hip and knee joints move through relatively larger range of motion (ROM) than the ankle, which acts almost isometrically during the support phase by virtue of reflex stiffness and stretch shortening cycle action. The ability to fully retract the leg during recovery is requisite to achieving a proper ground preparation position and subsequent ground strike. Athletes can develop mobility restrictions because of imbalanced training or adaptive shortening, for example, due to inactivity or immobilization. Regular flexibility training generally seems to have beneficial effects on athletic performance and equivocal effects in terms of injury prevention.

Athletes must develop power qualities to maximize their speed and agility performance. This does not imply, however, that they should perform only low-resistance, high speed movements in training. The ability to achieve high movement velocities requires skillful force application across a range of power outputs and muscle actions. For maximal transfer to athleticism, resistance-training programs should progressively address the entire force-velocity spectrum. This is achievable with mixed methods of training strategies.

f. Flexibility and Mobility

Maintaining high levels of flexibility and mobility will help prevent injuries as well as increasing strength. There are several factors that affect flexibility and mobility. These are known as unchangeable and changeable factors.

(1) Unchangeable Factors

When an individual attempts to make changes to core stability and flexibility, there are three factors that cannot be changed: joint structure, age, and gender.

(a) Joint Structure. This anatomical factor determines potential range of motion (ROM). For example, the shoulder joint (ball and socket) yields greater ROM than the wrist joint (ellipsoidal).

(b) Age. Young adults demonstrate more flexibility than older adults. This discrepancy is partly due to the process of fibrosis. As a person becomes older and less active, muscle fibers are replaced by less elastic fibrous connective tissue.

(c) Sex. Sex differences are often exhibited, with females possessing more flexibility than males.

(2) Changeable Factors

There are several factors that can be changed or controlled in one way or another to affect flexibility and mobility.

(a) Muscle Fibers. The elastic nature of muscle fiber allows itself to be stretched, followed by a return to the normal position. A large increase in muscle density may negatively affect ROM.

(b) Connective Tissue. Tendons, ligaments, and fascial sheaths affect flexibility in that they present a limiting influence on ROM. The role of most connective tissue mechanisms is to protect the body from injury by preventing over-stretching specific joints. When a relaxed muscle is stretched, most of the resistance to further lengthening is derived from the connective tissues around and throughout the muscle.

(c) Activity Level. Given the fact that an active person is apt to be more flexible than an inactive person, this factor has the most potential for change.

(3) Increasing Flexibility and Mobility

Flexibility and mobility are the combination of muscle flexibility, joint ROM, and neuromuscular control. Mobility can be decreased by injury, an inactive lifestyle, or muscle imbalances. The following methods can be used to increase flexibility and mobility.

(a) Static Stretching. Static stretching is a common mode of stretching by passively relaxing a specific muscle while holding a near-maximal stretch for an extended period. Static stretching should be performed as a post-session cool down.

(b) Dynamic Stretching. Dynamic stretching is the expression of ROM during movement and may be the most athletic-based method of flexibility training. The exercises consist of multi-joint, multi-muscle movements that are like sport movements while simultaneously extending the ROM of the joints.

(c) Muscle Activation. Muscle activation involves purposely activating muscle fibers to better facilitate the nervous and muscle systems interaction. This will help stabilize a joint and ensure that the correct muscle groups are used during exercise.

(d) Myofascial Release. Myofascial release is a manual therapy used to help break up adhesions by using foam rollers, lacrosse balls, or other implements. Myofascia is the connective tissue that covers the muscle. It can be restricted by injury, disease, inactivity, or inflammation, and can lead to adhesions or trigger points. These trigger points can lead to a decrease in ROM, muscle length, motor coordination.

g. Core Strength and Stability

The anatomical core can be defined as the trunk region, which includes parts of the skeleton system. These parts are the rib cage, vertebral column, pelvic girdle, shoulder girdle, and associated passive tissue being cartilage and ligaments. Core stability describes the ability to control the position and movement of the central portion of the body. Core stability training targets the muscles deep within the abdomen that connect to the spine, pelvis, and shoulders, which assist in the maintenance of good posture and provide the foundation for all arm and leg movements.

The core is the kinetic link between the upper and lower body, allowing force created in one-half of the body to transfer to another. An increase in core stability can help maximize running performance and prevent injury. Power is derived from the trunk region of the body, and a properly conditioned core helps control that power, allowing for smoother, more efficient, and better coordination within the limbs.

With the spine in its neutral curve (not flexed or extended) the muscles are able to provide most of the stabilization support most effectively. On the contrary, when the spine is not in its natural curve, the spinal musculature is inhibited from developing tension; thus, the passive tissue provides most of the stabilizing support, which greatly increases the risk of injury to these structures. The best approach for developing the core muscles is through a variety of different exercises that involve a combination of stabilizing (e.g. planks), as well as dynamic functions (e.g., crunches).

2. Performance Nutrition

Nutrition is a key enabler for successful military operations. Carefully planned and executed feeding practices in the field maintain and enhance operational performance and morale as well as significantly contribute to mission accomplishment. The overall goal of training is to optimize physiologic adaptations. Training goals can be met through appropriate nutritional strategies implemented before, during, and after training. Various strategies have been developed to maintain adequate energy stores, enhance recovery, stimulate muscle protein synthesis, maximize glycogen repletion, and minimize/protect against training injuries. Nutrient timing, combined with adequate rest and recovery periods are important components of any training program. Another consideration for optimal performance is familiarity: foods and fluids consumed before and during exercise should be consistent with those used in training to minimize the risk for an upset stomach from unfamiliar foods.

a. Energy Balance

A calorie (kcal) is the basic unit used to measure energy. Energy balance represents the balance between kcal intake and kcal output (expenditure). Maintaining energy balance is a critical aspect for optimizing performance in active populations, especially when energy expenditure can be extremely high. Three major contributors to total calorie needs are basal metabolic rate (BMR), physical activity, and energy for digesting foods. BMR is the amount of energy required just to maintain life (at rest) and can be estimated using a variety of equations and/or online calculators. The calories you burn while awake and moving (including exercising) are not included in the BMR. To calculate how many calories your body uses each day, you need to include your activity level. To estimate your total daily calorie needs, multiply your BMR by the activity factor appropriate for your lifestyle. Although not 100% accurate, such estimates may be useful to get an idea of energy needs.

According to the Military Dietary Requirement Index (MDRI), 50 to 55% of total calories consumed should come from foods and beverages with carbohydrate sources. The MDRI for protein is based on levels established in the recommended daily allowances (RDAs) and on the requirements during periods of intense physical activity. These range from 0.8 to 1.5 grams of protein per kilogram (kg) of body weight. Application of these standards to the reference measures for men and women equates to a protein range intake of 63 to 119 grams/day for men and 50 to 93 grams/day for women. The MDRI for protein is easily met when protein intake comprises 10 to 15 percent of the total energy intake. As far as fat is concerned, Marine Corps menu planners will establish menus with 30 percent or less of total calories from fat. Saturated fat should be limited to 10 percent of total calories and cholesterol should be limited to no more than 300 mg/day. Operational and restricted rations, however, may have a higher fat ratio to increase caloric density and reduce ration weight.

b. Pre-Activity Fuel

No consensus statement for optimal content and timing of a pre-exercise meal/snack has been published. Pre-activity meals provide the nutrition needed to fuel the body during light, moderate, or intense activity. During all out efforts (which may include intense resistance training or high intensity interval training, being fed with carbohydrates appears to be more beneficial than fasted training for the purpose of enhancing adaptations to training. Low levels of blood glucose and muscle glycogen can result in early exhaustion due to hypoglycemia or the depletion of muscle glycogen.

Consuming a meal or snack providing 200 to 300 calories three to four hours before heavy training or mission will allow sufficient time for the food to be digested and leave the stomach. Consuming up to 25 grams of protein before exercise may be important for those primarily engaged in strength/power training regarding maximizing protein synthesis. If a larger meal is desired, more time should be allowed for digestion. Unsaturated fat intake is an important source of essential fatty acids and carrier for fat soluble vitamins necessary for optimal physiological function. During prolonged, lower intensity training, fats are a major energy contributor and are stored in muscle as triglycerides for use during activity. Dietary intake is suggested to be between 20-35 percent of total daily caloric intake. Diets low in fat intake can negatively impact training, nutrient density of the diet, and the ability to consistently improve performance. Examples of good pre-workout foods are:

- Bananas
- Oats
- Greek yogurt
- Avocados
- Almonds and walnuts

c. During Activity

During exercise energy stores are being used to provide energy to the working muscles, and muscle protein is being broken down. Consuming small amounts of carbs at regular intervals can enhance athletic performance when the exercise duration is greater than one hour. For exercise less than one hour, water is generally sufficient. During exercise lasting longer than one hour, ingesting carbs in a fluid such as a sports drink can also help sustain hydration. Ingesting approximately 30–60 g carb/hour or 7–20 g carb every 15–20 minutes can extend endurance performance. When the exercise duration is greater than three hours, carb intakes of up to 110 g per hour may be needed, depending on the intensity of the exercise. For exercise longer than three hours, individuals typically eat both solid and liquid foods to meet carb needs. Marines should try various foods during training to determine which are most suitable. The amount of carbohydrates and fluid required and/or tolerated by any individual will be determined by the exercise intensity, duration, environmental conditions, and mode of exercise (running vs. marching vs. manual labor). The Marine should simulate mission events to determine his/her optimal fuel sources. Consuming protein during events does not appear to provide any benefit to performance. Examples of good intra-workout snacks are raisins and energy bars/gels. Sports drinks do well to replace lost fluids and electrolytes but do contain a higher caloric content. Coconut water has also been proven to enhance exercise capacity in warmer environments more so than water and sports drinks.

d. Recovery

After exercise, the body needs to transition from a catabolic (breakdown) state to an anabolic (build up) one to promote recovery and restore what was depleted during exercise. Recovery is dependent on food and beverage choices made throughout the day. The recommended time to begin recovery nutrition is no later than 60 minutes after activity, when glycogen stores and muscle protein synthesis are lowest; this provides the body with the necessary nutrients and fuels. To enhance muscle glycogen synthesis following prolonged, strenuous exercise (over 60 minutes), consuming approximately 50 grams of carbs within 60 minutes after the activity is recommended, followed by approximately 100-150 g carbs (as liquid, gel, or solid food) at two-hour intervals for up to six hours.

Although somewhat controversial, it is generally believed that ingesting some protein (at least 40g) with carbs during recovery can increase muscle protein synthesis and improve nitrogen balance more effectively than consuming carbs alone. Consuming foods with essential amino acids, especially leucine, will promote the post-exercise muscle protein synthesis needed for building and repairing muscle tissue. Foods containing leucine include eggs, dairy, and chicken. Without appropriate refueling after a hard training session or mission, performance may be compromised, especially if a second workout or mission is required the same day or in less than 24 hours. Replenishing the body with nutrients post workout is also vital to upregulate your body's insulin levels.

e. Hydration

Water is the most abundant component of the human body and makes at least half of your body weight. Fluid requirements can vary greatly depending on diet and health status, exercise intensity and duration, and environmental conditions. *The Institute of Medicine* established daily adequate intake values: 3.7 L/day or ~4 quarts/day for men and 2.7 L/day or ~2.9 quarts/day for women, but will increase based on the level of physical activity. Sweat rates have been reported to vary from 0.3 to 3.5 L per hour, depending on age, type of activity, physical fitness, acclimation status, clothing, and, of course, environmental conditions. Fluid losses can occur at higher altitudes, in the cold, in the heat, or even under conditions of low physical activity, and are especially greater during exercise of long duration. A ~2–3% loss of body weight due to dehydration can impair endurance performance and mood. Because of large differences among individuals, each person needs to determine his/her own sweat rate to estimate fluid losses and requirements more accurately across a variety of environmental conditions.

¹ <https://bodyrecomposition.com/training/categories-of-weight-training-11/attachment/forcevelocitypowernew>

Lesson 13: Sexual Harassment and Assault Awareness and Prevention

Lesson Hours: 1.00

1. Introduction

Sexual harassment and assault are criminal acts that erode unit cohesion and contradict the U.S. Marine Corps' core values. Enforcing a “zero tolerance” position, by itself, has proven ineffective at reducing the occurrence of these acts. The purpose of this lesson is to increase student understanding of how to create a command climate where behaviors and attitudes that may lead to sexual assault and harassment are identified and changed before they can result in criminal actions. Note that this lesson is not intended to replace or duplicate U.S. Marine Corps annual training, nor reduce the requirement for Marine officers to fully understand the policies and procedures related to sexual harassment and assault prevention, reporting, and response.

2. Educational Objectives

- a. Identify recent trends in incidents of sexual assault and sexual harassment in the U.S. military.
- b. Identify Department of Defense policy actions to address sexual assault and sexual harassment.
- c. Identify challenges the Department of Defense faces in reducing incidents of sexual assault and sexual harassment.
- d. Identify risk factors associated with increased risk for sexual assault.
- e. Identify behaviors associated with increased risk for sexual assault.
- f. Identify unit-level actions that may help reduce incidents of sexual assault and sexual harassment.

3. Key Terms

- Sexual Assault
- Sexual Harassment
- Prevention
- Command climate
- Risk factors
- Watch List Behaviors

4. Preparation

All quizzes, discussion forum posts, and assignments are executed in the Moodle course site.

- a. Lesson Reading: [Sexual Harassment and Assault Awareness and Prevention](#) (19 pages).

b. Other Requirements: None.

c. Supplemental References (optional):

- [RAND, *Countering Sexual Assault and Sexual Harassment in the U.S. Military*](#), RAND Corporation (Jun 2021); this material is protected by copyright; further distribution is prohibited.
- [Commander's Smart Pack, *Sexual Assault Prevention and Response*](#), Headquarters, U.S. Marine Corps (May 2022).
- [MCO 5354.1G with Administrative Change 1, *Marine Corps Prohibited Activities and Conduct \(PAC\) Prevention and Response Policy*](#) (11 Apr 2025).
- [MARADMIN 469/25, MCO5354.1G Prohibited Activities and Conduct Prevention and Response Policy Changes to Appeal Standard and Dismissal Authority](#) (3 Oct 2025).
- [MCO 1752.5C, *Sexual Assault Prevention and Response \(SAPR\) Program*](#) (3 Jun 2019).
- [NAVMC 1752.5, *Sexual Assault Prevention and Response Procedures*](#) (5 Jun 2019).

Lesson 13 Reading

SEXUAL HARASSMENT AND ASSAULT AWARENESS AND PREVENTION

References:

- a) RAND, *Countering Sexual Assault and Sexual Harassment in the U.S. Military*, RAND Corporation (Jun 2021); this material is protected by copyright; further distribution is prohibited.
- b) *Commander's Smart Pack, Sexual Assault Prevention and Response*, Headquarters, U.S. Marine Corps (May 2022).

1. Introduction

In 2021, the Secretary of Defense, Lloyd Austin, stated that sexual assault and sexual harassment prevention efforts within the Armed Services are “far short of what is required to make lasting change.”¹ It is not enough to react to sexual harassment or assault after the crime has already been committed and the damage done; the U.S. Marine Corps must maintain an environment within its commands that is inoculated against the behaviors and attitudes that may lead to these negative actions. This reading begins with a summary of the RAND Corporation institution-level study conducted in response to the Secretary of Defense’s request for a frank assessment of accountability measures and prevention approaches within the Department of Defense (DoD). This reading then presents some command-level recommendations on how to create an organizational culture that inherently rejects behaviors and attitudes that are associated with an increased risk for sexual harassment and assault.

The RAND study had the goal of informing the work of the Independent Review Commission (IRC) on Sexual Assault in the Military by summarizing the actions that DoD has taken to reduce sexual assault and sexual harassment behaviors, describing the challenges that DoD faces in reducing such behaviors, and offering recommendations to improve the prevention of sexual assault and sexual harassment in the military. The contents of this report are based on years of empirical study and direct support from hundreds of DoD installations and commands.

2. Trends

The estimated prevalence of sexual assault in the military has fluctuated, but in recent years, the prevalence has risen for women. Official reports of sexual assault and sexual harassment within the military have also significantly increased.

According to estimates from the Office of People Analytics (OPA), only one in three sexual assault victims within the military reports it to an authority.² In addition to data on reported offenses, the DoD has access to epidemiological estimates of sexual assault through biennial surveys administered by the OPA: the *Workplace and Gender Relations Survey of Active Duty Members* (WGRA), the *Service Academy Gender Relations Survey* (SAGR), and the *Workplace and Gender Relations Survey of Reserve Component Members* (WGRR).

Notably, these surveys also assess sexual harassment. The combination of data from official reports of sexual assault and survey estimates of population prevalence has allowed for a more complete picture of sexual assault and sexual harassment within the military. Rates of sexual assault have fluctuated across the six periods for which data are available (i.e., 2006, 2010, 2012, 2014, 2016, 2018; see Figure 1).

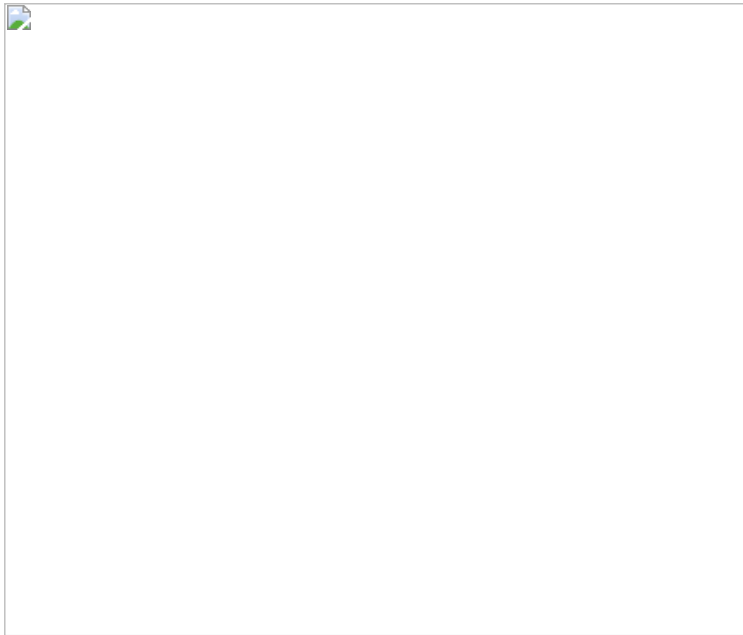


Figure 1: Estimated Sexual Assault Prevalence Rates, 2006-2018.

Rates of sexual assault against service men fell by more than 50 percent from the first year of data collection compared with the most recent (1.8 percent in 2006 to 0.7 percent in 2018). Unfortunately, the drop is considerably smaller for service women (6.8 percent in 2006 to 6.2 percent in 2018). Comparing the last two reporting periods (2016, 2018) shows there was a sizable increase in the rate of sexual assault among service women (from 4.3 percent in 2016 to 6.2 percent in 2018), and a marginal change for service men (0.6 percent in 2016 to 0.7 percent in 2018).

Available data on sexual harassment (2014-2018) are similarly concerning. Both service men and women experienced increased rates of sexual harassment in 2018 compared with the prior reporting period (see Figure 2), although the rate for men is similar to that in 2014.



Figure 2: Estimated Sexual Harassment Prevalence Rates, 2014-2018.

At the service academies, 2018 rates of sexual assault among cadets were higher than for the military as a whole. In 2018, across the service academies, the range of students experiencing unwanted sexual contact in the previous year was between 15.1 percent and 16.5 percent of women and between 1.8 percent and 3.4 percent of men (Davis et al., 2019).

In sum, one in 16 women and one in 143 men are estimated to experience sexual assault within the military. Estimates for sexual harassment are one in four women and one in 16 men. At the service academies, one in six women and one in 29 men experience sexual assault. These behaviors degrade military readiness. Many victims report that their performance suffered as a result of these behaviors, and units often experience decreased cohesion and stability.³ RAND research has documented that sexual harassment is associated with roughly four percent more service member separations from the military than would otherwise be expected.⁴ The DoD recognizes that without bold action, sexual assault and sexual harassment will continue to have negative consequences for the military.

3. DoD Policy Actions

The DoD has taken several steps to address sexual assault and harassment for over a decade.⁵ Notably, the Sexual Assault Prevention and Response Office (SAPRO) supports the services' prevention efforts by providing training to key service SAPR staff, collecting data via service-wide surveys, conducting original research into effective prevention programs, and issuing policy guidance. Although these efforts are good steps, the current state of sexual assault and sexual harassment prevention within the services and at individual installations should be improved.

In an effort to better address sexual assault, in 2005, the DoD established SAPRO and a process for sexual assault victims to file restricted reports, which allowed victims to receive services without notifying their commands and initiating an investigation.⁶ The intent behind these actions was to increase the likelihood that victims would report offenses, increase the care that is available to victims, and improve DoD's ability to track incidents that might otherwise go unreported.

The DoD established the *24/7 Safe Helpline*, a *Safe Help Room*, and the *Safe Helpline* mobile phone application, all of which are confidential and anonymous crisis support services specially designed for members of the DoD community who are affected by sexual assault. However, the DoD recognized that a more proactive, comprehensive, and coordinated approach to sexual assault prevention was needed. To this end, in 2016, the DoD launched the two-phased *Applied Prevention Project*.⁷ *Phase 1* included site visits to understand current prevention efforts among the services. *Phase 2*, launched in 2018 and is ongoing as of this writing, provides technical assistance to build the capacity of military installations to implement sexual assault and sexual harassment prevention programs. In addition to prevention focused trainings, the DoD implemented "Getting to Outcomes," RAND's evidence-based planning and implementation process, at eight military installations and service academies. Sites work with experts to select, implement, and evaluate sexual assault and sexual harassment prevention programs, informed by research evidence.⁸

In 2019, the SAPRO released the *Prevention Plan of Action (PPoA)*.⁹ Incorporating years of research on sexual assault and sexual harassment prevention and the science of implementation, the PPoA envisions that each installation will conduct sexual assault and sexual harassment prevention by choosing activities best suited for their community, implementing these activities with fidelity, and regularly evaluating the outcomes of these activities. The PPoA outlines the requirements for how to achieve this vision across several domains (i.e., infrastructure, leadership, and collaborations). PPoA requirements are critical not just for sexual assault and sexual harassment prevention, but also to support prevention across a broad spectrum of problem behaviors (e.g., alcohol abuse).

In addition to the PPoA, the DoD developed and implemented a *Retaliation Prevention and Response Strategy Implementation Plan* to prevent retaliatory behaviors and support victims.¹⁰ In 2005 and 2006, the DoD released policies that established comprehensive procedures for responding to sexual assault within the department; these policies have been updated regularly throughout the years to improve the department's approach and align with emerging requirements in law.¹¹ The 2019 *National Defense Authorization Act (NDAA)* included a provision (Section 543) that required DoD to develop an oversight plan for the implementation of sexual harassment prevention (Public Law 115-232). Moreover, recognizing the interconnectedness of various forms of interpersonal and self-directed violence (e.g., sexual harassment and assault, intimate partner violence, suicide), in September 2020, the DoD issued the *Policy on Integrated Primary Prevention of Self-Directed and Prohibited Abuse or Harm*.¹² This policy aims to integrate policies and responsibilities to mitigate abuse or harmful acts and establish a DoD-wide prevention system that makes data-informed decisions and implements research based policies and interventions.

4. Challenges

Significant limitations in the DoD's approach to sexual assault and harassment prevention have resulted in slow progress. The DoD faces challenges in reducing rates of sexual assault and sexual harassment in multiple areas: reporting and accountability, availability and use of evidence-based approaches, and prevention infrastructure.

a. Reporting and Accountability

DoD reporting and accountability mechanisms are not sufficient to deter perpetration. Most incidents of sexual assault and sexual harassment within the military go unreported. In 2018, there were 6,053 reported sexual assaults, compared with the estimated prevalence from surveys suggesting that over 20,000 service members were sexually assaulted.¹³ Sexual harassment is underreported by an even greater margin. In fiscal year 2019, the military services and the National Guard Bureau processed and investigated 1,021 formal and 591 informal complaints of sexual harassment.¹⁴ However, in a confidential survey of just active-duty service members in 2018, approximately 116,000 individuals reported experiencing sexual harassment in the previous 12 months (24 percent of women and 6 percent of men).¹⁵

There are multiple challenges related to sexual assault and sexual harassment reporting. DoD policy states that sexual harassment complaints "should be addressed at the lowest appropriate level."¹⁶ Thus, there might be reports to supervisors and managers that are not documented. Additionally, not all of the services have centralized reporting and tracking systems that give senior leaders the ability to monitor, intervene, or evaluate the processes, actions, or outcomes related to sexual assault and sexual harassment reporting. Furthermore, social and professional retaliation against victims is perceived to be common and is not deterred by military equal opportunity and command-directed investigation policies and procedures.¹⁷ The extent to which commanders are held accountable for their management of sexual assault and sexual harassment appears limited, and emphasis on the unacceptability of harassment and abuse may not reach all levels of command.¹⁸ All of these barriers reduce the likelihood that incidents are reported and documented, and this, in turn, reduces the potential that offenders will face consequences, thus weakening any deterrent effect.¹⁹

Higher rates of reporting and consistent consequences could have a deterrent effect because service members and commanders would know that they would be held accountable for sexual assault and sexual harassment behavior. Although deterrence can help contribute to prevention, deterrence alone is insufficient to prevent sexual assault and sexual harassment. This is because any deterrent effect after an incident does not directly address the underlying issues that precede these problem behaviors.

b. Use of Evidence-Based Approaches

DoD installations are not implementing prevention activities consistent with the best evidence. Prevention activities are needed to address sexual assault and sexual harassment risks and build skills among service members to prevent the occurrence of such behaviors.^{20 21} However, the services' annual sexual assault prevention and response training does not employ best practices documented in the prevention literature. Specifically, current training is not comprehensive or of sufficient length; it does not use varied, interactive teaching methods; it is not administered by well-trained prevention specialists; it does not foster positive relationships; it is not theory-driven; it does not build both skills and knowledge; and it does not include outcome evaluation.^{22 23 24} A recent review of U.S. Air Force sexual assault prevention programs found that there had been some progress in implementing best practices (e.g., tailoring prevention specifically to the Air Force context) but noted overall low rates of compliance with other practices (i.e., conducting prevention with sufficient length, based on theory, and with adequately trained personnel) and a lack of a systematic way to ensure compliance.²⁵ Furthermore, changes to programming do not appear to be governed by evidence-based reasoning or a theory of change. These issues are not unique to the Air Force and can be found throughout the services. In 2019, SAPRO asked each service branch to self-assess their prevention efforts and conducted 61 focus groups with 493 active-duty members and first responders at eight DoD installations. Those assessments found that the prevention activities focused more on building awareness than on building skills,²⁶ which is inconsistent with evidence-based prevention.

There are a limited number of evidence-based approaches to prevent sexual assault and sexual harassment available. The approaches that do exist target social norms acceptive of assault and sexual harassment, bystander intervention, and skills to reduce victimization.^{27 28} Experts advocate for multiple interventions that focus on risk reduction and promote positive social norms, teach healthy relationship skills, and create protective environments.²⁹ Few male-targeted sexual assault prevention programs have been shown to reduce sexual assault perpetration. Of the two programs that have demonstrated effectiveness, *Safe Dates* has only been tested in adolescents, and effects faded after one year;³⁰ and *Real Consent* had very high drop-out rates.³¹ However, these programs do tend to produce small but significant improvements in attitudes about sexual assault, self-reported likelihood of engaging in preventative behavior, and inclinations toward sexual aggressive behavior.³²

c. Prevention Infrastructure

Current DoD prevention infrastructure is insufficient. The services and individual installations lack adequate infrastructure to carry out evidence-based prevention.³³ In 2019, SAPRO and RAND created a *Prevention Evaluation Framework* tool that assesses critical prevention infrastructure elements at an installation or service academy consistent with the PPoA. Applying this tool to selected installations and to the service academies, SAPRO found gaps in most prevention infrastructure elements. The lack of a dedicated, well-trained prevention workforce and the lack of leadership support for prevention are especially noteworthy. Except for Violence Prevention Integrators in the Air Force, there are no personnel across DoD institutions or at the service academies whose sole job is to implement and evaluate sexual assault or sexual harassment prevention activities.

Services and installations often use response personnel (e.g., Equal Opportunity Advisors, SAPR Victim Advocates) to implement prevention activities, often as collateral duties. Implementing prevention programming well requires specialized skills, and these skills might be lacking among those who are assigned to carry out these programs.³⁴ A *Competency Assessment for Sexual Assault Prevention Practitioners* provides details on required specialized skills.³⁵ These competencies include skills suitable for any primary prevention effort and skills specific to sexual assault prevention.

In discussions with service members and leaders at hundreds of DoD installations, leadership support for high-quality prevention varies. Some commanders might not understand the value of prevention, or they might not recognize that specialized skills and established systems (e.g., for evaluation, professional development) are required. Also, instances of low leadership support for prevention might be because of high operational tempo and demands, crowding out consistent leadership support for effective prevention strategies.

Implementation science shows that while support from top leadership is critical, support ideally should come from all levels of leadership.³⁶ Support includes making time available for service members to participate in more robust prevention activities, publicly voicing support for the importance of the programming, setting specific benchmarks for performance, and regularly evaluating whether those benchmarks have been achieved and, if not, instituting improvements and tracking their implementation.³⁷ Although military-specific measures and tools exist that could be used to evaluate prevention approaches, the services and individual installations do not regularly collect or review evaluation data to drive strategic or operational prevention decisions.^{38 39 40} Those charged with conducting prevention activities might lack the resources or the skills to carry out evaluations of their prevention activities, and thus might select such activities based on preference rather than on past performance.

d. Effective Prevention Research

Research on effective prevention of sexual assault and harassment is limited. Although having adequate infrastructure to carry out prevention is important, ultimately, services and installations need to have a variety of evidence-based options from which to choose. Sexual assault and sexual harassment can occur in different settings and can be associated with many different factors. Thus, no single approach will address the problem universally. One setting (e.g., a military service academy) might experience high rates of alcohol misuse that contribute to rates of sexual assault. Another setting, with an older population, might have service members who endorse attitudes permissive of sexual harassment as a primary factor.

Furthermore, a multitude of factors can contribute to the likelihood of sexual assault or sexual harassment, increasing the need for a variety of simultaneous evidence-based approaches that can address specific factors. Unfortunately, there are a limited number of evidence-based approaches that have been shown to be effective in reducing unwanted behaviors. As described in the previous section, the approaches that do exist target social norms and the strengthening of certain skill sets, such as:

- Enhancing positive social norms around sexual assault and sexual harassment prevention;
- Degrading social norms that are tolerant of sexual assault and sexual harassment;
- Intervening as a bystander;
- Keeping oneself safe from victimization;
- Creating healthy relationships; and
- Fostering protective environments.

However, most of these approaches have not been developed or evaluated in a military context.

5. Recommendations for Institution-Level Changes

Preventing sexual assault and sexual harassment will require coordinated, multifaceted, and data driven actions. There is little reason to expect that minimally adjusting the current course of action will significantly reduce the rates of sexual assault and sexual harassment within the military. Instead, bold actions should be taken, from response-focused policies and programs that address victimization and perpetration to more proactive prevention policies and programs that promote healthy behaviors and force readiness. These actions include enhancing accountability, equipping service members and leaders with the knowledge and skills needed to prevent sexual assault and sexual harassment, and enacting a research agenda that will ensure that the DoD sets the standard for excellence in prevention.

a. Accountability of Perpetrators and Leaders

More consistent, fair, and transparent systems should be put in place to track when and where sexual assault and sexual harassment occur, prevent incidents from occurring or escalating, and hold leaders and service members accountable when such incidents do occur. When there is a low probability of being reported or held accountable, research suggests that sexual harassment flourishes.⁴¹ OSD should institute a regular auditing process to ensure that services and individual installations are taking steps to improve the current response systems for sexual assault and sexual harassment. This audit should, at a minimum, focus on whether services and installations are exercising the practices that are described below.

(1) Report Channels

The Armed Services should ensure that service members have multiple channels to report sexual harassment that will result in timely action, including confidential channels that exist outside their chain of command. Comparisons between official reports and estimated prevalence demonstrate that the vast majority of sexual harassment incidents go unreported. Thus, even if multiple reporting channels are technically

available, it appears that they are underutilized, perhaps because of several barriers (e.g., concern about career implications, fear of retaliation). Barriers to reporting need to be overcome, and one way to do so is to provide confidential reporting options. Evidence from civilian settings shows that barriers to reporting can be overcome by ensuring the existence of multiple functioning channels.⁴²

(2) Sexual Harassment Reporting

Commanders should document all incidents of sexual harassment, even if there is no official complaint made to an Equal Opportunity Office. Only a small number of incidents of sexual harassment are officially reported to an Equal Opportunity Office. When incidents of sexual harassment are reported, they are generally reported to the respective Equal Opportunity Office for most services, except for the Army. The Army is the only service that transferred the responsibilities of sexual harassment reporting from the Equal Opportunity Office to the Sexual Harassment/Assault Response and Prevention (SHARP) Program Office. Most are ostensibly resolved at the lowest level in accordance with policy guidance. There should be a mechanism in place to document these incidents, even if no official complaint is made to an equal opportunity office. These "lowest level" reports should contain sufficient information (e.g., time, location, presence of bystanders) to allow leaders to assess the timeliness, appropriateness, and success of the response to all complaints, not just the cases elevated to the Equal Opportunity Office. Including victims in the development of a more inclusive reporting process could further improve the process.^{43 44}

(3) Career-Length Allegation Tracking

The Armed Services should evaluate the use and effectiveness of systems for tracking allegations of inappropriate behaviors across service members' careers. DoDI 1020.03 states that secretaries of all the military departments should "mandate that substantiated complaints are annotated on fitness report(s) or Service-level reporting and tracking system(s). The Service-level tracking system must be reviewed for substantiated harassment incidents prior to service members' selection for promotion and other favorable personnel actions." Anecdotal evidence from subject-matter experts who review military personnel records suggests that commanders might not be complying with this accountability mechanism. A formal audit and evaluation of personnel records could help the services and the DoD determine how this accountability mechanism is being used and whether it has any impact. In addition to tracking substantiated findings, the DoD should also examine the legality and feasibility of tracking all allegations of sexual harassment and other inappropriate behaviors across service members' careers, regardless of whether the allegations were substantiated. Having commanders consistently document these behaviors and having data systems that consistently capture the occurrence of these behaviors could help inform future decisions about disciplinary action or intervention and decisions about selection for promotion or for key assignments where there might be more opportunities for abuse of authority. In addition, tracking these allegations could help identify the prevalence and patterns of inappropriate behaviors and better target prevention efforts.

(4) Address Unprofessional Conduct

Commanders should levy immediate, appropriate sanctions for low-level unprofessional conduct. Commanders should be using the full range of administrative disciplinary measures available (e.g., counseling, reprimands, Article 15 nonjudicial punishments, etc.) to address unprofessional conduct. Leadership plays an important role in establishing organizational climate. When leaders consistently discipline all instances of inappropriate conduct, they send a clear message about the norms and expectations for the command. This could also prevent the escalation of negative behaviors. Research shows that commands with this kind of approach (i.e., commands where service members believe that their commanders take sexual harassment seriously) have a lower risk of sexual assault and sexual harassment.⁴⁵

(5) Evaluate Report Management

The Armed Services should evaluate commanders, in part, by how they manage reports of sexual assault and sexual harassment within their commands. OSD should work with the Military Departments to ensure that the performance evaluation process for commanders includes an assessment of how they manage reports of sexual assault and sexual harassment. At a minimum, commanders should be tasked with ensuring that victims who report incidents do not experience retaliation or ostracization, and commanders should be held accountable when they fail to do so. Reviews should also consider how leadership style affects the safety climate. Commanders who take a proactive approach to upholding behavioral standards create a positive safety climate, which is associated with lower risk of sexual harassment and sexual assault.⁴⁶ Commanders who take a permissive or *laissez-faire* approach to upholding behavioral standards should be considered candidates for additional education or intervention. A review of commander responses to sexual assault and sexual harassment reports should be regularly included in performance evaluations. The Armed Service could also consider *360 performance evaluations*, in which all service members provide feedback to subordinates, peers, and supervisors.⁴⁷ This approach could create an additional channel for communication of concerns around command climate and unprofessional conduct.

(6) Separate Chains of Command

The Armed Services should consider removing Equal Opportunity Advisors (EOA) and Command Managed Equal Opportunity (CMEO) managers from the same chain of command as victims and offenders. OSD should work with the Military Departments to ensure that sexual harassment reports are able to be filed promptly and effectively to build service member trust in the response system. For example, EOAs and

CMEOs might not feel free to officially file reports of sexual harassment if they perceive that the commander prefers the matter remain unrecorded. Revisiting the command structure of equal opportunity staff members could reduce barriers to effectively reporting sexual harassment incidents.

b. Improved Tools

The DoD should better equip service members and leaders across the chain of command with the tools to prevent problem behaviors. Although it is essential, deterrence alone will be insufficient. Installations and commanders should choose prevention activities that are suited for their communities using the best evidence available, implement those activities with fidelity, and evaluate their impacts. To improve prevention, OSD should mandate and provide support for the Services and installations to take actions to align with the PPoA and implement the new DoD Instruction 6400.09 - *Policy on Integrated Primary Prevention of Self-Directed and Prohibited Abuse or Harm*. The following recommendations reflect first priorities for building momentum for stronger prevention activities.

(1) Infrastructure

The DoD should create an infrastructure that fosters the implementation of effective and lasting proactive prevention efforts. To tackle sexual assault and sexual harassment, the services should ensure that there are dedicated personnel whose primary duty is to conduct and evaluate prevention activities (i.e., not as a collateral duty). These personnel should be selected carefully to ensure that they have pertinent skills.

In addition, they should receive sufficient training, including sustained access to professional development. With these personnel in place, services should:

- Regularly monitor the implementation and impact of new and existing prevention efforts, including collecting data (before and after prevention efforts are instituted) about the participants, the fidelity of the implementation, and the short- and long-term outcomes;
- Share evaluation data so that leadership is informed about which efforts are working and which are not to ensure that appropriate collaboration is occurring by building relationships across the enterprise; and
- Maintain a consistent vision and set of shared goals, including the role that leadership plays in sexual assault and sexual harassment prevention.

The SAPRO tool could be used to monitor development of this infrastructure over time. This infrastructure could be used to target a cluster of related problem behaviors (e.g., alcohol abuse, suicide) in addition to sexual assault and sexual harassment.

(2) Training and Accountability

Train and hold leadership accountable for sound prevention practices. Leadership support for allocating sufficient resources to evidence-based prevention is crucial.⁴⁸ To establish leadership support, leaders should have a basic understanding of sound prevention practices as outlined in the *Prevention Evaluation Framework* (i.e., implementing prevention activities that are consistent with the best evidence and regularly evaluating those activities). Commanders should be required to use these prevention practices, and they, in turn, should hold subordinates accountable for using these prevention practices. One way to institutionalize expectations and best practices is to incorporate sexual assault and sexual harassment prevention training into leadership Professional Military Education.⁴⁹ DoD Instruction 6495.02, Volume 2 - *SAPR Program Training* sets professional military education standards for what is being described here. Curricula should be customized to the specific leadership context (recognizing that not all leadership roles are alike) and should emphasize the role of leadership in setting command climate. Accountability for the incidence of sexual assault and sexual harassment under a leader's command should be communicated openly to all ranks, integrated into performance evaluation, and regularly monitored for evidence of adherence.⁵⁰

(3) Prevention Practices

The DoD should implement prevention practices that reflect the best evidence available and that involve comprehensive planning and continuous evaluation. Although there are some evidence-based approaches currently available (though not yet demonstrated to be effective with a military population), much is known about prevention across several problem behaviors that could guide better programming.^{51 52 53} For example, research shows that the strongest prevention approaches are comprehensive and of sufficient length to yield change (not just one-time meetings); use varied, interactive teaching methods (not just didactic); are administered by well-trained staff who foster positive relationships with attendees; are culturally relevant and theory-driven; build both skills and knowledge; and include outcome evaluation. Thus, although the Services will need to develop and validate new evidence-based practices, they should also eliminate practices that clearly do not reflect the above factors.⁵⁴ For example, moving away from the widely used lecture-format trainings could free up resources for other approaches (such as those mentioned above) that have a stronger chance of improving outcomes. Once strong, data-driven prevention practices are identified, they should be disseminated widely. Also, the conceptualization of "conducting prevention" should be expanded to include planning (e.g., ensuring implementation runs according to plan), evaluation, and quality improvement. For example, the SAPRO is currently piloting the "Getting to Outcomes" process, which is training multiple DoD installations and service academies to engage in planning, evaluation, and quality improvement activities.

c. Comprehensive Prevention Research

The DoD should set the standard for excellence in comprehensive sexual assault and harassment prevention. Because of the small number of evidence-based prevention approaches that are currently available, DoD investment in developing and testing new approaches for effective prevention in the military context could be fruitful. The SAPRO is currently funding research trials of promising programs, but more could be done.⁵⁵ The OSD should prioritize future research on the following topics.

(1) Drivers of Risk

The DoD should investigate the circumstances and characteristics of units with unusually high or low documented cases of sexual assault and sexual harassment to better understand the drivers of risk that are within the military's control. The first step to developing new prevention interventions is to understand the variety of mutable risk factors that are linked to the problem behavior. These investigations could supplement the existing research by identifying drivers that might be unique to the military (e.g., differences in command climate, training, and prevention activities; differences in the roles and presence of civilians; differences in the surrounding community). A combination of surveys and interview tools might yield the most useful and actionable data.^{56 57}

(2) Prevention Approaches

The DoD should develop and rigorously test new prevention approaches that, based on the assembled research, target command climate and risk factors that contribute to multiple problem behaviors. Multiple studies show that sexual assaults do not happen in a vacuum. Rather, they are associated with other negative behaviors, such as a previous history of violence, sexual harassment, or excessive alcohol use.^{58 59 60} In particular, research has documented how a poor command climate that tolerates these behaviors is associated with sexual assault and what a change in culture and climate, in part, requires.^{61 62}

New preventative strategies should be robust, target a cluster of risk factors, and include commanders rather than just individual service members. Priority should be given to prevention efforts that aim to eliminate risk factors that contribute to multiple problem behaviors. For example, preventing binge drinking could reduce the risk of sexual assault and family violence.^{63 64} Because of the documented association between command climate and sexual harassment and assault, developing interventions that aim to improve the climate might be warranted. New approaches should be tested in focused pilots (i.e., across an installation rather than across an entire Service). Rigorous evaluation should be required for all new approaches prior to more widespread dissemination.^{65 66}

(3) LGBTQ Service Members

The DoD should study and report on sexual assault and sexual harassment of service members who describe themselves as lesbian, gay, bisexual, transgender, or queer (LGBTQ). Data from the 2016 WGRA shows that LGBTQ service members represent just 13 percent of the force but account for 45 percent of all sexual assaults (or 56 percent of all sexual assaults of people who identify as male and 37 percent of all sexual assaults of those who identify as female).⁶⁷ Those who identify as male and homosexual have more than nine times the risk of a sexual assault as heterosexual men. While this population is not responsible in any way, very little is known about what leads to sexual assaults in this population, the experience of victims, and how cases are addressed. Further study is required to understand how to effectively prevent them.

5. Unit-Level Actions

The DoD has invested a considerable amount of time and money to prevent sexual assault and sexual harassment, research the root causes of those behaviors, and develop robust response systems. Yet despite DoD efforts, close Congressional oversight, hundreds of new policies and initiatives, and attention from multiple DoD leaders over time, sexual assault and sexual harassment remain persistent problems in the services. Reversing this trend will require significant changes across the DoD enterprise and no single approach on its own will be sufficient. Progress is being made at the institution level, and Marine leaders can be at the forefront of those changes to have a positive impact on their Marines by employing the recommendations below.

a. Prevention

Sexual assault prevention starts with leaders. Protective factors are attitudes, beliefs, and behaviors associated with positive outcomes for the organization. The following ten characteristics are indicators of a healthy command and reduce the likelihood of sexual harassment, sexual assault, or suicide related incidents:

- Cohesion
- Connectedness
- Engagement and commitment
- Fairness
- Inclusion
- Leadership support
- Morale

- Transformational leadership
- Work-life balance
- Safe storage for lethal means

These elements are deeply embedded within Marine Corps leadership and all within a leader's ability to influence. There are many programs within the Marine Corps staffed with personnel and programs equipped to assist leaders in each of these categories.

- Embedded Preventive Behavioral Health Capability
- Family Advocacy Program– Prevention & Education
- Substance Abuse Program– Prime for Life
- Military Life Counselors
- Unit Marine Awareness and Prevention Integration Training
- Operational Stress Control and Readiness
- Semper Fit– Health Promotions, community fitness, unit recreation
- Single Marine Program
- Family Readiness Program (LINKS, LifeSkills)
- Chaplains
- Marine for Life
- Voluntary Education
- Leadership Scholar Program
- Skillbridge

It is also helpful to identify and address harmful behaviors that can lead to a higher likelihood of a sexual assault occurring in a unit. Knowing these behaviors can allow leaders to address them early before they escalate. The *Preventing Sexual Violence Risk and Protective Factors Annual Research Report* and the Department of the Navy's *Watch List Behaviors* are both useful tools to identify harmful behaviors in individual Marines and a unit's climate.

b. Risk and Protective Factors

Based on a review of academic and professional research, the following identifies risk and protective factors that have routinely been shown to be associated with sexual assault and other forms of sexual violence. Risk and protective factors for both victimization and perpetration are provided.

(1) Risk Factors

The risk factors identified below were selected because they are potentially modifiable or are factors that can be used to identify groups to target for sexual violence prevention. No single risk factor is sufficient to predict the occurrence of sexual violence. This list is not exhaustive but includes the most common risk factors associated with sexual violence.

(a) Perpetrators. The following are associated with an increased risk of perpetration of sexual violence:

- Strong adherence to traditional gender norms
- Aggressiveness/Acceptance of violence
- Alcohol and drug use
- Coercive sexual fantasies
- Exposure to sexually explicit media
- Hostility towards women
- Hyper-masculinity
- Prior sexual victimization or perpetration
- Lack of empathy
- Sexual risk-taking
- Association with sexually aggressive and hyper-masculine peers

- Childhood history of physical, sexual, or emotional abuse
- Family environment characterized by physical violence and conflict
- Involvement in a violent or abusive intimate relationship

(b) Victims. The following are associated with an increased risk of sexual violence victimization:

- Alcohol and drug use
- Being a woman
- Being young
- Maltreatment in childhood
- Previous victimization

(2) Protective Factors

Research on protective factors related to sexual violence continues to emerge and develop, but at present remains understudied when compared to risk factors. The protective factors identified below were selected because they are potentially modifiable or are factors that can be used to identify groups to target for sexual violence prevention. This list is not exhaustive but includes the most common protective factors associated with sexual violence.

(a) Perpetrators. The following are associated with a decreased risk of perpetration of sexual violence:

- Emotional health and connectedness
- Empathy and concern for how one's actions affect others
- Parental use of reasoning to resolve family conflict

(b) Victims. The following are associated with a decreased risk of sexual violence victimization:

- Social connectedness
- Coordination among community agencies

(3) Prevention Strategies

No one specific risk factor is the cause of sexual violence. Similarly, no one specific protective factor can prevent sexual violence. Any prevention strategy must take a multifaceted approach to:

- Build unit/community support (e.g., leadership support, command unity, unit/community resources);
- Encourage norms and behaviors that create protective environments; and
- Educate Marines on how to identify what to do in high-risk situations.

c. Watch List Behaviors

Analysis of the *2018 Workplace and Gender Relations Survey of Active Duty Members* revealed the "Top Five Signals" identifying when Marines may be at higher risk of sexual assault. Many of these same areas can be assessed using the Defense Organizational Climate Survey (DEOCS). If a Marine experiences any of the measured behaviors below, their risk for sexual assault may increase (see Table 1). Therefore, these are the behaviors that leaders need to look out for and immediately address.



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Table 1: Signals of Risk for Sexual Assault.

d. Recommendations for Marine Leaders

The following are some general tips for leaders at all levels to create command climates that are resistant to behaviors and attitudes that may lead to sexual harassment or assault:

- **Climate Briefs.** From all-hands to small discussion groups, commit to having routine, ongoing conversations about the organizational culture. Active listening increases your understanding of your unit's concerns and empowers you to adapt as needed.
- **Walk the Deck.** Daily engagement builds rapport, trust, and confidence between leaders and Marines. You are more likely to learn about potentially harmful behaviors when you take part in direct conversations with your Marines in real time. These personal interactions show that you respect your Marines and care about their well-being.
- **If You See Something, Say Something.** If you see or hear inappropriate conduct (e.g., *Watch List Behaviors*, etc.), respond immediately. On the spot corrections communicate standards of professional and appropriate conduct for all Marines and dispel any confusion about unacceptable behavior.
- **One Size Does Not Fit All.** A single set response (e.g., "zero tolerance") to every command climate issue is unlikely to work. In a learning institution, mistakes can be opportunities to learn and grow, and not all comments made are intended to be malicious. As a positive leader with your finger on the pulse of your unit, you can improve morale and *esprit de corps* by taking all reports seriously and addressing them in a swift manner.
- **Do Not Wait Until It Escalates.** Early intervention is important. *Watch List Behaviors* are indicators of eroding good order and discipline within your unit. Stopping these behaviors early and consistently can prevent more serious, destructive behaviors from occurring over time.
- **Take All Sexual Harassment Complaints Seriously.** Trust your Marines when they report sexual harassment, as they are trusting you to uphold the standards. Studies show that most people have tried everything they can to resolve the situation before making a sexual harassment complaint.
- **Look Beneath the Surface.** Complaints about *Watch List Behaviors* may indicate deeper issues. Use additional assessment tools, such as a command climate survey, to better understand how your Marines are experiencing the climate. Action what you know. Remain flexible and ready to pivot as climate information comes to your attention.
- **Develop a Reporting Culture.** Cultivate a command climate where Marines have confidence to report without fear of reprisal. A firm, fair, and consistent approach to reporting wrongdoing builds confidence in the system that supports Marines. Trivializing *Watch List Behaviors* could lead to a command climate that is more permissive to worse behaviors.
- **When in Doubt, Call in the Experts.** Most of the *Watch List Behaviors* are violations of Marine Corps policies and the Uniform Code of Military Justice. Routinely check in with the Judge Advocate officer or other relevant subject matter experts if you are unsure about how to respond to a potential issue.
- **Do Not Delegate It, Own It.** Make sure everyone understands your goal is to help and protect those in your unit. Hold yourself accountable. If health and welfare are at risk, then you absolutely want to be informed and involved.

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